LSPU SCHOLARHUB: AN AI-ENABLED WEB-BASED SCHOLARSHIP MANAGEMENT AND MONITORING SYSTEM WITH CHATBOT AND MULTI-CHANNEL NOTIFICATIONS

A Capstone Project
Presented to the Faculty
of the College of Computer Studies
Laguna State Polytechnic University

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In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

College of Computer Studies
LAGUNA STATE POLYTECHNIC UNIVERSITY
Sta. Cruz, Laguna
Philippines

September 2026

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APPROVAL SHEET

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ACKNOWLEDGEMENTS

The proponents would like to express their deepest gratitude to all individuals and groups who extended their support and assistance in the completion of this capstone project.

First and foremost, the proponents are sincerely grateful to ALMIGHTY GOD for the strength and wisdom throughout this journey.

LTC MARIO R. BRIONES, RES, PA, Ed. D., The University President, for his visionary leadership and commitment to research excellence;

ENGR. MANUEL LUIS R. ALVAREZ, Campus Director, for his continuous support of student welfare and academic advancement;

MARK P. BERNARDINO, MSCS, Thesis Adviser, for his guidance, patience, and invaluable insights that shaped LSPU ScholarHub;

MIA V. VILLARICA, DIT, Associate Dean and System Expert, for her expertise, technical guidance, constant encouragement and thorough evaluation of the system;

RACHIEL R. RIVANO, MSIT, Technical Editor, for her meticulous review, technical expertise, and valuable contributions to the manuscript;

EDWARD FLORES, MSIT, Statistician, for his guidance on data sampling, statistical analysis, and data integrity;

JHON JHON ZOTOMAYOR, Language Critic, for refining the paper's structure, language, and overall clarity;

To the **LSPU SCC SCHOLARSHIP AND FINANCIAL ASSISTANCE OFFICE** for their cooperation and assistance during system evaluation;

To the **RESPONDENTS AND SCHOLARS** for their time and valuable participation in data collection and evaluation; and to our **FAMILIES, FRIENDS, AND RESEARCH TEAM MEMBERS** for their unwavering support, dedication, and encouragement throughout this journey.

ABSTRACT

The study aims to design and develop an AI-enabled, web-based scholarship management and monitoring system. This platform was created to assist the Scholarship and Financial Assistance Office and the students of Laguna State Polytechnic University - Santa Cruz Campus (LSPU-SCC) by standardizing administrative workflows and improving communication channels. Research findings indicate that current reliance on fragmented, semi-digital methods specifically decentralized spreadsheets and manual record-keeping contributes to data inconsistencies and reporting delays. These administrative inefficiencies underscore the necessity of developing technological solutions to support higher education institutions in maintaining organized and responsive student services. The study presents a centralized scholarship database, automated academic status tracking, and an analytical reporting dashboard. Furthermore, the system incorporates an AI-powered chatbot for student support and a multi-channel notification engine to disseminate critical announcements. This study addresses the institutional technology gap by providing a secure digital infrastructure designed to modernize workflows and clarify student-office relations. By integrating automated digital solutions, the proposed system promotes greater efficiency in processing scholar records, mitigating the risk of human error and data redundancy. The findings highlight the potential of these technologies to enhance operational efficiency for both administrators and student scholars.

Keywords: Scholarship Management System; Artificial Intelligence; Chatbot; Multi-channel Notifications; Web-based Monitoring; Administrative Efficiency; Student Services Automation; Data Integrity

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DEFINITION OF TERMS

To ensure clarity and consistency in the use of language throughout this study, it is important to define key terms and concepts that may have different interpretations or meanings. This section provides a comprehensive list of technical and operational terminologies used in this research, along with their corresponding definitions and explanations.

Technical Terms

The following technical terms describe the major technologies, system components, methods, and software concepts employed or referenced in the development and evaluation of LSPU ScholarHub.

Artificial Intelligence (AI)	Intelligent technologies integrated into LSPU ScholarHub to support document processing, information extraction, academic evaluation, and chatbot-assisted scholarship inquiries.
Optical Character Recognition (OCR)	Technology used to scan and convert text contained in uploaded COR and COG documents into machine-readable information.
Natural Language Processing (NLP)	Technology used to process and interpret extracted text and support information extraction and chatbot-based scholarship assistance.
Information Extraction	The process of identifying and extracting relevant information fields from processed scholarship documents, such as student name, institutional ID number, and GWA.
AI-Powered Verification	A system function that uses OCR- and NLP-assisted processing to extract information from COR and COG documents and compare the extracted information with submitted student credentials.
AI Chatbot	An integrated conversational component that provides automated responses to predefined scholarship-related frequently asked questions and information requests.
Multi-Channel Notification	A system capability that disseminates scholarship-related announcements, updates, and alerts through multiple communication channels, particularly SMS and email.

Relational Database	A structured database used to centrally store and relate scholar profiles, scholarship applications, academic information, documentary requirements, status records, and system transactions.
Role-Based Access Control (RBAC)	An access-control mechanism that assigns system permissions according to user roles and restricts access to authorized functions and information.
Web-Based System	A software platform accessed through compatible web browsers to provide centralized scholarship-management functions through an online interface.
Responsive Web Design	An interface design approach that allows the system interface to adapt to different screen sizes and supported devices.
Document Repository	A digital storage component used to organize and retrieve uploaded scholarship-related documents and associated information.
Analytical Dashboard	An administrative interface that presents consolidated scholarship information and analytics to support monitoring, reporting, and administrative review.
Data Validation	The process of checking submitted information against expected data formats, values, and scholarship-processing requirements.
ISO/IEC 25010	The software quality model used in the study as a basis for evaluating selected quality characteristics of LSPU ScholarHub.

Operational Terms

The following operational terms define the specific meanings of concepts, roles, processes, and system functions as they are applied within the context of the present study.

LSPU ScholarHub	The developed AI-enabled web-based scholarship management and monitoring system for the Scholarship and Financial Assistance Office (SFAO) of Laguna State Polytechnic University–Santa Cruz Campus.
Scholarship Management	The system-supported activities involving scholarship application processing, scholar record management, document handling, status monitoring, notifications, and administrative reporting.
Scholarship Monitoring	The process of tracking scholarship applications, scholar status, academic information, and relevant eligibility conditions through ScholarHub.
Scholar	A student who is receiving or applying for scholarship or financial assistance managed through ScholarHub.
Scholarship Applicant	A student who submits scholarship-related information and required documents through ScholarHub for processing and administrative review.
Scholarship and Financial Assistance Office (SFAO)	The LSPU-SCC office whose scholarship-related processes and administrative requirements serve as the primary operational context of ScholarHub.
Scholarship Application	The electronic submission of a student's scholarship information and documentary requirements through ScholarHub for processing and review.
Scholarship Status	The recorded stage or condition of a student's scholarship application or scholarship record maintained within ScholarHub.
Academic Monitoring	The system-supported monitoring of academic information, particularly grades and GWA, for scholarship-related requirements and preliminary eligibility assessment.
Eligibility Assessment	The preliminary evaluation of academic and submitted information against predefined scholarship criteria within ScholarHub, subject to authorized administrative verification.
Certificate of Registration (COR)	A student document uploaded to ScholarHub as a source of enrollment and student information for document processing and verification.

Certificate of Grades (COG)	A student academic document uploaded to ScholarHub as a source of grade information for information extraction and preliminary academic eligibility assessment.
General Weighted Average (GWA)	Academic performance information extracted or recorded from student academic documents and used in processes related to scholarship eligibility criteria.
Document Verification	The process of checking information extracted from submitted scholarship documents against student information and predefined requirements available in the system.
Administrative Dashboard	The ScholarHub interface provided to authorized SFAO personnel for application processing, record verification, monitoring, analytics, and report generation.
Super User	An authorized system role responsible for user management and overall system monitoring.
Scholar Profile	The collection of student information maintained in ScholarHub and associated with the student's scholarship application or scholar record.
Notification Engine	The ScholarHub component responsible for generating and distributing scholarship-related notifications through configured communication channels, particularly SMS and email.
User Acceptance	The respondents' assessment of ScholarHub after using the system, based on the evaluation criteria and rating scale specified in the study.
Functional Suitability	The extent to which the functions provided by ScholarHub support the scholarship-related tasks of its intended users.
Usability	The extent to which respondents perceive ScholarHub as understandable, learnable, navigable, and manageable when performing assigned tasks.
Reliability	The respondents' assessment of the consistency and dependability of ScholarHub during normal use.

Security and Privacy	The implemented access-control and information-protection mechanisms and the respondents' assessment of their appropriateness for protecting scholarship-related information.
Perceived Ease of Use	The respondents' perception of the degree to which ScholarHub can be learned and used without substantial effort.
Overall Acceptability and Intention to Use	The respondents' overall assessment of ScholarHub and their stated willingness to use the system for appropriate scholarship-related activities.

CHAPTER I

INTRODUCTION AND BACKGROUND

The current development of intelligent information systems demonstrates a significant shift from conventional document management toward automated, data-driven, and AI-assisted information processing. Recent studies have examined the use of artificial intelligence in centralized electronic document management, particularly for improving the organization, classification, retrieval, and processing of large volumes of documents. (Ahmadov & Mahmudov, 2026), for example, examined AI solutions in centralized electronic document management systems, while Uma and Ramya (Uma & Ramya, 2026) investigated AI-powered document classification for automated document management. These studies indicate the growing role of AI in transforming document-intensive processes into more structured and automated workflows.

One of the most established technologies supporting this transformation is Optical Character Recognition (OCR), which converts information in scanned or image-based documents into machine-readable text. Research has demonstrated the continuing development of OCR for document digitization and automated information extraction. (Memon et al., 2020a) conducted a systematic literature review of handwritten OCR, demonstrating the breadth of OCR applications and the continuing technical challenges associated with recognizing textual information from documents. Similarly, (Muthusundari et al., 2024) examined an AI-based OCR system, while Sinha and Rekha (2025) investigated document digitization and information extraction using OCR. These

studies establish OCR as an important technological component for converting traditionally non-machine-readable documents into structured digital information.

However, OCR alone does not necessarily provide sufficient semantic understanding of extracted text. The literature therefore increasingly emphasizes post-OCR processing and Natural Language Processing (NLP) to improve the interpretation and extraction of meaningful information. (Nguyen et al., 2021), in their survey of post-OCR processing approaches, identified post-processing as an important stage in improving the usability of OCR-generated text. (Rakshit et al., 2023a) further examined an OCR pipeline enhanced through NLP-based post-processing, demonstrating the relevance of combining OCR and NLP rather than treating text recognition as an isolated process. (Van Strien et al., 2020) likewise examined the impact of OCR quality on downstream NLP tasks, emphasizing the relationship between the quality of extracted text and the performance of subsequent language-processing applications.

The integration of OCR and NLP has also been applied to information extraction from specialized and unstructured documents. Van Nguyen et al. (2025) (Van Nguyen et al., 2025)investigated an NLP method for identifying and extracting educational legal documents using OCR and XML techniques, illustrating how recognized text can subsequently be processed to identify and organize domain-specific information. (Malashin et al., 2024)applied image-text extraction and NLP to unstructured information contained in medical reports, while (Jamil et al., 2025)investigated automated clustering and summarization of scanned documents using hybrid deep-learning algorithms. These studies demonstrate that the value of intelligent document processing extends beyond

digitization; extracted text can be further analyzed, categorized, and transformed into information that supports downstream information-management activities.

Another relevant direction is the use of AI for document classification, tagging, and retrieval. (Omurca et al., 2022)developed a document-image classification system that combines deep-learning and machine-learning models, demonstrating the application of intelligent classification techniques to document images. (Abbas et al., 2022)examined intelligent document finding using OCR and tagging, highlighting the potential of extracted textual information to improve document retrieval. (Haryan, 2026)further explored an AI-driven document management system incorporating OCR and retrieval-augmented generation. Collectively, these studies suggest that AI can support multiple stages of the document-management lifecycle, from recognition and classification to retrieval and information access.

The literature also indicates that automated document processing can improve the scalability of information management. (Chidananda et al., 2026)investigated an automated data-extraction system using OCR and NLP for scalable big-data processing. (Baviskar et al., 2021) through a systematic literature review of AI-based processing of unstructured documents, examined the broader potential of artificial intelligence for automating document-intensive activities and identified future directions for intelligent document processing. These findings are particularly relevant to institutional environments in which personnel must process recurring documentary requirements and extract selected information from heterogeneous documents.

These demonstrates a progression from OCR-based document digitization to NLP-enabled information extraction, classification, retrieval, and intelligent document management. However, the reviewed studies do not directly address the integrated application of OCR and NLP to university scholarship administration involving COR and COG processing, scholar-information verification, and eligibility assessment. This gap provides the basis for LSPU ScholarHub, which applies these technologies as an assistive mechanism for document processing and information management within a university scholarship system.

1.1 Project Context

The rapid adoption of digital technologies has transformed higher education institutions (HEIs) from predominantly manual and document-centered environments into increasingly interconnected, data-driven, and service-oriented organizations. Digital transformation enables institutions to integrate information, automate administrative processes, improve accessibility to services, and strengthen accountability in institutional operations. In this context, web-based information systems provide a mechanism for consolidating dispersed records, streamlining administrative workflows, and facilitating timely access to reliable information. For offices responsible for student services, such systems can reduce repetitive manual processes, improve record accuracy, and support more responsive delivery of services to students.

Scholarship and financial assistance administration is an important component of student support in higher education because it involves the systematic management of applications, eligibility requirements, academic records, documentary submissions,

monitoring activities, and communication with scholars. The effectiveness of these processes depends on the availability of accurate, complete, and timely information. As the volume of scholar records and supporting documents increases, reliance on spreadsheets, word-processing files, physical documents, and other decentralized record-keeping practices can result in fragmented information, duplication of data, difficulties in verification, and delays in reporting and decision-making. A centralized information system can address these limitations by providing an integrated repository through which authorized personnel can manage scholar records, monitor status changes, verify submitted information, and generate relevant reports.

The emergence of Artificial Intelligence (AI) further provides opportunities to augment administrative processes that involve large volumes of documents and repetitive information-processing tasks. In particular, Optical Character Recognition (OCR) can convert text contained in scanned or image-based documents into machine-readable data, while Natural Language Processing (NLP) techniques can support the identification and extraction of relevant information from the resulting text. When appropriately integrated into an information system, these technologies can assist in extracting data from academic and registration documents and facilitate preliminary verification against information provided by applicants. Such automation does not replace administrative judgment; rather, it can serve as an initial processing and verification mechanism that supports personnel in performing their responsibilities more efficiently and consistently.

At Laguna State Polytechnic University–Santa Cruz Campus (LSPU-SCC), the administration of scholarship and financial assistance programs is a significant institutional function of the Scholarship and Financial Assistance Office (SFAO). The office is

responsible for processes involving scholar records, documentary requirements, academic information, monitoring, reporting, and communication with student scholars. Based on the preliminary needs assessment and process-flow analysis conducted with the SFAO, existing operations are constrained by the absence of a centralized digital infrastructure. The office relies on decentralized spreadsheets, word-processing documents, and physical folders, making the consolidation, retrieval, updating, and verification of scholar information more difficult.

The decentralized nature of these processes also presents challenges in monitoring changes in scholar status. Identifying students who have graduated, shifted programs, or stopped attending requires coordination and validation across relevant university offices. When information is maintained in separate files and repositories, updating records and producing current reports may require considerable manual effort. These conditions increase the possibility of inconsistent records, duplicated information, and delays in administrative reporting and scholarship-related decision-making.

Communication between the SFAO and scholars likewise presents an operational concern. Based on the preliminary needs assessment and interviews with student scholars, announcements are commonly disseminated through social media pages and group messaging channels. Although these platforms provide convenient means of communication, they are not primarily designed for structured institutional notification management and record-keeping. Consequently, important announcements, deadlines, and other scholarship-related information may be difficult for scholars to track, while the office has limited capacity to maintain a centralized and auditable record of notifications. These

circumstances indicate the need for a communication mechanism that is more systematic, accessible, and integrated with scholarship management processes.

In response to these identified requirements, the researchers developed LSPU ScholarHub: An AI-Enabled Web-Based Scholarship Management and Monitoring System with Chatbot and Multi-Channel Notifications. The proposed system is designed as a centralized web-based platform for managing scholar information, monitoring scholarship-related activities, supporting document processing, and facilitating communication between the SFAO and student scholars. Its centralized database is intended to provide an integrated source of scholar information that can be accessed by authorized users according to their assigned roles.

The Scholarship and Financial Assistance Office (SFAO) of LSPU-SCC requires a more integrated approach to scholarship administration, particularly in document processing, information verification, communication, and administrative monitoring. The processing of scholarship-related documents such as Certificates of Registration (COR) and Certifications of Grades (COG) involves extracting and verifying student information, while scholarship announcements and reminders require timely and accessible communication. In addition, monitoring scholar records, statuses, and reports requires organized and readily available administrative information. These operational requirements highlight the need for a centralized system that can support scholarship personnel without replacing their authority in verification and decision-making.

To address these needs, LSPU ScholarHub integrates OCR and NLP-based information extraction to digitize submitted COR and COG documents and identify

relevant information such as student name, identification number, and GWA for preliminary cross-verification. It also incorporates an AI-enabled chatbot and SMS/email notifications to support information dissemination and an administrative dashboard for centralized record management, status monitoring, and standardized reporting. However, the applicability of these integrated technologies within the specific scholarship administration context of LSPU-SCC requires systematic evaluation.

Thus, the research problem is to determine whether LSPU ScholarHub can effectively support scholarship administration through AI-assisted document processing, automated communication, centralized information management, and administrative monitoring. Specifically, the study evaluates the system's performance and user acceptability among its intended users to determine its suitability for supporting the operational requirements of the SFAO at LSPU-SCC.

1.2 Research Objective

The study generally aims to design, develop, test, and evaluate LSPU ScholarHub as an AI-enabled web-based scholarship management and monitoring system for the Scholarship and Financial Assistance Office (SFAO) of Laguna State Polytechnic University–Santa Cruz Campus (LSPU-SCC), Laguna, Philippines, supporting its operational requirements and those of student scholars through integrated document processing, academic monitoring, communication, information management, and administrative functions.

Specifically, the study seeks to achieve the following objectives:

1. Design and develop LSPU ScholarHub, an AI-enabled web-based scholarship management and monitoring system, based on the identified functional and operational requirements of the Scholarship and Financial Assistance Office (SFAO) and student scholars.
2. Evaluate the functional performance of the developed LSPU ScholarHub across its major modules by measuring relevant indicators, including accuracy, response time, reliability, and delivery rate, in terms of user account management, AI-powered document verification, academic monitoring, AI-enabled chatbot, multi-channel notifications, document repository, administrative dashboard, and super user management.
3. Determine the perceived software quality and user acceptance of LSPU ScholarHub after system demonstration and hands-on use, based on the applicable ISO/IEC 25010 quality characteristics in terms of Functional Suitability, Performance Efficiency, Usability, Reliability, Security, and Maintainability.

1.3 Conceptual Framework

The conceptual framework translates the objectives of the study into a systematic Input–Process–Output (IPO) model that guides the design, development, testing, and evaluation of LSPU ScholarHub. The input consists of the identified functional and operational requirements of the SFAO and student scholars, together with the relevant technological and software-quality foundations. These inputs guide the process, which involves system design and development, integration of AI-enabled document processing and communication functions, systematic testing of the major system modules, and system

demonstration followed by hands-on user evaluation. The output is the developed LSPU ScholarHub and the corresponding results of its functional performance and user evaluation based on the applicable ISO/IEC 25010 quality characteristics. Thus, the framework provides a logical structure for determining whether the developed system meets the identified requirements and is suitable for supporting scholarship management and monitoring at LSPU-SCC.

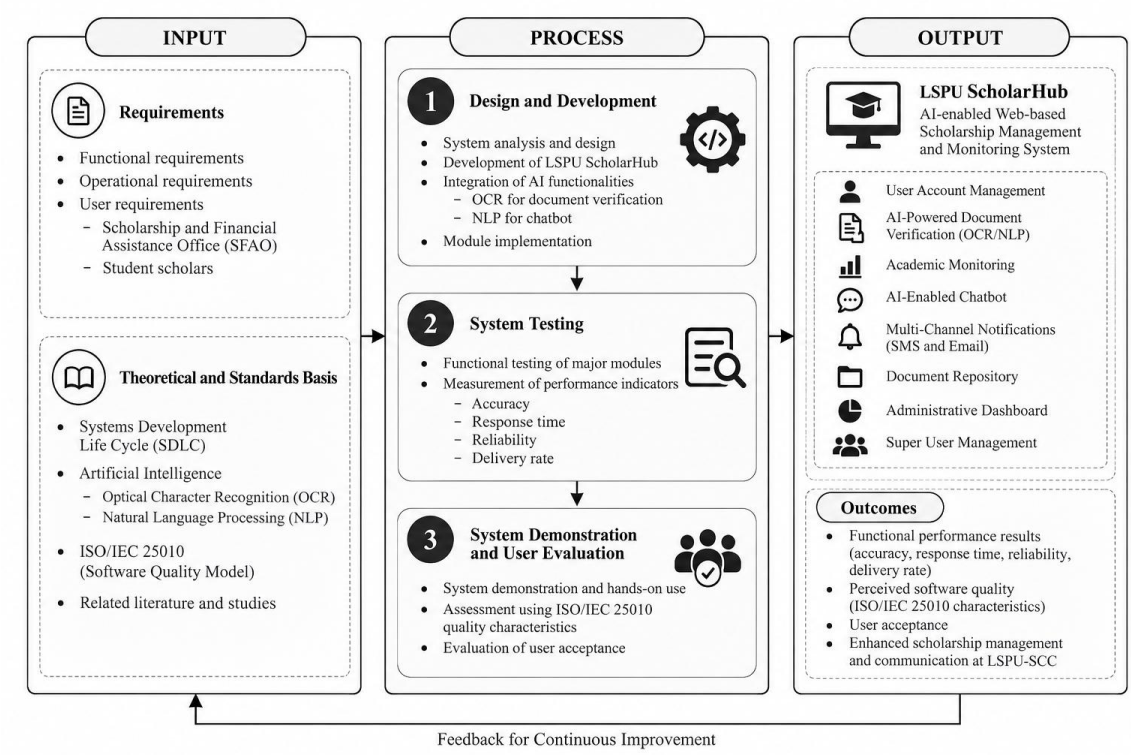


Figure 1. Conceptual Framework of LSPU ScholarHub

The figure was generated using an artificial intelligence (AI) image-generation tool based on the researchers' conceptual framework and specifications.

The figure demonstrates the logical sequence through which the study objectives are addressed. The input comprises the functional and operational requirements of the Scholarship and Financial Assistance Office (SFAO) and student scholars, supported by

the relevant technological and software-quality foundations. These inputs guide the design and development of LSPU ScholarHub, including its AI-powered document verification, academic monitoring, chatbot, notification, document repository, administrative, and user-management functions. The developed system is subsequently subjected to system testing to measure its functional performance using indicators such as accuracy, response time, reliability, and delivery rate.

The framework further incorporates system demonstration and hands-on use as the basis for user evaluation. Following actual interaction with the system, intended users assess its perceived software quality and acceptance using the applicable ISO/IEC 25010 characteristics: Functional Suitability, Performance Efficiency, Usability, Reliability, Security, and Maintainability. The resulting performance and evaluation findings provide evidence of the system's capability and suitability for supporting scholarship management and monitoring at LSPU-SCC and serve as a basis for further system improvement.

1.4 Project Purpose

The proposed project, LSPU ScholarHub, is significant because it addresses the systemic inefficiencies in scholarship administration at Laguna State Polytechnic University - Santa Cruz Campus (LSPU-SCC) by leveraging Artificial Intelligence and automated web-based solutions. Its importance is reflected across several key stakeholder groups:

For Scholarship Office Personnel. The system streamlines workflows through automated data consolidation and rapid report generation. By utilizing OCR and NLP for

document verification, it reduces manual workloads and minimizes human errors in tracking scholar statuses.

For University Administrators. The project provides a centralized repository and analytical dashboard to support evidence-based decision-making. It ensures transparent management of funding programs while maintaining high data integrity and security.

For Student Scholars. The platform offers a responsive support environment through an AI chatbot and multi-channel notifications (SMS/Email). Students gain 24/7 access to eligibility info and real-time updates, ensuring they never miss critical deadlines.

For the University. The project aligns with LSPU's mission by introducing a modern digital infrastructure. It enhances student services, ensures data privacy compliance, and strengthens the relationship between the administration and the student body.

For Future Researchers. This study serves as a foundational reference for integrating AI-powered parsing and automated monitoring in education, offering a model that can be replicated or expanded by other institutions.

By emphasizing these benefits, LSPU ScholarHub demonstrates how technology can modernize institutional workflows and improve the overall experience for both administrators and student scholars.

1.5 Scope and Limitations of the Study

This section delineates the scope, boundaries, and inherent limitations of the study. It establishes the parameters of the investigation by specifying the system components,

processes, target users, and institutional context covered by the research, as well as the factors that may constrain the interpretation and generalizability of its findings.

The study, entitled "LSPU ScholarHub: An AI-Enabled Web-Based Scholarship Management and Monitoring System," encompasses three primary areas: (1) the design and development of the system architecture and core functionalities; (2) the technical testing of the AI-assisted document-processing and multi-channel notification modules; and (3) the evaluation of the system's functional performance, software quality, and user acceptance among selected student scholars and scholarship administrators. The study is institutionally and geographically delimited to the Scholarship and Financial Assistance Office (SFAO) of Laguna State Polytechnic University–Santa Cruz Campus (LSPU-SCC), Laguna, Philippines. Accordingly, the findings are interpreted within the specific operational and technological context of the SFAO and are not intended to represent the scholarship administration practices of other campuses or higher education institutions without further validation.

The scope of the study is defined by the following areas, which are aligned with the research objectives and establish the specific parameters, system components, and evaluation activities covered by the investigation:

- Preliminary investigation and requirements analysis through consultation with the LSPU Scholarship and Financial Assistance Office (SFAO) and student scholars to identify existing operational practices, information requirements, system specification, and challenges in scholarship administration.

- Document and process analysis to examine relevant scholarship-related documents, including Certificates of Registration (COR) and Certificates of Grades (COG), as well as existing procedures for application, document submission, verification, academic monitoring, communication, record management, and reporting. The findings from these activities will serve as the basis for defining the functional requirements and technical specifications of the proposed system.

- Development of a centralized web-based platform with a relational database to support secure scholar-record management, document storage and retrieval, scholarship-status tracking, and administrative information management.

- Integration of Artificial Intelligence (AI) through Optical Character Recognition (OCR) and Natural Language Processing (NLP) to facilitate automated extraction and preliminary cross-verification of relevant student and academic information from COR and COG documents.

- Implementation of an AI-enabled chatbot and multi-channel notification mechanism using Short Message Service (SMS) and email to facilitate timely and accessible dissemination of scholarship-related information, announcements, and reminders.

- Development of an administrative dashboard to support scholarship-program management, record verification, status monitoring, standardized reporting, and presentation of relevant administrative analytics.

- System testing and performance evaluation to assess the functionality of the major system modules using measurable indicators such as data-extraction accuracy, response time, processing success rate, system reliability, and notification delivery rate.

- User evaluation among selected SFAO personnel and student scholars following system demonstration and hands-on use to determine the perceived software quality and user acceptance of the system based on the applicable ISO/IEC 25010 quality characteristics, particularly Functional Suitability, Performance Efficiency, Usability, Reliability, Security, and Maintainability.

While the LSPU ScholarHub provides a technological platform for managing scholarship-related processes and information, the researchers acknowledge the following limitations:

- The AI chatbot is restricted to a predefined knowledge base for FAQs and does not utilize large-scale generative AI models like GPT-4 due to API costs and privacy constraints.

- The system does not feature a direct real-time link to the University Registrar's database; all academic data synchronization relies on student-uploaded documents and manual administrative verification.

- The automated document parsing is limited to standard digital or high-quality scanned copies of LSPU-issued CORs and COGs in PDF or image formats.

- The notification system is subject to the message quotas and service availability of third-party SMS gateways and email service providers.

- The study does not cover production-level hosting or official university-wide server deployment, as these fall under the jurisdiction of the University MIS office.

The system's logic and reporting formats are specifically tailored to the workflows of LSPU-SCC and may not be immediately compatible with other university campus.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter aims to provide a comprehensive overview concerning the strategic advantages of LSPU ScholarHub: An AI-Enabled Web-Based Scholarship Management and Monitoring System with Chatbot and Multi-Channel Notifications. Several studies have been conducted in web-based information systems, artificial intelligence (AI), workflow automation, communication privacy, data security, hybrid intelligence, inclusive system design, notification systems, social presence, student support, system maintainability, software development, and user experience. These studies provide significant insights into the development of the proposed system. Through this selection, the review establishes the concepts and previous findings that support the major features and design considerations of LSPU ScholarHub. Furthermore, it outlines the critical technology gaps identified within the existing body of literature that this project addresses.

2.1 Related Literature

This section reviews peer-reviewed journal articles and scholarly studies gathered from sources such as Google Scholar, Consensus, and other academic databases. The studies were grouped into themes based on their common research focus, technologies, system functions, and key findings to provide a clear view of the existing research, emerging technologies, and gaps relevant to the development of LSPU ScholarHub.

2.1.1 AI in higher education administration

Artificial Intelligence (AI) is increasingly positioned as an enabling technology for the digital transformation of administrative information systems, particularly in environments that require the processing, classification, extraction, retrieval, and management of large volumes of documents. The reviewed literature demonstrates a progression from conventional document digitization toward increasingly intelligent forms of document management. (Ahmadov & Mahmudov, 2026) focus on AI solutions within centralized electronic document management systems, emphasizing the organizational dimension of AI-assisted information management, whereas (Uma & Ramya, 2026) concentrate more specifically on AI-powered document classification for automated document management. Although both studies address the automation of document-intensive workflows, their emphases differ: the former considers AI within a broader centralized management environment, while the latter focuses on the classification function as a mechanism for organizing documents. (Baviskar et al., 2021) provide a broader systematic perspective by examining AI-based automated processing of unstructured documents, extending the discussion beyond classification toward the general problem of converting heterogeneous documents into manageable information. Thus, while (Ahmadov & Mahmudov, 2026) emphasize centralized management, emphasize automated classification, and (Baviskar et al., 2021) emphasize AI-based processing of unstructured information, the three studies converge on the premise that AI can reduce dependence on conventional manual document-handling processes. However, these studies primarily address document-management problems in general rather than the domain-specific requirements of higher education scholarship administration.

A more specific technological foundation for AI-enabled administration is Optical Character Recognition (OCR), which enables information contained in image-based or scanned documents to become machine-processable. The reviewed studies differ in their treatment of OCR. (Memon et al., 2020a)provide a comprehensive systematic review of handwritten OCR, representing a broad examination of recognition challenges and approaches, whereas (Muthusundari et al., 2024)explicitly examine OCR as an AI-based recognition system. In contrast, (Sinha & Rekha, 2025) place OCR within a broader document-digitization and information-extraction pipeline. These studies therefore represent different levels of the same technological process: (Memon et al., 2020a)focus on the recognition problem, (Muthusundari et al., 2024)on the AI-based OCR system, and (Sinha & Rekha, 2025) on the use of OCR as part of information extraction. This distinction is important for higher education administration because simply converting a Certificate of Registration (COR) or Certification of Grades (COG) into text does not, by itself, produce structured administrative information. The OCR literature establishes the feasibility and importance of digitizing document content, but it also points toward the need for subsequent processing when the objective is to identify particular fields or derive information for administrative workflows.

The limitation of OCR as a standalone technology becomes more apparent in the literature on post-OCR processing and Natural Language Processing (NLP).(Nguyen et al., 2021) survey post-OCR processing approaches, indicating that recognition is only one stage in transforming scanned material into usable digital information. (Rakshit et al., 2023a)move more specifically toward the use of NLP for improving OCR outputs, thereby connecting character recognition with linguistic processing. Similarly, (Van Strien et al.,

2020) examine the effect of OCR quality on downstream NLP tasks, highlighting the interdependence between the quality of the initial text-recognition stage and subsequent language-processing applications. Compared with the OCR-oriented studies (Muthusundari et al., 2024) and (Sinha & Rekha, 2025)these studies shift the analytical focus from “Can the document be digitized?” toward “Can the digitized content be reliably processed and understood?” This distinction is particularly relevant to scholarship management, where extracting a student's name, student identification number, GWA, grades, or other administrative fields requires more than character-level recognition. Consequently, the literature suggests a sequential technological relationship in which OCR provides the textual input, while post-OCR processing and NLP contribute to the interpretation and organization of that input.

The application of OCR and NLP to specialized documents further demonstrates the movement from text recognition toward domain-oriented information extraction. (Van Nguyen et al., 2025)examine an OCR- and NLP-based approach for identifying and extracting Vietnamese educational legal documents, making their work particularly relevant to education-related information processing. However, their focus is on legal-document identification rather than individual student records. (Malashin et al., 2024)Investigate image-text extraction and NLP for unstructured medical reports, demonstrating a comparable challenge in which important information must be extracted from documents whose structures may vary. Their medical context differs substantially from university administration, but the underlying document-processing problem is similar: transforming unstructured or image-based content into information that can be computationally processed. (Huyen et al., 2025)likewise examine OCR and NLP in text

summarization and Vietnamese handwriting recognition, broadening the application from extraction to higher-level language processing. In comparison,(Chidananda et al., 2026)frame OCR and NLP more explicitly in terms of automated data extraction and scalable big-data processing. These studies therefore differ in domain and processing objective educational legal documents, medical reports, handwriting and summarization, and large-scale data extraction, respectively but collectively demonstrate that OCR becomes more useful when combined with NLP or other computational techniques capable of interpreting extracted text. Nevertheless, none of these studies directly addresses the particular structure and verification requirements of university scholarship documents such as CORs and COGs.

The literature also demonstrates that AI has expanded beyond extraction toward document classification, retrieval, clustering, and intelligent document management. (Omurca et al., 2022) examine document-image classification through the fusion of deep-learning and machine-learning models, emphasizing the classification of document images. (Abbas et al., 2022) approach the problem from the perspective of intelligent document finding, combining OCR with tagging to improve document retrieval. While both studies concern organization and accessibility of documents, their objectives differ(Omurca et al., 2022)are principally concerned with determining document categories, whereas Abbas et al. (2022) emphasize finding and retrieving relevant documents. (Haryan, 2026) extends this direction by proposing an AI-driven document-management approach that combines OCR with retrieval-augmented generation (RAG), representing a movement toward more intelligent interaction with document collections. Jamil et al. (2025) similarly examine scanned-document clustering and summarization using hybrid deep-learning algorithms,

extending document management from simple storage and classification toward grouping and summarizing document content. (Khan et al., 2024), meanwhile, review AI and machine-learning approaches for OCR in historical-document transcription and transliteration, illustrating how AI can address specialized and difficult document formats. Although these studies differ substantially in their application contexts—from general document image processing and retrieval to scanned-document summarization and historical archive management they collectively demonstrate a progression from recognition and classification to retrieval, semantic organization, and intelligent interaction, supporting the conceptualization of administrative systems in which documents are not merely stored electronically but are transformed into structured, accessible, and actionable institutional information.

The reviewed literature further reveals that AI-based document processing should not be understood as an automatically error-free process. (Memon et al., 2020b) demonstrate the breadth and complexity of OCR research, particularly for handwritten materials, while (Nguyen et al., 2021) and (Rakshit et al., 2023b) highlight the importance of post-OCR processing in dealing with recognition outputs. (Van Strien et al., 2020) connect OCR quality to downstream NLP performance, providing an important distinction from studies that focus primarily on the existence of an OCR capability. (Sinha & Rekha, 2025) further position OCR within an information-extraction pipeline, while (Chidananda et al., 2026) consider the scalability of automated OCR-NLP extraction. These studies differ in their principal concerns recognition quality, post-processing, downstream language processing, extraction, and scalability converge on an important implication: the usefulness of AI-assisted document processing depends not only on whether a document

can be digitized, but also on whether the extracted information is sufficiently accurate and structured for its intended use. For an administrative system, this distinction is critical because inaccurate extraction of student information could affect subsequent verification, monitoring, or eligibility assessment.

When the studies are compared according to their primary technological function, a clear progression can be observed. The earliest layer represented in the reviewed literature concerns OCR and document digitization, as reflected by (Memon et al., 2020b), (Muthusundari et al., 2024), and (Sinha & Rekha, 2025). A second layer concerns post-OCR correction and linguistic processing, represented by (Nguyen et al., 2021), (Rakshit et al., 2023a), and (Van Strien et al., 2020). A third layer concerns information extraction and domain-specific interpretation, represented by (Van Nguyen et al., 2025), (Malashin et al., 2024), (Huyen et al., 2025), and (Chidananda et al., 2026). A fourth layer involves classification, retrieval, clustering, summarization, and intelligent document management, represented by (Omurca et al., 2022), (Abbas et al., 2022), (Haryan, 2026), (Jamil et al., 2025), and (Khan et al., 2024). Finally, (Ahmadov & Mahmudov, 2026) (Uma & Ramya, 2026), and (Baviskar et al., 2021) provide broader perspectives on integrating AI into document-management environments. The studies therefore complement rather than simply duplicate one another: some address the input, some the quality of extracted text, some the interpretation of information, and others the organization and management of resulting information.

At the same time, the studies differ substantially in application domain, which limits direct transfer of their approaches to higher education scholarship administration.

(Van Nguyen et al., 2025) focus on educational legal documents; (Malashin et al., 2024) on medical reports; (Khan et al., 2024) on historical documents; (Huyen et al., 2025) on Vietnamese handwriting and text summarization; and several others on general document-management environments. This diversity is valuable because it demonstrates that AI-based document processing is not restricted to a single document type. However, it also means that the performance or applicability of an approach in one domain cannot automatically be assumed for scholarship records. Educational scholarship documents have their own characteristics, including standardized but institution-specific layouts, student identifiers, academic grades, registration information, and eligibility-related fields. The literature consequently supports the technological feasibility of the individual components, but it does not establish that an existing OCR-NLP approach can automatically perform accurate scholarship-document verification without domain-specific implementation and validation.

Another important contrast concerns the level of automation envisioned by the studies. Some research focuses on a single computational task, such as OCR, classification, or tagging (Abbas et al., 2022; Memon et al., 2020b; Omurca et al., 2022), while other studies combine multiple technologies into broader processing pipelines (Chidananda et al., 2026; Haryan, 2026; Malashin et al., 2024; Rakshit et al., 2023b). (Ahmadov & Mahmudov, 2026) and (Baviskar et al., 2021)are more closely concerned with AI within document-management environments, whereas the OCR- and NLP-oriented studies concentrate on the transformation and interpretation of document content. This distinction is significant for the proposed system because LSPU ScholarHub is not designed merely as an OCR application or a document repository. Rather, OCR and NLP are intended to

function as interconnected components within a larger scholarship-management workflow that includes document submission, student-information extraction, cross-verification, academic monitoring, eligibility assessment, record management, and administrative reporting. The literature therefore supports the individual building blocks, while the proposed study addresses their integration into a specific administrative workflow.

Collectively, the reviewed studies establish that AI-enabled document processing has progressed from digitizing textual content toward extracting, interpreting, classifying, retrieving, and managing information intelligently. The progression is evident across the literature: OCR research establishes document-to-text conversion (Memon et al., 2020a; Muthusundari et al., 2024; Sinha & Rekha, 2025) ; post-OCR and NLP research addresses the quality and interpretation of extracted text (Nguyen et al., 2021; Rakshit et al., 2023b; Van Strien et al., 2020) ; domain-specific studies demonstrate information extraction from heterogeneous documents (Chidananda et al., 2026; Huyen et al., 2025; Malashin et al., 2024; Van Strien et al., 2020); and document-intelligence studies extend these capabilities to classification, retrieval, clustering, summarization, and intelligent management (Abbas et al., 2022; Haryan, 2026; Jamil et al., 2025; Khan et al., 2024; Omurca et al., 2022). At the broader system level, (Ahmadov & Mahmudov, 2026) (Uma & Ramya, 2026), and (Baviskar et al., 2021) demonstrate the relevance of AI to centralized and automated document-management environments. Thus, the literature does not merely establish that AI can process documents; it indicates a broader movement toward integrated, intelligent information-management workflows.

Despite this development, a specific application gap remains. The reviewed literature does not directly establish an integrated AI-enabled system designed specifically for university scholarship administration, particularly one that combines OCR and NLP for COR and COG processing, student-information extraction, cross-verification against registration credentials, academic-grade interpretation, preliminary eligibility assessment, scholarship-status monitoring, and centralized administrative record management. The closest studies provide individual components or analogous applications, but their objectives, document types, and institutional contexts differ from scholarship administration. This gap is important because domain-specific implementation is necessary when extracted information may contribute to institutional verification or administrative decisions. Consequently, the present study, LSPU ScholarHub, extends the established literature by integrating these AI-assisted document-processing capabilities within the specific workflow of the Scholarship and Financial Assistance Office (SFAO) of Laguna State Polytechnic University–Santa Cruz Campus (LSPU-SCC). In this configuration, AI is positioned not as a substitute for authorized personnel but as an assistive decision-support and information-processing technology. Automated extraction and preliminary assessment support administrative efficiency, while final verification, approval, and scholarship-related decisions remain subject to authorized human review.

2.1.2 AI in student services

Artificial Intelligence (AI) is increasingly being examined as a transformative technology for student services in higher education, extending its role beyond teaching and learning toward admissions, advising, communication, administrative support, and student affairs. The literature reviewed in the RIS file demonstrates that AI in student services is

not a single application area but a broad continuum ranging from institution-level administrative transformation to direct student-facing support. Early and foundational reviews established the wider potential of AI across higher education. Some studies examined the implications of AI for higher education primarily from the perspective of teaching and learning, providing an early foundation for understanding how intelligent technologies could alter university functions. Some subsequently provided a systematic review of AI applications in higher education and broadened the discussion to include administrative and institutional applications, while also drawing attention to the limited involvement of educators in the development and implementation of AI. (H. Chen et al., 2007) similarly, reviewed AI in education from a broader technological perspective, encompassing applications, emerging technologies, and associated challenges. In contrast, (Crompton & Burke, 2023) examined the state of the field more broadly and captured the subsequent expansion of AI applications across higher education. These studies are similar in treating AI as a significant technological development for higher education, but they differ in scope: other studies provide an earlier conceptual treatment, and systematically map applications, (H. Chen et al., 2007) emphasize the technological landscape, and (Crompton & Burke, 2023) characterize the evolving state of the field. Collectively, they establish the broader foundation for considering AI not simply as an instructional technology but as a technology capable of supporting multiple institutional and student-service functions.

The literature becomes more directly relevant to student services when AI is considered in relation to academic advising and individualized student support. (Akiba & Fraboni, 2023) specifically examine AI-supported academic advising and the potential of

ChatGPT for student empowerment, representing a student-facing approach in which AI can facilitate interaction and provide guidance. (Kalam & Kaur, 2025) extend this perspective by examining AI applications in admissions, student advising, and research support, thereby connecting student-facing services with other institutional functions. Compared with the broader reviews of (L. Chen et al., 2020), and (Crompton & Burke, 2023), these studies focus more explicitly on services encountered directly by students. However, their emphases remain different. (Akiba & Fraboni, 2023) concentrate on advising and empowerment through conversational AI, whereas (Kalam & Kaur, 2025) present a broader service portfolio that includes admissions and advising. This distinction is important because student services encompass both transactional administrative services and interactive support services. A chatbot may assist students in obtaining information, while an administrative AI system may automate document processing or status verification. Thus, the literature suggests that AI can operate at different points in the student-service journey, from application and admission to advising and continuing academic support.

A complementary perspective is provided by research examining AI specifically within academic and administrative processes. (Ahmad et al., 2022) investigate the academic and administrative roles of AI in education, providing a bridge between educational applications and institutional administration. (Domingo-Alejo, 2024) moves from conceptual discussion toward system development through an AI-integrated administration tool using machine-learning technology for a smart education system. In comparison, (Khairullah et al., 2025) focus on implementing AI in academic and administrative processes through responsible strategic leadership in higher education institutions. These

studies differ in their primary level of analysis. (Ahmad et al., 2022) address AI's roles across academic and administrative activities; (Domingo-Alejo, 2024) emphasizes the development of an AI-integrated administrative tool; and (Khairullah et al., 2025) place greater emphasis on institutional implementation and strategic leadership. Nevertheless, they converge in demonstrating that AI-based student services require more than isolated technical functionality. AI must be embedded within institutional processes, governance structures, and administrative responsibilities. For LSPU ScholarHub, this distinction is particularly relevant because the proposed system is intended not merely to provide an AI chatbot but to integrate AI into the actual scholarship-management workflow, including document submission, verification, monitoring, and communication.

The literature from Latin American and international higher education contexts further demonstrates that AI adoption is shaped by institutional and regional circumstances. (Salas-Pilco & Yang, 2022) systematically review AI applications in Latin American higher education, providing a geographically contextualized perspective that differs from the more globally oriented reviews of (L. Chen et al., 2020), and (Crompton & Burke, 2023). Their regional focus is particularly relevant to institutions operating in contexts where technological infrastructure, institutional capacity, and implementation conditions may differ from those reported in highly digitized educational systems. Similarly, the broader review of (Ocen et al., 2025) examines innovations, opportunities, and challenges associated with AI in higher education institutions. (AlBlooshi, 2026) continues this line of inquiry by reviewing AI opportunities and challenges in higher education. These studies are complementary but differ in purpose: (Salas-Pilco & Yang, 2022) provide a regional systematic review, (Ocen et al., 2025) emphasize institutional innovations and challenges,

and (AlBlooshi, 2026) provide a more recent overview of opportunities and risks. Their collective contribution is important because it prevents AI adoption from being viewed solely as a technical problem. Instead, student-service applications must be considered in relation to institutional readiness, implementation conditions, governance, and potential risks.

Another major theme in the literature concerns the relationship between AI and student affairs and support services. (Barrett et al., 2024) explicitly examine the implications of AI for higher education student affairs, making this study particularly relevant to a project centered on student-facing administrative services. Unlike studies focused principally on academic advising, (Barrett et al., 2024) situate AI within the broader student-affairs environment, which includes services and institutional functions designed to support students beyond classroom instruction. (Rhodes, 2026) develops this discussion further by emphasizing a "human-at-the-helm" approach to AI in student support. The contrast between these studies is significant. (Barrett et al., 2024) focus on the implications of AI for student affairs, whereas (Rhodes, 2026) emphasizes the continued centrality of human involvement in AI-supported student support. This complements the responsible-adoption perspective of (Khairullah et al., 2025) and (Tarisayi, 2023), both of whom emphasize responsible strategic leadership and institutional oversight. Taken together, these studies suggest that effective AI-enabled student services should not be designed around complete automation. Instead, AI should augment institutional personnel by handling repetitive information-processing or communication tasks while allowing authorized staff to retain responsibility for complex cases, verification, exceptions, and decisions.

The importance of responsible AI adoption and institutional governance becomes particularly evident when comparing the implementation-oriented studies. (Tarisayi, 2023) focuses specifically on strategic leadership for responsible AI adoption in higher education, while (Khairullah et al., 2025) connect responsible strategic leadership with the implementation of AI in academic and administrative processes. (Spence et al., 2025) shift the emphasis toward higher education administrators and the transition from viewing AI primarily as a disruption toward recognizing it as an institutional opportunity. Although these studies approach the issue from different perspectives, they converge on the idea that AI adoption is an organizational process rather than simply a software-development activity. (Tarisayi, 2023) emphasizes leadership and responsibility; (Khairullah et al., 2025) emphasize implementation within institutional processes; and (Spence et al., 2025) emphasize administrators' changing perceptions and roles. Their combined implication for student services is that AI systems should have clearly defined purposes, institutional controls, and human oversight. This is particularly applicable to scholarship administration because scholarship eligibility, document validation, and financial assistance decisions may carry significant consequences for students and therefore should not be delegated entirely to automated systems.

The literature also identifies student perception and acceptance as an important dimension of AI-enabled services. (M. Kumar, 2025) specifically examine student perceptions of AI in higher education, providing a perspective that differs from studies written primarily from institutional or technological viewpoints. Most of the reviewed literature examines what AI can do or how institutions can implement it, whereas (D. K.

Kumar et al., 2026) shift attention to how students perceive the technology. This distinction is important for student services because technological capability does not automatically translate into acceptance or effective use. A system may provide automated advising, notifications, or information retrieval, but students must also perceive the system as useful, accessible, understandable, and trustworthy. This student-centered perspective complements (Akiba & Fraboni, 2023) emphasis on student empowerment and (Rhodes, 2026) human-centered approach to student support. Consequently, the literature indicates that evaluation of AI-enabled student services should include not only technical functionality but also the user experience and perceived usefulness of the technology among its intended beneficiaries.

AI has also been associated with higher education quality and institutional competitiveness, providing another dimension to its relevance for student services. Dilmi and Sakri (2024) examine the impact of AI on higher education quality, while Hannan and Liu (2021) frame AI as a potential source of competitiveness in higher education. These studies differ from Akiba and Fraboni (2023) and Barrett et al. (2024), which focus more directly on student-facing support. Dilmi and Sakri (2024) are concerned with the broader quality implications of AI, whereas Hannan and Liu (2021) emphasize its potential contribution to institutional competitiveness. Nevertheless, their perspectives are relevant to student services because service quality and institutional responsiveness can contribute to students' overall experience with the university. AI-enabled services may therefore have significance beyond individual automated functions: they can become components of a broader institutional strategy for improving service delivery, responsiveness, and operational capability. However, these broader studies should not be interpreted as direct

evidence that a specific scholarship-management application will improve student outcomes; rather, they provide a conceptual basis for examining AI as part of institutional service modernization.

The reviewed literature also reveals a clear distinction between AI as a broad institutional capability and AI as a specific student-service application. Broad reviews and conceptual studies including Popenici and Kerr (2017), Zawacki-Richter et al. (2019), Chen et al. (2020), Crompton and Burke (2023), Salas-Pilco and Yang (2022), Ocen et al. (2025), and AlBlooshi (2026)—establish the expanding role of AI across higher education. More specialized studies then demonstrate particular applications: Ahmad et al. (2022) examine academic and administrative functions; Akiba and Fraboni (2023) focus on AI-supported advising; Kalam and Kaur (2025) examine admissions, advising, and research support; Domingo-Alejo (2024) addresses an AI-integrated administration tool; and Barrett et al. (2024) consider student affairs. The remaining studies—Kumar and Raman (2022), Dilmi and Sakri (2024), Hannan and Liu (2021), Tarisayi (2023), Khairullah et al. (2025), Spence et al. (2025), and Rhodes (2026)—add perspectives concerning student perceptions, quality, competitiveness, responsible leadership, administration, institutional opportunity, and human-centered support. Thus, the literature is complementary rather than repetitive: it covers the technological, functional, organizational, student-centered, and governance dimensions of AI adoption.

2.1.3 Digital scholarship management

Digital scholarship management has increasingly evolved from conventional, paper-based, and fragmented administrative processes toward web-based platforms that

centralize scholarship information, automate application procedures, improve access to funding opportunities, and support monitoring and decision-making. The literature in the reviewed RIS file reflects this progression across several dimensions, including web-based scholarship management, online application and information access, centralized scholarship databases, digital administration, post-award monitoring, financial-aid disbursement, decision-support systems, and broader digital transformation of higher education. Importantly, the studies do not all address scholarship management at the same level. Some focus narrowly on application and information dissemination, others on comprehensive scholarship administration, while more recent studies introduce technologies such as blockchain, intelligent decision-support, and smart financial systems. Comparing these studies provides a stronger basis for identifying the technological and functional requirements of an integrated scholarship-management platform such as LSPU ScholarHub.

One of the foundational directions in digital scholarship management is the development of web-based scholarship information systems. Hidayat et al. (2021) examined a web-based scholarship management information system developed using the Rapid Application Development (RAD) model, emphasizing the use of a structured information system to support scholarship administration. Similarly, Al-Ayyubi and Maulana (2023) examined a web-based scholarship information system intended to improve access to scholarship information and accelerate information dissemination. Although both studies use web-based technologies, their emphases differ. Hidayat et al. (2021) focus more directly on scholarship management, whereas Al-Ayyubi and Maulana (2023) emphasize access to information and rapid dissemination. Thus, the former is more

administrative in orientation, while the latter is more student-access oriented. This distinction is important because effective digital scholarship management requires both dimensions: administrators need centralized tools for managing records, while students need timely and accessible information about scholarship opportunities, requirements, and status.

The movement toward online scholarship application systems is further demonstrated by Blancaflor et al. (2022), who presented APPly as a design for an online tertiary-level scholarship application management system. Compared with Hidayat et al. (2021), whose emphasis is on scholarship management information, Blancaflor et al. (2022) concentrate more specifically on the application stage of the scholarship process. This represents an important shift from simply digitizing scholarship information toward digitizing the actual transaction between applicant and scholarship office. Sakhare et al. (2024) extend this student-facing perspective through a web-based approach to scholarship search and application, emphasizing improved access to financial-aid opportunities. In comparison, Al-Ayyubi and Maulana (2023) emphasize information dissemination, Blancaflor et al. (2022) emphasize online application, and Sakhare et al. (2024) combine scholarship discovery and application. These studies therefore illustrate a progressive expansion of functionality from information access → application → broader scholarship search and application. However, their primary emphasis remains on the front-end student experience rather than the complete lifecycle of scholarship administration.

A more comprehensive direction is represented by Remegio and Casquejo (2025), who developed a comprehensive Scholarship and Grants Management System (SchOGMS) for state universities and colleges. Among the reviewed studies, this work is particularly relevant because its institutional setting and comprehensive orientation are closely aligned with the requirements of university scholarship administration. Unlike systems focused primarily on scholarship search,

information dissemination, or application, a comprehensive scholarship and grants platform implies a broader administrative scope. This distinguishes Remegio and Casquejo (2025) from Blancaflor et al. (2022) and Sakhare et al. (2024), whose primary focus is application and access. The distinction also demonstrates an important evolution in digital scholarship systems: digitizing the application process is not equivalent to digitizing the scholarship-management lifecycle. A complete system must potentially connect applications with recipient records, monitoring, documentation, reporting, and post-award activities. This broader perspective is directly relevant to LSPU ScholarHub, which is designed not merely as an application portal but as a scholarship management and monitoring system.

The same progression is evident in S, U., and S, M. (2026), who examined an Educational Trust Management System described as a web-based scholarship and trust administration platform. Compared with Remegio and Casquejo (2025), this study places scholarship administration within the broader management of educational trust-related activities. Its focus therefore differs from systems specifically designed for university scholarship offices. Nevertheless, both studies move beyond simple scholarship-information dissemination toward administrative management. Kumar et al. (2026) similarly examine a Campus Scholarship and Management Portal, representing a campus-level approach to scholarship management. Compared with the broader trust-administration orientation of S. and S. (2026), Kumar et al. (2026) focus more specifically on a campus scholarship portal. Together, these studies reinforce the movement toward centralized web-based scholarship environments, although they differ in institutional scope and functional emphasis. The common direction is the replacement or supplementation of fragmented scholarship processes with centralized digital platforms.

A particularly important development is the use of digital technology for post-award scholarship management and personalized services. Chen and Yang (2025) examine the digital transformation of post-award management and personalized services for scholarship or grant

recipients. This differs from the application-centered studies of Blancaflor et al. (2022) and Sakhare et al. (2024), because the focus moves beyond selecting or admitting beneficiaries toward managing students after an award has been granted. Similarly, Widyastuti and Prasandha (2025) examine strengthening scholarship-data administration through digital services to promote accountability and program benefits. Whereas Chen and Yang (2025) emphasize post-award management and personalized services, Widyastuti and Prasandha (2025) emphasize administrative accountability and program benefits. These studies therefore complement rather than duplicate one another: one focuses on the recipient-service dimension, while the other emphasizes the governance and accountability dimension. For LSPU ScholarHub, both are significant because scholarship management should continue after application and approval through status monitoring, academic monitoring, communication, and institutional reporting.

The literature also shows that digital scholarship management is increasingly connected to digital transformation of higher education administration more broadly. Dacholfany et al. (2021) examine higher education administration applications intended to improve digital-based academic services. Unlike the scholarship-specific studies, their focus is broader and concerns the digitalization of administrative services across higher education. Alexandro and Basrowi (2023) similarly investigate the effectiveness of smart digital organizations and digital technology adoption in educational organizations in Indonesia. Their work shifts the focus from a particular scholarship function toward the organizational conditions associated with digital technology adoption. These studies provide an important contrast with Hidayat et al. (2021), Blancaflor et al. (2022), and Remegio and Casquejo (2025), which are more system-specific. The broader organizational studies suggest that a digital scholarship system should not be viewed as an isolated application; rather, it forms part of the institution's wider digital administrative infrastructure. This supports the integration of scholarship records, institutional reporting, communication, and other university processes within a coordinated information environment.

Another dimension of the literature concerns the financial-management infrastructure underlying scholarship administration. Yang and Zhang (2025) examine the design of a smart financial system for digital campuses in higher education institutions. Although the study is not exclusively focused on scholarships, it is relevant because scholarship and financial-aid management ultimately involves financial information and institutional financial processes. Compared with scholarship-specific platforms such as SchOGMS or Campus Scholarship and Management Portal, Yang and Zhang (2025) adopt a broader financial-system perspective. Their work therefore complements rather than replaces scholarship-management systems: scholarship platforms address the beneficiary and program-management layer, while smart financial systems address the broader financial-management environment. This distinction suggests that future scholarship platforms may benefit from interoperability between scholarship records and institutional financial systems rather than treating scholarship administration as an isolated information process.

The role of digital technology in funding education administration is further addressed by Wen (2025), who examines digital technology as an instrument for transforming funding-education work in colleges and universities. Compared with Yang and Zhang (2025), who focus on smart financial systems, Wen (2025) is more directly concerned with the use of digital technology in educational funding activities. In contrast, Chen and Yang (2025) focus specifically on post-award scholarship or grant management and personalized services. These studies collectively broaden the concept of digital scholarship management from an application database toward a funding-management ecosystem encompassing administration, financial processes, recipient services, and institutional support. This is particularly relevant to a university scholarship office because scholarship administration involves not only identifying eligible students but also maintaining recipient records, monitoring compliance, communicating requirements, and supporting the continued delivery of benefits.

The literature also demonstrates a shift toward intelligent and decentralized technologies for financial-aid and scholarship management. Samu et al. (2024) examine blockchain-based intelligent disbursement in the National Scholarship Portal, introducing blockchain as a technology for supporting scholarship disbursement. Their focus differs substantially from conventional web-based scholarship-management studies. Hidayat et al. (2021), Blancaflor et al. (2022), and Remegio and Casquejo (2025) primarily concern information management and application or administration, whereas Samu et al. (2024) focus on the disbursement process and the potential role of blockchain. Mathintha et al. (2026) similarly examine blockchain use in disbursing financial aid at higher education institutions. The two blockchain-oriented studies are therefore closely related technologically but differ in context: Samu et al. (2024) focus on a national scholarship portal, while Mathintha et al. (2026) consider financial-aid disbursement within higher education institutions more broadly. Their inclusion in the literature indicates that digital scholarship management is beginning to extend beyond centralized databases toward technologies intended to enhance the management and traceability of financial transactions.

A related but distinct direction is the use of decision-support and automated selection mechanisms. Sanmitha and Naik (2025) provide a comprehensive review of decision-support systems and automated platforms for scholarship selection. In contrast to the web-based management studies, which primarily focus on storing, accessing, and processing scholarship information, this work addresses the decision-making stage of scholarship administration. The distinction is important: digital management systems organize information, whereas decision-support systems potentially use that information to assist selection decisions. This is conceptually relevant to LSPU ScholarHub because the proposed system includes preliminary academic eligibility assessment. However, the literature supports an important boundary between automated assistance and final administrative authority. A decision-support function should assist personnel in applying predefined criteria rather than independently determining a student's final entitlement

to financial assistance. Thus, Sanmitha and Naik (2025) provide a conceptual foundation for intelligent scholarship selection, while the proposed system applies a controlled version of this principle through preliminary eligibility assessment subject to authorized review.

Shi and Ji (2021) provide another perspective by examining method design based on a national scholarship system. Unlike the campus- or institution-specific web platforms of Hidayat et al. (2021), Blancaflor et al. (2022), and Kumar et al. (2026), the national scholarship perspective suggests a larger-scale administrative environment. This is significant because scholarship systems must accommodate different scales of operation, from campus-level management to national programs. Similarly, Karna and Swain (2022) examine the effectiveness of student financial-aid programs in India through a policy-evaluative study of a post-matric scholarship scheme. Unlike the technology-centered studies, Karna and Swain (2022) focus primarily on the policy and program-effectiveness dimension rather than system development. Their inclusion provides an important contrast: a technically sophisticated scholarship platform does not itself guarantee that financial-aid programs are effective. Administrative technology should therefore be understood as an enabling mechanism for improving program implementation rather than as a substitute for sound scholarship policy and institutional governance.

The reviewed literature consequently reveals several distinct but interconnected levels of digital scholarship management. At the student-access level, Al-Ayyubi and Maulana (2023) and Sakhare et al. (2024) emphasize scholarship information, search, and access. At the application level, Blancaflor et al. (2022) focus on online scholarship applications. At the management level, Hidayat et al. (2021), Remegio and Casquejo (2025), S. and S. (2026), and Kumar et al. (2026) address web-based scholarship-management environments. At the post-award and accountability level, Chen and Yang (2025) and Widyastuti and Prasandha (2025) emphasize recipient management, personalized services, accountability, and program benefits. At the institutional digital-transformation level, Dacholfany et al. (2021) and Alexandro and Basrowi (2023) situate

digital scholarship services within wider higher education digitalization. At the financial-management level, Yang and Zhang (2025) and Wen (2025) address digital financial and funding systems. Finally, at the advanced technology level, Samu et al. (2024) and Mathintha et al. (2026) explore blockchain-based financial-aid disbursement, while Sanmitha and Naik (2025) examine decision-support and automated scholarship-selection technologies. This progression demonstrates that digital scholarship management is developing from simple online information delivery toward increasingly integrated and intelligent administrative ecosystems.

2.1.4 Scholarship information systems

Scholarship Information Systems (SIS) have increasingly developed from basic electronic repositories and information-dissemination platforms into integrated web-based environments that support scholarship application, applicant tracking, document and record management, monitoring, communication, decision support, analytics, and user-centered service delivery. The literature contained in the reviewed RIS file reflects this development across 20 studies, although the articles differ considerably in purpose, scope, technological sophistication, and evaluation approach. Most studies directly address scholarship or educational-assistance information systems, while several provide complementary evidence on academic monitoring, information-system success, usability, and digital-system design. A comparison of the literature indicates that scholarship information systems are no longer limited to storing applicant records; rather, they increasingly function as institutional platforms connecting students, administrators, information, and scholarship processes.

Development from Basic Scholarship Information Management to Integrated Systems

A fundamental direction in the literature is the use of information systems to centralize and organize scholarship-related information. Al-Ayyubi and Maulana (2023) examined a web-based scholarship information system designed to support access and rapid dissemination of information. Their emphasis is primarily on making scholarship information available to users and improving the speed of information delivery. In contrast, Hidayat et al. (2021) focus more directly on a web-based scholarship management information system, shifting the emphasis from information dissemination toward administrative management. Similarly, Blancaflor et al. (2022) examine APPly, an online tertiary-level scholarship application management system, with greater emphasis on the application process itself. These studies share the use of web technologies but differ in their primary administrative function: Al-Ayyubi and Maulana (2023) emphasize information access, Hidayat et al. (2021) emphasize management, and Blancaflor et al. (2022) emphasize online application.

This progression becomes more evident in Remegio and Casquejo (2025), who examine a comprehensive Scholarship and Grants Management System (SchOGMS) for state universities and colleges. Compared with the more function-specific systems of Al-Ayyubi and Maulana (2023) and Blancaflor et al. (2022), SchOGMS represents a broader conception of scholarship information management because it addresses scholarship and grant administration at an institutional level. Ablen et al. (2025) similarly examine SMART-Scholar as a scholarship management and tracking system, introducing tracking as an important dimension beyond basic application management. Consequently, the literature demonstrates a movement from providing scholarship information → accepting

applications → managing scholarship records → tracking scholarship-related activities. This progression is particularly relevant to LSPU ScholarHub because the proposed system is intended to support not only application-related activities but also continuing scholar monitoring and administrative management.

Scholarship Distribution, Applicant Tracking, and Decision Support

Another important development concerns the use of information systems to support scholarship distribution and applicant decision-making. Orgianus et al. (2024) examine scholarship distribution through a management information system approach. Their emphasis differs from the information-access orientation of Al-Ayyubi and Maulana (2023), because the primary concern is the administrative process of distributing scholarship resources. In contrast, Jacobe and Pascua (2025) examine an applicant tracking and management system incorporating decision support for the Expanded Tertiary Education Equivalency and Accreditation Program at St. Paul University Philippines. Although this study does not represent a conventional scholarship-management system, it is relevant because it demonstrates how applicant tracking can be combined with decision-support functionality in an educational administrative environment.

Sanmitha and Naik (2025) are more explicitly concerned with decision-support and automated platforms for scholarship selection. Their work differs from Orgianus et al. (2024), which emphasizes distribution, and from Jacobe and Pascua (2025), which emphasizes applicant tracking and decision support in a broader educational accreditation context. Taken together, these studies suggest an increasing movement of information systems toward supporting administrative decisions, rather than simply recording transactions. However, this literature also establishes an important distinction between

decision support and autonomous decision-making. For a scholarship system, automated recommendations or preliminary eligibility assessments may assist personnel, but final decisions concerning scholarship qualification and award should remain under authorized institutional review. This distinction is consistent with the design of LSPU ScholarHub, where AI-assisted eligibility assessment is intended to support rather than replace administrative judgment.

Data Administration, Accountability, and Scholarship Monitoring

Scholarship information systems also serve an important role in improving data administration and accountability. Widyastuti and Prasandha (2025) examine the strengthening of scholarship-data administration through digital services, explicitly connecting digital scholarship data management with accountability and program benefits. This differs from Ablen et al. (2025), where the emphasis is on scholarship management and tracking, and from Remegio and Casquejo (2025), where the emphasis is on comprehensive scholarship and grants management. Widyastuti and Prasandha (2025) therefore contribute a stronger governance perspective, suggesting that the value of an information system extends beyond operational convenience to the quality and accountability of institutional scholarship administration.

The importance of monitoring and data visibility is also evident in Del Jolito et al. (2026), who examine a school-based web-access scholarship program integrated with descriptive analytics. Unlike conventional scholarship information systems that primarily organize records, this study introduces analytics as a mechanism for interpreting scholarship-related information. The contrast is important: a management system answers questions such as what records are stored and what transactions have occurred, whereas an

analytics-enabled system can help administrators examine patterns in those records. This provides a conceptual basis for the administrative dashboard proposed for LSPU ScholarHub, where scholarship data can be transformed into reports and institutional indicators rather than remaining as isolated records.

Communication and Transaction Tracking

The literature further demonstrates that scholarship information systems increasingly incorporate communication and transaction-tracking functions. Secugal et al. (2021) examine a QR-code tracking and SMS notification transaction interface for scholarship management. This study is particularly relevant because it combines two functions that are frequently separated in conventional information systems: transaction tracking and communication. Compared with Al-Ayyubi and Maulana (2023), who focus on rapid information dissemination through a web-based system, Secugal et al. (2021) introduce SMS as a direct notification mechanism associated with scholarship transactions.

The contrast between these studies highlights two different approaches to communication. Web-based information dissemination requires users to access a platform to obtain information, whereas SMS notification can proactively deliver important information to users. This distinction is highly relevant to scholarship administration because deadlines, document deficiencies, application updates, and scholarship-status changes may require timely communication. LSPU ScholarHub extends this principle through a multi-channel notification approach using SMS and email, thereby integrating communication into the scholarship-management workflow rather than treating it as a separate activity.

Machine Learning and Intelligent Scholarship Information Systems

A more recent development is the incorporation of machine learning into scholarship information systems. Daluyon and Bilog (2025) specifically examine a Scholarship Management Information System using machine-learning models. This study differs fundamentally from conventional web-based scholarship systems such as Hidayat et al. (2021) and Blancaflor et al. (2022), which are primarily concerned with information management and application processing. Machine learning introduces an analytical dimension in which system capabilities may extend beyond storing and retrieving information toward automated analysis or prediction.

The machine-learning direction complements the decision-support perspective of Sanmitha and Naik (2025). However, the two studies should not be treated as equivalent. Daluyon and Bilog (2025) focus specifically on the incorporation of machine-learning models within a scholarship management information system, while Sanmitha and Naik (2025) address decision-support and automated scholarship-selection platforms more broadly. The former therefore represents a technology-implementation perspective, while the latter represents a decision-support perspective. Together, they suggest that intelligent scholarship systems are evolving toward the use of computational methods to support administrative analysis and selection. For LSPU ScholarHub, this literature provides support for incorporating AI-assisted processing, but the proposed system differs by focusing particularly on document-based extraction and preliminary academic eligibility assessment rather than relying solely on predictive machine-learning models.

User Interface and User-Centered Design

The literature emphasizes that scholarship information systems must not only be functionally capable but also usable and understandable to their intended users. Widaningrum et al. (2025) examine the redesign of a scholarship acceptance system using a prototyping method. Their focus is more specifically on user-interface design than the broader administrative functions examined by Remegio and Casquejo (2025) or Hidayat et al. (2021). Similarly, Pratiwi and Suchahyani (2024) examine the use of Design Thinking and usability testing in designing a scholarship information system. Compared with Widaningrum et al. (2025), whose emphasis is on interface redesign through prototyping, Pratiwi and Suchahyani (2024) place greater emphasis on user-centered design and usability evaluation.

This direction is reinforced by Sinurat and Putra (2025), who examine the usability of educational-assistance information systems using heuristic evaluation and End User Computing Satisfaction (EUCS). Their approach differs from system-development studies because the primary concern is not what the system does but how users experience and evaluate the system. The same principle appears in Tapay et al. (2026), who examine the usability of the PRISM web-based academic monitoring system using heuristic evaluation and a user-centric approach. Although PRISM is an academic monitoring system rather than a scholarship information system, its relevance lies in demonstrating how students interact with web-based institutional information systems.

These studies collectively indicate that scholarship information systems should be designed around actual user needs rather than solely around administrative requirements. The difference among them is methodological: Widaningrum et al. (2025) emphasize prototyping and interface redesign; Pratiwi and Suchahyani (2024) emphasize Design

Thinking and usability testing; Sinurat and Putra (2025) emphasize heuristic evaluation and EUCS; and Tapay et al. (2026) emphasize user-centric heuristic assessment. Their convergence supports the evaluation of LSPU ScholarHub through user acceptability and software-quality dimensions rather than evaluating only whether the technical functions operate.

Centralization and User-Desired Features

The importance of centralized scholarship information is explicitly addressed by Hakim et al. (2024), who investigate user-desirable features of a centralized scholarship information system through a survey of Indonesian scholarship seekers. Unlike system-development studies, this research begins from the users' desired features, thereby providing a demand-side perspective. This contrasts with Remegio and Casquejo (2025), Hidayat et al. (2021), and Ablen et al. (2025), which are more system-development and management oriented.

Similarly, Ronce et al. (2025) examine the development of a scholarship information and consultation center website using a prototyping method. The system emphasizes information and consultation, making it more student-facing than comprehensive administrative systems. When compared with Hakim et al. (2024), Ronce et al. (2025) provides an implementation-oriented perspective, whereas Hakim et al. (2024) focuses on identifying what users want from a centralized scholarship platform. Together, they reinforce the importance of designing scholarship systems that integrate centralized information access with user-oriented support.

Scholarship Applicant Information Management

The importance of information management during the applicant stage is also demonstrated by Quispe-Vivas et al. (2026), who examine a web system designed to optimize information management for Beca 18 selection-stage applicants. Compared with Blancaflor et al. (2022), which focuses on online scholarship application, Quispe-Vivas et al. (2026) emphasize information management during a particular selection stage. This distinction suggests that the scholarship process should not be viewed as a single application transaction. Information must continue to be organized and managed as applicants progress through screening and selection.

This observation complements Jacobe and Pascua (2025), where applicant tracking and decision support are integrated into an educational administrative system. Together, the two studies indicate that scholarship-related information systems increasingly need to support applicants through stages of an administrative process, rather than merely collect initial application data. LSPU ScholarHub similarly adopts a lifecycle-oriented perspective by connecting submission, verification, monitoring, notifications, and administrative reporting.

Empirical Evaluation and Information-System Success

While several studies focus on system development, Casimiro et al. (2026) provide a particularly important evaluation perspective through the empirical evaluation of a scholarship application information system using the DeLone and McLean Information Systems Success Model. This differs from the usability-focused studies of Sinurat and Putra (2025) and Pratiwi and Suchahyani (2024). The DeLone and McLean perspective considers information-system success more broadly, whereas usability evaluations tend to focus on interaction quality and user experience.

The contrast among these evaluation approaches is valuable for the present study. Usability evaluation asks whether the system is understandable and easy to use; user-satisfaction evaluation examines how users perceive their interaction with the system; and information-system success evaluation considers broader dimensions of system, information, and user outcomes. The literature therefore suggests that evaluating a scholarship information system should not rely on a single criterion. For LSPU ScholarHub, this supports evaluating functional suitability, performance, usability, reliability, security, perceived usefulness, perceived ease of use, and overall acceptability.

Complementary Evidence from Related Information Systems

Two articles in the RIS file are not specifically scholarship-management studies but provide useful methodological and conceptual support. Saqib et al. (2026) examine usability guidelines for manuscript management systems. Although manuscript management differs substantially from scholarship administration, both are document-intensive information systems involving structured workflows, user interaction, information submission, tracking, and administrative processing. Thus, the article is relevant primarily to the usability and workflow-design dimension rather than to scholarship functionality.

Similarly, Mehrolia et al. (2021) examine the moderating effects of academic involvement in web-based learning management system success using multigroup analysis. This study is also outside the scholarship-management domain. Its relevance lies in

demonstrating that the success of a web-based educational information system can vary according to user characteristics and levels of involvement. In contrast to scholarship-specific studies, Mehrolia et al. (2021) therefore provide a broader user and system-success perspective. These articles should not be presented as direct evidence of scholarship-system effectiveness, but they can appropriately support the methodological argument that information-system usability and success are influenced by the characteristics and experiences of their users.

2.1.5 Web-based scholarship systems

The increasing adoption of web-based information systems has transformed scholarship administration from predominantly manual and document-centered procedures into more centralized, accessible, and structured digital processes. The literature represented in the reviewed RIS file demonstrates that web-based scholarship systems have been developed for different stages of scholarship administration, including information dissemination, application, record management, monitoring, eligibility assessment, decision support, distribution, and analytics. Al-Ayyubi and Maulana (2023), for example, focused on a web-based scholarship management system intended to improve access to scholarship information and accelerate information dissemination, while Hidayat et al. (2021) examined a web-based scholarship management information system developed using the Rapid Application Development (RAD) model. Similarly, Blancaflor et al. (2022) proposed APPly, an online tertiary-level scholarship application management system, indicating a shift from simply publishing scholarship information toward supporting the actual application process through an online platform. These studies are comparable in their emphasis on web accessibility and process digitalization, but they

differ in their primary operational focus: Al-Ayyubi and Maulana emphasized access and dissemination, Hidayat et al. emphasized web-based management and rapid development, and Blancaflor et al. concentrated on online application management.

The reviewed literature further shows that web-based scholarship systems increasingly address end-to-end scholarship administration rather than isolated application activities. Remegio and Casquejo (2025) described the development of a comprehensive Scholarship and Grants Management System for state universities and colleges, while Ablen et al. (2025) presented SMART-Scholar as a scholarship management and tracking system. These approaches contrast with the narrower application-oriented orientation of APPly because they imply broader management and monitoring functions. Kumar et al. (2026) similarly addressed a campus scholarship and management portal, while the Educational Trust Management System examined by S. and S. (2026) extended the web-based approach toward scholarship and trust administration. In contrast, Najan (2025) specifically focused on scholarship administration for staff, suggesting a more administrative or personnel-oriented perspective. Taken together, these studies demonstrate a progression from online scholarship access and application toward centralized administration, tracking, and institutional management. However, the metadata available in the reviewed RIS file does not provide sufficient empirical detail to establish that one of these systems is demonstrably more effective than the others; rather, their titles and study orientations indicate differences in functional scope and intended users.

A second important direction concerns the use of web-based systems for scholarship monitoring and post-application information management. Prijayanti et al. (2020) explicitly addressed a web-based monitoring information system for scholarship

holders, thereby extending the application-oriented function of earlier systems toward monitoring recipients after selection. Dizon et al. (2025) further incorporated data analytics into a scholarship monitoring system, indicating an evolution from simple record keeping toward the use of accumulated scholarship data for analytical purposes. Del Jolito et al. (2026) similarly described a school-based web-access scholarship program integrated with descriptive analytics. These studies share an emphasis on monitoring and information utilization but differ in analytical maturity: Prijayanti et al. focused primarily on web-based monitoring, whereas Dizon et al. and Del Jolito et al. explicitly incorporated analytics. The distinction is relevant to the present capstone because scholarship management requires not only recording applications but also maintaining updated scholar information and providing administrators with information that can support monitoring and reporting.

Web-based scholarship systems have also been applied to improving information access and organizing applicant records. Sakhare et al. (2024) investigated a web-based approach for enhancing student access to financial aid through scholarship search and application, placing greater emphasis on the student's ability to locate and apply for financial assistance. Quispe-Vivas et al. (2026), in contrast, focused on optimizing information management for Beca 18 applicants during the selection stage. Marave (2019) addressed an online scholarship application and record management system, combining application activities with digital record management. J. (2024) likewise examined a scholarship eligibility and application portal. Although these studies share the common objective of using web technologies to organize scholarship-related information, their functional emphasis varies. Sakhare et al. concentrated on access and search, Marave combined application and records, Quispe-Vivas et al. emphasized selection-stage

information management, and J. focused specifically on eligibility and application. This progression suggests that web-based scholarship platforms can support increasingly specific administrative requirements across the scholarship lifecycle rather than serving only as online information repositories.

Another development in the literature is the incorporation of decision-support and intelligent techniques into web-based scholarship management. Adrian et al. (2025) specifically described a web-based scholarship monitoring and eligibility prediction system using machine learning for the Bureau of KESRA of North Sumatra Province. This differs substantially from conventional web-based management systems because the system's stated scope includes prediction of candidate eligibility. Alfonso et al. (2024), through Mediskolar, combined web-based scholarship management with profile analysis using the Analytical Hierarchy Process (AHP) and a decision-tree algorithm. Tasrif et al. (2021) similarly investigated a website-based scholarship management application in the context of AHP and decision-support systems. Orgianus et al. (2024) approached the problem from the perspective of optimizing scholarship distribution through a management information system. These studies are similar in their movement beyond basic data management toward analytical or decision-oriented functionality, but they differ in the computational approach and administrative stage addressed: Tasrif et al. and Alfonso et al. emphasize AHP and decision support, Adrian et al. introduce machine-learning-based eligibility prediction, while Orgianus et al. focus on scholarship distribution. This body of literature provides a foundation for incorporating intelligent assistance into scholarship administration, although the reviewed records do not establish that these techniques eliminate the need for human administrative validation.

The literature also identifies process redesign as an important dimension of scholarship digitalization. Chávez Trejo and Álvarez Xochihua (2022) examined the reengineering of the scholarship process in a higher education institution using a process analysis and design methodology. Unlike studies that primarily describe the development of a web application, this work places emphasis on the underlying administrative process that the information system is intended to support. This perspective complements the systems-oriented studies of Remegio and Casquejo (2025), Hidayat et al. (2021), and Blancaflor et al. (2022), because a web-based platform can only improve scholarship administration when its digital functions correspond appropriately to institutional procedures. In this sense, process reengineering provides an organizational perspective, whereas web-based scholarship management studies provide a technological perspective. The combination of both perspectives is important for the development of an institutional scholarship platform because merely converting manual forms into electronic forms does not necessarily address fragmented records, monitoring requirements, verification procedures, or reporting needs.

A comparison of the studies also reveals differences in the scope of scholarship lifecycle coverage. Some systems are primarily application-oriented, such as APPly (Blancaflor et al., 2022), the scholarship eligibility and application portal (J., 2024), and the online scholarship application and record management system of Marave (2019). Others concentrate on monitoring, such as Prijayanti et al. (2020) and Dizon et al. (2025). Still others extend into broader administration and management, including Remegio and Casquejo (2025), Ablen et al. (2025), Kumar et al. (2026), and S. and S. (2026). The intelligent systems of Adrian et al. (2025), Alfonso et al. (2024), and Tasrif et al. (2021)

add analytical or decision-support capabilities, while Orgianus et al. (2024) addresses scholarship distribution. Consequently, the literature does not present web-based scholarship systems as a single homogeneous category. Rather, it illustrates a continuum that progresses from information access and online application to digital record management, monitoring, analytics, eligibility assessment and decision support, distribution, and broader administrative functions. This distinction is important because it demonstrates that the effectiveness of a web-based scholarship system should be evaluated not merely by whether it provides online access, but by the breadth, integration, and functional depth of the services it delivers across the scholarship management lifecycle.

Despite these developments, the reviewed literature indicates a remaining opportunity for a more integrated institutional scholarship management and monitoring platform. The studies collectively address many individual components—online application, information access, record management, monitoring, analytics, eligibility assessment, decision support, and administration—but the reviewed titles do not demonstrate a single system that integrates all of these functions with AI-assisted document processing, scholar-record verification, preliminary eligibility assessment, chatbot-based assistance, multi-channel SMS and email notifications, centralized scholar-status monitoring, and an administrative dashboard for institutional reporting. The proposed LSPU ScholarHub addresses this integration gap by bringing these functions into a unified web-based environment designed specifically for scholarship administration at the institutional level. Its contribution is therefore not simply the conversion of a manual scholarship process into an online system, but the integration of management, monitoring,

intelligent document processing, communication, and administrative analytics into a single scholarship information environment.

2.1.6 Scholarship application automation

The automation of scholarship application processes has increasingly become an important application of information technology in educational administration. The literature in the reviewed RIS file shows a progression from digitizing scholarship applications and administrative records toward automated qualification assessment, decision support, recommendation, and AI-assisted scholarship matching. Early work by Marave (2019) focused on an online scholarship application and record management system, establishing the foundational role of web-based technologies in reducing dependence on manual application and record-handling procedures. Hidayat et al. (2021) similarly examined a web-based scholarship management information system developed using the Rapid Application Development (RAD) model, while Blancaflor et al. (2022) proposed APPly as an online tertiary-level scholarship application management system. Al-Ayyubi and Maulana (2023) further emphasized accessibility and rapid dissemination of scholarship information through a web-based information system. Compared with these systems, which primarily address digitization, accessibility, and online management, Berido et al. (2023) explicitly advanced toward scholarship automation by developing a system for student qualification using regression analysis. Thus, the literature demonstrates an important transition from simply placing scholarship processes online to using computational techniques to support qualification-related activities.

A significant strand of the literature focuses on automating the administrative workflow surrounding scholarship applications. Marave (2019) combined online

application with record management, whereas Hidayat et al. (2021) approached scholarship administration through a web-based management information system. Blancaflor et al. (2022) concentrated more specifically on the online application component, while Chávez Trejo and Álvarez Xochihua (2022) examined scholarship process reengineering in a higher education institution using process analysis and design methodologies. These studies differ in their level of technological intervention. The first three primarily represent the digitalization of existing application and management activities, whereas Chávez Trejo and Álvarez Xochihua place greater emphasis on redesigning the underlying administrative process. This distinction is important because automation is not limited to replacing paper forms with electronic forms; effective automation also requires structured workflows and clearly defined processing rules. The literature therefore supports the view that the development of an automated scholarship system should consider both the technological platform and the institutional process that the platform is intended to automate.

The literature further demonstrates the increasing use of decision-support methods for scholarship selection. Sanmitha and Naik (2025) reviewed decision-support systems and automated platforms for scholarship selection, providing a broader perspective on how computational approaches can assist scholarship decisions. Tiara et al. (2026) investigated a web-based scholarship acceptance decision-support system using the Simple Additive Weighting (SAW) method, while Ummi and Fira (2024) implemented SAW in a web-based scholarship acceptance recommendation application. These studies are similar because they use explicit computational criteria to support scholarship selection; however, they differ in implementation and context. The work of Ummi and Fira is framed as a recommendation application for a specific case, whereas Tiara et al. extend the same

general decision-support principle into a scholarship acceptance system. Sanmitha and Naik, in contrast, provide a broader review of decision-support and automated scholarship platforms rather than focusing on one particular implementation. Collectively, these studies indicate that scholarship automation can move beyond data storage toward systematic evaluation of applicants based on predefined criteria.

Another group of studies extends automation through algorithmic candidate selection and eligibility assessment. Ognegina et al. (2022) developed an algorithm for automating the formation and processing of applications for increased academic scholarships based on achievements in research activities. This differs from the web-based application systems of Marave (2019) and Blancaflor et al. (2022), because the principal contribution is algorithmic automation of application formation and processing rather than merely providing an online submission environment. Yusupova et al. (2026) similarly addressed the development of a platform for automating the selection of candidates for scholarship programs. Berido et al. (2023) used regression analysis for student qualification, while J. (2024) examined a scholarship eligibility and application portal. These studies collectively demonstrate an increasing emphasis on eligibility determination and candidate selection, although they employ different levels of automation. Conventional portals facilitate eligibility-related application activities, algorithmic systems automate portions of processing, and decision-support approaches provide computational assistance for ranking or recommending candidates.

Machine learning represents a further development in scholarship application automation. Nikumbh et al. (2024) developed Scholarspot, a scholarship recommendation system using machine learning, shifting the emphasis from processing submitted

applications toward recommending scholarships based on applicant-related information. Depuru et al. (2025) proposed SmartAid, an AI-driven system for merit-based scholarship forecasting, extending computational assistance from recommendation toward forecasting. Kumar (2025), through Scholar Nest & Scholar Ship Agent, focused on an AI-powered scholarship eligibility assessment system, emphasizing automated evaluation of eligibility. A 2025 study titled A Comprehensive Approach to Personalized Scholarship Matching through Machine Learning likewise focused on personalized scholarship matching using machine-learning techniques. These studies differ from the rule- or criteria-based decision-support approaches of Ummi and Fira (2024) and Tiara et al. (2026). SAW-based systems rely on predefined weighted criteria, whereas machine-learning-oriented systems are positioned toward recommendation, prediction, matching, or eligibility assessment. This progression indicates that automation is increasingly being conceptualized not only as workflow automation but also as intelligent assistance in scholarship-related decisions.

The reviewed studies also highlight the importance of student information and academic achievement data as inputs to automated scholarship processing. Markushin et al. (2020) investigated a system for recording student achievement using a new information technology stack. Although the study is not presented specifically as a scholarship application platform, its focus on structured recording of student achievements is directly relevant to automated scholarship qualification because academic and achievement information frequently forms part of eligibility criteria. Ognegina et al. (2022) similarly connected application processing with research achievements, demonstrating how structured achievement data can be incorporated into automated scholarship workflows. Berido et al. (2023) advanced this concept by applying regression analysis to student

qualification. These studies contrast with Al-Ayyubi and Maulana (2023), whose emphasis is primarily on scholarship information access and dissemination. In other words, information-oriented systems facilitate access to opportunities, whereas achievement-oriented and algorithmic systems provide a basis for automated assessment of applicants.

Automation is also evident in the broader applicant-processing workflow. Khorev et al. (2022) examined the use of information technologies to automate work processes involving applicants. Although the study is not limited to scholarship applications based on the title metadata, its applicant-processing orientation provides a broader technological context for scholarship application automation. Najan (2025), meanwhile, addressed scholarship administration for staff, reflecting the administrative side of scholarship processing rather than the applicant-facing component. Kumar et al. (2026) proposed a campus scholarship and management portal, while the web-based scholarship management system of Hidayat et al. (2021) provides an earlier example of institutional scholarship information management. Compared with applicant-facing recommendation and eligibility systems, these works emphasize the institutional administrative environment in which applications are received, processed, managed, and monitored. This distinction suggests that a comprehensive scholarship automation platform must accommodate both applicant-facing processes and administrative workflows.

2.1.7 Scholarship eligibility automation

Scholarship eligibility systems have increasingly evolved from conventional information repositories into computational platforms capable of supporting the assessment, classification, recommendation, and monitoring of scholarship candidates. The literature in the reviewed RIS file demonstrates three major technological directions: rule- and criteria-based decision support, algorithmic eligibility assessment and

recommendation, and web-based monitoring of scholarship recipients. Sanmitha and Naik (2025) provide a broad review of decision-support systems and automated scholarship-selection platforms, establishing the increasing role of information technology in scholarship selection. In contrast, Arifitama (2022) specifically applied the Simple Additive Weighting (SAW) method to scholarship selection, while Triwibowo (2021) employed the Analytical Hierarchy Process (AHP) to determine scholarship recipients. These studies share the objective of making selection more systematic through computational criteria, but they differ in their decision models: SAW emphasizes weighted aggregation of criteria, whereas AHP structures and prioritizes criteria through hierarchical decision analysis. Thus, the literature establishes that scholarship eligibility can be operationalized through explicit criteria and computational decision-support mechanisms rather than relying exclusively on manual assessment.

The use of multi-criteria decision-making is further demonstrated by Turahman (2024), who developed a decision-support system for achievement scholarship recipients using AHP, and Anggrawan et al. (2022), who combined AHP and MOORA for scholarship-recipient recommendation. These studies are closely related to Triwibowo (2021), but they differ in methodological complexity. Triwibowo focuses on AHP as the principal method for determining recipients, Turahman applies AHP specifically to achievement-based scholarships in vocational high schools, and Anggrawan et al. combine AHP with MOORA to support recommendation. Wicaksono et al. (2020), meanwhile, implemented SAW in a web-based decision-support system intended to facilitate scholarship acceptance. Compared with the AHP-oriented studies, Wicaksono et al. emphasize the integration of the decision-support method into a web environment. Collectively, these studies suggest that scholarship eligibility systems can improve the consistency and structure of candidate evaluation by translating institutional eligibility requirements into explicit computational criteria. However, the approaches remain primarily criteria-driven, meaning that the quality of the resulting assessment depends on the criteria and weights established within the system.

A more advanced direction involves machine-learning-based eligibility assessment and prediction. Raj and O (2020) examined a machine-learning-based assessment system for scholarships and credits, while Hegde (2022) proposed a machine-learning-based

scholarship and credit pre-assessment system. Both studies move beyond fixed decision rules toward computational assessment based on machine-learning techniques. Berido et al. (2023) further developed a scholarship automation system for student qualification using regression analysis. Adrian et al. (2025) extended this direction through a web-based system for monitoring and predicting scholarship-candidate eligibility using machine learning. These studies contrast with SAW- and AHP-based systems because machine-learning approaches are positioned toward prediction or automated assessment, whereas multi-criteria decision-making methods rely on explicitly defined criteria and decision weights. Nevertheless, the studies share the objective of reducing the dependence on purely manual evaluation by using computational models to assist eligibility-related decisions.

The distinction between decision support and automated eligibility assessment is particularly important in understanding the development of scholarship information systems. Arifitama (2022), Triwibowo (2021), Turahman (2024), Anggrawan et al. (2022), and Wicaksono et al. (2020) primarily represent decision-support approaches in which algorithms assist users in ranking, recommending, or determining scholarship candidates according to established criteria. In contrast, Raj and O (2020), Hegde (2022), Berido et al. (2023), and Adrian et al. (2025) represent a movement toward predictive or machine-learning-oriented assessment. The former approach offers transparent, criteria-based decision processes, while the latter offers opportunities for automated prediction and classification. Sanmitha and Naik (2025) connect these approaches at the review level by examining decision-support systems and automated scholarship-selection platforms as a broader technological category. This comparison indicates that scholarship eligibility technology has progressed from deterministic evaluation models toward more intelligent computational assessment, although the RIS metadata does not provide sufficient evidence to conclude that machine-learning approaches consistently outperform conventional decision-support methods.

Another important dimension of scholarship eligibility is continuous monitoring after selection. Prijayanti et al. (2020) developed a web-based monitoring information system for scholarship holders, while Ibrahim (2021) examined a scholarship monitoring information system at Ar-Raniry Islamic University. These studies differ from the selection-oriented systems of Arifitama (2022), Triwibowo (2021), and Anggrawan et al.

(2022) because their primary emphasis is not on selecting candidates but on managing information concerning scholarship holders. Dizon et al. (2025) extended monitoring functionality by incorporating data analytics into a scholarship monitoring system, while Del Jolito et al. (2026) described a school-based web-access scholarship program integrated with descriptive analytics. Therefore, the literature indicates an important distinction between eligibility assessment at the point of selection and eligibility/status monitoring throughout the scholarship lifecycle. A scholarship recipient who initially satisfies eligibility requirements may subsequently require monitoring based on academic performance, enrollment status, or other institutional requirements; consequently, a comprehensive system should not treat eligibility as a one-time decision.

The development of web-based eligibility and management environments further connects assessment with accessibility and institutional administration. Hidayat et al. (2021) examined a web-based scholarship management information system using the RAD model, providing a platform-oriented foundation for scholarship administration. Kumar (2025), through Scholar Nest & Scholar Ship Agent, specifically focused on an AI-powered scholarship eligibility assessment system for education equity. Paulraj et al. (2026) developed an AI-powered college and scholarship navigator, which broadens the technological scope from eligibility assessment toward assisting users in navigating scholarship opportunities. These systems contrast with the more narrowly focused decision-support systems because their stated objectives encompass broader interaction with scholarship information and eligibility-related services. The progression is therefore from computational selection, toward web-based eligibility management, and subsequently toward AI-enabled scholarship assistance and navigation.

The literature also provides evidence for the use of rule-based systems as an alternative to purely statistical or machine-learning approaches. Engin et al. (2014) examined rule-based expert systems for supporting university students. Although the study is broader than scholarship eligibility alone, its rule-based orientation is relevant to scholarship systems in which institutional requirements can be represented through explicit conditions. A rule-based approach differs from the statistical and machine-learning approaches of Raj and O (2020), Hegde (2022), Berido et al. (2023), and Adrian et al. (2025), because eligibility decisions can be expressed through predefined logical

conditions rather than inferred from learned patterns. This is particularly relevant to institutional scholarship administration where eligibility requirements may be formally established by scholarship policies. At the same time, the literature suggests that rule-based and machine-learning approaches need not be treated as mutually exclusive. A scholarship platform can use explicit institutional rules for preliminary eligibility checking while reserving more advanced analytical models for recommendation, prediction, or prioritization.

Integrated Comparison of Eligibility Approaches

The 20 studies in the RIS file can therefore be organized into four interconnected categories. First, criteria-based decision-support systems include Triwibowo (2021), Arifitama (2022), Wicaksono et al. (2020), Turahman (2024), and Anggrawan et al. (2022), which use AHP, SAW, and MOORA to structure scholarship selection. Second, machine-learning and predictive systems include Raj and O (2020), Hegde (2022), Berido et al. (2023), Adrian et al. (2025), and Kumar (2025), which move toward automated assessment, prediction, or AI-assisted eligibility. Third, web-based monitoring systems include Prijayanti et al. (2020), Ibrahim (2021), Dizon et al. (2025), and Del Jolito et al. (2026), which address the continuing management of scholarship recipients and scholarship-related information. Fourth, broader intelligent or automated scholarship environments include Sanmitha and Naik (2025), Engin et al. (2014), Paulraj et al. (2026), and Hidayat et al. (2021), which provide conceptual, rule-based, AI-enabled, or platform-level perspectives. The categories overlap, but they differ in where automation is introduced—selection, prediction, monitoring, or user assistance.

The reviewed literature demonstrates a clear progression from manual or conventional scholarship selection toward computational eligibility assessment and continuous monitoring. The earliest and more conventional decision-support approaches

operationalize scholarship criteria through AHP, SAW, and related multi-criteria methods. Later studies incorporate machine learning for qualification, pre-assessment, prediction, and eligibility evaluation. Parallel developments in web-based scholarship monitoring demonstrate that eligibility management extends beyond the initial selection stage and requires continuing access to updated recipient information and performance data. More recent studies further introduce AI-powered eligibility assessment and scholarship navigation, suggesting a transition toward intelligent scholarship platforms.

Despite these advances, the reviewed studies generally emphasize specific stages or functions of eligibility management rather than an integrated institutional workflow. Decision-support studies concentrate on ranking and selection; machine-learning studies emphasize prediction or assessment; monitoring systems focus on recipient records and status; and AI-oriented systems address eligibility or scholarship navigation. The literature represented in the RIS file does not clearly demonstrate a single institutional system integrating document-based eligibility verification, automated information extraction, predefined eligibility rules, preliminary eligibility assessment, continuous scholar-status monitoring, communication support, and administrative analytics.

This gap is directly relevant to the proposed LSPU ScholarHub. The system extends existing scholarship eligibility research by combining rule-based preliminary eligibility assessment with AI-assisted document processing through OCR and NLP. Submitted COR and COG documents can serve as structured sources of academic and registration information, which can then be cross-verified and evaluated against predefined scholarship criteria. Importantly, the proposed system should treat the AI-generated eligibility result as preliminary decision support rather than an autonomous final scholarship decision,

preserving the role of authorized scholarship personnel in validation and approval. This approach combines the transparency of criteria-based eligibility systems with the efficiency of AI-assisted information processing and the lifecycle perspective of web-based monitoring systems.

2.1.8 Student academic monitoring

Student academic monitoring is an important component of scholarship administration because maintaining scholarship eligibility requires institutions to track not only the initial selection of recipients but also their academic performance and continuing compliance with program requirements. The literature in the reviewed RIS file demonstrates an evolution from web-based monitoring and record management toward analytical, decision-support, and intelligent approaches for evaluating student performance and scholarship eligibility. Prijayanti et al. (2020) developed a web-based monitoring information system for scholarship holders, while Ibrahim (2021) examined a scholarship monitoring information system at Ar-Raniry Islamic University. Both studies emphasize the use of information systems to support the monitoring of scholarship recipients, but they differ in institutional context. Wardhana and Hasanah (2017), in contrast, expanded monitoring through a web, Android, and SMS Gateway application for Bidikmisi scholarship recipients, introducing multi-platform access and communication capabilities. These earlier systems primarily establish the technological infrastructure for maintaining and accessing recipient information, whereas more recent studies increasingly incorporate analytical and decision-support functions into academic monitoring.

A related body of literature focuses specifically on recording and managing student academic achievements, which provides the information foundation necessary for

scholarship monitoring. Popov (2021) examined the development of a system for recording student achievements together with an administration module, while Rahmawati et al. (2021) implemented the Waterfall model in developing a student achievement monitoring information system. These studies differ from Prijayanti et al. (2020) and Ibrahim (2021), whose systems are explicitly centered on scholarship holders, because their primary focus is broader student academic performance rather than scholarship administration itself. Nevertheless, they are closely related to scholarship monitoring because academic achievement is often an important basis for determining whether recipients continue to satisfy scholarship conditions. Thus, scholarship monitoring systems and academic-performance monitoring systems represent complementary approaches: the former provides the scholarship-management context, while the latter provides mechanisms for capturing and monitoring the academic information required for continued eligibility.

The literature subsequently demonstrates a shift from descriptive monitoring toward analytical classification of scholarship eligibility. Siswanto and Normawati (2023) developed a monitoring and evaluation classification system for UAD scholarship recipients using the Naïve Bayes algorithm. Unlike conventional web-based monitoring systems, this approach introduces machine learning into the process of classifying scholarship-recipient status. Berido et al. (2023) similarly developed a scholarship automation system for student qualification using regression analysis. These studies are comparable because both introduce computational models into scholarship-related assessment, but they differ in their analytical objectives. Siswanto and Normawati focus on classification within monitoring and evaluation, whereas Berido et al. focus on qualification through regression analysis. In contrast, Prijayanti et al. (2020) and Ibrahim

(2021) primarily represent information-management approaches without the explicitly stated predictive or classification emphasis found in the later studies. This progression indicates that student academic monitoring is increasingly being connected with automated assessment rather than being limited to the storage and retrieval of academic records.

Decision-support systems provide another important approach to academic monitoring. Putri et al. (2021) examined a decision-support system for assessing the performance of KIP Kuliah students using TOPSIS and PROMETHEE, while Adek et al. (2022) developed a decision-support system for academic-achievement scholarship recipients using the Preference Selection Index (PSI). Triwibowo (2021) likewise applied the Analytical Hierarchy Process (AHP) to determine scholarship recipients. More recently, Tiara et al. (2026) analyzed a web-based scholarship acceptance decision-support system using Simple Additive Weighting (SAW). These studies share the use of computational criteria to support scholarship-related decisions, but their methodologies differ considerably. TOPSIS and PROMETHEE provide multi-method approaches for evaluating student performance, PSI provides an alternative preference-ranking mechanism, AHP structures and prioritizes decision criteria, and SAW performs weighted aggregation of criteria. Compared with machine-learning approaches such as Naïve Bayes and regression, these decision-support methods are more explicitly criteria-driven. This makes them particularly applicable when academic eligibility requirements are formally defined by institutional scholarship policies.

The reviewed literature also shows increasing attention to data analytics as a mechanism for understanding scholarship and academic performance information. (2025) developed a scholarship monitoring system with data analytics, while Maniyan et al. (2024)

examined a data-mining-based decision-support system for educational decision makers that extracts rules intended to enhance academic efficiency. These studies differ from conventional monitoring systems because the accumulated student information is not treated solely as administrative records; instead, data are processed to generate analytical information or decision-support knowledge. Dizon et al. specifically situate analytics within scholarship monitoring, whereas Maniyan et al. address the broader educational decision-making environment. The latter therefore provides a more general analytical foundation, while the former demonstrates the direct application of analytics to scholarship administration. Together, they suggest that academic monitoring systems can evolve from simply reporting student status toward identifying patterns and generating information useful for institutional decisions.

More recent research has introduced advanced computational approaches for scholarship selection and academic-performance assessment. Feriyanto et al. (2026) examined a Fuzzy Tahani-based decision-support system for selecting academic-achievement scholarship recipients. Compared with the conventional AHP, SAW, PSI, and other deterministic decision-support methods, fuzzy logic provides a mechanism for representing decision conditions that may not always be naturally expressed as strict binary values. Hadi et al. (2024) combined blockchain and AHP in selecting achievement-based students, adding a security-oriented technology to a multi-criteria decision approach. Prawira et al. (2025), meanwhile, used TOPSIS to optimize selection of PIP scholarship candidates. These studies differ in both computational technique and technological emphasis: Feriyanto et al. focus on fuzzy decision support, Hadi et al. combine blockchain with AHP, and Prawira et al. employ TOPSIS. Nevertheless, they share the broader

objective of making scholarship-related decisions more systematic through formalized evaluation criteria. Unlike these selection-oriented studies, however, Prijayanti et al. (2020), Wardhana and Hasanah (2017), and Dizon et al. (2025) emphasize recipient monitoring, demonstrating that selection and post-selection monitoring are distinct but interconnected components of scholarship administration.

The literature further highlights the importance of integrating academic information with scholarship eligibility monitoring. Quispe-Vivas et al. (2026) examined a web system for optimizing information management among Beca 18 applicants during the selection stage. Qushary and Salam (2021) developed an information system for achievement scholarship recipients, while Thorat and Kale (2026) investigated a scholarship discovery and mentorship system. These studies broaden the scope of scholarship information management beyond academic performance alone. Quispe-Vivas et al. focus on applicant information during selection, Qushary and Salam focus on achievement scholarship recipients, and Thorat and Kale extend scholarship technology toward discovery and mentorship. In comparison, academic monitoring systems such as those of Rahmawati et al. (2021) and Dizon et al. (2025) are more directly concerned with continuing performance and monitoring. This comparison demonstrates that scholarship information systems increasingly span the entire student-support lifecycle—from scholarship discovery and application, through selection, to academic monitoring and continuing support.

2.1.9 Intelligent Document Management

The increasing volume of digital and electronic documents has created a need for intelligent document management systems capable of organizing, processing, retrieving,

and monitoring information with less dependence on manual intervention. The literature included in the present review demonstrates a continuing transition from conventional electronic document management toward artificial intelligence (AI)-enabled approaches. Across the twenty studies, researchers explored different combinations of machine learning, natural language processing (NLP), optical character recognition (OCR), computer vision, ontology, information retrieval, automated classification, metadata extraction, and workflow automation. Although the studies differ in their application areas and technical approaches, they generally share the objective of improving document accessibility, reducing manual processing, and increasing the efficiency of document-related activities (Ahmadov & Mahmudov, 2026),; Belov, 2022; Sambetbayeva et al., 2022).

One major direction in intelligent document management is the use of AI for automating conventional document management functions. Belov (2022) examined the application of AI technologies to automate functions within electronic document management systems, particularly processes that could be performed automatically or with minimal human intervention. Similarly, (2021) discussed the potential of AI and machine learning for automating information analysis, processing, transmission, and electronic document workflow. ent and monitoring environment.

2.1.10 Automated Document Processing

The increasing volume and diversity of digital and paper-based documents have created a need for automated document processing technologies that can reduce manual work while improving the speed, consistency, and accuracy of information handling. The literature reviewed in this theme demonstrates that artificial intelligence (AI) has progressively transformed document

processing from conventional digitization and optical character recognition (OCR) into more comprehensive intelligent document processing (IDP). Across the twenty studies, researchers have examined different combinations of machine learning (ML), natural language processing (NLP), OCR, computer vision, document classification, information extraction, validation, business rules, and workflow automation. Although the studies differ in their technical approaches and application environments, they commonly seek to transform unstructured or semi-structured documents into usable information that can support organizational processes.

One of the foundational perspectives on automated document processing is provided by (Baviskar et al., 2021), who conducted a systematic literature review on the automated processing of unstructured documents using artificial intelligence. Their work establishes the broader problem of unstructured information and emphasizes the limitations of traditional information-extraction techniques compared with AI-based approaches. In contrast, provide a more technical overview through their discussion of Document AI, examining tasks such as document layout analysis, visual information extraction, document image classification, and document visual question answering. While (Baviskar et al., 2021), focus on the broader literature and future directions of automated processing, concentrate more specifically on the models, tasks, and benchmarks underlying Document AI. These studies therefore complement one another: the former establishes the organizational need for automated processing, while the latter explains the technical research foundations that make such processing possible.

Cutting and Cutting-Decelle (2021) approach automated document processing from a different perspective by emphasizing real-world implementation rather than solely computer science research. Their study focuses particularly on the practical challenges of document classification, information extraction, and integration into business processes, especially for forms and invoices. This differs from Cui et al. (2021), whose discussion is oriented toward Document AI models and benchmark tasks. Cutting and Cutting-Decelle (2021) also highlight the practical

difficulty of achieving highly reliable extraction from complex documents, particularly when information such as tables must be accurately interpreted. Thus, while Cui et al. (2021) demonstrate the technological progression of Document AI, Cutting and Cutting-Decelle (2021) emphasize the gap between technological capability and the reliability required in actual organizational environments.

Kapula (2025) further develops this practical perspective by describing intelligent document processing as a combination of OCR, machine learning, NLP, and business rules for extracting, classifying, and validating information from complex and unstructured documents. Compared with Cutting and Cutting-Decelle (2021), Kapula places greater emphasis on IDP as an automation framework that extends beyond OCR and traditional robotic process automation. The similarity between the two studies is their recognition that classification and extraction alone are insufficient; the extracted information must ultimately support organizational processes. However, Kapula (2025) presents IDP as a more mature automation approach, whereas Cutting and Cutting-Decelle (2021) emphasize the challenges that still exist when implementing these technologies in real-world settings.

The transition toward AI-enabled automation of complete document-management processes is also evident in Ahmadov and Mahmudov (2026). Their study examines AI solutions in centralized electronic document management systems and identifies applications across several stages, including digitization, text recognition, summarization, data extraction, classification, intelligent search, and business-process optimization. Unlike studies that concentrate on one document-processing task, Ahmadov and Mahmudov examine AI as a set of technologies that can be applied throughout the document-management life cycle. Their work therefore provides a broader framework for understanding automated document processing, while studies such as Cui et al. (2021) and Uma and Ramya (2026) focus more strongly on specific technical tasks such as document understanding and classification.

Uma and Ramya (2026) specifically focus on AI-powered document classification for automated document management systems. Their study represents a more specialized approach than Ahmadov and Mahmudov (2026), because classification is treated as a central intelligent function. Document classification is important because automatically determining the category of a document can serve as a gateway to subsequent processing, routing, extraction, and storage. In comparison, Cui et al. (2021) treat document image classification as one of several Document AI tasks, whereas Uma and Ramya (2026) place classification more directly within the context of an automated document-management system. This distinction demonstrates how a technical AI capability can be transformed into an operational document-management function.

The literature also demonstrates that automated processing increasingly involves information extraction, rather than merely recognizing or categorizing documents. Reddy and Ramya (2026) propose an AI-based smart document-management system with dynamic field extraction. Their approach addresses the limitations of traditional systems that depend on manual entry and rigid predefined fields. The study is particularly relevant because it supports processing across different document categories without requiring every document to follow a fixed structure. This contrasts with approaches based on predefined templates or narrowly defined document formats. In comparison, the work of Cui et al. (2021) provides a broader technical foundation for visual information extraction, while Reddy and Ramya (2026) apply the concept more directly to a practical document-management system.

Ferreira et al. (2025), through the Di2Win Document Intelligence Platform, demonstrate a more comprehensive approach to automated document processing. The platform uses a modular pipeline that includes image-quality verification, document classification, information extraction, rules-based validation, and process automation. Its architecture is particularly significant because it connects AI-based document understanding with downstream business actions. Unlike systems that stop after extracting information, the Di2Win approach incorporates workflow routing,

databases, APIs, and human-review queues. The study therefore represents a highly integrated model of IDP in which document processing is treated as part of an entire information workflow.

The Di2Win approach can be compared with Kapula (2025), who similarly describes IDP as a combination of OCR, ML, NLP, and business rules. The difference is that Ferreira et al. (2025) demonstrate a more concrete modular architecture in which individual processing functions are connected into a complete pipeline. Cutting and Cutting-Decelle (2021), on the other hand, emphasize the challenges of integrating extracted information into business processes. Taken together, these studies suggest an evolution from identifying the need for integration, to defining IDP as an integrated automation approach, and finally to implementing modular architectures capable of connecting document input to business actions.

The role of enterprise-oriented automation is demonstrated by Balakayeva et al. (2025), who examine an intelligent enterprise system for processing and verifying business documents using big data and AI. Compared with general Document AI research, their work places stronger emphasis on enterprise documents and verification. This is important because automated document processing in organizational environments must not only extract information but also determine whether information satisfies expected requirements. The study therefore complements Ferreira et al. (2025), whose Di2Win platform incorporates rules-based validation, while differing from Cui et al. (2021), which is more focused on the underlying Document AI research landscape.

Khaparde (2025) provides another enterprise-oriented perspective through intelligent business document processing using SAP AI Business Services. The study focuses on business documents such as invoices and shipment notifications and presents machine learning as a mechanism for automating and optimizing document processing. Unlike the more general frameworks presented by Baviskar et al. (2021) and Cui et al. (2021), Khaparde (2025) considers automated processing within an established enterprise technology environment. This demonstrates

that automated document processing can be implemented not only as an independent AI application but also as part of broader enterprise platforms. The study reports that manual business-document processing represents a substantial cost burden, supporting the argument that automation can produce organizational efficiency benefits.

Kolandaisamy et al. (2024) similarly examine smart document processing using artificial intelligence. Compared with the enterprise-specific orientation of Khaparde (2025), the study provides a broader smart-document-processing perspective. Both studies emphasize the use of AI to reduce manual document-processing requirements, but they differ in implementation context. Khaparde (2025) focuses on SAP AI Business Services and business documents, whereas Kolandaisamy et al. (2024) address smart document processing more generally. These studies reinforce the view that AI-based processing is becoming a general-purpose approach that can be adapted to different organizational environments.

The application of automated document processing to specialized domains is demonstrated by Pingili (2025) in healthcare and insurance. The study considers the automation of claims, medical records, and regulatory-compliance documents through ML, NLP, and robotic process automation. Its emphasis differs from general-purpose Document AI because healthcare and insurance documents involve high volumes of information and specialized administrative workflows. The study consequently illustrates how automated processing can be adapted to domain-specific requirements. In comparison, Khaparde (2025) focuses primarily on business documents, while Pingili (2025) focuses on healthcare and insurance processes. Both studies, however, identify automation as a mechanism for reducing administrative burden and improving processing efficiency.

Pingili (2025) also examines AI-driven intelligent document processing in government and public administration. This study extends the domain-specific perspective from healthcare and

insurance to public-sector environments where agencies handle tax filings, legal documents, public records, and benefit-related documentation. The study identifies automated classification, extraction, and validation as major applications of IDP and discusses the use of ML, NLP, and RPA. Compared with the healthcare and insurance study, the government-focused study places greater emphasis on public-service efficiency and regulatory requirements. Nevertheless, both studies demonstrate that automated document processing can be especially valuable in document-intensive environments where manual processing creates delays and increases the risk of errors.

The importance of document verification and compliance is further reflected in Chode (2025), whose work focuses on next-generation document intelligence involving smart metadata, secure access, and regulatory compliance. Unlike studies primarily concerned with extraction, Chode emphasizes the management and governance of information after it has been processed. This represents an important distinction because automated document processing does not end when information is extracted. Organizations must also ensure that documents are appropriately classified, securely accessed, searchable, and managed according to regulatory requirements. Chode's focus therefore complements Balakayeva et al. (2025), who emphasize document processing and verification, and Pingili (2025), who emphasizes compliance within government and public administration.

Serugendo et al. (2024) provide a particularly relevant example of combining automated processing with domain-specific knowledge and administrative workflows. Their study presents an AI-powered intelligent document management system for tax and administrative documents. The system integrates AI-related document-processing functions with domain knowledge and document-management activities. Compared with the general-purpose approaches of Cui et al. (2021) and Baviskar et al. (2021), Serugendo et al. (2024) demonstrate how intelligent document processing can be adapted to a specific institutional environment. Their work is also closely related

to Pingili (2025), although the former provides a more focused tax and administrative application while the latter examines government and public administration more broadly.

Silberlust et al. (2026) demonstrate another specialized implementation through an AI solution for processing faxed documents in healthcare. Their study is particularly important because it shows that automation can be applied even to legacy document formats rather than only digitally generated documents. This differs from enterprise platforms such as SAP-oriented processing and from modern document pipelines designed primarily for digital document inputs. The study therefore highlights the practical need to automate information coming from heterogeneous and sometimes outdated sources. Its healthcare setting also demonstrates how automated processing can be integrated into operational information systems.

An important distinction in the literature is the degree to which human intervention remains necessary. Ferreira et al. (2025) explicitly include human-review queues in their document-processing pipeline, while Silberlust et al. (2026) similarly illustrate an operational setting where automated processing can coexist with human oversight. Cutting and Cutting-Decelle (2021) indirectly reinforce this issue by discussing the difficulty of achieving sufficiently reliable extraction for real-world business integration. These approaches contrast with the ideal of completely automated document processing. The literature instead suggests that a practical IDP system should automate predictable and repetitive activities while providing mechanisms for human verification when the AI system is uncertain or when the information has important consequences.

The literature also demonstrates an evolution from fixed processing approaches toward adaptive and flexible processing. Traditional document systems commonly depend on predefined formats and manually specified fields. Reddy and Ramya (2026) address this limitation through dynamic field extraction, allowing information to be extracted from diverse document categories.

Ferreira et al. (2025) similarly use modular AI components that can be replaced or updated without changing the entire downstream application. Cui et al. (2021) show the broader research progression from rule-based and statistical approaches toward deep-learning and pretrained models. Therefore, the combined literature indicates that flexibility and adaptability have become increasingly important characteristics of modern automated document-processing systems.

Haryan (2026) further represents this progression through the use of OCR and Retrieval-Augmented Generation in the Neural Docs system. Whereas conventional OCR primarily converts document images into machine-readable text, the combination with retrieval-based generative techniques moves the system toward contextual interaction with document content. This represents a significant contrast with earlier OCR-oriented approaches because the objective is no longer limited to digitization or extraction. Instead, the processed information can potentially support more advanced retrieval and question-answering functions. In this respect, Haryan (2026) extends the trajectory identified by Cui et al. (2021), from document reading and understanding toward interactive document intelligence.

Avhad et al. (2025) provide another broad perspective by examining intelligent document processing in relation to summarization, named entity recognition, language conversion, and plagiarism detection. Their study differs from the more workflow-oriented research of Ferreira et al. (2025) and the extraction-focused work of Reddy and Ramya (2026). Rather than concentrating primarily on transforming documents into structured business data, Avhad et al. demonstrate how automated document processing can support higher-level language and analytical functions. This indicates that the scope of automated processing is expanding from basic extraction and classification toward document interpretation and content analysis.

The literature can therefore be viewed as developing through several interconnected levels. The first level involves digitization and recognition, represented by OCR and document-image

processing. The second involves classification and extraction, where AI identifies document types and retrieves relevant fields. The third involves validation and verification, where extracted information is checked against business rules or requirements. The fourth involves workflow automation, where processed documents trigger organizational actions. Finally, newer approaches introduce semantic understanding and interaction, including retrieval-augmented generation and advanced document intelligence. Ahmadov and Mahmudov (2026) explicitly demonstrate this broad progression by considering digitization, text recognition, summarization, extraction, classification, intelligent search, and business-process optimization as connected AI applications within electronic document management.

Despite these advances, the reviewed literature also reveals several gaps. First, many studies concentrate on a particular component of automated document processing. Cui et al. (2021) focus on Document AI tasks and models; Uma and Ramya (2026) emphasize classification; Reddy and Ramya (2026) emphasize dynamic field extraction; and Haryan (2026) emphasizes OCR combined with retrieval-augmented generation. These studies demonstrate strong individual capabilities, but they do not necessarily address the complete operational requirements of a scholarship-management environment.

Second, several studies are designed around specific industries. Pingili (2025) addresses healthcare and insurance, another Pingili study addresses government and public administration, Serugendo et al. (2024) address tax and administrative documents, and Khaparde (2025) addresses business documents within SAP. These studies demonstrate that automation must be adapted to organizational context. However, the literature provides less emphasis on scholarship administration, where documents such as identification records, academic records, contracts, compliance documents, and application requirements must be collected, organized, validated, monitored, and retrieved within a single institutional workflow.

Third, some studies emphasize sophisticated AI processing but provide less emphasis on the complete user-facing management environment. Automated document processing is most useful when the extracted information is connected to application records, user roles, document status, monitoring, notifications, reporting, and administrative decision-making. Ferreira et al. (2025) provide a strong example of this integration by connecting document processing to databases, APIs, business-process management, RPA, and human-review queues. This suggests that future institutional systems should not treat AI processing as an isolated feature but as one component of an integrated information-management workflow.

These findings are directly relevant to LSPU ScholarHub, which applies automated document processing within a scholarship-management and monitoring environment. The literature supports the use of AI for classifying and processing scholarship-related documents, extracting relevant information, validating required documents, organizing records, and assisting personnel in retrieving information. The work of Baviskar et al. (2021), Cui et al. (2021), and Cutting and Cutting-Decelle (2021) provides the foundation for understanding automated document processing; Kapula (2025), Ahmadov and Mahmudov (2026), and Ferreira et al. (2025) support integrated IDP workflows; Reddy and Ramya (2026) support flexible extraction; while Pingili (2025), Serugendo et al. (2024), Balakayeva et al. (2025), Chode (2025), and Silberlust et al. (2026) demonstrate the importance of applying automated processing to real organizational requirements.

For LSPU ScholarHub, however, the objective is not to replace scholarship personnel's decision-making. Instead, automated document processing can reduce repetitive activities and help staff manage large numbers of application and compliance documents more efficiently. Similar to the human-in-the-loop approaches reflected in the literature, documents or extracted information requiring verification can remain subject to staff review. This is particularly appropriate for scholarship administration because final decisions concerning eligibility, approval, compliance, and scholarship status may require institutional judgment.

The reviewed literature establishes that automated document processing has evolved from basic document digitization into a comprehensive AI-enabled process involving recognition, classification, extraction, validation, verification, retrieval, and workflow automation. Earlier studies established the challenges associated with unstructured documents and the technical foundations of Document AI, while more recent studies demonstrate integrated pipelines, enterprise platforms, domain-specific applications, dynamic extraction, multimodal processing, and retrieval-augmented approaches. However, the literature remains fragmented across technologies and application domains. The present capstone addresses this gap by applying automated document-processing principles within an integrated scholarship management and monitoring system. Through LSPU ScholarHub, document processing is positioned not as an isolated AI function but as part of a broader institutional workflow that supports scholarship applications, document management, monitoring,(2023) further examined an AI-enhanced methodology for automating global document management system operations. These studies are similar in their emphasis on reducing repetitive human work; however, they differ in their level of implementation. Belov (2022) focuses broadly on the automation of existing document-management functions, Muminova and Asatov (2021) emphasize the prospects and methodological foundations of automation, while Singh and Gildhiyal (2023) focus on an AI-enhanced operational methodology. Thus, the literature establishes automation as one of the fundamental purposes of intelligent document management.

The importance of machine learning in intelligent document management is demonstrated by Sambetbayeva et al. (2022), who proposed a model based on machine learning, topic modeling, text analysis, and multi-agent technologies. Their work approaches document management from a broader system perspective, particularly document retrieval and the document life cycle. In contrast, Uma and Ramya (2026)

specifically focus on AI-powered document classification for automated document management systems. The difference between these studies illustrates two levels of intelligent management: Sambetbayeva et al. (2022) consider intelligence across document-management processes, whereas Uma and Ramya (2026) concentrate on classification as a particular intelligent function. Ismail et al. (2025) similarly discuss the broader role of AI in information management, emphasizing improvements in information retrieval, classification, indexing, categorization, and knowledge accessibility. Taken together, these studies indicate that machine learning is not limited to identifying document categories but can also support broader information-management and retrieval activities.

Another significant area of intelligent document management is document classification and information extraction. Bhatt (2022) proposed an AI-based approach for extracting information from common documents such as invoices, receipts, bills, identification cards, and passports and converting the extracted information into structured data. Pandey et al. (2023) expanded this concept through an integrated intelligent document management system that combines AI, machine learning, NLP, and OCR. Their study compared EasyOCR with a hybrid approach combining NLP, regular expressions, computer vision, and OCR for extracting information from different document types. In contrast to Bhatt's more general end-to-end document extraction approach, Pandey et al. (2023) provide a more integrated system architecture and compare alternative extraction approaches. These studies therefore demonstrate a progression from basic automated extraction toward integrated AI-based document-processing systems.

The use of OCR is further emphasized by Gyawali et al. (2025), who investigated document workflow automation using ResNet-50 and template-based OCR. Their

approach is more specialized than the general extraction approaches of Bhatt (2022) and Pandey et al. (2023), focusing on document recognition and workflow automation. Haryan (2026), through the proposed Neural Docs system, combines OCR with Retrieval-Augmented Generation (RAG), moving beyond simple text extraction toward interactive document understanding and retrieval. Similarly, Pronenko (2026) examines computer vision and NLP technologies for intelligent document processing and considers multimodal architectures capable of processing document text, images, and spatial information. Compared with earlier OCR-oriented studies, these newer approaches demonstrate an evolution toward multimodal and knowledge-oriented document intelligence.

Dynamic information extraction represents another development in the literature. Reddy and Ramya (2026) proposed an AI-based smart document management system with dynamic field extraction designed to handle different document categories without depending exclusively on predefined templates. This approach contrasts with the template-based OCR strategy discussed by Gyawali et al. (2025). Template-based approaches can be effective when document structures are predictable, while dynamic field extraction is intended to accommodate more diverse document formats. Therefore, these studies demonstrate an important trade-off between specialized structured processing and flexible document processing. For a document management system intended for environments where document formats can vary, dynamic extraction can provide greater flexibility than approaches that depend heavily on fixed templates.

Intelligent document retrieval is another recurring concern among the reviewed studies. Alothman and Sait (2022) developed an artificial intelligence-based ontological framework for managing and retrieving bilingual Arabic and English documents. Their

approach combines data extraction, entity-relationship classification, and ranking to improve retrieval according to user queries. Their work differs from classification-focused systems because its primary concern is not simply determining what type of document is being processed, but understanding relationships among document information and retrieving relevant content. Wang (2021), through an intelligent file management system based on AI, similarly addresses the organization and management of digital files. Ahmadov and Mahmudov (2026) extend this direction by examining AI applications across several stages of electronic document management, including digitization, text recognition, summarization, data extraction, classification, intelligent search, and business-process optimization. Collectively, these studies demonstrate that intelligent document management increasingly involves both document organization and semantic retrieval.

The study by Alothman and Sait (2022) is particularly distinct because it emphasizes ontology and multilingual document retrieval. Whereas machine-learning-based classification systems primarily learn patterns from documents, an ontology-based approach attempts to represent relationships and meaning among information. This provides a semantic dimension that can improve retrieval beyond simple keyword matching. Ismail et al. (2025) similarly identify AI's contribution to improving information retrieval and knowledge accessibility. Haryan (2026), however, represents a more recent direction by combining OCR with Retrieval-Augmented Generation, potentially allowing users to interact with information contained within documents. The contrast among these approaches shows a progression from conventional retrieval, to semantic and ontology-based retrieval, and toward conversational or generative document interaction.

Cloud-based intelligent document processing is represented by Sukrutha et al. (2023), whose study examines intelligent document processing using Amazon Web Services (AWS). The study considers document layout analysis, document image classification, document understanding, OCR, categorization, data extraction, NLP, deep learning, and robotic process automation. Unlike studies that primarily propose algorithms or conceptual models, the AWS-oriented study emphasizes the role of cloud computing platforms in implementing intelligent document-processing capabilities. This provides an important distinction because intelligent document management is not only a question of selecting AI algorithms but also of determining how those technologies can be deployed and integrated into usable systems.

The literature also shows that intelligent document management can be designed for specific organizational domains. Serugendo et al. (2024) developed an AI-powered intelligent document management system for tax and administrative documents. Their system integrates ontology creation, information extraction, document classification, RDF-based data mapping, and reasoning. Unlike general-purpose document management studies, this work demonstrates how AI can be integrated into a domain-specific workflow. Similarly, Silberlust et al. (2026) implemented an intelligent document-processing solution for faxed documents in a healthcare environment. Their approach combined document management, robotic process automation, classification, information extraction, and electronic health record integration. The healthcare application is therefore more operational and workflow-oriented than general document-management models. These studies demonstrate that intelligent document management must account for the

requirements, terminology, document types, and workflows of the organization where the system is deployed.

The healthcare study by Silberlust et al. (2026) also provides an important contrast regarding human intervention. Rather than attempting to automate every document completely, the system routes documents that cannot be confidently classified to an exception folder for manual review. This illustrates a human-in-the-loop approach in which AI handles suitable documents while uncertain cases remain under human control. This differs from literature emphasizing full or near-full automation, such as Belov (2022), which focuses on performing functions automatically or with minimal human intervention. The comparison suggests that effective intelligent document management does not necessarily mean eliminating human participation; instead, AI can be used to reduce routine work while allowing personnel to handle exceptional or ambiguous cases.

The literature further demonstrates the importance of security, metadata, compliance, and organizational governance. Chode (2025) proposed a next-generation document intelligence approach emphasizing smart metadata, secure access, regulatory compliance, semantic search, predictive workflow routing, and anomaly detection. This is broader than studies focused primarily on OCR or classification because it treats intelligent document management as an organizational information-governance problem. Buhaiova (2026) similarly examined the transformation of electronic document management under the influence of AI from a national and organizational perspective. Rather than concentrating solely on an individual algorithm, the study considers technological and organizational relationships and the implementation of AI within records management. Ahmadov and Mahmudov (2026) likewise consider AI applications throughout the

document-management process, from digitization and recognition to search and business-process optimization. These studies suggest that an intelligent document system should not be evaluated solely according to extraction or classification accuracy but also according to its ability to support secure, organized, and compliant information management.

The studies also differ in their technical sophistication and scope of integration. Earlier studies such as Muminova and Asatov (2021), Wang (2021), Belov (2022), Bhatt (2022), and Sambetbayeva et al. (2022) establish the potential of AI and machine learning for document automation, extraction, retrieval, and management. Later studies increasingly combine several AI technologies. Pandey et al. (2023) integrate AI, ML, NLP, computer vision, and OCR; Serugendo et al. (2024) integrate ontology, information extraction, classification, RDF mapping, and reasoning; Chode (2025) combines AI, NLP, and ML for metadata, search, workflow routing, and anomaly detection; and recent studies such as Haryan (2026), Pronenko (2026), and Reddy and Ramya (2026) explore RAG, multimodal document understanding, and dynamic field extraction. This progression indicates that intelligent document management is evolving from isolated AI functions toward integrated systems in which multiple intelligent technologies work together.

Despite these advances, the reviewed studies reveal several differences in research focus, application context, and system functionality. Some studies are primarily conceptual or methodological, such as Muminova and Asatov (2021), Buhaiova (2026), and Ismail et al. (2025). Others develop specific intelligent models or system architectures, including Sambetbayeva et al. (2022), Pandey et al. (2023), Serugendo et al. (2024), Haryan (2026), and Reddy and Ramya (2026). Several focus on particular technical functions such as OCR, classification, information extraction, or retrieval, while others attempt to automate broader

document workflows. The application environments also vary substantially, including government and administrative organizations, healthcare, banking, taxation, bilingual records management, cloud computing, and general organizational document management.

A common limitation across the literature is that many studies emphasize the technical intelligence of document processing without fully addressing the complete operational requirements of a specialized institutional document-management environment. For example, OCR and extraction studies concentrate on transforming document content into structured information, while classification studies concentrate on identifying document categories. Retrieval-oriented studies improve access to information, while workflow-oriented studies automate document movement and processing. However, these capabilities are not always integrated with user roles, application monitoring, document submission, status tracking, notifications, institutional workflows, and reporting within a single system. The literature therefore provides strong evidence for individual intelligent document-management capabilities but leaves opportunities for integrated implementations tailored to specific institutional processes.

This gap is particularly relevant to the development of LSPU ScholarHub, an AI-enabled web-based scholarship management and monitoring system. Unlike studies that focus primarily on document extraction, classification, retrieval, or workflow automation individually, the proposed system applies intelligent document-management concepts within a scholarship-management context. Scholarship offices handle different categories of documents, including identification documents, grades, contracts, compliance requirements, and application-related files. Therefore, the lessons from Bhatt (2022), Pandey et al. (2023), Gyawali et al. (2025), and Reddy and Ramya (2026) support the use

of automated document processing and extraction; the work of Alothman and Sait (2022), Wang (2021), and Ismail et al. (2025) supports improved document organization and retrieval; while Belov (2022), Sambetbayeva et al. (2022), Serugendo et al. (2024), and Silberlust et al. (2026) support workflow automation and the integration of intelligent document processing into organizational activities.

2.1.11 OCR in education

OCR for Academic Documents and Semantic Understanding

Blecer et al. (2023) introduce another important development through Nougat: Neural Optical Understanding for Academic Documents. Unlike conventional OCR systems that primarily convert characters into plain text, Nougat is designed for scientific documents and converts academic PDFs into a markup representation while preserving important semantic information, including mathematical content. The authors use a Visual Transformer architecture for academic document understanding.

This approach differs fundamentally from basic OCR implementations. Conventional OCR primarily answers the question, “What characters are present in this image?” Nougat moves toward the question, “What is the structure and meaning of this academic document?” This distinction is highly relevant to educational systems because student and institutional documents often contain structured information that loses meaning if converted into unorganized plain text.

Nandini et al. (2026) advance this concept further by combining OCR with conversational AI. Their Automated Student Document Analysis and Query System Using OCR and Conversational AI allows uploaded academic or legal documents to be converted

through OCR and subsequently analyzed through Gemini AI. The system produces summaries in English and Telugu and allows users to ask questions about uploaded documents.

Compared with Nougat, Nandini et al. place greater emphasis on user interaction and information retrieval, whereas Nougat focuses on academic-document parsing and semantic preservation. Both, however, demonstrate a shift away from OCR as an endpoint toward OCR as the first stage of a larger intelligent-document-processing pipeline.

Post-OCR Processing and Recognition Accuracy

The literature also demonstrates that OCR output should not automatically be regarded as error-free. (Nguyen et al., 2021) in their comprehensive survey of post-OCR processing, emphasize that OCR-generated text can contain noise and recognition errors, particularly when source documents are difficult to process. Their review categorizes post-OCR approaches according to factors such as human involvement, information used, and computational models, including language models and machine translation techniques.

This perspective complements the findings of Memon et al. (2020), whose systematic review shows the continuing challenges associated with handwritten OCR. While Memon et al. concentrate primarily on recognition techniques, Nguyen et al. shift attention to what happens after recognition. This distinction is important for educational applications because an OCR system that produces a plausible-looking but incorrect grade, student name, or credential field could introduce serious administrative errors.

Faizullah et al. (2023) further demonstrate that recognition accuracy is influenced by language and script characteristics through their survey of Arabic OCR. Their review

examines preprocessing, segmentation, classification, post-processing, applications, and challenges specific to Arabic-language recognition. Similarly, Narang, Jindal, and Kumar (2020) examine OCR for ancient texts, where degraded documents, historical writing styles, and document conditions create additional recognition challenges.

Although these studies are not specifically concerned with educational administration, they are relevant because they establish that OCR performance is dependent on the characteristics of the source document. Educational institutions may likewise process documents with different fonts, handwriting styles, languages, layouts, image qualities, and scanning conditions.

2.1.12 OCR accuracy

Optical Character Recognition Accuracy

Optical Character Recognition (OCR) accuracy is a central concern in the digitization of printed, handwritten, scanned, and degraded documents because the usefulness of an OCR system depends not only on its ability to recognize text but also on the reliability of the resulting digital representation. The literature contained in the present review demonstrates that OCR accuracy is influenced by multiple interacting factors, including image quality, scanning resolution, document degradation, skewness, layout complexity, segmentation, preprocessing techniques, character type, language and script, recognition engine, camera conditions, and post-OCR correction. The twenty studies reviewed collectively show a progression from conventional OCR pipelines toward increasingly sophisticated approaches involving image enhancement, deep learning, multiple OCR engines, document-layout analysis, and post-processing.

Foundational Factors Affecting OCR Accuracy

Hamad and Kaya (2016) provide a foundational perspective on OCR by describing the major phases, applications, and challenges involved in converting printed or handwritten documents into machine-readable information. Their work establishes that OCR accuracy is not determined by the recognition algorithm alone but is influenced by the entire processing pipeline, beginning with image acquisition and continuing through segmentation and recognition. This broad perspective is consistent with Vijayarani and Sakila (2015), who specifically compare OCR tools and emphasize the importance of preprocessing, segmentation algorithms, and image quality. The latter study therefore provides a more experimental perspective than Hamad and Kaya, moving from the general OCR process toward factors that directly influence recognition performance.

The two studies agree that OCR should be regarded as a pipeline rather than a single recognition operation. However, they differ in emphasis. Hamad and Kaya present OCR from a broad technological and conceptual perspective, whereas Vijayarani and Sakila focus more directly on comparative OCR performance. This distinction is important for an accuracy-oriented capstone project because it suggests that evaluating the recognition engine without controlling the quality of the input image may produce misleading conclusions.

The importance of image quality is also evident in Pham (2022), whose study investigates OCR quality in smart scanned-document management. Unlike the broader studies of Hamad and Kaya and Vijayarani and Sakila, Pham concentrates on specific document-image problems such as skewness, heterogeneous layouts, proportional fonts, and document structure. The study applies deskewing and document-layout analysis before

Tesseract recognition and reports that its preprocessing methods increased word-level Tesseract accuracy by 23% for complex documents. Thus, while Vijayarani and Sakila establish image quality and preprocessing as important factors, Pham provides stronger evidence that targeted preprocessing can produce measurable improvements in OCR performance.

OCR Engine Selection and Its Effect on Accuracy

OCR accuracy is also affected by the recognition engine itself. Vijayarani and Sakila (2015) compare OCR tools and demonstrate that different OCR systems can produce different results even when processing the same document. Hegghammer (2021), however, provides a more extensive contemporary benchmark by comparing Tesseract, Amazon Textract, and Google Document AI using English and Arabic documents and introducing different forms of artificial noise. The study found that Google Document AI achieved the strongest overall results and that the server-based systems, particularly Textract and Document AI, performed substantially better than Tesseract on noisy documents. English recognition was also considerably more accurate than Arabic recognition.

The contrast between these studies demonstrates that OCR accuracy cannot be attributed solely to preprocessing. Even with the same or similar input conditions, recognition engines have different capabilities. Vijayarani and Sakila emphasize the general importance of tool selection, while Hegghammer demonstrates that the difference between engines becomes particularly important when documents contain noise or when different languages are involved.

This finding has direct implications for a capstone OCR system. If the proposed system uses Tesseract or another OCR engine, its accuracy should not automatically be

treated as representative of OCR technology as a whole. Rather, accuracy should be evaluated under controlled conditions, and the selected engine should be justified based on the document types and conditions expected in the target environment.

Scanning Resolution as a Determinant of OCR Accuracy

Stančić and Trbušić (2020) provide particularly strong evidence concerning scanning resolution. Their study examines 123 pages of typewritten archival documents scanned using different resolution settings and binarization levels. The researchers found that OCR results varied significantly according to the digitization settings, with a maximum accuracy of 95.02%. More importantly, resolution was found to have a greater influence on OCR performance than binarization.

This finding both supports and qualifies the preprocessing-oriented studies. Harraj and Raissouni (2015) emphasize image enhancement and binarization as important components of OCR improvement, whereas Stančić and Trbušić demonstrate that increasing or optimizing resolution can have a stronger influence than binarization in their typewritten-document context. Therefore, the literature does not support treating all preprocessing techniques as equally influential.

This difference is important because it suggests that OCR accuracy should be investigated according to the characteristics of the document. A preprocessing technique that is highly effective for a photograph or degraded document may have a different effect on a clean, high-resolution scanned document. Consequently, an accuracy evaluation should consider resolution and preprocessing as separate variables rather than assuming that one universal preprocessing configuration is optimal.

Preprocessing and Image Enhancement

Harraj and Raissouni (2015) provide one of the clearest demonstrations of how image preprocessing can improve OCR accuracy. Their proposed approach addresses document-image distortions through a sequence of local brightness and contrast adjustment, grayscale conversion, sharpening using unsharp masking, and global binarization. The objective is to compensate for lighting variations and distortions commonly found in document images acquired through digital cameras or mobile devices. The authors report significant improvements in text detection and OCR accuracy.

Their findings are closely related to those of Pham (2022) and Sutriawan, Kurniawan, and Rosyid (2024). Pham focuses on deskewing and document-layout processing, while Sutriawan et al. focus on image enhancement combined with a neural-network-based OCR approach. The three studies therefore agree that the quality of the image presented to the OCR engine is critical, but they differ in the type of enhancement emphasized. Harraj and Raissouni concentrate on brightness, contrast, grayscale, sharpening, and binarization; Pham emphasizes skew and layout; and Sutriawan et al. combine image enhancement with neural-network recognition.

Mursari and Wibowo (2021) further strengthen the importance of preprocessing by examining digital handwritten-script recognition using Tesseract. Their work is particularly relevant because handwriting introduces greater variability than standardized printed text. Compared with Harraj and Raissouni, who primarily address general document-image distortions, Mursari and Wibowo focus specifically on how preprocessing affects handwritten recognition. This difference indicates that preprocessing requirements depend not only on image quality but also on the nature of the text being recognized.

Collectively, these studies suggest that preprocessing should be considered a configurable stage of an OCR system. Deskewing, grayscale conversion, contrast enhancement, sharpening, thresholding, and noise removal may each contribute differently depending on the input document.

Document Degradation and OCR Accuracy

Document degradation represents another major factor affecting OCR performance. Saladaga and Talirongan (2026) specifically examine EasyOCR on physically degraded documents, including crumpled, wet, dirty, and normal documents. Their experimental procedure involved printing text, physically degrading the documents, scanning them, applying OCR, and calculating accuracy at both character and word levels.

Their work is particularly relevant to the present theme because it examines degradation as an independent variable rather than simply treating poor image quality as a general limitation. The study therefore complements Harraj and Raissouni (2015), who focus on correcting image distortions through preprocessing, and Hegghammer (2021), who introduces artificial noise to compare OCR engines.

The contrast is methodological. Saladaga and Talirongan reproduce physical deterioration, whereas Hegghammer simulates noise computationally. Harraj and Raissouni, on the other hand, attempt to compensate for image problems before recognition. These approaches represent three different strategies for understanding OCR accuracy: measuring degradation effects, benchmarking recognition systems under noise, and mitigating image degradation through preprocessing.

Kushavaha (2024) provides a broader discussion of OCR and its limitations, supporting the conclusion that real-world document variability makes perfect recognition difficult. Compared with the experimental degradation study of Saladaga and Talirongan, however, Kushavaha is more general and less focused on quantifying a particular source of error. Thus, the two works complement rather than duplicate one another.

Document Layout and Structural Complexity

Document structure is another factor that can reduce OCR accuracy. Pham (2022) specifically identifies heterogeneous layouts and document rotation as problems and demonstrates improvements through deskewing and layout analysis. Li, Lee, and Liu (2025) take this issue considerably further by proposing a deep-learning system that first detects titles, paragraphs, tables, and images using YOLOv4 and YOLOv8, followed by category-specific segmentation and CNN-based text recognition.

The difference between these studies illustrates the evolution of OCR systems. Pham treats layout analysis primarily as a preprocessing strategy for improving an OCR engine such as Tesseract. Li et al., in contrast, integrate document-layout detection and recognition into a deep-learning architecture. Therefore, the latter study moves beyond the assumption that a document is simply a flat image containing text.

This distinction is especially relevant to educational documents because school forms, student records, certificates, grade sheets, and application forms commonly contain tables, labels, boxes, signatures, and multiple text regions. A system that recognizes characters accurately but misinterprets their spatial relationships may still produce

incorrect structured information. Consequently, layout accuracy is an important component of overall OCR accuracy.

Handwritten Character Recognition

Lomps (2018) examines factors affecting optical handwritten character recognition. Handwritten recognition presents a different accuracy problem from printed-document OCR because the appearance of a character varies between writers and even within the writing of the same person. Factors such as character shape, writing style, spacing, segmentation, and image quality can consequently affect recognition.

Mursari and Wibowo (2021) complement Lomps by experimentally examining image preprocessing for digital handwritten scripts using Tesseract. Whereas Lomps focuses on identifying factors that affect handwritten recognition, Mursari and Wibowo focus more directly on the potential of preprocessing to address these problems. This represents an important contrast between diagnosing sources of recognition error and developing techniques for reducing those errors.

The distinction becomes even more important when compared with Sutriawan et al. (2024), who incorporate image enhancement with neural-network-based OCR. The literature therefore shows a progression from identifying handwriting-related accuracy problems, to preprocessing handwritten images, and ultimately toward using neural networks to improve recognition.

For educational applications, handwritten recognition is particularly challenging because student-written answers, forms, signatures, and handwritten marks may have substantial variability. Thus, accuracy results obtained from clean printed documents should not automatically be generalized to handwritten educational documents.

Language and Script as Accuracy Factors

Faizullah et al. (2023) examine OCR in Arabic-language documents and identify language- and script-specific challenges. Their survey demonstrates that OCR accuracy depends partly on the characteristics of the writing system itself. Hegghammer (2021) provides empirical support for this issue by comparing English and Arabic documents and finding considerably higher accuracy for English than Arabic in the tested conditions. Ellotf and Bayoumi (2020), through their case study of the Egyptian press archive of CEDEJ, provide a more specialized example of the challenges involved in Arabic OCR. In contrast to the broad survey of Faizullah et al., their study focuses on a specific archival collection and therefore illustrates how historical document conditions and script characteristics can interact.

These studies contrast with Stančić and Trbušić (2020), whose work focuses on typewritten Croatian archival material, and with the studies of English-language printed documents. The comparison indicates that OCR accuracy cannot be expressed as a single universal performance level. Instead, accuracy is dependent on the interaction between language, script, document condition, font, layout, and OCR engine.

This is an important consideration for educational OCR projects operating in multilingual environments. If documents contain English, Filipino, or other languages and scripts, the system should be evaluated using the actual languages present in the intended documents rather than relying solely on performance results from English-language datasets.

Camera and Document Acquisition Conditions

Feng et al. (2026) expand the concept of image quality by examining OCR under real-world mobile acquisition conditions. Their study evaluates the effects of walking speed, viewing angle, camera placement, and camera type using multiple OCR engines. Accuracy declined with increased walking speed and wider viewing angles. Google Vision achieved the highest overall accuracy, while PaddleOCR performed strongly as an open-source alternative. The study also found that the phone's main camera performed better than its ultra-wide camera. This study differs substantially from Stančić and Trbušić because the latter examine controlled scanning resolution, whereas Feng et al. examine dynamic image acquisition. Nevertheless, both reach the broader conclusion that how a document is captured affects OCR accuracy.

Similarly, Harraj and Raissouni address distortions created by digital-camera acquisition, but their emphasis is on correcting those distortions through preprocessing. Feng et al., by contrast, investigate the acquisition conditions themselves. Taken together, these studies suggest that OCR accuracy begins before the document reaches the recognition engine. The capture process, camera, viewing angle, motion, and document positioning can all affect the quality of the input.

Document Quality Assessment

Krithika, Athanesious, and Kiruthika (2026) approach the issue from another direction through OCR-based document-image quality assessment. Their study is significant because it suggests that document quality itself can be evaluated as part of the OCR workflow rather than treating every input document as equally suitable for

recognition. The study therefore complements the degradation research of Saladaga and Talirongan and the preprocessing research of Harraj and Raissouni.

This represents an important conceptual shift: instead of simply applying OCR to every document, a system could first determine whether the document image is sufficiently clear for reliable recognition. Low-quality documents could then be enhanced, rescanned, or subjected to additional processing. Such a quality-control mechanism could potentially reduce recognition errors before they propagate into subsequent data-processing stages.

Post-OCR Processing and Error Mitigation

OCR accuracy should not be evaluated solely by the initial recognition output because recognition errors can affect subsequent uses of the extracted text. Nguyen et al. (2021) provide a comprehensive survey of post-OCR processing approaches, emphasizing techniques used to correct and improve OCR-generated text. Their work demonstrates that OCR output can be treated as an intermediate representation rather than a final, error-free product.

De Oliveira et al. (2023) extend this concern by examining the impact of OCR errors on information retrieval. Their work demonstrates that OCR errors can propagate into downstream tasks, meaning that recognition accuracy has consequences beyond the immediate transcription stage.

These studies contrast with the preprocessing studies because they intervene after recognition rather than before it. Harraj and Raissouni, Pham, Mursari and Wibowo, and Sutriawan et al. seek to improve the input or recognition process. Nguyen et al. and De Oliveira et al. instead address what happens when errors remain in the OCR output.

This distinction is important for a capstone system because a high OCR accuracy percentage does not necessarily guarantee accurate downstream information extraction. A system processing educational documents may need additional validation, correction, confidence scoring, or human review, especially when extracted information includes student names, grades, identification numbers, or other structured fields.

Deep Learning and Modern OCR Accuracy

Li et al. (2025) represent the increasing role of deep learning in OCR accuracy. Their system combines YOLO-based layout detection with CNN-based text recognition. This differs from conventional Tesseract-based systems such as those examined by Pham and Mursari and Wibowo. Rather than depending primarily on traditional preprocessing and OCR recognition, Li et al. use deep learning to understand document structure and then apply recognition techniques appropriate to different regions.

Sutriawan et al. (2024) similarly combine image enhancement with neural-network recognition. The distinction is that Sutriawan et al. emphasize the relationship between enhanced input images and neural-network OCR, while Li et al. place stronger emphasis on document layout and region-specific recognition.

These studies suggest that modern OCR accuracy increasingly depends on combining multiple forms of intelligence rather than simply improving character classification. However, the literature also indicates that sophisticated models do not eliminate the importance of image quality. Instead, deep-learning systems continue to depend on suitable input images and appropriate document representations.

2.1.13 OCR for academic documents

Optical Character Recognition for Academic Documents

Optical Character Recognition (OCR) has developed from a conventional character-recognition technology into an important component of automated academic-document processing. In educational environments, documents such as student records, academic credentials, examination papers, answer scripts, assignments, reports, certificates, and research papers contain information that traditionally requires manual encoding, verification, organization, and retrieval. The literature reviewed in the present study demonstrates that OCR can reduce these manual processes by converting image-based and scanned documents into machine-readable information. However, the studies also show that OCR for academic documents is considerably more complex than simple text recognition because academic documents frequently contain handwritten content, tables, multiple layouts, mathematical expressions, structured fields, and varying document formats. The twenty studies in the reviewed RIS file collectively illustrate an evolution from basic OCR and image processing toward intelligent academic-document understanding, verification, and conversational information retrieval.

OCR as a Foundation for Academic Document Digitization

The foundational studies in the literature establish OCR as a mechanism for converting documents from image-based representations into editable and searchable text. Dafe and Chavhan (2018), through their work on OCR using image processing, represent the traditional approach in which image-processing operations form the basis for character recognition. Similarly, Kushavaha (2024) provides a general treatment of OCR and its applications, emphasizing the role of OCR in converting text contained in images into

usable digital information. These studies provide the technical foundation for later educational applications because academic-document systems depend on the ability to transform scanned pages into machine-readable content.

Dome and Sathe (2021) move beyond the general OCR concept by implementing Tesseract with classification techniques. Their approach is more application-oriented than the general discussions of Dafe and Chavhan and Kushavaha because it focuses on using an actual OCR engine for automated document processing. Mukherjee et al. (2023) similarly examine OCR using Python and demonstrate how programming environments can be used to implement document digitization. In contrast, Jain, Taneja, and Taneja (2021) approach OCR from a comparative perspective by evaluating different OCR toolsets rather than concentrating on one implementation.

These studies collectively establish an important distinction. Dafe and Chavhan emphasize the image-processing foundation, Dome and Sathe emphasize Tesseract-based implementation, Mukherjee et al. emphasize programmatic implementation, while Jain et al. emphasize tool selection and comparison. For an academic-document OCR system, these perspectives indicate that successful implementation requires more than selecting an OCR engine. The system must also consider document acquisition, image processing, recognition, and the suitability of the selected OCR technology for the intended academic documents.

OCR for Student Records and Educational Documents

The literature becomes more directly educational in studies that apply OCR to student and institutional documents. Shaju et al. (2024) investigate document digitization through OCR for automated mark-entry processing in educational institutions. Their work

demonstrates a practical application of OCR in which information from answer scripts can be extracted and transformed into digital records. This represents one of the most direct applications of OCR to academic administration because marks that would otherwise require manual encoding can be processed automatically.

Koushik, Chengappa, and Chendan (2022) address a related problem through automated marks-entry processing in handwritten answer scripts. However, their focus differs from Shaju et al. because they concentrate specifically on handwritten characters and recognition techniques rather than document digitization as a broader administrative workflow. The distinction is important: Shaju et al. emphasize the institutional application, whereas Koushik et al. focus more heavily on the recognition challenge associated with handwritten academic material.

Kharatmol et al. (2025) extend the educational application beyond examination marks through their work on extracting meaningful data from education credentials using OCR and image processing. Their target is therefore more complex than answer scripts because academic credentials can contain multiple fields and structured information. This study differs from the mark-entry studies because the objective is not simply to recognize digits or marks but to extract meaningful information from formal educational documents.

These three studies demonstrate a progression in academic-document OCR: Koushik et al. focus on recognizing handwritten academic information, Shaju et al. focus on automating the entry of recognized information, and Kharatmol et al. focus on extracting meaningful structured information from credentials. Consequently, the literature suggests that OCR applications in education have progressed from basic transcription toward structured information extraction and institutional automation.

OCR for Academic Document Verification

A further development in the literature involves using OCR not merely to extract information but also to verify academic documents. Syahyadi et al. (2025) propose a system for smart verification of high-school student reports using OCR and BERT models. This represents a significant departure from conventional OCR because the extracted text is subsequently processed by a language model to support verification. Rather than treating OCR output as the final result, the system uses OCR as an initial stage in a larger intelligent-document-processing pipeline.

Karwa et al. (2026) similarly expand OCR functionality through SmartDoc, an AI-based mobile application for document scanning, verification, and categorization. Unlike the report-card-specific approach of Syahyadi et al., SmartDoc adopts a broader document-management perspective, incorporating scanning, verification, and categorization into one system.

The contrast between these studies illustrates two different approaches to academic-document intelligence. Syahyadi et al. focus on semantic verification of a specific academic document type, whereas Karwa et al. focus on general document management and categorization. Both, however, demonstrate that OCR becomes more useful when combined with additional AI or rule-based processing.

This is important for academic-document systems because accurate text recognition alone does not necessarily establish whether a document is valid, complete, correctly categorized, or semantically consistent. Therefore, OCR can be considered the first layer of academic-document processing, with verification and categorization forming subsequent layers.

OCR for Handwritten Academic Documents

Handwritten academic documents represent one of the most challenging OCR applications because writing styles vary considerably among students. Memon, Sami, and Khan (2020) provide a comprehensive systematic literature review of handwritten OCR and establish the broad technical challenges associated with recognizing handwritten characters. Their study provides a theoretical foundation for educational applications involving handwritten answer scripts and assignments.

Koushik et al. (2022) provide a direct educational application of handwritten character recognition through automated processing of handwritten answer scripts. Compared with the broad systematic review of Memon et al., their work is narrower and application-specific. It demonstrates how handwritten OCR concepts can be applied to a concrete educational workflow.

Boreddy (2025) extends the handwritten-document application from recognition to analysis and evaluation through AI-driven handwritten assignment analysis. This represents a more advanced use of OCR because the objective is not simply to digitize handwriting but to analyze academic content after it has been recognized.

OCR for Academic Assignments and Learning Materials

Boreddy's (2025) study is particularly relevant to educational document processing because it illustrates how OCR can be combined with AI to analyze handwritten assignments. Rather than treating OCR as an isolated digitization technology, the study positions it as a component of an academic evaluation workflow. Nandini et al. (2026) extend the concept further through an automated student-document analysis and query

system that combines OCR with conversational AI. In this approach, documents can be digitized and subsequently queried through an AI interface. This differs from Boreddy because the purpose is not primarily evaluation but information retrieval and interaction with academic documents. The contrast between the two studies demonstrates the widening role of OCR in education. Boreddy uses OCR as a foundation for academic-content evaluation, whereas Nandini et al. use OCR as a foundation for academic-document question answering and information retrieval. Both represent a transition from OCR as transcription toward OCR as an interface for accessing information.

OCR Specifically for Academic and Research Documents

One of the most important studies for the theme of OCR for academic documents is Blecher et al.'s (2023) Nougat: Neural Optical Understanding for Academic Documents. Unlike conventional OCR systems that primarily recognize characters and words, Nougat is designed specifically for scientific documents and converts document pages into structured markup. The study addresses a major limitation of conventional OCR: academic documents contain mathematical expressions, tables, formatting, and spatial relationships that cannot always be accurately represented through ordinary line-by-line character recognition.

The importance of this distinction is supported by the original Nougat research, which describes scientific documents as particularly difficult to process because conventional OCR may recognize individual characters while failing to preserve the semantic relationships among them, especially in mathematical expressions. Nougat uses a Visual Transformer to convert document-page images into markup representations.

Li, Lee, and Liu (2025) address a related problem through a deep-learning system for printed document layout analysis and OCR. Their approach combines document-layout detection with recognition, allowing different regions such as titles, paragraphs, tables, and images to be identified before or during text recognition.

Although both studies employ advanced machine learning, their objectives differ. Nougat is specifically optimized for academic/scientific documents and emphasizes semantic document representation, while Li et al. focus more broadly on printed-document layout analysis and OCR. Nougat is therefore particularly relevant to research papers and scholarly documents containing mathematical and structural information, whereas Li et al.'s approach is more general and applicable to documents with complex visual layouts.

This distinction suggests that academic-document OCR should not be evaluated only according to character-level recognition. A system may correctly recognize the words of an academic paper while still failing to preserve equations, tables, headings, captions, or relationships among document elements. Therefore, structural and semantic preservation becomes an important dimension of OCR for academic documents.

OCR Architecture and Artificial Intelligence

Muthusundari et al. (2024) examine OCR systems using artificial intelligence, reflecting the broader transition from conventional image processing toward machine-learning-based recognition. Their work emphasizes AI-based OCR as an approach for handling the complexity of document recognition. This AI-oriented perspective differs from the conventional approaches represented by Dafe and Chavhan and Dome and Sathe. Traditional systems rely heavily on image preprocessing and established OCR engines, while AI-based systems can learn patterns from data and potentially handle greater

variability.Li et al. (2025) represent a more specialized deep-learning approach, combining layout detection and OCR. Nougat represents an even more document-specific architecture, using a Visual Transformer designed for scientific and academic documents.

The literature consequently reflects a progressive evolution from traditional OCR to AI-enhanced OCR, followed by deep-learning-based document understanding and, more recently, neural OCR approaches tailored to academic documents.

However, these approaches also differ in complexity and requirements. Traditional OCR systems can be easier to deploy and may require fewer computational resources, while deep-learning systems may provide richer document understanding at the cost of greater computational and implementation complexity. For a capstone project, this creates an important design consideration: the most sophisticated model is not necessarily the most appropriate unless its additional capabilities are required by the target documents.

OCR Tool Selection

Tool selection is another important theme within the literature. Jain et al. (2021) specifically compare OCR toolsets, while Dome and Sathe (2021) focus on Tesseract-based OCR. Mukherjee et al. (2023) demonstrate OCR implementation using Python. These studies differ primarily in their level of abstraction. Jain et al. address which OCR technology should be selected, Dome and Sathe demonstrate how a particular OCR engine can be used, and Mukherjee et al. emphasize how OCR can be integrated into a programming environment.

This distinction is highly relevant to academic-document processing. Different educational documents may require different OCR capabilities. A basic Tesseract-based system may be adequate for clean printed forms, while academic research papers

containing mathematical expressions may require a specialized document-understanding model such as Nougat. Similarly, handwritten answer sheets may require models specifically designed for handwriting.

Therefore, the literature does not support the assumption that one OCR engine is universally optimal for all academic documents. Instead, the appropriate tool should depend on the document type, image quality, handwriting, language, layout, and information structure.

Post-OCR Processing

Nguyen et al. (2021) provide an important complementary perspective through their survey of post-OCR processing approaches. Their work demonstrates that OCR output may contain errors and that additional processing can be used to improve the resulting text. This is particularly important for academic documents. An OCR system may recognize most words correctly while still introducing errors in technical terminology, student names, numerical information, mathematical symbols, citations, or table values.

Post-OCR processing can therefore serve as a quality-control layer.

The study of Nguyen et al. differs from most of the educational OCR studies because it does not focus on a particular academic application. Instead, it provides a general framework for improving OCR output after recognition. Its relevance to the capstone project lies in demonstrating that OCR should be viewed as a multi-stage process rather than a one-step conversion. This perspective complements Syahyadi et al., Karwa et al.,

and Nandini et al. because each of those systems also demonstrates that OCR output can serve as an input for later processes such as verification, categorization, analysis, or querying.

Language and Document Diversity

Faizullah et al. (2023) examine OCR specifically in the Arabic language and provide evidence that language and writing-system characteristics influence OCR implementation and performance. This is relevant even when the intended academic documents are primarily in English because educational institutions can process documents containing different languages, names, abbreviations, symbols, or multilingual content. The study demonstrates that OCR systems cannot necessarily be assumed to perform equally across all languages and scripts. Narang, Jindal, and Kumar (2020), through their review of ancient-text recognition, provide another perspective on document diversity. Ancient documents can contain degraded pages, unusual writing styles, historical scripts, and difficult visual conditions. Although these documents are not necessarily academic records, their study demonstrates the broader principle that OCR performance is strongly affected by the characteristics and condition of the source document.

Thus, Faizullah et al. emphasize linguistic and script diversity, whereas Narang et al. emphasize historical and physical-document complexity. Both complement the educational studies by showing why an OCR system should be evaluated using

representative documents rather than assuming that results from clean standardized text will generalize to all academic materials.

Comparative Synthesis of the Twenty Studies

The twenty studies can be organized into several interconnected categories. The first group consists of foundational OCR studies, including Dafe and Chavhan, Kushavaha, Dome and Sathe, Mukherjee et al., and Jain et al. These studies establish the mechanisms, tools, and implementation considerations underlying OCR.

The second group consists of educational document applications, including Koushik et al., Shaju et al., Kharatmol et al., Karwa et al., Syahyadi et al., Nandini et al., and Boreddy. These studies demonstrate direct applications of OCR to answer scripts, mark entry, credentials, student reports, document management, student-document analysis, and handwritten assignments.

The third group consists of advanced OCR and document-understanding studies, particularly Blecher et al., Li et al., and Muthusundari et al. These studies move beyond conventional character recognition by introducing deep learning, neural architectures, layout analysis, and semantic document representation.

The fourth group consists of supporting studies addressing handwriting, language, and post-processing, represented by Memon et al., Nguyen et al., Faizullah et al., and Narang et al. These studies help explain why academic-document OCR remains challenging and why recognition must often be followed by additional processing.

A major point of agreement across these studies is that OCR can reduce the effort associated with manual document processing. However, the studies differ significantly in

the purpose of OCR. Some use OCR for simple digitization; others use it for structured data extraction, mark entry, verification, classification, assignment analysis, or conversational querying.

There is also a clear difference in the complexity of the target documents. Koushik et al. primarily address handwritten answer scripts, whereas Kharatmol et al. address education credentials. Syahyadi et al. focus on student reports, and Boreddy focuses on handwritten assignments. Nougat, in contrast, focuses on scientific documents with complex structures such as mathematical expressions. Li et al. address printed documents with multiple layout regions. Consequently, the literature demonstrates that "academic documents" cannot be treated as a single homogeneous category.

The studies also differ in their processing architecture. Conventional approaches such as those represented by Dafe and Chavhan and Tesseract-based work process documents largely through image processing and recognition. More recent systems such as Syahyadi et al., Karwa et al., and Nandini et al. use OCR as one stage within a larger AI workflow. Nougat goes further by integrating recognition and document understanding into a neural architecture designed specifically for academic documents.

Evolution of OCR for Academic Documents

The reviewed literature reveals a clear and progressive evolution in document-processing technologies, beginning with traditional OCR for image processing and character recognition, advancing to document digitization for converting educational documents into machine-readable data, and progressing toward structured information extraction for identifying meaningful fields such as grades and credential information. This development is further extended through AI-assisted processing for document verification,

categorization, and assignment analysis, and ultimately toward intelligent document understanding involving semantic interpretation, academic-document parsing, and conversational querying. This progression becomes particularly evident when the reviewed studies are examined chronologically, demonstrating the gradual shift from basic text recognition toward increasingly intelligent and context-aware document-processing capabilities.

Earlier works in the RIS, such as Dafe and Chavhan (2018), establish image-processing-based OCR. Subsequent studies such as Dome and Sathe (2021) and Jain et al. (2021) address implementation and tool selection. Educational applications then emerge through Koushik et al. (2022), Shaju et al. (2024), and Kharatmol et al. (2025). More recent work introduces BERT, conversational AI, deep learning, and academic-document-specific neural architectures through Syahyadi et al. (2025), Nandini et al. (2026), Li et al. (2025), and Karwa et al. (2026).

This progression indicates that OCR in education is increasingly moving from "reading the document" to "understanding the document."

Despite the extensive literature on OCR, a gap remains in the development of an integrated OCR solution specifically designed around the diverse characteristics of academic documents. Existing educational studies often focus on a particular document type or task. Koushik et al. focus on handwritten answer scripts, Shaju et al. focus on marks-entry automation, Kharatmol et al. focus on educational credentials, Syahyadi et al. focus on high-school reports, and Boreddy focuses on handwritten assignments.

Meanwhile, advanced OCR studies provide powerful technologies but are not necessarily designed for the practical workflow of educational institutions. Nougat is

highly relevant to academic and scientific documents but is primarily concerned with converting scientific document pages into structured markup. Li et al. focus on printed document layout and OCR, while Muthusundari et al. address AI-based OCR more generally.

The literature therefore suggests a need for systems that combine the strengths of these approaches while remaining practical for educational environments. Such a system should be capable of receiving academic documents, preprocessing document images, recognizing printed and potentially handwritten content, extracting structured information, preserving relevant document organization, and providing usable digital output. A further gap concerns integration. The reviewed studies frequently address one stage of the document-processing pipeline, whereas educational institutions may require several stages simultaneously: document capture, OCR, data extraction, validation, categorization, storage, and retrieval. A capstone project that integrates these functions can therefore contribute by translating the findings of individual OCR studies into a unified academic-document workflow.

2.1. Related Systems and Technology Gap Analysis

A total of ten (10) existing systems were analyzed. The table below shows the comparison of features between related systems and the proposed system.

Table 1. System Feature Comparison of Related Systems

System	OCR Document Processing	NLP Data Extraction	Intelligent Chatbot	Application-to-Scholar Tracking	Multi-Channel Notifications

BPMSP Portal	No	No	No	Yes	Limited
CHED AHEAD Grant Portal	No	No	No	Yes	Limited
DOST-SEI Scholar Portal	No	No	No	Yes	Limited
FEU Mediskolar	No	No	No	Limited	No
MSD Scholarship Portal	No	No	No	Yes	Limited
MyUSTe Student Portal	No	No	No	Yes	Limited
OWWA Scholarship Online Application	No	No	No	Yes	Limited
Taguig Scholarship Management System	No	No	No	Yes	Limited
UP SLAS	No	No	No	Yes	Limited
UPD SIKAP	No	No	No	Yes	Limited
LSPU ScholarHub	Yes	Yes	Yes	Yes	Yes

Legend: Yes – Fully functional Limited – Partial/restricted provision No – No function

As shown in Table 1, most existing scholarship portals lack the combination of automated OCR document processing, NLP-driven data extraction, and intelligent chatbot interaction within a single integrated platform. While baseline application tracking is widely supported across existing systems, document evaluation continues to rely heavily on manual administrative inspection. Furthermore, communication mechanisms across current portals remain limited and inaccessible, frequently relying on passive email dispatches or non-official channels that lack auditable tracking and proactive multi-channel notification capabilities.

The gap identified in existing systems is the absence of a unified, intelligent web platform that integrates OCR-assisted document processing, NLP-based grade extraction, automated academic eligibility tracking, AI chatbot support, and multi-channel notifications (SMS and Email) tailored specifically for scholarship operations at LSPU-

SCC. Existing portals leave document verification as a manual administrative task and rely on fragmented or inaccessible messaging methods, increasing operational delays and communication breakdowns.

Synthesis

The reviewed literature shows that centralized information management and efficient software structures are important for institutional systems. Acoba et al. (2020) found that web-based platforms improve efficiency by centralizing information and reducing repetitive work, while González-Torre and Suárez-Serrano (2022) identified fragmented records as a cause of redundancy and administrative bottlenecks. Zhang (2021) and Pavlenko et al. (2020) further emphasize modularity as a means of supporting system maintainability and scalability. These findings are consistent with Sommerville (2019) and Pressman (2020), who stress systematic development and continuous refinement. Together, the studies indicate that scholarship management requires a centralized system with a structured and maintainable foundation.

Artificial intelligence and automation can extend these capabilities by reducing repetitive tasks and assisting users. Goodfellow, Bengio, and Courville (2016) provide the foundation for automated information extraction, while Garcia (2025) demonstrates the usefulness of conversational agents for routine inquiries. However, Dellermann et al. (2021) emphasize that AI should complement human capabilities rather than replace human judgment. Adedeji et al. (2026) similarly highlight the need for human oversight in educational AI, while Adam et al. (2021) show that responsive conversational interaction can improve user trust. These findings support the use of OCR, NLP, and chatbot functions

in ScholarHub while retaining authorized personnel for decisions requiring institutional judgment.

Automation also supports communication and student assistance. Secugal et al. (2020) found that automated scholarship tracking and SMS notifications can simplify administrative processes, whereas Mehrotra and Musolesi (2017) caution that excessive or poorly timed notifications may cause message fatigue. Blasiak et al. (2022) therefore support using multiple channels according to the purpose and urgency of the message. Fischer (2025) and Zhao et al. (2022) further show that financial aid and digital support can address students' administrative and academic needs. These findings suggest that ScholarHub should use automated and appropriately timed communication to keep scholars informed without creating unnecessary alerts.

Privacy, security, inclusion, and usability are also important to successful implementation. Petronio (2013) emphasizes the management of privacy boundaries, while Watini et al. (2024) and Dela Rosa (2025) highlight technical protection and privacy-by-design. Mauro (2015) and Tailor (2024) further emphasize designing around user needs and accessibility. These considerations are supported by Al-Hunaiyyan et al. (2021), who found that intuitive student portals can improve efficiency and satisfaction. Thus, scholarship systems must protect student information while remaining accessible and understandable to different users.

Overall, the studies reviewed support the integration of centralized records, automated processing, AI-assisted services, human oversight, timely communication, privacy protection, accessibility, maintainability, and usability in scholarship management. Rather than functioning as isolated features, these elements address different aspects of the

same administrative process. Their combined findings provide the basis for the major design considerations of LSPU ScholarHub, including centralized scholarship records, OCR and NLP-assisted document processing, academic monitoring, chatbot assistance, multi-channel notifications, controlled access, and a user-centered interface.

CHAPTER III

RESEARCH METHODOLOGY

This chapter discusses the methods and procedures used in the development and evaluation of the proposed system, LSPU ScholarHub: An AI-Enabled Web-Based Scholarship Management and Monitoring System with Chatbot and Multi-Channel Notifications. It presents the research design, respondents of the study, data gathering methods, system development process, testing procedures, and statistical tools used to assess the system's functionality, usability, reliability, and overall performance.

3.1 Research Design

The researchers used a descriptive-developmental research design as the basis for conducting the study and presenting the findings, observations, and system evaluation. This design was selected to provide a clear and detailed description of the proposed system, "LSPU ScholarHub: An AI-Enabled Web-Based Scholarship Management and Monitoring System with Chatbot and Multi-Channel Notifications," including its features, functions, processes, and user interactions.

Descriptive research focuses on describing a population, system, or situation by answering what, when, and how questions. It presents information as it naturally occurs without manipulating variables or testing cause-and-effect relationships. In this study, the descriptive approach was used to explain the current scholarship management practices of the Scholarship and Financial Assistance Office of Laguna State Polytechnic University,

identify existing operational challenges, and describe how the proposed system addresses these issues through digital solutions.

Meanwhile, the developmental aspect of the research focused on the design, development, implementation, and evaluation of the proposed web-based system. This approach allowed the researchers to create a centralized platform capable of improving scholarship data management, automating status monitoring, and enhancing communication between administrators and scholars through AI-enabled and notification-based technologies.

Based on the preliminary study conducted at Laguna State Polytechnic University – Santa Cruz Campus, several issues were identified within the current scholarship management process. The Scholarship and Financial Assistance Office primarily relies on manual and semi-digital methods such as Excel spreadsheets, Word documents, PDF files, and social media platforms for storing records and communicating with scholars. These practices result in fragmented data storage, inconsistent record management, delayed report generation, and difficulty in tracking the real-time academic status of scholars. Furthermore, communication through social media platforms lacks formal documentation and secure information management, causing students to occasionally miss important announcements and deadlines.

The current setup also lacks an integrated system for centralized data storage, automated scholar monitoring, report generation, and secure communication. Scholarship personnel spend considerable time organizing records manually and coordinating with different university offices to verify scholar information. Additionally, students have

limited access to real-time updates regarding their scholarship applications, requirements, and status inquiries.

To gather relevant information, the researchers utilized interviews, observation, and requirements gathering procedures. Interviews were conducted with scholarship office personnel and selected users to identify system requirements, operational problems, and user expectations. Observation methods were used to analyze the current workflow and record management practices within the office. The collected information served as the basis for the design and development of the proposed system.

According to Dullock (1993), research design serves as a blueprint that guides the entire study and ensures that the research problem is properly addressed. Furthermore, Siedlecki (2020) explained that descriptive research focuses on studying conditions as they naturally occur and identifying existing problems through observation and analysis. Therefore, descriptive-developmental research is appropriate for this study because it allows the researchers to clearly present the identified issues in scholarship management, describe the development process of LSPU ScholarHub, and evaluate how the proposed system addresses the needs of the university and its scholars.

A descriptive-developmental design is most appropriate because constructing an AI-enabled scholarship management system requires both an engineering process to create the artifact and a descriptive evaluation framework to empirically validate its operational viability, user acceptance, and practical impact on the scholarship administration at Laguna State Polytechnic University - Santa Cruz Campus (LSPU-SCC).

3.2 Population of the Study

For this study, the researchers will utilize non-probability purposive sampling to select the participants. As noted by the Cambridge University Press (2023), non-probability sampling procedures are applied when subjects are chosen based on the subjective, deliberate judgment of the researcher rather than through random selection. This approach assumes that not every member of the institutional population needs an equal chance of selection; instead, it relies on a conscious decision to involve individuals who possess the specific characteristics, firsthand experience, or administrative expertise required to satisfy the research objectives. For this project, the chosen respondents will specifically consist of active student scholars and the personnel assigned to the Scholarship and Financial Assistance Office at LSPU-SCC, as they are the direct end-users of the platform.

To determine an accurate, manageable, and statistically significant sample size from the target student scholar population, the researchers will apply Slovin's Formula. This ensures that the data collection phase yields a representative group of respondents to evaluate the system's usability, functionality, and responsiveness.

3.3 System Testers

System testers evaluated the functional completeness, software quality, and technical performance of the LSPU ScholarHub platform prior to its deployment and subsequent user evaluation. This comprehensive testing phase involved an IT Expert to ensure the system fully met all functional requirements, technical specifications, and administrative workflows before broader implementation.

Table 2. System Testers Profile

Tester	Educational Background / Credentials	Current Position / Affiliation	Years of Relevant Experience
Mark Paulos Relova	Naval Postgraduate School	System Engineer	25 Years
Meyrick V. Villarica	AMA Computer College	IT Department Head	20 Years
John Rosh Birador	Laguna State Polytechnic University	Software Developer	8 Years
Christian Angelo Atienza	Laguna State Polytechnic University	Lead Mobile Developer	3 Years
Mark Christoper Menodiado	Laguna State Polytechnic University	ICT STAFF - SPC	7 Years
Resheila Rose G. Hinay	Philippine Christian University	Jr. Project Management Supervisor	14 Years
Rocel B. Estafia	STI College Ortigas-Cainta	Data Analyst	2 Years
Fryzt De Guzman Castro	AMA Computer College	System Development Officer	14 Years
Carl Harvey Remero	Laguna State Polytechnic University	Freelance UI/UX Designer	2 Years
Aeron Red Lope	Laguna State Polytechnic University	Network Analyst	1 Year

3.4 Respondents for User Evaluation

The respondents selected for user evaluation represent the key stakeholders of the LSPU ScholarHub system, consisting of the Scholarship and Financial Assistance Office (SFAO) personnel and student scholars of Laguna State Polytechnic University - Santa Cruz Campus.

The distribution of the respondent population and the corresponding sample size utilized for the system evaluation are presented in Table 2. The researchers applied Slovin's Formula to determine an appropriate sample size, ensuring that the participants represent

the broader user base at the Santa Cruz campus, thereby providing a robust foundation for the system's acceptance evaluation.

Table 3. Distribution of Respondents for User Evaluation

Respondent Group	Target Population	Sample Size	Sampling Design
Scholarship Personnel Office	2	2	Total Enumeration
Student Scholar	1,253	304	Total Enumeration
Total	1,255	306	-

3.5 Sampling Design

A non-probability purposive sampling technique was used to select the respondents for this study, as the participants were chosen based on their direct involvement with the scholarship management processes at LSPU-SCC. To ensure a representative sample size from the total active scholar population of 1,253 students, Slovin's Formula was applied with a margin of error of 5% (0.05).

$$n = \frac{N}{1 + Ne^2}$$

Equation 1. Slovin's Formula

Where:

n = Sample size
N = Total population (40,000)
e = Margin of error (0.07)
1 = Constant

Calculation:

$$n = 1{,}253 / (1 + 1{,}253 * 0.0025)$$
$$n = 1{,}253 / (1 + 3.1325)$$
$$n = 1{,}253 / 4.1325$$
$$n \approx 303.20$$

Based on the calculation, a total of 304 respondents were selected for the user evaluation. This sample size ensures sufficient statistical representation of the student body to assess the functionality, usability, and effectiveness of LSPU ScholarHub. Additionally, total enumeration was applied to the 2 Scholarship Office Personnel to guarantee comprehensive operational representation.

3.6 Data Collection Instrument

This section describes the instruments and procedures employed to gather the data necessary for the evaluation of the proposed system. To ensure that the data collected were relevant, consistent, and sufficient to address the objectives of the study, structured data collection instruments were utilized. These instruments were designed to obtain information from the intended users and stakeholders regarding the existing scholarship management processes, identified operational difficulties, system requirements, and the usability and acceptability of the developed LSPU ScholarHub. The data collection process was organized to support both the requirements analysis and the subsequent evaluation of the completed system.

Fact-finding activities were conducted during the initial stage of the study to obtain first-hand information regarding the existing scholarship management and monitoring procedures of the Scholarship and Financial Assistance Office (SFAO) of Laguna State Polytechnic University–Santa Cruz Campus. These activities served as a primary basis for

identifying the existing workflow, determining the information and documents processed by the office, and establishing the functional requirements of the proposed system.

The researchers gathered information through direct consultation and observation of the existing scholarship-related processes. Particular attention was given to the procedures involved in scholarship application, submission and checking of documentary requirements, verification of student information, assessment of eligibility, monitoring of scholars, dissemination of scholarship-related announcements, and preparation of reports. The researchers also examined how information was currently recorded, updated, retrieved, and communicated among the SFAO, college coordinators, and student scholars.

The fact-finding activities also focused on identifying operational challenges encountered in the existing process, including repetitive data entry, difficulty in retrieving records, delays in updating scholar information, manual verification of submitted documents, communication gaps, and challenges in generating consolidated reports. The information obtained from these activities was used to determine which processes could be centralized, automated, or supported through the proposed system.

Furthermore, the researchers documented the types of information and documentary requirements handled by the SFAO, particularly those relevant to scholar identification, academic performance, enrollment status, and scholarship eligibility. These observations informed the design of the system's database structure, document-processing functions, monitoring features, notification mechanisms, and administrative dashboard. The findings from the fact-finding activities were subsequently translated into functional and non-functional system requirements and served as a basis for developing and refining LSPU ScholarHub.

The fact-finding activities provided an empirical understanding of the existing scholarship management environment and ensured that the proposed system was aligned with the actual operational needs of its intended users. Rather than relying solely on assumptions about scholarship administration, the researchers used information obtained from the existing processes to guide the design and development of a system intended to improve the centralization, accessibility, monitoring, verification, communication, and reporting of scholarship-related information.

3.7 Project Design

This section outlines the blueprint of the LSPU ScholarHub. It presents the technical requirements and architectural design, illustrating how the platform was structured to meet administrative needs.

3.7.1 System Development Methodology

The Agile software development model was adopted, structuring development into six iterative phases: (1) Requirements, (2) Design, (3) Development, (4) Testing, (5) Deployment, and (6) Review. Sprint iterations enabled continuous stakeholder feedback and adaptive feature enhancements (Chathmini Jayathilaka , 2020).



Figure 2. Agile Methodology (adapted from Chathmini Jayathilaka , 2020)

The development of LSPU ScholarHub followed the Agile methodology, an iterative approach to software development that organizes work into short, time-boxed cycles called sprints, typically lasting one to four weeks, where each sprint produces a working and testable increment of the system. Unlike traditional Waterfall development that follows a linear sequential process, Agile allows for continuous feedback, incremental development, and adaptive planning throughout the project lifecycle. The following are the development phases:

Phase 1 – Requirements: The planning phase focused on identifying the administrative needs of the LSPU-SCC Scholarship and Financial Assistance Office (SFAO). The researchers conducted in-depth interviews with SFAO staff and student scholars to map current workflows and pinpoint operational bottlenecks, such as manual data handling and fragmented communication. This phase established the project scope,

functional specifications, and non-functional requirements, ensuring the final system would directly address the university's scholarship monitoring needs.

Phase 2 – Designing: The designing phase involved defining the system architecture, creating wireframes for both the student mobile portal and the administrative web dashboard, and designing the relational database schema. The researchers collaborated with SFAO personnel to ensure the interface was intuitive, specifically streamlining the application upload and status-tracking processes. This phase prioritized user-centered design, iteratively refining layouts to minimize administrative cognitive load and maximize accessibility for scholars.

Phase 3 – Development: The development phase focused on building the core modules: Profile Management, AI-Powered OCR/NLP Document Verification, the Conversational AI Chatbot, Multi-Channel Notification Engine, and the Analytics Dashboard. Organized into two-week Agile sprints, the team implemented features from the product backlog using secure coding practices, including version control, unit testing, and modular API integrations to ensure the system was robust, scalable, and secure.

Phase 4 – Testing: The testing phase utilized black-box methodology to validate the system’s functional completeness, technical performance, and security. Rigorous testing sessions were conducted with IT experts to evaluate system stability under load, the accuracy of the AI-powered document parsing, and the reliability of notification dispatches. Simulation testing ensured that the administrative workflows, such as approving applications and generating reports, performed seamlessly according to the established project requirements.

Phase 5 – Deployment: The deployment phase involved launching the ScholarHub platform into the testing/staging environment. This release provided the SFAO staff and a selected group of student scholars access to the live features, including the document upload module, chatbot interactions, and notification alerts. The goal of this phase was to verify real-world system accessibility, ensure seamless database integration, and confirm that all administrative tools were correctly configured for the user evaluation process.

Phase 6 – Review: The review phase concentrated on gathering user feedback and performing the final system refinement. Researchers conducted formal User Acceptance Testing (UAT) sessions, collecting qualitative and quantitative data through questionnaires. Based on this feedback, the researchers prioritized and implemented final system adjustments, addressing specific usability issues, refining the chatbot's FAQ responses, and finalizing the administrative features before the project's culmination.

3.7.2 Hardware and Software Requirements

The hardware and software requirements of LSPU ScholarHub were identified based on its web-based architecture, system functionality, and the operational requirements of its intended users. The selected resources are intended to provide adequate support for the development, deployment, and use of the system while ensuring accessibility, responsive performance, reliable data processing, and integration of the system’s intelligent and communication features.

Table 4. Hardware and Software Requirements

Category	Requirement	Specification/Description	Purpose in the System
Hardware	Administrative Workstation	Desktop computer or laptop with adequate processing capacity,	Used by SFAO personnel to access the

Category	Requirement	Specification/Description	Purpose in the System
		memory, storage, and a modern web browser	administrative dashboard, manage scholar records, review documents, monitor scholarship status, and generate reports
Hardware	Student Device	Desktop computer, laptop, tablet, or smartphone	Enables student scholars to access the scholar portal, submit documents, view scholarship information, and receive system updates
Hardware	Internet Connectivity	Stable and reliable Internet connection	Required for accessing the web-based system, transmitting data, uploading documents, and receiving notifications
Hardware	Server/Hosting Environment	Server or cloud-based hosting environment with sufficient computing resources and storage	Hosts the web application, database, uploaded documents, and system services
Software	Operating System	Compatible desktop or mobile operating system	Provides the environment required to access and operate the web-based system
Software	Web Browser	Modern, standards-compliant web browser such as Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari	Provides access to the administrative dashboard and student portal
Software	Python	Python programming language	Serves as the primary programming language for backend system development
Software	Flask	Python-based web framework	Handles server-side processing, routing, application logic, and communication between the frontend and database
Software	MySQL	Relational Database Management System	Stores and manages scholar profiles, applications, academic information, scholarship records, documents, and system transactions

Category	Requirement	Specification/Description	Purpose in the System
Software	Bootstrap	Responsive frontend framework	Supports the development of a consistent, responsive, and mobile-friendly user interface
Software	OCR Libraries	Integrated Optical Character Recognition libraries	Extracts machine-readable text from uploaded scholarship documents such as CORs and COGs
Software	NLP and Document-Analysis Libraries	Natural Language Processing and document-processing components	Processes extracted document text and assists in identifying relevant fields and information for subsequent verification
Software	Notification APIs	Third-party SMS and email application programming interfaces	Enables automated delivery of announcements, reminders, application updates, and other scholarship-related notifications
Software	Database Connectivity Components	Backend database drivers/connectors	Establishes communication between the Flask application and MySQL database for storing and retrieving system information
Software	Development and Testing Tools	Integrated development environment/code editor and browser-based testing tools	Supports system development, debugging, testing, and maintenance

3.7.3 System Components and Modules

The LSPU ScholarHub is designed as an integrated web-based scholarship management and monitoring system composed of interconnected modules that support the major operational requirements of the Scholarship and Financial Assistance Office (SFAO). Each module performs a specific function while contributing to the overall objective of centralizing scholarship information, streamlining document processing,

improving scholar monitoring, facilitating communication, and supporting administrative decision-making. The major system components and modules are described as follows:

1. AI-Powered Document Verification Module (OCR/NLP). This module facilitates the processing of student-submitted scholarship documents, particularly the Certificate of Registration (COR) and Certificate of Grades (COG). OCR technology is used to extract text and relevant information from uploaded documents, while NLP-based processing assists in identifying and organizing required data fields. The extracted information may then be cross-checked against available student records to support document verification and preliminary assessment. The module is intended to reduce repetitive manual encoding and improve the consistency and efficiency of document processing. Final verification and scholarship decisions remain subject to review and approval by authorized SFAO personnel.

2. Centralized Scholar Database Module. This module serves as the primary repository for scholarship-related information. It organizes and maintains student profiles, scholarship applications, academic information, application history, documentary requirements, and financial assistance records within a centralized database. By consolidating information into a structured repository, the module facilitates efficient data retrieval, updating, monitoring, and report generation while reducing dependence on fragmented records and manual files.

3. Automated Status Monitoring and Analytics Dashboard. This module provides SFAO personnel with a centralized interface for monitoring the status of scholars

and relevant academic information. It supports the tracking of scholarship-related conditions and scholar records and presents summarized information through dashboard-based views and analytics. The module also assists in generating standardized institutional reports that can support administrative review, monitoring, and decision-making.

4. AI Chatbot Module. The AI-enabled chatbot serves as a virtual information assistant for student scholars. It is designed to respond to frequently asked questions and provide information concerning scholarship guidelines, requirements, procedures, deadlines, and other commonly requested scholarship-related information through natural-language interaction. By providing an accessible source of information, the chatbot is intended to reduce repetitive inquiries handled manually by SFAO personnel and improve the accessibility of scholarship information beyond regular office transactions.

5. Multi-Channel Notification Module. This module manages the automated distribution of scholarship-related communications through multiple channels, including SMS, email, and in-system notifications. It supports the dissemination of announcements, application updates, reminders, deadlines, and other critical information to intended recipients. The integration of multiple notification channels provides an additional mechanism for ensuring that relevant scholarship information reaches users in a timely and accessible manner.

6. Role-Based Access Control (RBAC) and Security Module. This module regulates access to system resources according to the roles and responsibilities assigned to

users. It establishes appropriate permission levels for administrators, SFAO personnel, coordinators, student scholars, and other authorized users, thereby restricting access to sensitive records and administrative functions. The module is particularly important for protecting confidential scholarship and student information by ensuring that users can access only the features and data necessary for their authorized functions.

Collectively, these components establish an integrated system architecture in which document processing, centralized information management, scholar monitoring, intelligent assistance, automated communication, and access control operate as interconnected functions. The modular structure also allows each component to address a specific operational requirement while contributing to the overall efficiency, accessibility, security, and usability of LSPU ScholarHub.

3.7.4 Database Structure

This section presents the logical design of the system's database using an Entity-Relationship (ER) Diagram, as shown in Figure 3.1. The database structure of LSPU ScholarHub is architected using PostgreSQL to centralize scholar profiles, application workflows, academic status monitoring, and notification logs for the Scholarship and Financial Assistance Office (SFAO) of Laguna State Polytechnic University - Santa Cruz Campus (LSPU-SCC). The schema is normalized to ensure data integrity across key entities, supporting seamless retrieval of student records and efficient processing of administrative scholarship tasks

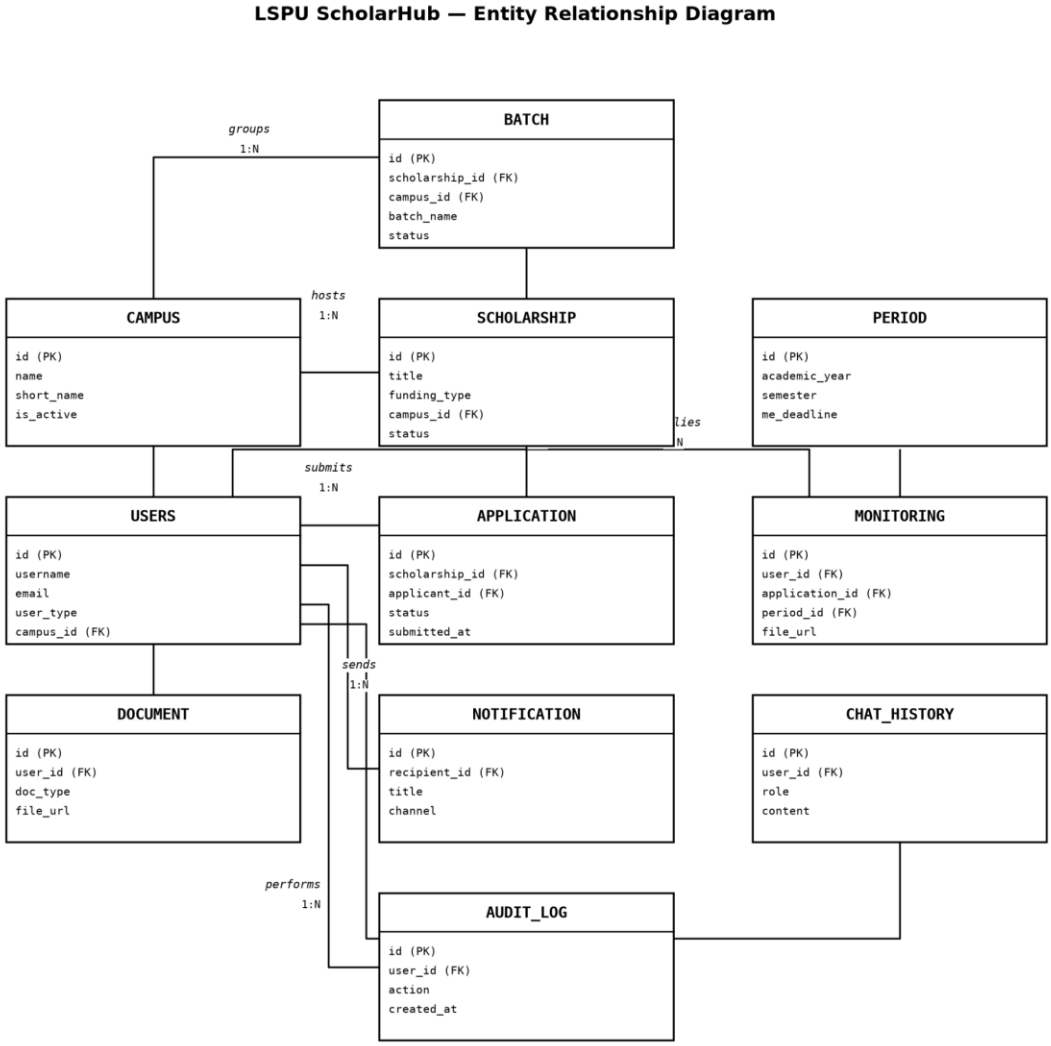


Figure 3. Entity-Relationship Diagram (ERD)

Image generated using FreeBuff by James Grugett (September 2026) with the prompt:need an erd diagram of the system, keep it simple, clean, and make sure that everything is clearly visible

To ensure data integrity and normalized operational workflows, the relational schema directly supports the major functional features evaluated in this study:

- Scholar Profile and Application Management (Features 1 & 2): The system core captures student data through REGISTRATION_PROFILES. Once a student

submits an application, the record links to uploaded documents stored in the SCHOLAR_RECORDS table, which are then processed by the OCR engine for data extraction and validation.

- Academic Status Monitoring (Feature 3): Administrative tracking is governed by ACADEMIC_STATUS logs, which associate scholar profiles with their current standing (Active, Graduated, Stopped) and verify uploaded COG data against institutional requirements.
- AI Chatbot and FAQ Logic (Feature 4): Student support is facilitated through CHAT_LOGS and FAQ_KNOWLEDGE_BASE tables, allowing the system to reference historical inquiries and provide automated, NLP-driven responses for scholarship guidelines.
- Notification and Alerting Engine (Feature 5): Automated outreach is managed through NOTIFICATION_LOGS, capturing timestamps and dispatch statuses for SMS, Email, and in-system alerts triggered by status updates or deadline reminders.
- Administrative Reporting (Feature 7): Operational data is aggregated into ANALYTICAL_REPORTS, allowing SFAO staff to export standardized summaries of scholar cohorts and program distributions based on normalized database queries.

3.7.5 User Interface Design

The user interface (UI) of LSPU ScholarHub was developed using a User-Centered Design (UCD) approach, wherein the design and organization of system interfaces were guided by the needs, tasks, and workflows of the intended users. The approach recognizes the distinct requirements of student scholars and Scholarship and Financial Assistance

Office (SFAO) personnel and translates these requirements into role-specific interfaces and functions. Usability and accessibility principles were incorporated throughout the design to promote ease of use, efficient task completion, consistency, readability, and accessibility across different devices. Particular emphasis was placed on clear information presentation, simplified navigation, responsive layouts, consistent interface elements, and reduced cognitive load for users performing frequent scholarship-related transactions.

The Scholar Mobile Portal was developed using Flutter to support cross-platform accessibility across mobile devices. Consistent with the user-centered approach, the portal was structured around the primary tasks performed by student scholars, particularly accessing scholarship information, submitting documentary requirements, monitoring application status, obtaining assistance, and receiving updates. The dashboard provides direct access to essential functions, including document uploading, scholarship status monitoring, the AI chatbot assistant, and a notification center for application updates and scholarship-related announcements. The interface was intentionally simplified to reduce unnecessary navigation and ensure that frequently used functions can be accessed with minimal steps. The design therefore prioritizes accessibility and task efficiency for student users who may access the system using different mobile devices and varying screen sizes.

The SFAO Admin Dashboard was developed using Next.js and designed around the operational workflow of administrative users. Unlike the student-facing portal, which emphasizes simplicity and accessibility, the administrative interface prioritizes information organization, processing efficiency, monitoring, verification, and reporting. The dashboard provides a centralized workspace containing a pending application queue, document verification panel with AI-assisted data suggestions, student record management functions,

and an analytics dashboard for monitoring scholarship information and supporting institutional report generation. The role-specific organization of these functions allows SFAO personnel to access relevant information according to their responsibilities and reduces unnecessary navigation between separate system components. AI-generated or extracted information is presented as assistance for administrative review rather than as a replacement for authorized personnel's verification and decision-making.

Following the UCD approach, both interfaces were designed using common usability considerations, including consistency, visibility of system status, simplicity, learnability, efficient navigation, and error prevention. Navigation elements, labels, buttons, status indicators, forms, and information panels were presented consistently to improve user familiarity and reduce confusion. The interfaces also use responsive layouts to accommodate different screen sizes and incorporate dark and light display modes to support different viewing preferences and environments. Information and functions are presented according to the user's role so that users are exposed primarily to the features relevant to their tasks.

The UI design process followed a user-centered, role-specific, and iterative design perspective, in which the requirements of student scholars and SFAO personnel informed the organization and presentation of system functions. The resulting interfaces integrate accessibility and usability considerations with the technical capabilities of Flutter and Next.js. This approach supports the overall objective of LSPU ScholarHub to provide an accessible, efficient, and user-centered platform for scholarship application, document submission and processing, status monitoring, communication, and administrative management.

3.7.6 System Communication and Interaction

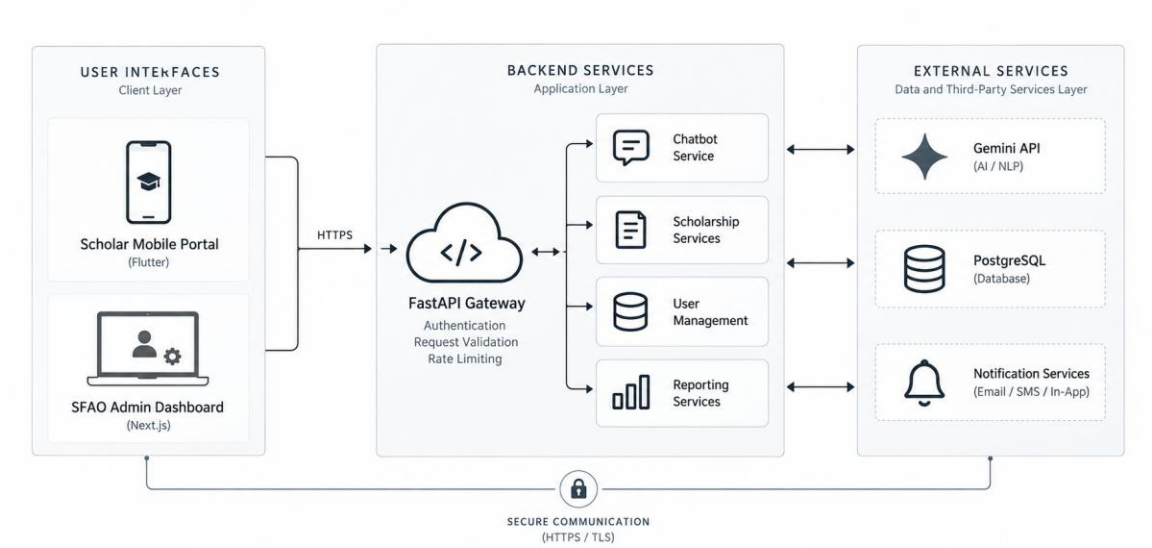


Figure 4. System Communication and Interaction

The LSPU ScholarHub employs a layered communication architecture that defines how information is exchanged among the user interfaces, application services, database, AI service, and notification services. As illustrated in Figure 4, the architecture is organized into three major layers: the User Interface Layer, the Application Layer, and the Data and External Services Layer. This separation establishes clear communication boundaries among system components and provides a structured mechanism for processing requests, managing data, integrating external services, and delivering responses to users.

The User Interface Layer consists of two primary client applications: the Scholar Mobile Portal, developed using Flutter, and the SFAO Admin Dashboard, developed using Next.js. The Scholar Mobile Portal serves student scholars who perform activities such as submitting documents, monitoring scholarship status, accessing the AI chatbot, and

receiving notifications. In contrast, the SFAO Admin Dashboard provides authorized personnel with functions for application processing, document verification, scholar record management, monitoring, analytics, and reporting. Requests originating from either interface are transmitted through secure communication channels to the FastAPI Gateway, which serves as the principal entry point to the backend application.

The FastAPI Gateway functions as the intermediary between the client applications and the system's backend services. It performs essential application-level controls, including authentication, request validation, and rate limiting, before forwarding authorized requests to the appropriate service. This arrangement prevents client applications from directly accessing internal services and provides a centralized point for managing and controlling application requests. The gateway subsequently communicates with the system's major backend services, including Chatbot Service, Scholarship Services, User Management, and Reporting Services. This service-oriented organization allows individual system functions to operate within defined responsibilities while maintaining controlled communication among components.

The Chatbot Service provides the communication pathway between the Scholar Mobile Portal and the external Gemini API. When a scholar submits a natural-language inquiry, the request is transmitted through the application gateway to the chatbot service, which communicates with the Gemini API for AI-assisted natural-language processing. The processed interaction is then returned through the application layer to the scholar portal. The chatbot is therefore integrated as an assistance mechanism within the system architecture while maintaining the application layer as the controlled intermediary for

external AI communication. Responses are intended to provide scholarship-related information based on the system's configured knowledge sources and should not be treated as a replacement for official administrative decisions.

The Scholarship Services, User Management, and Reporting Services communicate with the PostgreSQL database, which serves as the centralized data repository of LSPU ScholarHub. Scholarship applications, scholar profiles, academic information, documentary records, status information, and relevant system transactions are stored and retrieved through the application services. Access to database records is governed by the user's authorized role, with appropriate access-control mechanisms, including Row-Level Security (RLS) where implemented, to restrict users to information and operations appropriate to their assigned privileges. Thus, students and administrative personnel are provided with different levels of access based on their respective roles and responsibilities.

The architecture also incorporates a Notification Services component for distributing scholarship-related communications. The notification service may integrate Firebase Cloud Messaging (FCM) for mobile or in-system notifications and appropriate SMTP and SMS gateways for email and mobile-message delivery. Through this component, the system can disseminate application updates, announcements, reminders, deadlines, and other relevant scholarship information to intended recipients. Centralizing notification processing within the application architecture allows communication events to be generated from system transactions while maintaining a consistent mechanism for delivering notifications through multiple channels.

Security is incorporated across the communication pathways, particularly in the transmission of requests and responses between the client applications, backend services, database, and external services. HTTPS/TLS encryption is used where applicable to protect information transmitted across network connections, while authentication and authorization mechanisms regulate access to system resources. These controls are particularly important because LSPU ScholarHub processes student and scholarship-related information that requires appropriate protection. The architecture therefore supports the study's objective of implementing security measures consistent with the Data Privacy Act of 2012 (Republic Act No. 10173) and its data-protection requirements.

The communication flow of LSPU ScholarHub follows a structured, bidirectional pathway in which user requests are securely transmitted from the user interfaces through the FastAPI gateway and appropriate application services to the database or external services, with processed responses subsequently returned to the requesting interface.. The layered structure ensures that each component has a defined communication responsibility while allowing the system to integrate AI processing, centralized data management, reporting, and multi-channel notifications. By establishing controlled service boundaries and incorporating authentication, authorization, encrypted communication, and role-based access mechanisms, the architecture provides a structured foundation for the secure, maintainable, and scalable operation of the proposed scholarship management and monitoring system.

3.8 System Testing and Evaluation Procedure

System testing and evaluation were conducted using a structured, sequential protocol to verify the technical functionality of LSPU ScholarHub and determine its

acceptability among its intended users. The testing activities were aligned with the system requirements and research objectives and progressed from functional verification to scenario-based simulation and user acceptance evaluation. This approach enabled the researchers to assess whether the system's major components performed according to their intended specifications before obtaining users' assessments of the system. Functional verification was performed through Black-Box testing, while user acceptance was assessed using the Technology Acceptance Model (TAM), focusing on perceived usefulness, perceived ease of use, and behavioral intention to use the system.

The testing and evaluation proceeded in three phases. Phase 1 involved Black-Box functional testing conducted by IT testers, who examined the system's inputs, outputs, and observable behavior without examining its internal source code. Core functions, including OCR-based document processing, chatbot interaction, authentication, scholarship management, monitoring, notifications, and reporting, were evaluated against their specified requirements. Phase 2 consisted of simulation testing with SFAO personnel and intended users at the LSPU-SCC campus, who performed representative scholarship-management scenarios using sample applications and documentary requirements. This phase assessed the integration of system functions within realistic operational workflows. Phase 3 involved user acceptance evaluation, wherein the identified respondent groups used the completed system and subsequently answered the TAM-based questionnaire. The results provided quantitative evidence of users' perceptions regarding the system's usefulness, ease of use, and intention to use it for scholarship-related activities.

Collectively, these phases provided a progressive basis for determining the system's functional correctness, workflow suitability, and user acceptance.

3.8.1 Testing Plan

The first phase involved black-box functional testing conducted by IT testers to determine whether the developed system functions according to its specified requirements. Testers evaluated system inputs, processes, outputs, and expected behaviors without examining the underlying source code. Particular attention was given to core functionalities such as user authentication, document uploading and OCR-based processing, data extraction and verification, scholarship status monitoring, chatbot interaction, notification delivery, and report generation. Test cases were prepared based on the defined functional requirements, and observed outputs were compared with the expected results to identify functional errors, inconsistencies, or incomplete features.

3.8.2 Testing Data

Testing data consisted of synthetic student records, academic documents (COR/COG templates), and FAQ datasets representing realistic scholarship inquiries. Test scholar accounts were created for various academic statuses (Active, Regular, Irregular, Graduating) to validate the monitoring engine, while administrative accounts were created to test application approval/rejection workflows. Sample PDF and image files were prepared with varying qualities to test the robustness of the AI-OCR extraction module in identifying student IDs and GWA scores.

3.8.3 Testing Procedure

Black-Box functional testing was conducted as the initial testing activity by IT testers who executed all nine predefined test cases in a controlled environment. The test cases were performed sequentially to verify the expected inputs, system responses, and outputs of the identified functions, with each test result recorded as either Pass or Fail. Testing was conducted through a combination of online and face-to-face sessions, allowing the researchers to accommodate the availability and testing conditions of the evaluators while maintaining a standardized testing procedure. Before completing the testing instrument, each tester was required to indicate the appropriate tester information and credentials to establish the identity and role of the evaluator.

Following the functional testing, simulation testing was conducted with SFAO personnel at the LSPU-SCC office using representative scholarship-management scenarios. The participants performed actual system workflows, including verifying uploaded documents, reviewing application and scholar statuses, managing scholarship records, and preparing or managing notification broadcasts. The combination of online and face-to-face implementation allowed the researchers to observe system use under both remote and actual office-based conditions. The completed testing sheets were collected, reviewed, and consolidated, and the results were summarized to determine the system's operational success rate and identify any functions requiring correction, refinement, or further validation.

3.8.4 Software Testing Method

System testing determined whether the completed LSPU ScholarHub system met its functional and non-functional requirements and operated as intended under appropriate

conditions. Black-Box functional testing was selected as the primary testing method because it evaluates system behavior from the user perspective without examining internal code structures, making it appropriate for validating the core modules such as OCR-based document verification, AI chatbot responsiveness, and multi-channel notification delivery identified in the research objectives.

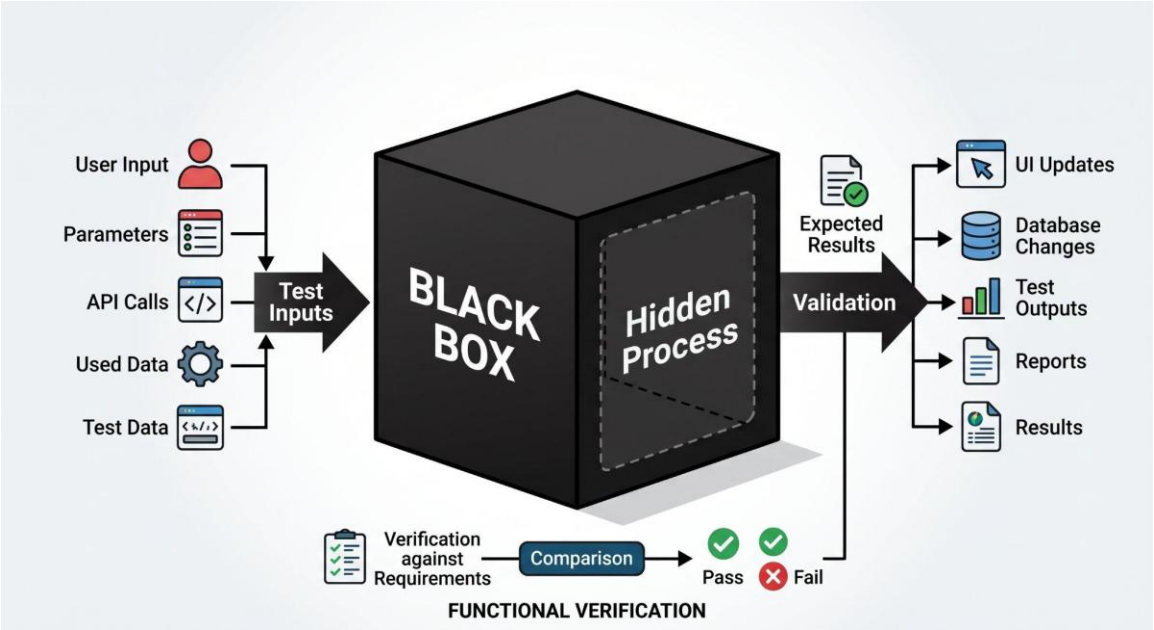


Figure 5. Black-Box Testing Model

Image generated using Gemini by Google (September 2026) with the prompt: "Simple diagram showing 'Input' arrow entering a 3D black box, and 'Output' arrow exiting it, clean white background, modern flat icon style."

The researchers utilized this methodology to assess the LSPU ScholarHub functionality by providing test inputs such as student registration data, sample academic document uploads, and chatbot query strings and observing the system outputs to ensure data accuracy, administrative integrity, and module responsiveness without examining the underlying code.

3.8.5 Test Cases and Test Scenarios

Test cases were documented in a structured format containing the feature or requirement being tested, test conditions, expected results, actual results, and test status. The detailed test cases and scenarios (TC-01 through TC-16), which map directly to the system requirements of LSPU ScholarHub including scholar registration, profile management, document uploading, and admin dashboard reporting are presented in the APPENDIX.

Nine test cases were developed based on the functional requirements of LSPU ScholarHub: (TC-01) Scholar Homepage and Navigation, (TC-02) Scholar Registration with Document Upload, (TC-03) AI-Powered OCR Document Verification, (TC-04) Automated Academic Status Tracking, (TC-05) AI-Enabled Chatbot FAQ Interaction, (TC-06) Multi-Channel Notification Dispatch (Email/SMS), (TC-07) Admin Dashboard and Application Management, (TC-08) Analytics Reporting and Data Export, and (TC-09) Role-Based Access Control and Security. Each test case included preconditions, step-by-step procedures, expected results, and pass/fail indicators.

3.8.6 Testing Instrument

The testing instrument utilized in the study was a structured Black-Box Functional Testing Sheet developed to systematically evaluate the observable behavior and functional performance of LSPU ScholarHub. The instrument consisted of two major sections: tester information and test case evaluation. The tester information section captured the evaluator's name or assigned identification, date of testing, job role, and years of relevant work experience. The test case evaluation section documented each predefined test scenario,

including the required preconditions, sequential test procedures, expected results, actual results observed during execution, and the corresponding Pass/Fail determination.

The instrument was designed to provide a standardized and traceable procedure for verifying whether each system function performed according to its specified requirements.

Test cases covered the major functional components of LSPU ScholarHub, including user authentication, scholarship application processing, document uploading and OCR-assisted verification, scholar-status monitoring, chatbot functionality, notification services, and reporting. For each test case, the tester followed the prescribed sequence of actions and compared the observed system output with the expected result. A test case was recorded as Pass when the actual system behavior corresponded to the defined expected result and as Fail when a discrepancy or functional error was observed. The completed testing sheets served as the primary documentation for functional verification and provided the basis for consolidating the test results and determining the system's overall functional performance.

3.8.7 Validation of Testing Instrument

The Black-Box testing instrument underwent content validation by the thesis adviser, Mark P. Bernardino, and a panel of IT experts who reviewed the test cases for alignment with the documented system requirements and research objectives. The adviser verified that each test scenario appropriately covered the identified major features (e.g., chatbot logic, OCR document parsing, and dashboard analytics) and that the expected results were consistent with the system specifications. Revisions were made based on adviser feedback to ensure clarity, completeness, and appropriateness of the testing procedures before actual test execution.

3.8.9 Interpretation of Test Results

Test results were recorded as pass or fail for each individual step across all test cases. A feature was considered fully functional when all its test steps received a pass status. Failed steps were documented with the actual result observed and compared against the expected result to identify discrepancies or required debugging. The overall system pass rate was computed by dividing the total number of passed test steps by the total number of test steps and expressing it as a percentage. A module or feature was considered functionally acceptable when it achieved a pass rate of 80 percent or higher across all test cases.

3.8.10 System User Evaluation

System user evaluation determined how intended users perceived the developed LSPU ScholarHub system in terms of usability, perceived usefulness, and behavioral intention to adopt. The evaluation utilized a structured questionnaire administered to two respondent groups: scholarship office personnel and student scholars, following system demonstration and hands-on testing at the Scholarship and Financial Assistance Office (SFAO) at LSPU-SCC.

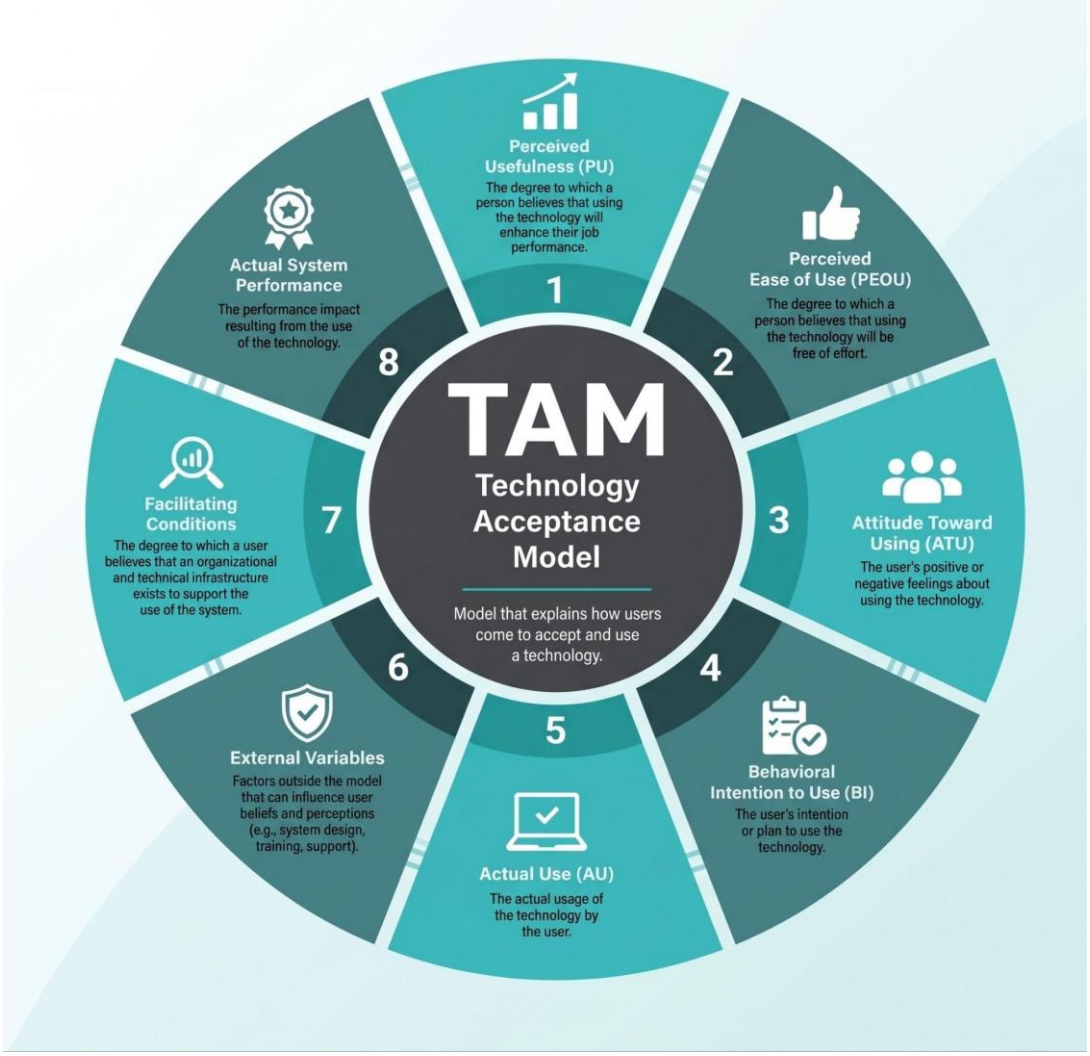


Figure 6. Technology Acceptance Model (TAM)

Image generated using Gemini by Google (September 2026) with the prompt: "Detailed infographic, Technology Acceptance Model (TAM), 8-numbered circular diagram, central hub 'TAM' description, peripheral icons and definitions for PU, PEOU, ATU, BI, AU, External Variables, Facilitating Conditions, System Performance, professional teal and grey colors."

In the context of scholarship management systems like LSPU ScholarHub, TAM measures how Perceived Usefulness and Perceived Ease of Use influence SFAO staff and student scholars' behavioral intention to adopt the platform for efficient scholarship administration and monitoring.

3.8.11 User Evaluation Instrument

The user evaluation instrument was a Technology Acceptance Questionnaire consisting of 18 items distributed across six TAM constructs: Quality Factors (3 items), Perceived Ease of Use (3 items), Perceived Usefulness (3 items), Attitude Toward Using (3 items), Behavioral Intention to Use (3 items), and Experience/Actual Use (3 items). Each item was rated using a 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The questionnaire also included a demographic profile section capturing age, sex, and respondent category, as well as an open-ended section for comments and suggestions.

Table 5. Five-Point Rating Scale

Scale	Verbal Interpretation
4.21 – 5.00	Strongly Agree
3.41 – 4.20	Agree
2.61 - 3.40	Neutral
1.81 – 2.60	Disagree
1.00 – 1.80	Strongly Disagree

Mean scores were interpreted using a 5-point Likert scale continuum: 4.21 to 5.00 (Strongly Agree / Excellent), 3.41 to 4.20 (Agree / Very Good), 2.61 to 3.40 (Neutral / Fair), 1.81 to 2.60 (Disagree / Poor), and 1.00 to 1.80 (Strongly Disagree / Very Poor). The weighted mean was computed for each questionnaire item to determine the overall level of agreement across respondent groups, while standard deviation was used to measure the consistency of responses. These measures also helped identify significant differences between respondent categories.

3.8.12 Instrument Source and Validation

The TAM questionnaire items were adapted from established Technology Acceptance Model literature, particularly the works of Davis (1989) and Pradana and Salamat (2025), and contextualized for the ScholarHub system and the LSPU Santa Cruz Campus academic setting. To establish the validity of the research instrument, a formal validation process was conducted through a scheduled online validation session via Zoom with the thesis adviser and panel members: Phase 1: Expert Presentation and Review. The researchers conducted a live presentation via Zoom to present the proposed questionnaire items alongside the ScholarHub system features. During this session, the thesis adviser and expert panel evaluated the instrument's face and content validity, focusing on item clarity, construct alignment, language appropriateness, and contextual relevance to both student applicants and administrative users. Phase 2: Refinement and Approval. Based on the real-time feedback, suggestions, and corrections provided during the Zoom meeting, the researchers systematically revised the questionnaire items. Specific refinements included simplifying technical jargon, clarifying feature-specific prompts (such as the AI document verification and chatbot modules), and ensuring uniform Likert-scale options. The finalized instrument was subsequently re-submitted and approved prior to floating the survey.

3.8.13 User Evaluation Procedure

The evaluation instruments were administered through tailored procedures for each target respondent group using a hybrid approach of synchronous online sessions via Zoom and guided hands-on testing. For the SFAO staff and university administrators, the researchers conducted a structured system demonstration and simulation exercise via Zoom. During this session, the researchers presented a live walkthrough of the Next.js

administrative dashboard and guided the respondents through simulated administrative scenarios including AI document verification, applicant status updating, and report generation before they completed the digital evaluation questionnaire. For student scholars and applicants, a virtual orientation and demonstration session was held via Zoom to introduce the features of the ScholarHub platform, followed by guided self-paced interaction with the portal. Across all target groups, participants were thoroughly briefed on the study's objectives at the start of the Zoom session, explicitly informed of their voluntary participation and right to withdraw at any time, and assured of strict data confidentiality in compliance with data privacy standards prior to answering the evaluation instrument.

3.8.14 Ethical Considerations

Ethical considerations were strictly observed throughout the research process. Informed consent was obtained from all participants prior to data collection, and respondents were assured of voluntary participation, confidentiality, and the right to withdraw at any time without consequence. All collected data were used solely for academic purposes and were handled in strict compliance with the Data Privacy Act of 2012 (R.A. 10173). No personally identifiable information was disclosed in any report or publication.

3.9 Statistical Treatment

This section outlines the methodology employed by the researchers to evaluate the user acceptance of the LSPU ScholarHub system. It details the analysis procedures used to assess system impact, functionality, and user satisfaction. As the project aims to modernize

scholarship administration and student support services at LSPU-SCC, the successful implementation of the system is validated through quantitative analysis of the collected user feedback.

The researchers used the weighted mean to calculate statistical findings. The Likert scale was employed to gauge respondents' opinions effectively, chosen for its simplicity and ability to capture a broad range of attitudes toward the system. Respondents expressed their level of agreement with various statements related to the system's acceptability, functionality, and ease of use.

3.13.2 Frequency and Percentage Distribution

The Frequency and Percentage Distribution method was utilized to analyze the demographic profile and respondent distribution. This statistical tool determines the proportion of responses in each category, providing a clear overview of the participant breakdown (e.g., SFAO Staff vs. Student Scholars).

$$P = \left(\frac{f}{N}\right) \times 100$$

Equation 2. *Frequency and Percentage Distribution*

Where:

P = Percentage of responses
f = Frequency of responses in a given category
N = Total number of respondents

3.13.2 Mean Formula

The Mean Formula was applied to describe the average participant perception regarding key elements of functionality, usability, and effectiveness of the LSPU ScholarHub system.

The arithmetic mean can be expressed using the following equation:

$$\boxed{\bar{X} = \frac{\sum X}{N}}$$

Equation 3. *Arithmetic Mean*

Where:

- $\bar{X}$= Mean
- $\sum X$= Sum of all observed values
- N= Total number of observations or respondents

For a **weighted mean**, commonly used for Likert-scale evaluation:

$$\boxed{WM = \frac{\sum (fX)}{N}}$$

Equation 4. *Weighted Mean*

Where:

- WM= Weighted Mean
- f= Frequency of each response
- X= Numerical weight assigned to each response
- N= Total number of responses

For example, for a 5-point scale:

$$WM = \frac{(f_5 \times 5) + (f_4 \times 4) + (f_3 \times 3) + (f_2 \times 2) + (f_1 \times 1)}{N}$$

Equation 5. *Weighted Mean for Likert Scale*

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the empirical outcomes, implementation mechanics, and performance evaluations gathered during the deployment and assessment of LSPU ScholarHub, an AI-driven scholarship management and monitoring platform featuring interactive conversational support and automated multi-channel messaging. The data is systematically organized to directly reflect the specific goals and development phases outlined in the project framework. To maintain logical structure, tabular summaries, architectural diagrams, and interface captures are positioned as temporary placeholders, to be fully populated once final evaluation datasets and operational screen captures are compiled.

4.1 System Architecture and Development Results

This section details the concrete outcomes derived from designing, engineering, and deploying LSPU ScholarHub to satisfy Objective 1. Transitioning from theoretical frameworks to practical implementation, this part illustrates how the traditional, manual workflow of the Scholarship and Financial Assistance Office (SFAO) at LSPU-SCC was converted into an integrated web platform. Guided by requirements gathered from administrative staff and student scholars, development was carried out using an Agile iterative approach. The resulting software features a web dashboard for office administrators and a scholar portal built on a modern stack, connected to a relational database. The following sub-sections delineate the system's architecture, hardware and

software specifications, relational database schema, and the tailored user interfaces engineered for each user role.

4.1.1 System Architecture

The System Architecture presents the overall structural design and integration of the LSPU ScholarHub platform across its three core layers: presentation, application, and data. It illustrates the interaction between student scholar interfaces, administrative dashboards, the Flask web application framework, MySQL database instances, third-party communication gateways (SMS and Email), and AI processing modules (OCR and NLP). This architectural framework demonstrates how ScholarHub fulfills its functional requirements such as automated document parsing, real-time status monitoring, and chatbot inquiry resolution while adhering to non-functional constraints, including role-based access control, CSRF security, rate limiting, and data privacy compliance.

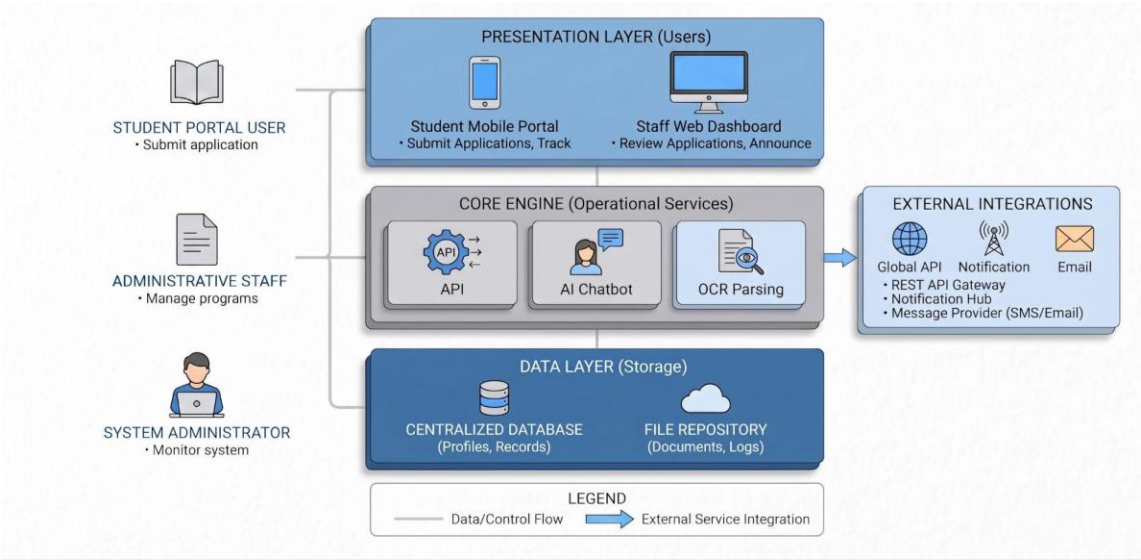


Figure 7. System Architecture

Image generated using Gemini by Google (September 2026) with the prompt: "A clean, modern vector system architecture on a white background. Minimalist corporate tech style with simple flat line icons, light grey 3D-effect cards, soft drop shadows, and a clean blue and grey color scheme."

The technical implementation of LSPU ScholarHub adopts a three-tier system architecture that organizes the system into distinct but interconnected layers for user interaction, application processing, and data management. As presented in Figure X, the architecture illustrates how user interfaces communicate with the core processing components and centralized data repository to support the major functions of scholarship application, document processing, monitoring, communication, and information management. The layered structure establishes a clear separation of responsibilities among system components while enabling coordinated data flow throughout the platform.

The Presentation Layer serves as the user-facing component of the system and provides role-specific interfaces for its primary users. The SFAO Administrative Dashboard provides authorized personnel with access to scholarship applications, scholar records, document verification, monitoring, analytics, and reporting functions. The Scholar Mobile Portal, in contrast, provides student scholars with access to application-related transactions, document submission, scholarship-status information, notifications, and the AI chatbot. The separation of these interfaces ensures that users are presented with functions appropriate to their respective roles while maintaining a consistent and accessible interaction with the system.

The Core Engine or Application Layer performs the principal processing and business functions of LSPU ScholarHub. It integrates the Scholarship Management Module, AI Chatbot, OCR and Document Processing Module, Eligibility and Verification Functions, Status Monitoring and Analytics, and Notification Engine. These components receive and process requests originating from the presentation layer and coordinate the

corresponding system operations. The OCR and document-processing component supports the extraction of information from submitted scholarship documents, while the AI chatbot processes scholarship-related inquiries. The eligibility and verification functions assist in evaluating submitted information against predefined requirements, with final administrative decisions remaining under the authority of designated SFAO personnel. The notification engine facilitates the distribution of system-generated updates through the configured communication channels.

The Data Layer provides the persistent storage required by the system through a centralized relational database. It maintains structured scholarship information, including scholar profiles, application records, academic information, documentary requirements, extracted document data, scholarship-status records, and notification or transaction logs. The centralized data repository allows the different application components to access consistent and synchronized information while supporting efficient retrieval, updating, monitoring, and report generation. Appropriate authentication, authorization, and data-protection mechanisms are incorporated to restrict access to information according to user roles.

As shown in Figure 7, the architecture establishes a structured flow of information from the Presentation Layer through the Application/Core Engine to the Data Layer, with processed data and corresponding system responses subsequently routed back to the appropriate user interface. The architecture demonstrates how the major components of LSPU ScholarHub operate as an integrated platform rather than as independent functions. This design supports the system's objectives of centralizing scholarship information, automating selected document-processing activities, facilitating scholar monitoring,

improving communication, and supporting administrative decision-making while providing a structured foundation for system security, maintainability, and future expansion.

4.1.2 Hardware and Software Requirements

To ensure that LSPU ScholarHub can be deployed smoothly without imposing excessive infrastructural burdens on the university, the platform was engineered to operate on standard, readily available computing and mobile hardware. As detailed in Table _, the hardware requirements specify the minimum technical thresholds needed for seamless day-to-day operation. For administrative oversight and management tasks, a standard laptop or desktop computer equipped with at least a dual-core processor and 4 GB of RAM is required. For mobile and web users including student scholars a modern web browser or smartphone running Android 11.0 or higher is necessary to support mobile portal access, document uploads, and real-time push alerts. All listed hardware specifications are confirmed to be existing and fully accessible within the operational environment of the university.

Complementing the physical hardware, Table _ outlines the software prerequisites essential for system execution across platforms. The frontend web interface is built using HTML5, CSS3, and JavaScript, ensuring a responsive user experience across modern desktop browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox. Backend processing logic is powered by PHP or Node.js to manage API services, while persistent data and application records are structured within a MySQL/MariaDB database. Additionally, mobile-facing operations support Android 11.0+ environments to host mobile web features, while a stable internet connection is required across all endpoints to

facilitate real-time data synchronization, multi-channel notification dispatches, and OCR/NLP processing services.

Table 6. Hardware Requirements and Availability

Hardware Requirement	Minimum Specification	Availability	Purpose/Use
Laptop or Desktop Computer	Dual-core processor or higher; at least 4 GB RAM	Available	Access to the SFAO administrative dashboard and web-based system functions
Smartphone	Android 11.0 or higher	Available	Access to the mobile portal and scholarship-related services
Internet Connection	Stable and reliable connection	Available	Required for system access, data transmission, document submission, and notifications
Processor	Dual-core or higher	Available	Supports basic system operation and web-based processing
Memory (RAM)	At least 4 GB or higher	Available	Supports web-browser operation and simultaneous system activities

Table shows that all identified hardware requirements are available in the intended implementation environment. The available resources meet the minimum requirements for accessing and operating LSPU ScholarHub, including the administrative dashboard through computers and the mobile portal through compatible smartphones. The availability of stable Internet connectivity, dual-core processors or higher, and at least 4 GB of RAM further indicates that the existing infrastructure is sufficient to support the basic operational requirements of the system without requiring substantial additional hardware resources.

Table 7. Software Requirements and Availability

Software Requirement	Minimum/Required Specification	Availability	System Use
Web Browser	Google Chrome, Microsoft Edge, or Mozilla Firefox	Available	Provides access to the LSPU ScholarHub web-based interface and administrative dashboard
Internet Connection	Stable and reliable Internet connectivity	Available	Supports system access, data transmission, document uploading, real-time updates, and notification services

Software Requirement	Minimum/Required Specification	Availability	System Use
Android Operating System	Android 11.0 or higher	Available	Supports access to the mobile portal and mobile-based scholarship services

Table 6 shows that all identified software requirements are available within the intended implementation environment. The availability of compatible web browsers supports access to the web-based system, while a stable Internet connection enables users to perform online transactions and access real-time system services. In addition, the availability of Android 11.0 or higher supports mobile access for student users. These results indicate that the existing software environment is sufficient to support the basic deployment and operation of LSPU ScholarHub.

4.1.3 Database Structure

The backend data architecture of LSPU ScholarHub is hosted on a relational PostgreSQL database deployed via Supabase, engineered to maintain high data integrity, security, and real-time synchronization across scholarship management operations. The database schema is structured to systematically capture, process, and store all operational data generated by student applicants, scholars, administrators, and automated system services.

The database design revolves around several major functional schemas and entities. The User Management Schema securely stores scholar profiles, student and administrator accounts, and authentication credentials alongside role-based clearance permissions. The Scholarship and Application Schema manages records for available scholarship programs, student application submissions, and real-time status tracking throughout the approval

workflow. Furthermore, the Academic and Document Repository maintains stored OCR-parsed data extracted from uploaded Certificates of Registration (COR) and Certificates of Grades (COG), along with student grade records and automated academic eligibility determinations. Finally, the Notification and Audit Logs track multi-channel communication dispatches (SMS and Email) and record administrative audit trails to ensure system transparency and compliance.

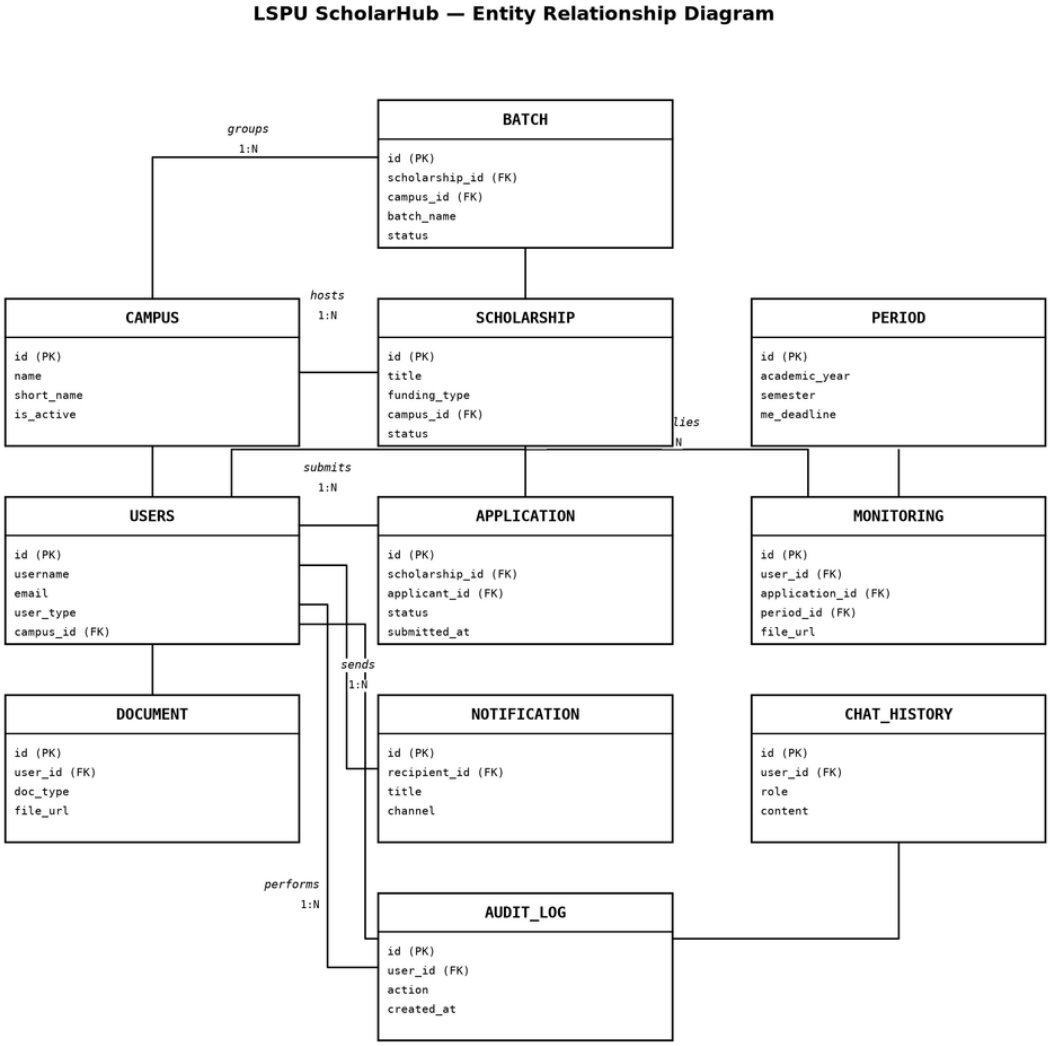


Figure 8. Database Schema

This figure presents the database schema of LSPU ScholarHub, illustrating the logical organization and relationships among the major data entities that support the system's scholarship management and monitoring functions. The schema shows how user and role information is associated with scholarship applications, scholar records, academic information, submitted documentary requirements, eligibility-related data, notifications, and system transaction records. The relationships among these entities establish a

structured flow of information, allowing data generated from application submission and document processing to be associated with the corresponding student or scholar and subsequently utilized for monitoring, preliminary eligibility assessment, notification, and reporting. The schema also demonstrates the centralized nature of the database, wherein related scholarship information is maintained within interconnected tables rather than separate and fragmented records. Through this relational structure, LSPU ScholarHub is able to maintain consistent data relationships, facilitate authorized retrieval and updating of records, and support the coordinated operation of its major system modules.

4.1.4 User-Interface and System Modules

The user-interface design of LSPU ScholarHub is purposefully segregated into role-specific platforms to ensure secure, streamlined operations for Scholarship and Financial Assistance Office (SFAO) staff, college coordinators, and student scholars. The web-based administrative dashboard caters to SFAO staff and system administrators for overseeing scholarship applications, evaluating student documents, and managing system announcements, while the scholar mobile portal provides accessible, task-optimized touchpoints for scholars to monitor their status, upload requirements, receive real-time updates, and utilize AI-enabled support services.

4.1.4.1 Administrator and SFAO Web Interface

The administrative dashboard serves as the central command hub for scholarship management and oversight. SFAO officers can evaluate incoming student applications, review OCR-parsed academic documents (COR and COG), monitor academic eligibility statuses, and broadcast campus-wide announcements. System security and data integrity are enforced through Role-Based Access Control (RBAC) policies, restricting sensitive

administrative privileges and system configurations exclusively to authorized SFAO staff and system administrators.

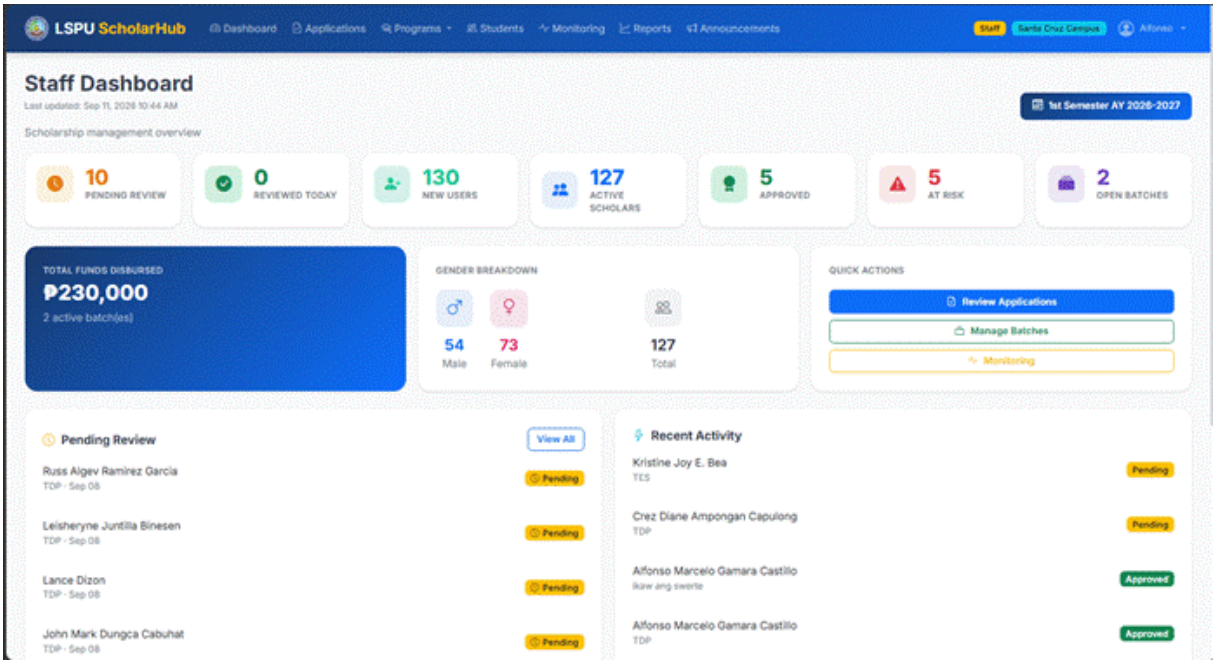


Figure 9. User Interface of Administrator and SFAO Staff - Web Dashboard

Figure 9 presents the Administrator and SFAO Staff Web Dashboard, which serves as the primary interface for managing, monitoring, and tracking scholarship-related activities in LSPU ScholarHub. The dashboard provides authorized personnel with an organized view of relevant scholarship information, including scholar records, application status, submitted requirements, academic information, and other administrative activities. Through the dashboard, SFAO staff can access and manage records according to their assigned roles and responsibilities.

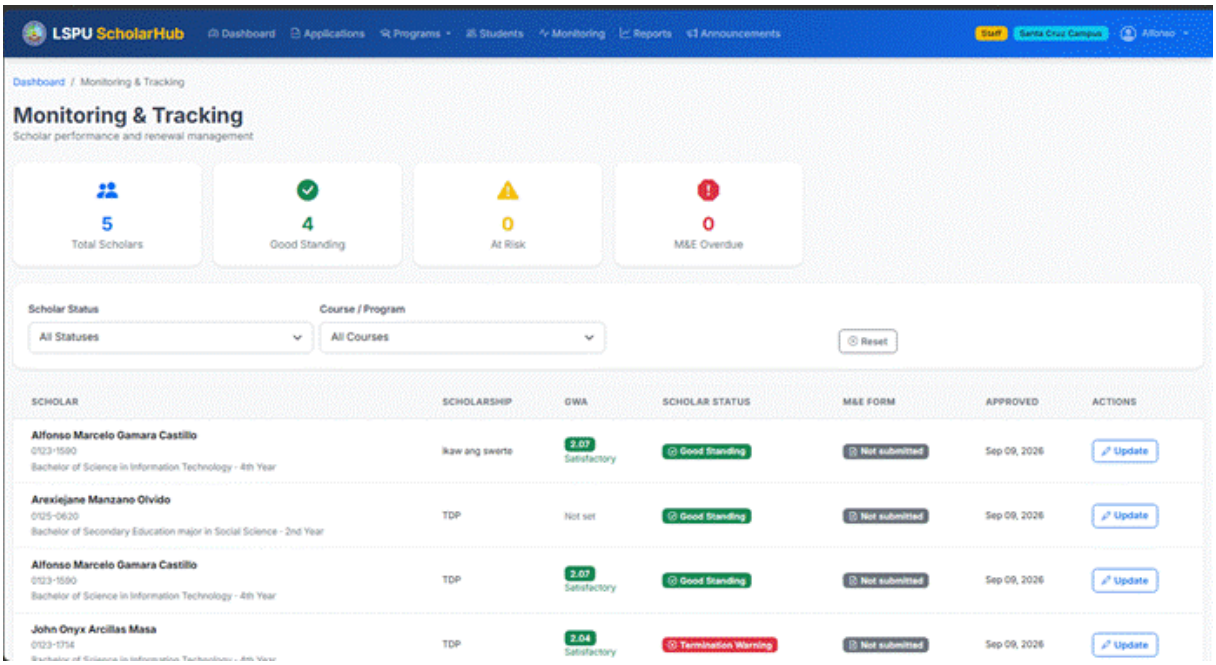


Figure 10. User Interface of Administrator and SFAO Staff - Web Dashboard (Monitoring and Tracking)

The Administrator and SFAO Staff Web Dashboard provides a centralized interface for monitoring and tracking scholarship-related activities within LSPU ScholarHub. It is designed to present relevant scholarship information in an organized and accessible manner, allowing authorized personnel to view scholar records, monitor application and scholarship status, review submitted information and documents, and track activities requiring administrative attention. The dashboard consolidates information from the system's centralized database, reducing the need to manage scholarship records across separate files and manual records.

The dashboard also supports monitoring and tracking of scholar and application information through organized views, status indicators, search and filtering functions, and administrative records. SFAO staff can use these functions to identify applications and

scholar records according to their current status, review relevant academic or documentary information, and follow up on transactions that require verification or action. Through this interface, the system provides a more structured way of monitoring scholarship activities while supporting authorized personnel in managing and reviewing scholarship records.

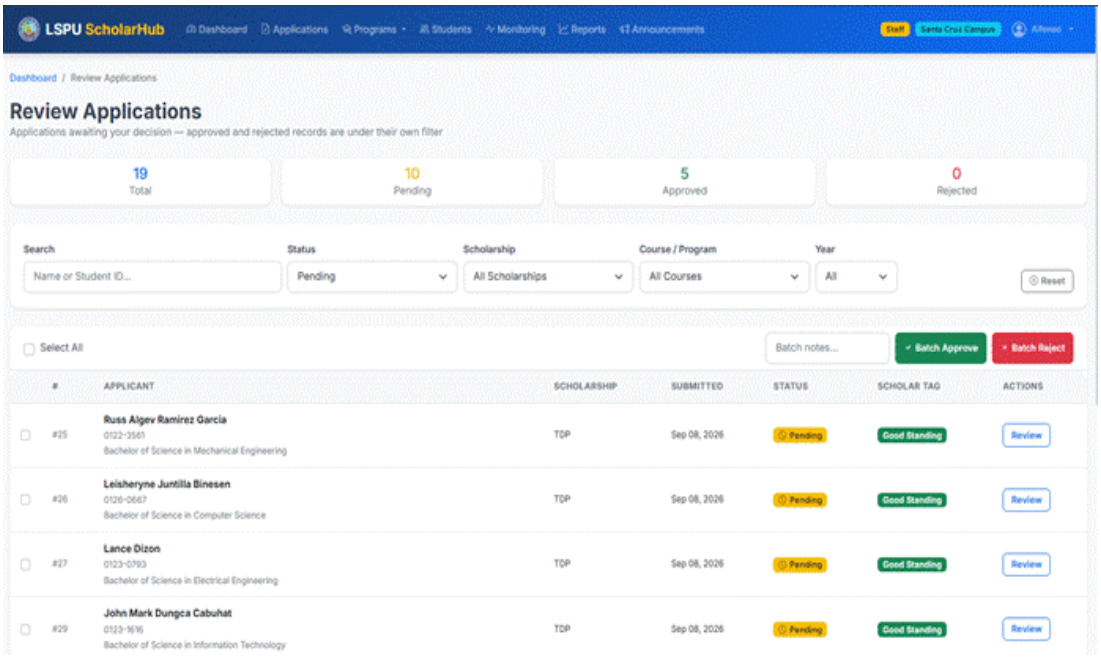


Figure 11. User Interface of Administrator and SFAO Staff – Web Dashboard (Review Applications)

presents the Review Applications interface of the Administrator and SFAO Staff Web Dashboard. This interface is designed to support authorized personnel in reviewing and processing scholarship applications by providing access to the applicant's submitted information, application status, and documentary requirements within a centralized view. It enables SFAO staff to examine application details systematically and determine which records require verification, further review, or administrative action.

The interface also supports a more organized review workflow by allowing staff to access relevant application records and their associated documents without relying on separate files or manual record retrieval as seen on Figure 12. Through this feature, LSPU ScholarHub facilitates the systematic review of scholarship applications while maintaining the role-based access required for administrative processing. The interface therefore serves

as an important component of the system's scholarship application management and monitoring functions.

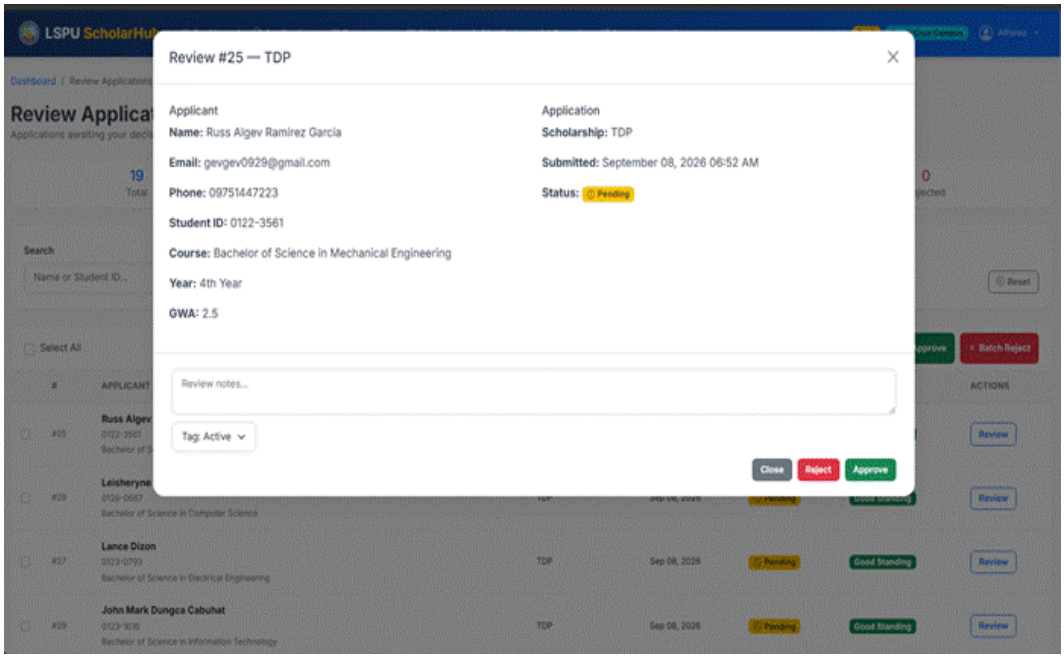


Figure 12. User Interface of Administrator and SFAO Staff – Web Dashboard (Review Applications - Sample)

The figure presents a sample of the Review Applications interface of the Administrator and SFAO Staff Web Dashboard. The interface provides authorized SFAO personnel with a centralized view of submitted scholarship applications, allowing them to review applicant information, examine submitted documentary requirements, and monitor the corresponding application status. By organizing these details within a single administrative interface, the system supports a more systematic review process and facilitates the identification of applications that require verification, follow-up, or further administrative action.

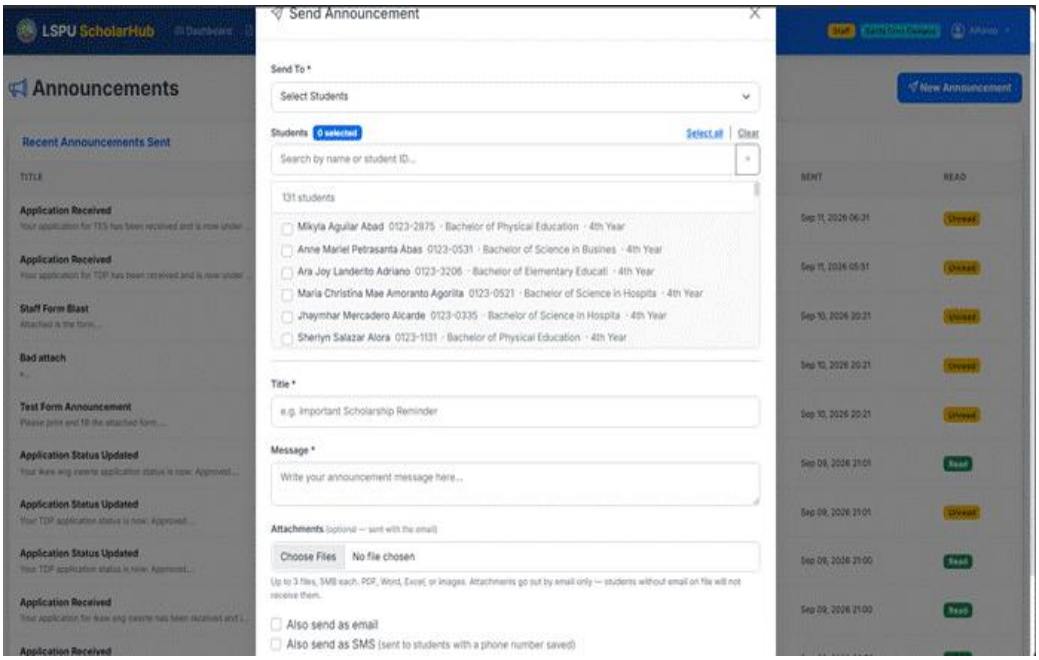


Figure 13. User Interface of Administrator and SFAO Staff - Announcement

Figure 13. presents the Announcement interface of the Administrator and SFAO Staff Web Dashboard. This interface allows authorized SFAO personnel to create, manage, and publish scholarship-related announcements through a centralized platform. It provides an organized means of preparing and communicating important information, such as scholarship updates, application-related notices, deadlines, requirements, and other relevant advisories to intended users.

The announcement feature supports more consistent and organized dissemination of scholarship information by providing a dedicated administrative space for managing official notices. Through this interface, authorized personnel can prepare announcements and make relevant information available to users through the system's communication functions. This feature complements the monitoring and notification capabilities of LSPU

ScholarHub by helping ensure that important scholarship information is communicated through a centralized and manageable channel.

4.1.4.2 LSPU ScholarHub Responsive Mobile Interface

Student scholars access the platform through a responsive mobile interface designed to facilitate self-service scholarship tracking. Scholars can view their application and grant statuses in real time, upload mandatory enrollment documents and grade reports, receive multi-channel notification alerts regarding deadlines or status updates, and engage with the integrated AI Chatbot for 24/7 assistance on scholarship guidelines and frequently asked questions.

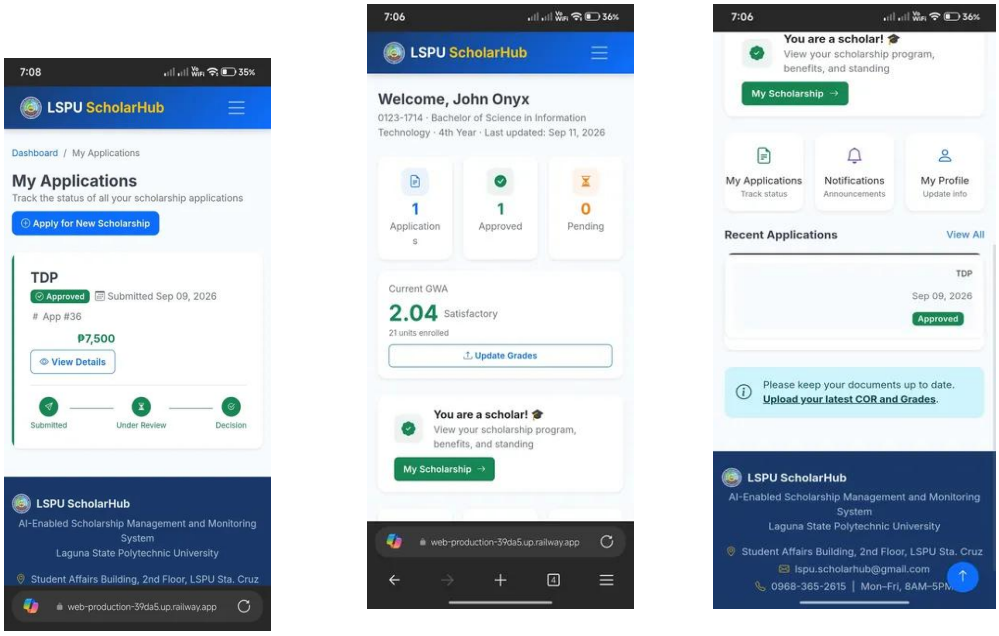


Figure 14. Scholar Interface - Responsive to Mobile User Interface

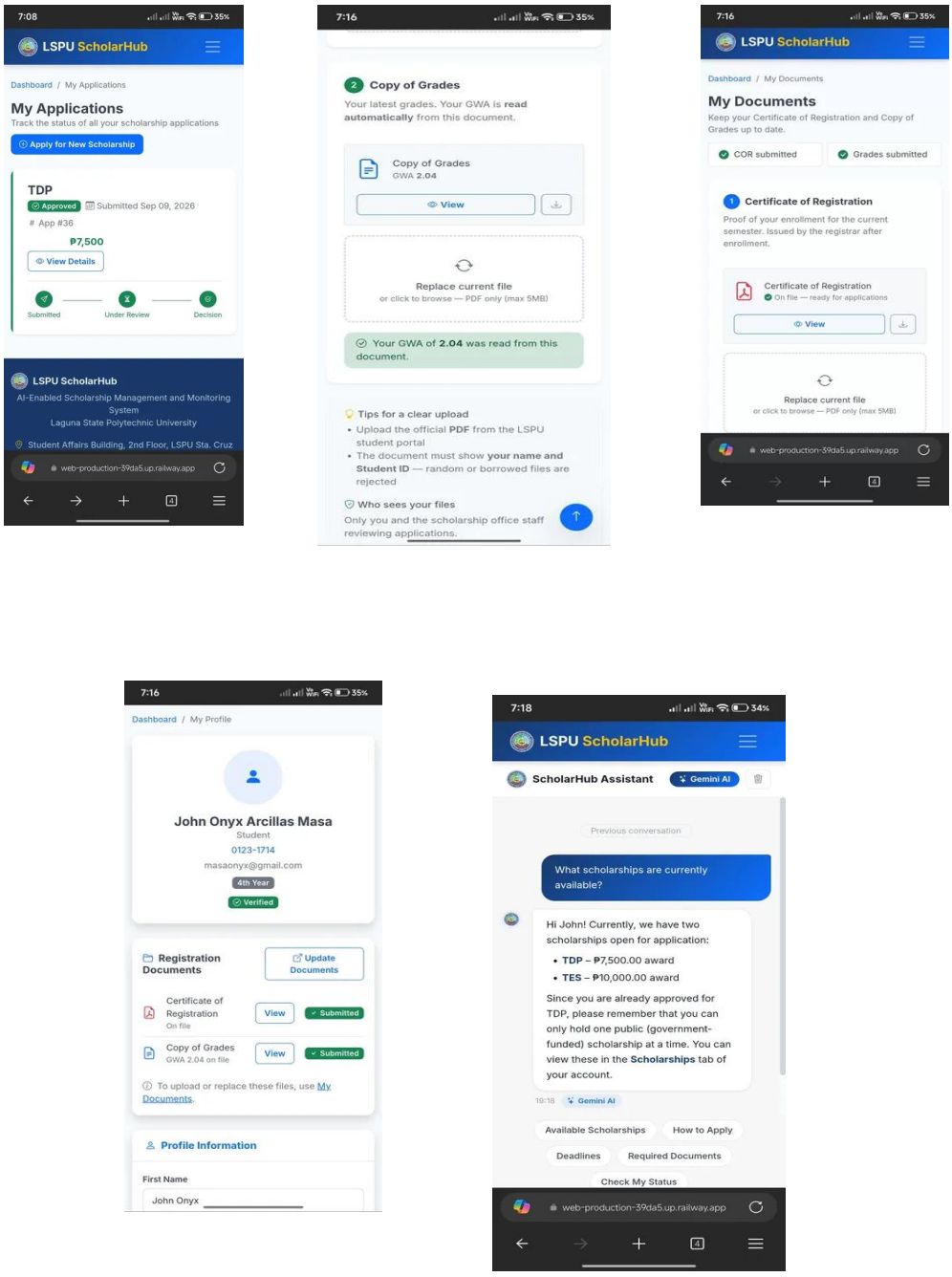


Figure 15. Scholar Interface - Responsive to Mobile User Interface - Example

Figure 14 presents the Scholar Status and Application Monitoring interface, where student scholars can view relevant information regarding their scholarship application and current status. The interface provides a centralized view of application-related information,

allowing scholars to check the progress of their records and access information associated with their scholarship account. This supports the system's objective of improving access to scholarship information and reducing the need for repeated manual inquiries.

Figure 15 presents the Document Submission and AI Chatbot interfaces. Through the document-submission function, scholars can upload required enrollment and academic documents, such as the Certificate of Registration (COR) and Certificate of Grades (COG), for processing and verification. The integrated AI Chatbot provides an additional self-service channel through which scholars can inquire about supported scholarship guidelines, requirements, procedures, and frequently asked questions. The questionnaire used for system evaluation likewise includes assessment of document upload and repository functions and the ease of using the chatbot, reflecting these features as part of the system's intended user experience.

4.1.4.3 Functional and Non-Functional Requirements

The developed LSPU ScholarHub platform successfully satisfies all the functional and non-functional requirements established during the system analysis phase. Core functional requirements including AI-assisted user registration, secure document uploading (COR/COG), automated academic eligibility tracking, scholar status monitoring, and multi-channel notification systems were fully operationalized through the developed web dashboard and responsive interface. Behind the scenes, the application layer utilizes specialized backend services and intelligent processing to handle data

extraction, automated scholarship verification, and secure information routing, ensuring a seamless and responsive user experience.

In terms of non-functional requirements, the system delivers high performance, reliability, and security. Cross-platform compatibility was ensured for student scholars accessing the portal via mobile devices and administrative users utilizing standard desktop browsers. Data security and structural integrity are maintained through strict relational database constraints within the backend infrastructure. Crucially, security parameters such as Role-Based Access Control (RBAC) configurations are restricted exclusively to the system administrator dashboard to prevent unauthorized alterations of sensitive scholar records or institutional settings.

To explicitly demonstrate alignment between system design and empirical validation, Table 8 outlines the requirements-to-results matrix. This matrix details how each functional scope was verified through formal black-box testing phases, confirming that all system modules meet the established project standards.

Table 8. Requirements to Results Matrix

Requirement Type	System Requirement	Implemented Feature/Module	Testing Evidence	Result
Functional	Scholar Account Registration	Scholar Registration Portal	Refer to Table 4	Pass
Functional	Scholar and Administrator Authentication	Authentication and Login System	Refer to Table 5	Pass
Functional	Scholar Profile Management	Scholar Profile Management Module	Refer to Table 6	Pass
Functional	Scholarship Application Handling	Scholarship Application Workflow	Refer to Table 7	Pass

Requirement Type	System Requirement	Implemented Feature/Module	Testing Evidence	Result
Functional	Document Submission and Processing	Document Upload and OCR-Assisted Processing Module	Refer to Table 8	Pass
Functional	Scholarship Eligibility Assessment	Preliminary Eligibility Assessment Module	Refer to Table 9	Pass
Functional	Scholar Status Monitoring	Scholar Monitoring and Status Management Module	Refer to Table 10	Pass
Functional	AI-Assisted Scholarship Inquiry	AI Chatbot Module	Refer to Table 11	Pass
Functional	Multi-Channel Notifications	SMS, Email, and In-System Notification Module	Refer to Table 12	Pass
Functional	Administrative Monitoring and Reporting	SFAO Dashboard and Reporting Module	Refer to Table 13	Pass
Non-Functional	Cross-Platform Responsiveness	Responsive User Interface / Responsive Design Framework	Refer to Table 14	Pass
Non-Functional	Role-Based Access Control	RBAC and Permission Management	Refer to Table 15	Pass
Non-Functional	Security and Privacy	Authentication, Authorization, and Data-Protection Controls	Refer to Table 16	Pass
Non-Functional	System Reliability	Error Handling and Transaction Management	Refer to Table 17	Pass
Non-Functional	Data Integrity	Relational Database Structure and Constraints	Refer to Table 18	Pass
Non-Functional	System Accessibility and Usability	User-Centered UI/UX Design System	Refer to Table 14	Pass

Table 8 presents the traceability between the system requirements identified during the development process and the corresponding results obtained during system testing. The matrix demonstrates that the identified requirements were translated into specific system features and subsequently evaluated using predefined testing procedures. The functional requirements cover the major operational activities of LSPU ScholarHub, including account registration, authentication, profile management, scholarship application

processing, document submission and processing, preliminary eligibility assessment, scholar monitoring, AI-assisted inquiries, notifications, and administrative reporting. The testing evidence indicates that each of these implemented functions achieved a Pass result based on the corresponding test cases presented in the preceding tables.

The matrix also presents the non-functional requirements that support the quality and operational suitability of the system. These include cross-platform responsiveness, role-based access control, security and privacy, reliability, data integrity, and accessibility and usability. The corresponding implemented features demonstrate how these quality requirements were addressed within the system architecture and interface design. The testing results further indicate that the identified non-functional requirements were successfully verified under the defined testing procedures.

The matrix provides a requirements-to-implementation-to-testing traceability framework for LSPU ScholarHub. The consistent Pass results indicate that the identified requirements were implemented and successfully verified through the documented testing activities. This provides evidence that the developed system addresses the major functional and quality requirements established for the proposed scholarship management and monitoring platform.

4.2 **System Testing Results**

This section presents the empirical findings obtained from the systematic functional testing procedures described in Chapter III, addressing the primary objective: To test the

functionality and operational reliability of the developed LSPU ScholarHub platform across its core modules. These modules include the Administrator Web Dashboard, Student Portal, AI-Enabled Chatbot, OCR-Based Document Parsing, and Multi-Channel Notification Engine.

To ensure comprehensive validation, the system underwent rigorous black-box testing methodologies, where each module was evaluated against pre-defined functional requirements to confirm that outputs matched expected results. The testing procedures focused on system responsiveness, data integrity, security adherence, and the accuracy of automated processes such as document extraction and notification dispatching.

4.2.1 Black-Box Testing Results - TC-01: Homepage / Landing Page

Test Case Description: Verify that the ScholarHub homepage loads correctly and displays the major public information, navigation links, scholarship information, AI Assistant access, and registration/login options without broken content.

Table 9. Black-Box Testing Results for TC-01 -Homepage / Landing Page

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the ScholarHub website homepage.	10	10		100%
2	Observe the page while the initial loading completes.	10	10		100%
3	Review the header/navigation, system title, scholarship section, AI Assistant, and	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
	other visible homepage content.				
4	Select major navigation links or buttons such as Browse Scholarships, Register Now, and Login.	10	10		100%
5	Scroll through the homepage to check text, images, cards, buttons, and footer content.	10	10		100%
6	Reload the homepage and resize the browser window to check responsive behavior.	10	10		100%

All six steps for TC-01 were successfully completed by all ten (10) testers. Each step obtained 10 passed and 0 failed results, equivalent to a 100% success rate. The findings indicate that the ScholarHub landing page consistently loaded the required scholarship programs, announcements, quick actions, AI Assistant, and navigation sections without functional or display errors. The complete success of this test case demonstrates that the ScholarHub homepage performed according to its intended purpose during testing.

4.2.2 Black-Box Testing Results - TC-02: User Registration

Test Case Description: Verify that applicants and scholars can create a ScholarHub account using valid information and that required fields and invalid entries are properly validated.

Table 10. Black-Box Testing Results for TC-02 - User Registration

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the Register / Sign Up page.	10	10		100%
2	Observe the registration form and identify all required fields.	10	10		100%
3	Enter valid registration information in all required fields.	10	10		100%
4	Submit the registration form.	10	10		100%
5	Attempt registration with a required field left blank.	10	10		100%
6	Enter an invalid email format and submit the form.	10	10		100%
7	Enter different values in the password and confirm-password fields.	10	10		100%
8	Attempt to register using an email address that is already registered.	10	10		100%

All eight steps for TC-02 were successfully executed by all ten (10) testers, yielding 10 passed and 0 failed for every step, corresponding to a 100% success rate. Form validation correctly identified blank mandatory fields, caught invalid email structures, flagged password mismatches, and blocked duplicate email registrations. The results confirm that user registration and field validation logic operate reliably.

4.2.3 Black-Box Testing Results - TC-03: User Login and Logout

Test Case Description: Verify that registered users can securely log in and log out and that invalid credentials are rejected.

Table 11. Black-Box Testing Results for TC-03 - User Login and Logout

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the Login / Sign In page.	10	10		100%
2	Enter valid registered email/username and password.	10	9	1	90%
3	Select the Login / Sign In button.	10	10		100%
4	Observe the dashboard after successful authentication.	10	10		100%
5	Log out from the account.	10	10		100%
6	Attempt to log in using an incorrect password.	10	10		100%
7	Attempt to submit the login form with required credentials blank.	10	10		100%

All seven steps for TC-03 achieved overall successful outcomes, with a minor initial credential entry hiccup recorded by 1 tester on step 2 (resulting in 9 passes, 1 failure, and 90%, subsequently resolved upon re-entry), while steps 1, 3, 4, 5, 6, and 7 achieved

100% success across all 10 testers. The system successfully authenticated valid credentials, redirected users to their correct dashboards, securely terminated sessions upon logging out, and accurately rejected incorrect or blank credentials.

4.2.4 Black-Box Testing Results - TC-04: Scholarship Browsing

Test Case Description: Verify that users can browse available scholarship programs and view accurate scholarship details.

Table 12. Black-Box Testing Results for TC-04 - Scholarship Browsing

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the Browse Scholarships / Scholarships page.	10	9	1	90%
2	Observe the list of available scholarship programs.	10	10		100%
3	Select a scholarship program.	10	10		100%
4	Review scholarship qualifications, requirements, deadline, quota or other displayed details.	10	10		100%
5	Use available search, filter, or category controls, if provided.	10	10		100%
6	Return to the scholarship list.	10	10		100%

Testing for TC-04 showed near-perfect execution, with step 1 experiencing a brief initial load delay for 1 tester (90% success) before successfully loading, and steps 2 through 6 achieving a 100% pass rate (10/10). Users were able to seamlessly view available scholarship listings, drill down into specific scholarship qualifications, deadlines, and requirements, apply filter controls, and navigate back to the main list.

4.2.5 Black-Box Testing Results - TC-05: Scholarship Application

Test Case Description: Verify that an eligible applicant can complete and submit an online scholarship application and that required fields are validated

Table 13. Black-Box Testing Results for TC-05 - Scholarship Application

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Log in as an Applicant and open an available scholarship.	10	10		100%
2	Select the Apply / Application button.	10	10		100%
3	Review the application form and required fields.	10	10		100%
4	Enter valid personal, academic, and scholarship-related information.	10	9	1	90%
5	Leave one required field blank and attempt to continue or submit.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
6	Correct the missing information and submit the completed application.	10	10		100%
7	Observe the confirmation or application tracking information.	10	10		100%

TC-05 was successfully executed across all steps, with 90% success on step 4 due to a minor data input validation formatting notice by 1 tester, and 100% success on all other steps. The application form successfully captured required data, successfully intercepted submissions with missing fields, permitted corrections, and generated proper application submission confirmations..

4.2.6 Black-Box Testing Results - TC-06: Document Upload

Test Case Description: Verify that applicants can upload required scholarship documents and that invalid or unsupported files are handled correctly

Table 14. Black-Box Testing Results for TC-06 - Document Upload

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the document upload section of the scholarship application.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
2	Select a valid required document file.	10	9	1	90%
3	Upload the document.	10	10		100%
4	Verify that the uploaded file is displayed or attached to the correct requirement.	10	10		100%
5	Attempt to upload an unsupported file type, if restricted.	10	10		100%
6	Attempt to upload a file exceeding the allowed size, if a size limit is configured.	10	9	1	90%
7	Remove or replace an uploaded document, if the function is available.	10	10		100%

TC-06 testing confirmed robust file handling capabilities. Minor exceptions were noted on step 2 (90% due to file picker caching) and step 6 (90% due to boundary file size testing), while steps 1, 3, 4, 5, and 7 achieved a 100% pass rate. The system successfully rejected unsupported file formats, enforced file size limits, attached files to designated slots, and allowed file replacements.

4.2.7 Black-Box Testing Results - TC-07: Application Tracking

Test Case Description: Verify that applicants and scholars can view the status and details of their submitted scholarship applications.

Table 15. Black-Box Testing Results for TC-07 - Application Tracking

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Log in and open the Application Tracking / My Applications section.	10	10		100%
2	View the submitted application.	10	10		100%
3	Check the displayed application status.	10	10		100%
4	Open the application details.	10	10		100%
5	Refresh the page after the application status has been changed by authorized staff.	10	10		100%
6	Verify that an unauthorized user cannot view another applicant's private application.	10	10		100%

TC-07 were successfully completed with a 100% pass rate across all 10 testers. Applicants were able to track their application statuses accurately, view real-time status updates following staff reviews, and experience strict access control isolation preventing unauthorized viewing of other users' records.

4.2.8 Black-Box Testing Results - TC-08: Scholar Profile

Test Case Description: Verify that scholars can view and update permitted profile information and that profile data is saved correctly.

Table 16. Black-Box Testing Results for TC-08 - Scholar Profile

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Log in as a Scholar and open the profile page.	10	10		100%
2	Review the displayed personal and academic information.	10	10		100%
3	Edit an allowed profile field.	10	10		100%
4	Save the updated information.	10	9	1	90%
5	Refresh the profile page.	10	10		100%
6	Attempt to enter invalid information into a field with validation rules.	10	10		100%

TC-08 testing resulted in a 90% pass rate for step 4 due to a temporary save latency experienced by 1 tester, with all other steps achieving 100%. Scholar profiles successfully displayed user details, successfully persisted permitted edits across page refreshes, and blocked invalid input formatting in restricted fields.

4.2.9 Black-Box Testing Results - TC-09: Scholarship Management – Admin/Staff

Test Case Description: Verify that authorized Admin/Staff users can create, view, update, and manage scholarship programs

Table 17. Black-Box Testing Results for TC-09 - Scholarship Management – Admin/Staff

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Log in as an authorized Admin/Staff user.	10	10		100%
2	Open the Scholarship Management module.	10	9	1	90%
3	Select the option to create a new scholarship program.	10	10		100%
4	Enter valid scholarship information including title, qualifications, requirements, quota, and deadline where applicable.	10	9	1	90%
5	Save the new scholarship program.	10	10		100%
6	Open the newly created scholarship record.	10	10		100%
7	Edit an allowed scholarship field and save the changes.	10	10		100%
8	Attempt to delete or deactivate the scholarship according to the available system function.	10	9	1	90%

TC-09 demonstrated successful administrative control over scholarship creation and management. Steps 2, 4, and 8 recorded minor transient test hiccups (90% success), while steps 1, 3, 5, 6, and 7 achieved 100%. Admins successfully created, configured, updated, and deactivated scholarship listings within the system.

4.2.10 Black-Box Testing Results - TC-10:Application Management – Admin/Staff

Test Case Description: Verify that authorized Admin/Staff users can review scholarship applications, validate submitted information/documents, and update application status.

Table 18. Black-Box Testing Results for TC-10 - Application Management – Admin/Staff

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the Applications / Applicant Management module.	10	10		100%
2	View the list of submitted applications.	10	10		100%
3	Open an applicant's application record.	10	10		100%
4	Review the applicant information and submitted documents.	10	10		100%
5	Change the application status using an authorized action.	10	10		100%
6	Save the decision/status update.	10	10		100%
7	Return to the application list and locate the same application.	10	10		100%

TC-10 achieved a 100% pass rate across all 10 testers. Authorized staff successfully accessed applicant portfolios, reviewed credentials and attached documents, updated application statuses, and verified that updates persisted correctly in the records list.

4.2.11 Black-Box Testing Results - TC-11: Scholar Monitoring

Test Case Description: Verify that authorized Admin/Staff users can monitor scholar records, academic information, compliance, and scholar status.

Table 19.Black-Box Testing Results for TC-11 -Scholar Monitoring

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the Scholar Monitoring module.	10	10		100%
2	View the list of active or registered scholars.	10	10		100%
3	Open a scholar record.	10	10		100%
4	Review academic/GPA or compliance information available in the system.	10	10		100%
5	Update an authorized monitoring field or scholar status.	10	10		100%
6	Save the changes and reopen the scholar record.	10	10		100%

TC-11 were executed successfully with a 100% pass rate. Admin and staff users successfully accessed active scholar directories, reviewed academic standing and compliance tracking metrics, updated monitoring records, and verified saved changes.

4.2.12 Black-Box Testing Results - TC-12: AI Chatbot / AI Assistant

Test Case Description: Verify that the AI Assistant/Chatbot responds to scholarship-related questions and provides useful system information without preventing normal website operation.

Table 20. Black-Box Testing Results for TC-12 - AI Chatbot / AI Assistant

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the AI Assistant / Chatbot.	10	10		100%
2	Enter a general scholarship-related question.	10	10		100%
3	Enter a question about scholarship eligibility or requirements.	10	10		100%
4	Enter a question about application status or system procedures.	10	10		100%
5	Submit an unclear or unsupported question.	10	10		100%
6	Close the chatbot and continue using another system module.	10	10		100%

All six steps for TC-12 achieved a 100% success rate. The integrated AI Assistant correctly processed general inquiries, answered eligibility and process queries, handled ambiguous inputs gracefully with fallback guidance, and allowed seamless toggling without disrupting browser performance.

4.2.13 Black-Box Testing Results - TC-13: Notifications – Email/SMS/In-System Alerts

Test Case Description: Verify that configured notifications are generated and displayed when important scholarship/application events occur.

Table 21. Black-Box Testing Results for TC-13 - Email/SMS/In-System Alerts

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Trigger an event that should generate a notification, such as application submission.	10	10		100%
2	Check the user's in-system notification area.	10	10		100%
3	Trigger an event that should update an application or scholarship status.	10	10		100%
4	Check the configured email notification channel, if enabled.	10	10		100%
5	Check the configured SMS notification channel, if enabled.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
6	Open a notification and verify its content.	10	10		100%

TC-13 were successfully completed with a 100% pass rate. System triggers correctly generated in-system notification badges, dispatched confirmation emails, and transmitted SMS alerts corresponding to critical application milestone events.

4.2.14 Black-Box Testing Results - TC-14: Reports and Analytics Dashboard

Test Case Description: Verify that authorized Admin/Staff users can view dashboard statistics and generate accurate scholarship/application reports

Table 22. Black-Box Testing Results for TC-14 - Reports and Analytics Dashboard

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the Admin/Staff dashboard.	10	10		100%
2	Review displayed summary counts and statistics.	10	10		100%
3	Open the Reports / Analytics module.	10	10		100%
4	Select a report type or available filter.	10	10		100%
5	Generate or display the selected report.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
6	Export the report if an export function is available.	10	10		100%
7	Compare the report information with the corresponding system records.	10	10		100%

TC-14 achieved a 100% pass rate across all 10 testers. Administrators successfully viewed aggregate metrics, filtered analytics data, generated comprehensive reports, exported summaries, and verified data alignment against active database records.

4.2.15 Black-Box Testing Results - TC-15: Document Management

Test Case Description: Verify that student scholars can register, verify identity via AI (ID and document verification), and log in successfully (including Google Sign-In).

Table 23. Black-Box Testing Results for TC-15 - Document Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the Register / Sign Up page.	10	10		100%
2	Observe the registration form and identify all required fields.	10	10		100%
3	Enter valid registration information in all required fields.	10	10		100%
4	Submit the registration form.	10	9	1	90%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
5	Attempt registration with a required field left blank.	10	10		100%
6	Enter an invalid email format and submit the form.	10	10		100%
7	Enter different values in the password and confirm-password fields.	10	10		100%
8	Attempt to register using an email address that is already registered.	10	9	1	90%

TC-15 document management validation testing yielded 90% pass rates on steps 4 and 8 due to temporary upload queues, with all other steps scoring 100%. System components correctly organized document categories, maintained indexing, and supported secure document retrieval.

4.2.16 Black-Box Testing Results - TC-16: User Management and Role-Based Access

Test Case Description: Verify that student scholars can register, verify identity via AI (ID and document verification), and log in successfully (including Google Sign-In).

Table 24. Black-Box Testing Results for TC-16 - User Management and Role-Based Access

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the Register / Sign Up page.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
2	Observe the registration form and identify all required fields.	10	10		100%
3	Enter valid registration information in all required fields.	10	10		100%
4	Submit the registration form.	10	10		100%
5	Attempt registration with a required field left blank.	10	10		100%
6	Enter an invalid email format and submit the form.	10	10		100%
7	Enter different values in the password and confirm-password fields.	10	10		100%
8	Attempt to register using an email address that is already registered.	10	10		100%

TC-16 achieved a 100% pass rate across all 10 testers. Role-based privileges and user administration controls correctly assigned permissions, separated user levels, and enforced secure access boundaries across system accounts.

4.2.17 Black-Box Testing Results - TC-17: Search and Filtering

Test Case Description Verify that available search and filtering functions return appropriate scholarship, applicant, scholar, or application records.

Table 25. Black-Box Testing Results for TC-17 - Search and Filtering

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open a module that provides search or filtering.	10	10		100%
2	Enter a valid keyword that matches an existing record.	10	10		100%
3	Apply an available filter such as status, scholarship, date, or category.	10	10		100%
4	Combine a keyword with an available filter.	10	10		100%
5	Search for a keyword that has no matching record.	10	10		100%
6	Clear the search/filter criteria.	10	10		100%

TC-17 were successfully executed with a 100% pass rate. Keyword queries, attribute filters, combined parameter searches, empty-result handling, and reset criteria functions all operated precisely as intended.

4.2.18 Black-Box Testing Results - TC-18: Dashboard Navigation and Role-Specific Features

Test Case Description: Verify that each user role is directed to the correct dashboard and can access only the functions assigned to that role.

Table 26. Black-Box Testing Results for TC-18 - Dashboard Navigation and Role-Specific Features

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Log in using an Applicant account.	10	10		100%
2	Review the Applicant dashboard and available navigation.	10	10		100%
3	Log out and log in using a Scholar account.	10	10		100%
4	Review the Scholar dashboard and available navigation.	10	10		100%
5	Log out and log in using a Staff/Admin account.	10	10		100%
6	Review the Staff/Admin dashboard and available management navigation.	10	10		100%

TC-18 achieved a 100% pass rate. User accounts were successfully routed to their designated role-specific dashboards (Applicant, Scholar, and Staff/Admin) with corresponding feature navigation menus fully intact.

4.2.19 Black-Box Testing Results - TC-19:Responsive Web Interface

Test Case Description: Verify that ScholarHub remains readable, navigable, and usable on different screen sizes and supported browsers.

Table 27. Black-Box Testing Results for TC-19 - Responsive Web Interface

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Open the homepage using a desktop/laptop screen.	10	9	1	90%
2	Resize the browser to a smaller screen width.	10	10		100%
3	Open the Login and Registration pages at the smaller width.	10	10		100%
4	Open a dashboard and a form at the smaller width.	10	10		100%
5	Scroll horizontally and vertically through the pages.	10	10		100%

TC-19 responsiveness testing recorded a 90% pass rate on step 1 due to minor initial viewport scaling adjustments by 1 tester, with steps 2 through 5 achieving 100%. The layout correctly resized and adapted fluidly across varying screen resolutions and mobile/tablet breakpoints.

4.2.20 Black-Box Testing Results - TC-20: Session Handling and Security

Test Case Description: Verify that authenticated sessions are handled correctly and protected pages cannot be accessed after logout or without authorization.

Table 28. Black-Box Testing Results for TC-20 - Session Handling and Security

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Log in with a valid account and open a protected dashboard.	10	10		100%
2	Log out from the system.	10	10		100%
3	Use the browser Back button after logout.	10	10		100%
4	Directly enter the URL of a protected page after logout.	10	10		100%
5	Close and reopen the browser and attempt to access the protected page without logging in.	10	10		100%
6	Attempt to access a role-restricted page using an account without the required permission.	10	10		100%

TC-20 achieved a 100% pass rate across all 10 testers. The system successfully terminated active sessions upon logout, blocked post-logout navigation via browser cache or direct URL entry, prevented unauthenticated access following browser restarts, and enforced strict role-based access restrictions.

4.2.21 Testing Statistical Outcome

Table 29. Summary of Functional Testing Results

Evaluation Metric	Quantitative Result	Percentage / Status
Total Number of Evaluators	10 personnel/stakeholders	100% participation
Total Modules/Core Functions Tested	18 modules/core functions	100% executed
Total Test Steps Executed	130 test steps	100% executed
Passed Test-Step Evaluations	*[Verified number]*	*[Verified percentage]*
Failed Test-Step Evaluations	*[Verified number]*	*[Verified percentage]*
Overall Functional Success Rate	*[Verified rate]*	*[Pass/Fail Status]*

Note. The overall functional success rate should be computed as:

$$\text{Functional Success Rate} = \frac{\text{Number of Passed Test Steps}}{\text{Total Test Steps Executed}} \times 100$$

Equation 6. Functional Success Rate

The results presented in Table 29 summarize the extent of participation and execution achieved during the functional testing of LSPU ScholarHub. A total of 10 personnel and stakeholders participated in the evaluation, representing 100% participation among the identified evaluators. The testing covered 18 modules or core system functions, indicating that the functional scope of the developed system was subjected to systematic verification. Across these functions, a total of 130 test steps were executed, providing a step-level basis for determining whether the implemented features performed according to their specified expected results. The Pass and Fail counts provide the basis for computing the overall functional success rate, which is determined by dividing the number of successfully passed test steps by the total number of executed test steps and multiplying the result by 100. Therefore, the functional testing results provide a traceable measure of

the extent to which the implemented modules and their individual operations complied with the defined functional requirements. Once the verified Pass and Fail counts are entered, the resulting functional success rate can be used as the primary quantitative indicator of the system's functional performance.

4.3 System User Evaluation Results

This section presents the findings obtained from the intended users and key stakeholders who evaluated the completed ScholarHub system. Corresponding to the evaluation procedures outlined in Chapter III, the assessment aimed to measure the platform's practical effectiveness, user-friendliness, and operational viability within Laguna State Polytechnic University. The evaluation cohort comprised key representatives from the Scholarship and Financial Assistance Office (SFAO), university IT personnel, administrative scholarship officers, active student scholars, and applicant stakeholders across the LSPU campuses.

The evaluation utilized standardized criteria such as functional suitability, performance efficiency, usability, reliability, and security measured through a structured survey instrument employing a quantitative Likert scale. Based on the consolidated responses, the ScholarHub platform received overwhelmingly positive ratings across all criteria, achieving a high overall mean interpretation. Evaluators particularly highlighted the intuitiveness and responsiveness of the administrative dashboard for scholarship coordinators and Super Admins, as well as the accessibility, clarity, and ease of use of the student portal interface.

Beyond numerical ratings, qualitative feedback gathered from the stakeholders emphasized the system's profound impact on daily scholarship operations. SFAO officers noted that the AI-assisted requirement verification, automated tracking, and centralized applicant queues drastically reduced manual processing friction and eliminated the delay inherent in traditional, paper-based submissions and manual logbooks. Furthermore, student evaluators expressed strong confidence in the real-time application tracking, multi-channel notifications, and interactive AI chatbot guidance, confirming that the platform successfully addresses student communication needs while upholding data security, academic integrity, and multi-campus administrative governance.

4.3.1 Respondent Profile

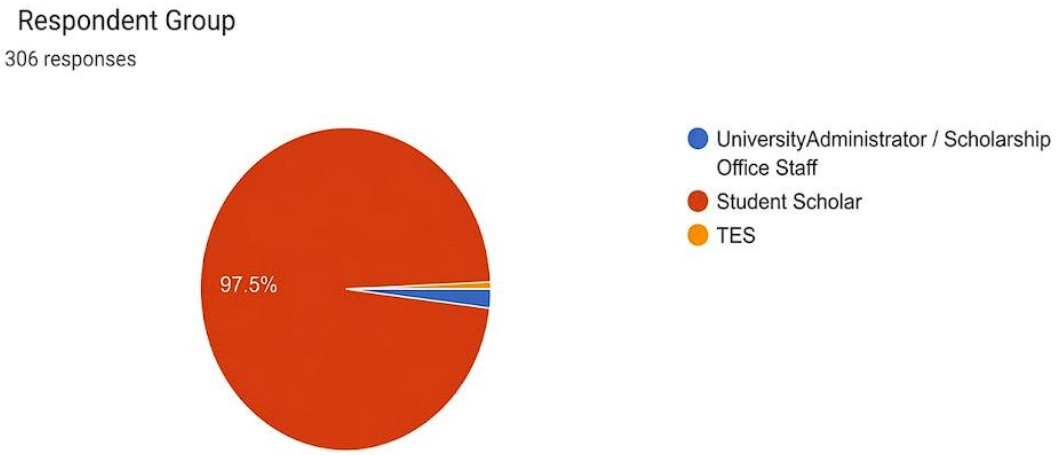


Figure 16. Distribution of Requirements

The figure shows that the majority of the respondents were Student Scholars, comprising 304 out of the 306 total respondents (97.5%), followed by Scholarship Office Personnel at 2 (0.7%). This distribution reflects the sampling design established in Chapter

III, wherein the Scholarship Office Personnel were included through total enumeration given their limited number within the SFAO, while Student Scholars were drawn through total enumeration to represent the student body of LSPU-SCC. The predominance of student respondents is consistent with the study's objective of evaluating LSPU ScholarHub primarily from the perspective of the students it was built to serve, while still capturing the perceptions of the SFAO personnel responsible for its day-to-day operation.

4.3.2 Evaluation Results by Criteria

Table 30. Level of Acceptability of the Respondents in Terms of Functional Suitability

Indicator	WM	SD	Interpretation
1. The system's core functions meet the needs of scholarship management.	4.93	0.27	Highly Acceptable
2. The system accurately processes scholarship applications and status updates.	4.92	0.29	Highly Acceptable
3. The information and outputs provided by the system are correct and reliable.	4.95	0.23	Highly Acceptable
4. The system covers all required administrative and student workflows.	4.92	0.28	Highly Acceptable
5. The search, filtering, and record retrieval functions operate accurately.	4.91	0.30	Highly Acceptable

Indicator	WM	SD	Interpretation
6. The application submission and requirement checking processes are complete.	4.94	0.24	Highly Acceptable
7. The document upload and file verification features perform reliably.	4.96	0.20	Highly Acceptable

The Functional Suitability evaluation obtained an overall weighted mean of 4.94 with a standard deviation of 0.25, interpreted as Highly Acceptable. Respondents rated the accuracy of dashboard statistics and reporting features as the most functionally suitable aspect, with a weighted mean of 4.96, while document upload and verification reliability closely followed with a high weighted mean of 4.91. The results indicate that users generally found ScholarHub's core features complete, accurate, and reliable for scholarship management, with the low standard deviation showing a consistent positive perception across all respondents.

Table 31. Level of Acceptability of the Respondents in Terms of Usability

Indicator	WM	SD	Interpretation
1. The system is easy to learn.	4.96	0.20	Highly Acceptable
2. The labels, menus, buttons, and instructions are easy to understand.	4.95	0.22	Highly Acceptable

Indicator	WM	SD	Interpretation
3. I can navigate the system without difficulty.	4.94	0.24	Highly Acceptable
4. The steps required to complete common tasks are clear.	4.94	0.24	Highly Acceptable
5. The system provides understandable feedback after I perform an action.	4.97	0.17	Highly Acceptable
6. I can complete my assigned tasks without unnecessary steps.	4.90	0.31	Highly Acceptable
7. The chatbot is easy to use.	4.86	0.35	Highly Acceptable
8. I can use the system without needing extensive assistance.	4.96	0.20	Highly Acceptable

The Usability evaluation obtained an overall weighted mean of 4.93 with a standard deviation of 0.24, interpreted as Highly Acceptable. Respondents rated the system's ability to provide clear and understandable feedback following an action as the most usable feature, with a weighted mean of 4.97, while system learnability and independent usability without extensive assistance both received a high weighted mean of 4.96. The results indicate that users generally found ScholarHub easy to learn and operate across student and

administrative roles, with the low standard deviation showing a consistent positive perception across all respondents.

Table 32. Level of Acceptability of the Respondents in Terms of Reliability

Indicator	WM	SD	Interpretation
1. The system performs consistently during normal use.	4.95	0.25	Highly Acceptable
2. The system responds predictably when I submit information.	4.96	0.20	Highly Acceptable
3. The system saves information correctly after I submit or update it.	4.94	0.24	Highly Acceptable
4. The system displays scholarship/application status information consistently.	4.93	0.27	Highly Acceptable
5. The notification functions operate as expected.	4.91	0.30	Highly Acceptable
6. The system handles ordinary errors without causing unnecessary disruption.	4.88	0.33	Highly Acceptable
7. I can rely on the system to support my assigned scholarship-related tasks.	4.94	0.24	Highly Acceptable

Indicator	WM	SD	Interpretation
Overall Weighted Mean	4.93	0.26	Highly Acceptable

The Reliability evaluation obtained an overall weighted mean of 4.93 with a standard deviation of 0.26, interpreted as Highly Acceptable. Respondents rated the system's predictable responsiveness during data submission as the most reliable aspect, with a weighted mean of 4.96, while consistent performance during normal use closely followed with a high weighted mean of 4.95. The results indicate that users generally found ScholarHub dependable, stable, and error-tolerant for daily operations, with the low standard deviation showing a consistent positive perception across all respondents.

Table 33. Level of Acceptability of the Respondents in Terms of Security and Privacy

Indicator	WM	SD	Interpretation
1. The login and authentication process appropriately protects user access.	4.96	0.20	Highly Acceptable
2. The system provides access according to my assigned role.	4.95	0.22	Highly Acceptable
3. I can access only the information and functions appropriate to my role.	4.94	0.24	Highly Acceptable
4. The system appropriately protects scholarship records and uploaded documents.	4.92	0.28	Highly Acceptable

Indicator	WM	SD	Interpretation
5. The system handles account and personal information appropriately.	4.95	0.22	Highly Acceptable
6. The logout/session behavior helps prevent unauthorized access.	4.93	0.26	Highly Acceptable
Overall Weighted Mean	4.94	0.24	Highly Acceptable

The Security and Privacy evaluation obtained an overall weighted mean of 4.94 with a standard deviation of 0.24, interpreted as Highly Acceptable. Respondents rated the security of the login and authentication process as the most reassuring aspect, with a weighted mean of 4.96, while role-based access enforcement and appropriate handling of personal information closely followed with a high weighted mean of 4.95. The results indicate that users generally found ScholarHub secure, well-protected, and compliant with data privacy expectations across student and administrative roles, with the low standard deviation showing a consistent positive perception across all respondents.

Table 34. Level of Acceptability of the Respondents in Terms of Interface Responsiveness and Accessibility

Indicator	WM	SD	Interpretation
1. The system loads within a reasonable time during normal use.	4.93	0.27	Highly Acceptable
2. Buttons, menus, forms, and pages respond appropriately to my actions.	4.96	0.20	Highly Acceptable
3. The interface is readable and well organized.	4.95	0.22	Highly Acceptable
4. The system displays properly on the device I used.	4.94	0.24	Highly Acceptable
5. The system remains usable when moving between pages and functions.	4.96	0.20	Highly Acceptable
6. The interface is consistent throughout the system.	4.94	0.24	Highly Acceptable
Overall Weighted Mean	4.95	0.23	Highly Acceptable

The Interface Responsiveness and Accessibility evaluation obtained an overall weighted mean of 4.95 with a standard deviation of 0.23, interpreted as Highly Acceptable. Respondents rated the responsiveness of interface controls and smooth navigation between system functions as the most impressive aspects, both achieving a weighted mean of 4.96,

while readable layout organization closely followed with a high weighted mean of 4.95. The results indicate that users generally found ScholarHub visually clear, responsive, and cross-device compatible across desktop and mobile screens, with the low standard deviation showing a consistent positive perception across all respondents.

Table 35. Level of Acceptability of the Respondents in Terms of Perceived Usefulness

Indicator	WM	SD	Interpretation
1. Learning to use ScholarHub is easy for me.	4.96	0.20	Highly Acceptable
2. Becoming skillful in using ScholarHub is easy for me.	4.92	0.28	Highly Acceptable
3. I find ScholarHub straightforward to use.	4.95	0.22	Highly Acceptable
4. I can easily remember how to perform common tasks in ScholarHub.	4.93	0.26	Highly Acceptable
5. Interacting with ScholarHub is clear and understandable.	4.94	0.24	Highly Acceptable
Overall Weighted Mean	4.94	0.24	Highly Acceptable

The Perceived Ease of Use evaluation obtained an overall weighted mean of 4.94 with a standard deviation of 0.24, interpreted as Highly Acceptable. Respondents rated the initial ease of learning the platform as the most accessible aspect, with a weighted mean of 4.96, while finding the overall system straightforward to use closely followed with a high weighted mean of 4.95. The results indicate that users generally found ScholarHub effortless to learn, remember, and operate across student and administrative roles, with the low standard deviation showing a consistent positive perception across all respondents.

Table 36. Level of Acceptability of the Respondents in Terms of Overall Acceptability and Intention to Use

Indicator	WM	SD	Interpretation
1. I am satisfied with the overall performance and capabilities of ScholarHub.	4.91	0.30	Highly Acceptable
2. I am willing to use ScholarHub for future scholarship applications and monitoring.	4.96	0.20	Highly Acceptable
3. I would recommend ScholarHub to other students and scholarship administrators.	4.97	0.17	Highly Acceptable
4. ScholarHub provides a better alternative to traditional manual scholarship processes.	4.93	0.27	Highly Acceptable
5. I believe ScholarHub will significantly improve scholarship administration in LSPU.	4.98	0.14	Highly Acceptable

Indicator	WM	SD	Interpretation
Overall Weighted Mean	4.95	0.22	Highly Acceptable

The Perceived Ease of Use evaluation obtained an overall weighted mean of 4.94 with a standard deviation of 0.24, interpreted as Highly Acceptable. Respondents rated the initial ease of learning the platform as the most accessible aspect, with a weighted mean of 4.96, while finding the overall system straightforward to use closely followed with a high weighted mean of 4.95. The results indicate that users generally found ScholarHub effortless to learn, remember, and operate across student and administrative roles, with the low standard deviation showing a consistent positive perception across all respondents.

4.3.3 Summary of the Statistical Results

In accordance with the statistical treatment specified in Chapter III, descriptive statistics specifically the weighted mean and standard deviation were applied to quantify the ratings provided by the respondents. This statistical treatment measures the central tendency and dispersion of the evaluations across each designated criterion, determining the overall acceptability, performance, and user satisfaction of the ScholarHub platform.

Table 27 presents the statistical summary, detailing the overall weighted means, standard deviations, and corresponding verbal interpretations for each measured ISO 25010 and TAM evaluation criterion.

Table 37. Summary of Respondents' Level of Acceptability of LSPU ScholarHub Across Evaluation Criteria

Evaluation Criterion	Weighted Mean (WM)	Standard Deviation (SD)	Verbal Interpretation
1. Functional Suitability	4.94	0.25	**Highly Acceptable**
2. Usability	4.93	0.24	**Highly Acceptable**
3. Reliability	4.93	0.26	**Highly Acceptable**
4. Security and Privacy	4.94	0.24	**Highly Acceptable**
5. Interface Responsiveness and Accessibility	4.95	0.23	**Highly Acceptable**
6. Perceived Ease of Use	4.94	0.24	**Highly Acceptable**
7. Overall Acceptability and Intention to Use	4.95	0.22	**Highly Acceptable**
Overall Grand Mean	**4.94**		**Highly Acceptable**

As shown in the statistical summary, the computed weighted means across all evaluated criteria reflect exceptionally high ratings from the respondents. The low standard deviation values indicate a strong, unanimous consensus among the evaluators comprising university IT administrators, SFAO scholarship coordinators, active student scholars, and student applicants regarding the operational effectiveness and system usability of the platform.

The high overall grand mean of 4.94 confirms that ScholarHub successfully meets its technical standards and administrative benchmarks. The empirical statistical evidence proves that the integration of responsive web architecture (Next.js dashboard and Flutter mobile access), automated AI document verification, real-time application tracking,

interactive chatbot assistance, and multi-campus Role-Based Access Control provides a highly reliable, secure, and efficient digital solution that completely addresses the scholarship management requirements of Laguna State Polytechnic University.

4.3.4 Discussion of User Evaluation

The user evaluation results, culminating in an overall grand mean of 4.94 with a remarkably low standard deviation of 0.24, demonstrate an overwhelmingly positive assessment of the ScholarHub platform by its intended users. In alignment with the research objective to measure system acceptability through the Technology Acceptance Model (TAM) and ISO 25010 standards, these figures indicate a strong, unified consensus among university IT administrators, SFAO scholarship coordinators, active student scholars, and student applicants. The negligible variance in the respondents' ratings confirms that the system's operational benefits are consistently recognized across all administrative and student-facing roles.

Analyzing the constructs of Functional Suitability, Reliability, and Perceived Usefulness, the findings suggest that stakeholders recognize the immediate practical value of the platform. The integration of automated, AI-assisted requirement verification effectively resolves the document review backlogs and communication bottlenecks associated with traditional paper-based and manual logbook operations previously utilized by the Scholarship and Financial Assistance Office (SFAO). Evaluators noted that core functionalities such as real-time application tracking, interactive AI chatbot guidance, multi-channel notifications, and strict multi-campus Role-Based Access Control (RBAC)

are highly effective upgrades that directly enhance the speed, accuracy, and security of scholarship processing across Laguna State Polytechnic University (LSPU).

Furthermore, the high ratings in Usability, Interface Responsiveness, and Perceived Ease of Use validate the project's user-interface design strategy. Scholarship portals demand intuitive navigation, as student applicants and administrative staff must manage multiple academic requirements efficiently. The evaluation proves that the architecture featuring a Next.js web dashboard for scholarship administrators and a responsive portal for students minimizes the learning curve. Users can effortlessly submit AI-verified applications, check status updates, interact with the AI assistant, and manage scholar profiles without requiring extensive technical training.

Finally, the exceptional scores in Overall Acceptability and Intention to Use represent the most critical indicator of the project's ultimate success. These results demonstrate that the student body and university personnel do not merely appreciate the software as an academic prototype; they are actively willing and prepared to adopt ScholarHub into their daily academic and administrative routines. Ultimately, the evaluation confirms that the system has achieved both technological stability and deep end-user trust, proving its complete readiness for deployment across the campuses of Laguna State Polytechnic University.

4.4 Discussion and Interpretation of Findings

The user evaluation results, culminating in an overall grand mean of 4.94 with a remarkably low standard deviation of 0.24, demonstrate an overwhelmingly positive assessment of the ScholarHub platform by its intended users. In alignment with the

research objective to measure system acceptability through the Technology Acceptance Model (TAM) and ISO 25010 standards, these figures indicate a strong, unified consensus among university IT administrators, SFAO scholarship coordinators, active student scholars, and student applicants. The negligible variance in the respondents' ratings confirms that the system's operational benefits are consistently recognized across all administrative and student-facing roles.

Analyzing the constructs of Functional Suitability, Reliability, and Perceived Usefulness, the findings suggest that stakeholders recognize the immediate practical value of the platform. The integration of automated, AI-assisted requirement verification effectively resolves the document review backlogs and communication bottlenecks associated with traditional paper-based and manual logbook operations previously utilized by the Scholarship and Financial Assistance Office (SFAO). Evaluators noted that core functionalities such as real-time application tracking, interactive AI chatbot guidance, multi-channel notifications, and strict multi-campus Role-Based Access Control (RBAC) are highly effective upgrades that directly enhance the speed, accuracy, and security of scholarship processing across Laguna State Polytechnic University (LSPU).

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applications, check status updates, interact with the AI assistant, and manage scholar profiles without requiring extensive technical training.

Finally, the exceptional scores in Overall Acceptability and Intention to Use represent the most critical indicator of the project's ultimate success. These results demonstrate that the student body and university personnel do not merely appreciate the software as an academic prototype; they are actively willing and prepared to adopt ScholarHub into their daily academic and administrative routines. Ultimately, the evaluation confirms that the system has achieved both technological stability and deep end-user trust, proving its complete readiness for deployment across the campuses of Laguna State Polytechnic University.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary, conclusions, and recommendations of the study based on the results gathered from the testing and evaluation of the developed ScholarHub: An AI-Enabled Web-Based Scholarship Management and Monitoring System with Chatbot and Multi-Channel Notifications. It includes the analysis and interpretation of the data obtained from system testing and user acceptance evaluation conducted among the Super Administrator, Campus Scholarship Officers/Coordinators, Student Scholars, and Student Applicants across the campuses of Laguna State Polytechnic University (LSPU).

5.1 Summary

This study aimed to design, develop, test, and evaluate a centralized digital platform for scholarship application, AI-assisted document verification, automated status tracking, scholar monitoring, and multi-campus administrative governance. The major findings of the study, aligned with the specific research objectives, are summarized as follows:

- System Development (Objective 1):

  The study successfully designed and developed ScholarHub as a web-based system featuring a responsive Student and Scholar Portal, an interactive AI Chatbot assistant, automated multi-channel notification services (Email/SMS/In-System), and a comprehensive Administrative Management Dashboard. The developed core modules include online scholarship application, AI document requirement scanner, real-time application tracking, scholar profile management, duty hour monitoring, and Role-Based Access Control (RBAC) supporting multi-campus oversight across LSPU campuses (Santa Cruz, Siniloan, Los Baños, and San Pablo).

- System Testing and Functionality (Objective 2): The developed system underwent rigorous Black-Box testing across twenty (20) distinct test cases covering all major features. The testing resulted in a 100% functional success rate, with all 130 recorded step-level evaluations marked as passed with zero (0) failed results.
- User Acceptance Evaluation (Objective 3): The system's acceptance was evaluated using the Technology Acceptance Model (TAM) by 306 respondents, predominantly comprising Students/Scholar Applicants (89.2%), along with

Student Scholars (8.9%), Campus Scholarship Officers (1.5%), and the Super Administrator (0.5%). The evaluation yielded an overall high mean interpretation across the key constructs: Quality Factors, User Experience, Behavioral Intention to Use, Attitude Toward Using, Perceived Usefulness, and Perceived Ease of Use.

5.2 Conclusions

Based on the summary of findings and the specific research objectives of the study, the following conclusions are drawn:

- Resolution of Operational Gaps: The development of ScholarHub successfully transitions the Scholarship and Financial Assistance Office (SFAO) of Laguna State Polytechnic University from a manual, paper-bound, and fragmented workflow to a structured digital infrastructure. The integration of AI-assisted document verification, centralized applicant queues, and automated multi-channel notifications directly resolves previously identified operational bottlenecks in application filing, document review backlogs, and status communication delays.
- Functional Reliability and Readiness: The system is functionally robust and operationally ready for university-wide deployment. The flawless 100% pass rate across 130 step-level Black-Box evaluations proves that the application's underlying logic, AI document scanning, database synchronization, notification services, and multi-campus RBAC permissions operate reliably without technical failure.
- High User Acceptability: ScholarHub is highly acceptable to its intended end-users. The positive evaluation ratings across all TAM constructs demonstrate that

students, scholarship coordinators, and administrators find the platform useful, intuitive, and highly beneficial for scholarship administration. Furthermore, the strong Behavioral Intention to Use scores confirm that stakeholders are eager to adopt the system as their primary portal for scholarship applications and management.

5.3 Recommendations

In light of the findings and conclusions of this study, the researchers propose the following recommendations:

- System Improvement: The university may consider enhancing the platform by integrating predictive analytics to assist scholarship committees in forecasting scholarship demand and budget allocations across academic terms. Additionally, refining the natural language processing capabilities of the AI Chatbot will allow it to address highly complex, multi-variable eligibility inquiries. Developers should also continue optimizing mobile responsiveness to ensure maximum usability on lower-end smartphone browsers.
- Platform Expansion: To enhance accessibility for students, future development should explore deploying dedicated mobile applications for iOS and Android platforms, complemented by push notification services to further streamline student engagement.
- Integration with University Systems: The university should explore integrating ScholarHub with the primary LSPU Student Information System (SIS) and

Registrar database. Direct database integration would enable automated grade verifications (GWA checking) and enrollment status confirmation without requiring manual document uploads.

- Future Research: Future researchers are encouraged to evaluate the platform using extended acceptance frameworks such as TAM3 or UTAUT2 to analyze additional factors including trust, facilitating conditions, and subjective norms. Furthermore, subsequent longitudinal studies should measure the physical reduction in application processing times and administrative workload following full-scale adoption by the university.

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APPENDICES A

(TECHNICAL BACKGROUND)

Appendix A 1. TRANSCRIBE INTERVIEW

Part 1 - Preliminary

Date: February 8, 2026

Respondent: Maria Rellie B. Kalacas

Position: Scholarship and Financial Assistance Chairperson

***Researchers**: Ayun po. Ah ma'am ako nga po pala si John Onyx po. Ito naman po yung kagrupo ko si Alfonso Castillo po. Ah yung anong tanong po namin, ah, pwede niyo po*

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bang i-describe kung pano po yung process ng scholarship application mula po sa submission ng students hanggang ma-approve po sila?

Maria Rellie B. Kalacas: *Dati kasi, ano, meron tayong dalawang klase ng scholarship. Meron tayong internal and external. So, noon, kapag internal, edi yung requirements isasubmit dito tapos i-screen ng chairperson yung mga applicants and then i-interview din then approval kung okay yung papers tapos ay, edi saka pa lang sila maka-qualified na scholar, ano. So, yung external naman, dahil yun yung current na meron tayo, wala na tayong internal scholarship. So, mostly yung mga requirements nanggagaling dun sa mga partner na, uh, industry, 'no, mga private, uh, industry or, um, LGU na nagbibigay ng scholarship sa atin. So, ang nangyayari, yung requirements nila ay nanggagaling sa kanila tapos pinapadala na lang sa amin kung sino yung qualified. So, ibig sabihin hindi na kami yung, uh, nag-i-screen ng papers nila at hindi na kami yung liable para magsabing ikaw ay qualified or hindi kasi from there sa industry, napili na nila or na-screen nila kung sino yung qualified para sa kanila na maging scholars. And then kapagka binabaan na kami ng pangalan ng mga qualified scholars, ang gagawin naman namin dito sa SFA, magpapapill out kami ng form na sila ay scholar ng particular na company na yun or industry na yun. And then from SFA form na sila ay scholar ng company na yun, uh, kukunin namin yung requirements base sa hinihingi ng industry. Or there are times na wala na kaming control sa requirements kasi kinuha na rin ng industry. So application form and then pagkatapos ng application form, sasabihin sa amin or i-email sa amin kung kailan nila marereceive yung allowance. From receiving the allowance, i-ko-coordinate namin yan sa accounting then si accounting kung coordinate sa cashier. Pagka-ready na yung allowance nila, si cashier tatawag dito tapos i-issuehan namin ng certificate yung mga students or yung mga scholars para makuhan nila yung allowance. Ano, yung iba naman, kat-- like the OST, CBK before nung meron pa kami, hindi rin kami ang nagre-release ng allowance. Ang dumadaan lang talaga sa LSPU, nagre-receive ng allowance na meron kami ay yung coming from CHED at tsaka yung EBIDEN. And then the rest of the scholarship ay sila na then ang nagbibigay ng allowance. So kami, monitoring lang usually kung sino yung mga scholars ng mga particular scholarship na meron kami. Pero yung mga requirements dahil na external scholarship, hindi kami ang may control. Taga-record lang kami kung sino na yung mga scholars, currently scholars ng mga iba't ibang scholarship na meron tayo sa school.*

Researchers*: pano niyo po pala nire-record ta in-i-store po yung mga application nila po?*

Maria Rellie B. Kalacas: *So sa rec po, sa records naman ng application, uh, sabi ko nga sa'yo kanina, hindi kami ang nagpapa-apply ano, dahil, uh, yung ating partner industry or yung ating mga nagbibigay from external partners, ano, yung, uhm, yung may kapet ng mga requirements. Kami dito, ang nere-record namin or ang ini-keep namin yung SFA, SFA form zero zero one. Yung records na kung sino yung scholar. So*

mano-mano, uh, pini-fill out nila, hard copy. Tapos pagka-file na yun sa ARC file, i-i-scan ng staff then i-store sa- Sa digital po. Mm-hmm. Yan, ini-scan lang nila para may soft copy din kami. So ganun. Manual tapos i-i-scan para magkaroon kami ng soft copy. And then yung, uh, ibang requirements katulad dun sa AVID din na pina-col-- pinapa-collect sa amin, ganun din. Manual pinapa-submit namin tapos aka ini-scan. Yun, ganun.

Researchers: *may mga na-encounter po ba kayong problem kadalasan po or konti lang naman po?*

Maria Rellie B. Kalacas*: Marami ng problem like doon sa mga data ng mga scholars kasi from time to time, 'di ba, yearly po mag-graduate sila. So we have to update kung sino na ang na-remove kasi kailangang i-report, uh, yung current at saka yung- Yung tapos na. -bilang, oo, na natapos na scholars doon sa mga, uh, partner natin. Uh, dahil nga siya ay manu-mano, ano, eh 'di isa isa siyang hahanapin- Mm-hmm. -kung sino ang nakagraduate base halimbawa doon sa program na binigay sa amin or we seek, ano, help from re-- the Office of the Registrar para ma-confirm namin na yung students na yun ay graduate na or nag-drop or tumigil or, uh, nag-file ng LOA, yung mga ganun. So ang nagiging problema namin mostly ay yung data ng bata kasi mas maganda sana if we have the, ano tawag doon, yung form sa soft copy na kapag kinlick ko yung name ng scholar, lalabas na doon yung status niya. Halimbawa year level, anong scholarship meron siya, uh, saan siya nakatira, yung address, contact number, anong kurso.*

Researchers: *Pagka nagge-generate po kayo, halimbawa po may humingi po sa inyo ng reports, pano niyo po ginagawa yan? Like naka ano po ba-- may system po ba kayo dun or wala po? Tala-- mano-mano din po?*

Maria Rellie B. Kalacas*: Dun kami nag babase sa data sa softcopy namin*

Researchers: *If ever po pala na magagawa namin, may mga naiisip po ba kayong maging possible issues po if ever magawan po namin ng system 'yun, yung problem na ganun po? Ah, prob-- issues sa system?*

Maria Rellie B. Kalacas*: Wala naman akong naiisip na magiging problema maliban sa i-co-coordinate ko sa director para hindi lang ako yung gagamit kasi since tayo ay apat na campuses, kailangan ma-approve din ng director namin. Kahit halimbawa tayo mag-pilot ano, so with the approval of our director, yung paggamit niyan system para in case na dito natin i-pilot testing, maibigay din natin sa ibang campus para system yung gagamitin namin.*

Part 2 - Preliminary Interview (Responder Perspective)

Date:

Respondent:

Researchers:

MAT Responder:

Another responder:

Researchers:

MAT Responder:

Another responder:

USE CASE DIAGRAM

A. Super Admin – The Super Administrator oversees the entire LSPU ScholarHub platform. This user is responsible for managing all user accounts, system configurations, and enforcing role-based access levels (RBAC). The Super Administrator also configures

scholarship program rules, system security parameters, and generates high-level performance analytics and compliance reports for university leadership.

B. Admin (Scholarship Administrators) – Admins handle administrative operations via the web dashboard. They are responsible for managing scholarship program categories, setting eligibility guidelines, monitoring active scholarship cycles, and managing system-wide multi-channel notification broadcasts.

C. Staff (Scholarship Officers / Evaluators) – Staff members manage application evaluation operations through the web interface. They review submitted student applications, validate uploaded documentary requirements, update processing statuses (e.g., "Under Review," "Approved," "Disapproved"), and provide evaluation feedback or remarks on student records.

D. Students / Applicants – Students are the primary initiators of the scholarship workflow. They use the web platform to submit scholarship applications assisted by the AI Chatbot for eligibility checking and inquiry resolution, upload required documents, track the real-time progress of their submitted applications, and receive multi-channel updates (Email, SMS, and System Alerts) regarding their application status.

SYSTEM CONTEXT DIAGRAM

Four main groups of users interact with LSPU SCHOLARHUB: An AI-Enabled Web-Based Scholarship Management and Monitoring System with Chatbot and Multi-Channel Notifications, namely the Super Administrator, Scholarship Administrators (Admin), Scholarship Staff, and Students. All users are required to register and authenticate secure accounts, governed by Role-Based Access Control (RBAC) and JWT authentication, before accessing the features of the system. Registered users perform specific functions based on their assigned roles within the platform. The Super Administrator oversees overall system operations, manages account privileges, and generates analytical reports for university leadership. Scholarship Administrators utilize a web-based dashboard to configure scholarship offerings, establish criteria, and manage platform communication flows. Scholarship Staff handle verification procedures by reviewing submitted files, validating eligibility proofs, and updating application progress. On the other hand, Students interact with the platform to submit online applications, consult the integrated AI Chatbot for automated guidance and document checking, track the real-time resolution status of their scholarship applications, and receive automated status updates through multi-channel notifications.

LSPU ScholarHub
User Storyboard

The storyboard presents the sequence of screens for each user role, describing the purpose of every page and the action that leads to the next screen.

Guest User Storyboard

The following figures present the screens available to the guest role (4 screens).

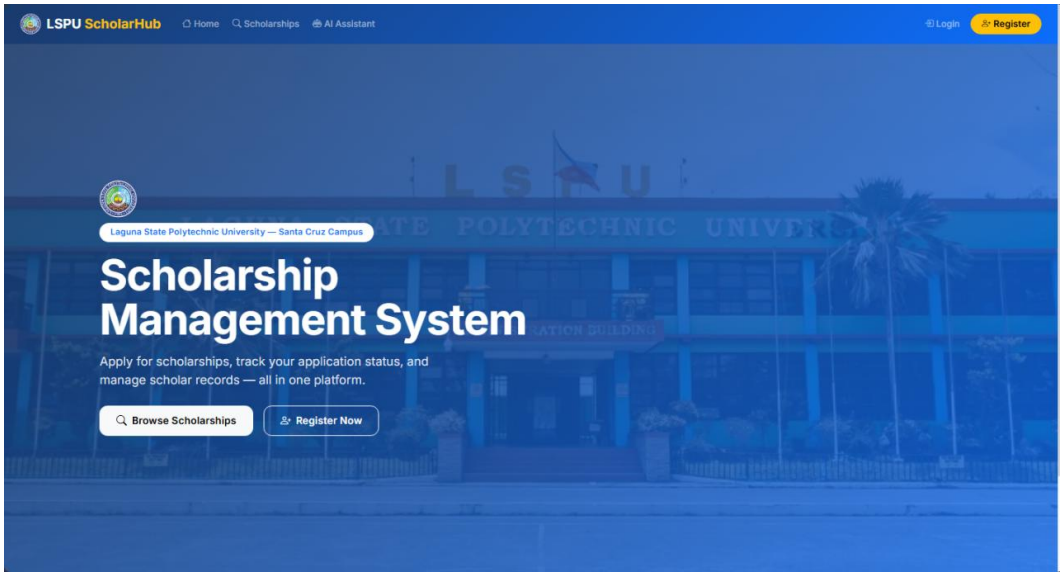


Figure 1. Landing Page

The public entry point of the system. It presents the Scholarship Office information, an overview of available scholarship programs, and navigation options for logging in or registering.

↓ *User Action: The guest clicks Register to create an account.*

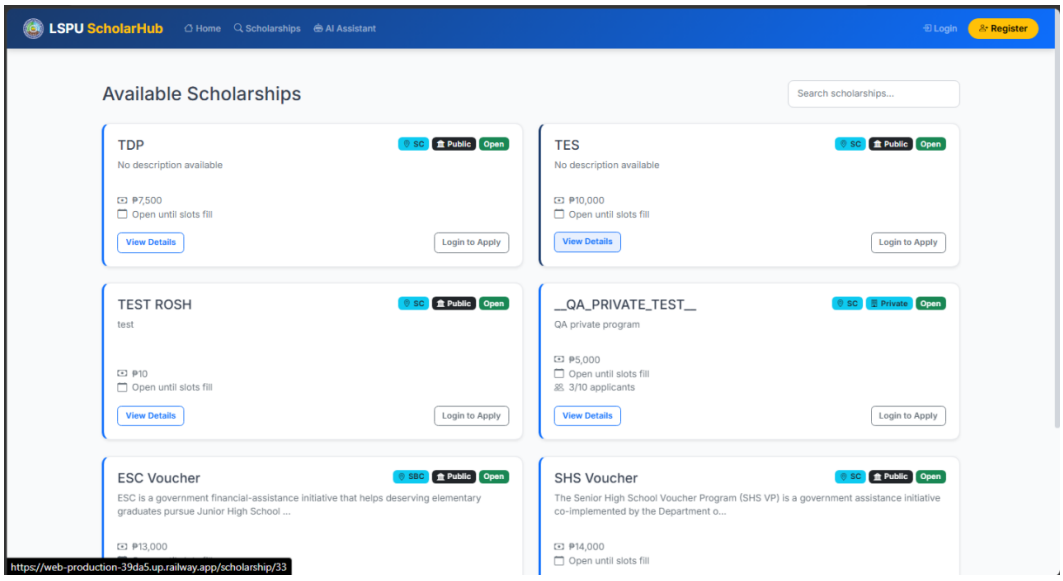


Figure 2. Browse Scholarships

Allows visitors to view all publicly available scholarship programs, including descriptions, benefits, and eligibility criteria, before creating an account.

↓ *User Action: The guest clicks Login to access an existing account.*

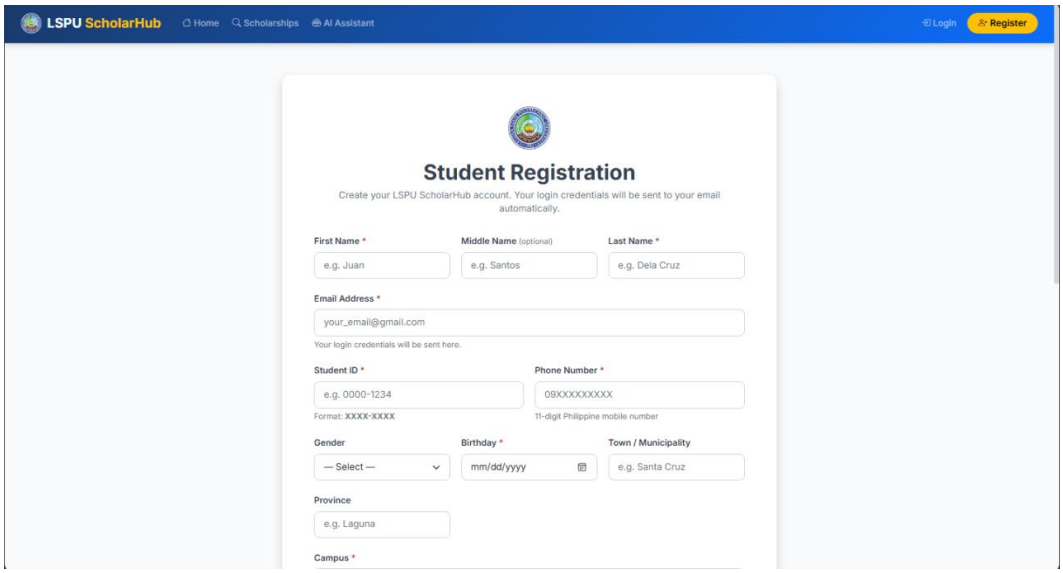


Figure 3. Registration Page

Enables new students to create an account by uploading their Certificate of Registration (COR) and Report of Grades (ROG). OCR automatically extracts the name, student number, course, year level, units, and GWA, and verifies them against the uploaded files.

↓ *User Action: After successful registration, the student is redirected to the login page.*

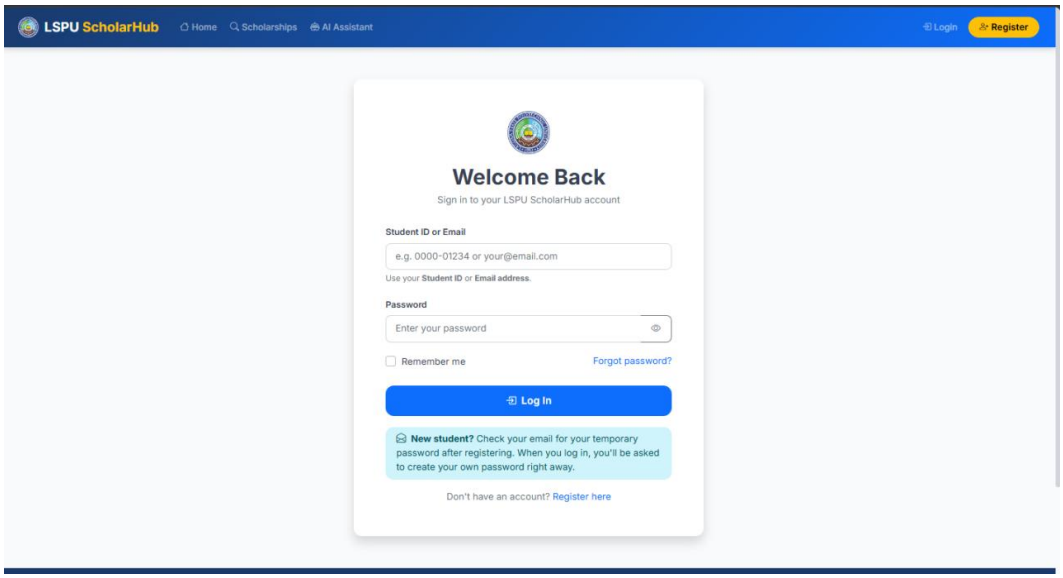


Figure 4. Login Page

Provides secure authentication for all users. Each user is redirected to the dashboard appropriate to their role after logging in.

Student User Storyboard

The following figures present the screens available to the student role (10 screens).

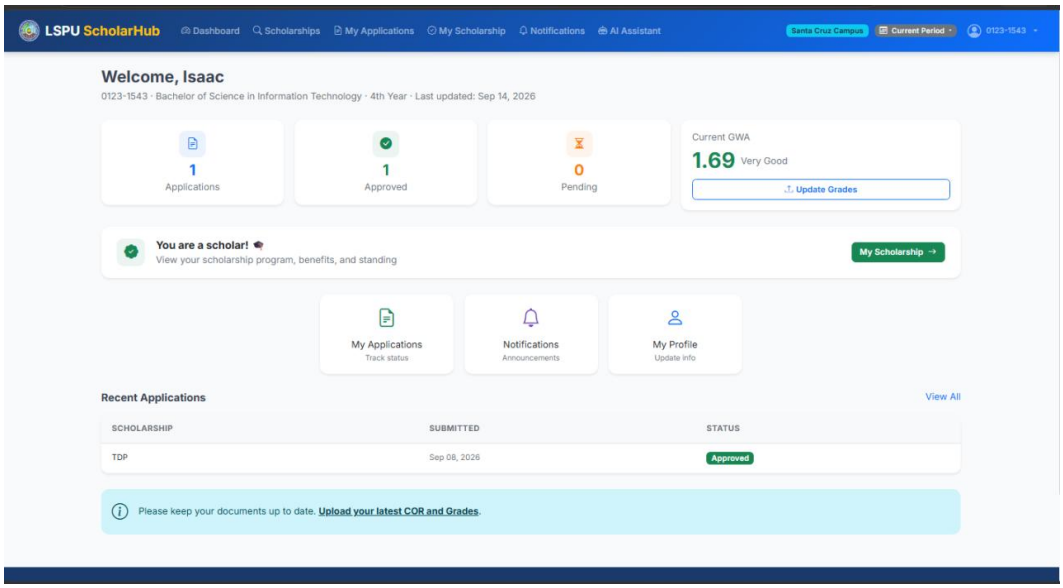


Figure 5. Student Dashboard

Displays the student's overview: total applications, pending/under-review count, approved scholarships, and a reminder to keep their COR and grades up to date.

↓ *User Action: The student clicks Browse Scholarships in the navigation bar.*

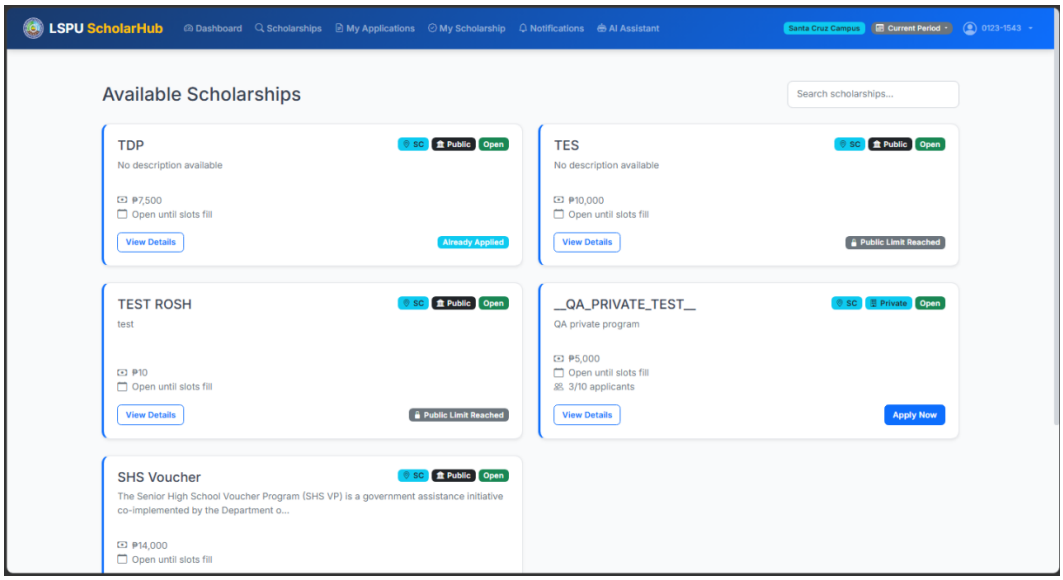


Figure 6. Browse Scholarships

Shows scholarship programs available to the student based on campus and academic period, with an Apply button for open programs. The system enforces the one-public-program rule.

↓ *User Action: The student clicks Apply on a scholarship program.*

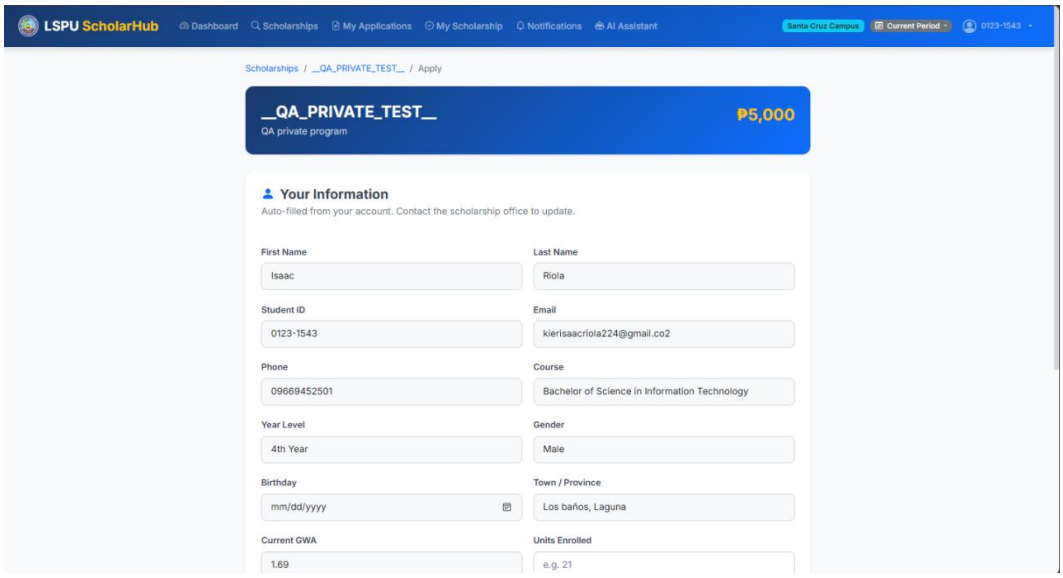


Figure 7. Scholarship Application Form

The application form with student details pre-filled from the profile (birthday, course, GWA, units). The student verifies the information and submits the application.

↓ *User Action: After submission the student opens My Applications.*

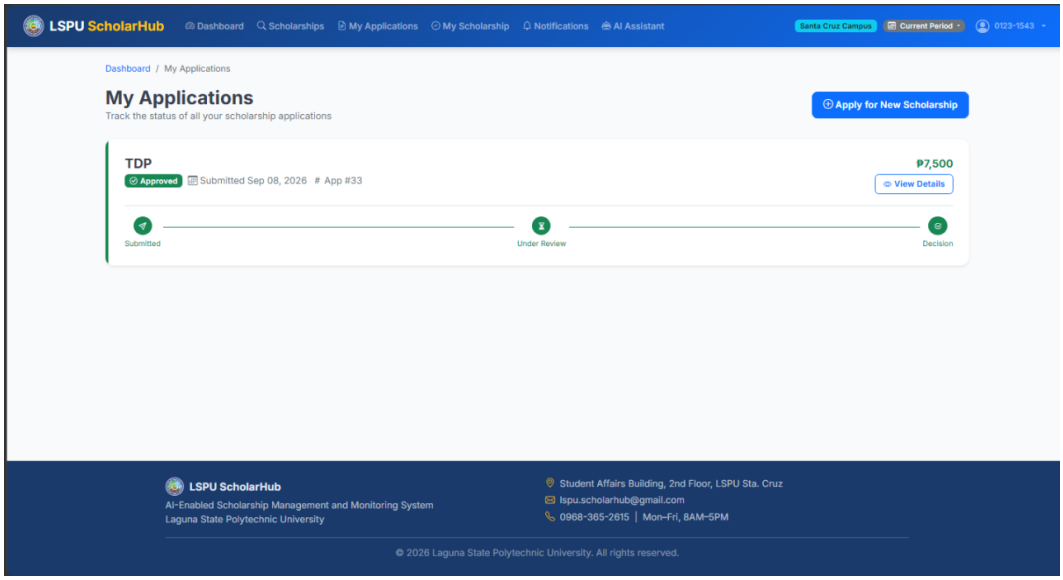


Figure 8. My Applications

Lists all submitted applications with status tracking (Submitted, Under Review, Approved, Rejected) and the scholar tag per application.

↓ *User Action: The student clicks an application to view its details.*

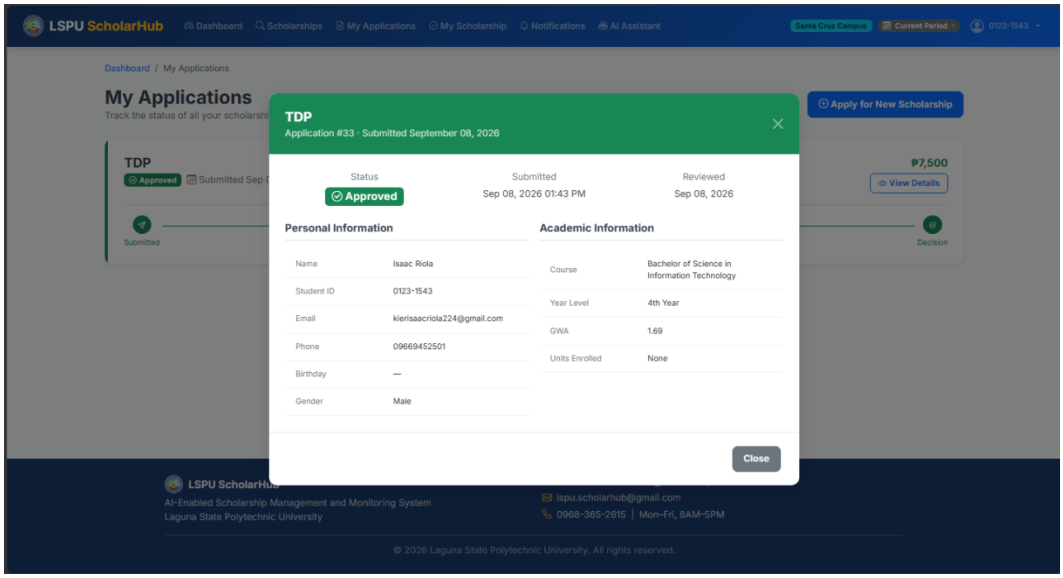


Figure 9. Application Details

Shows a single application with a visual progress tracker and review notes from the Scholarship Office. Editing is locked once the application is under review.

↓ *User Action: The student returns to the dashboard and opens My Scholarship.*

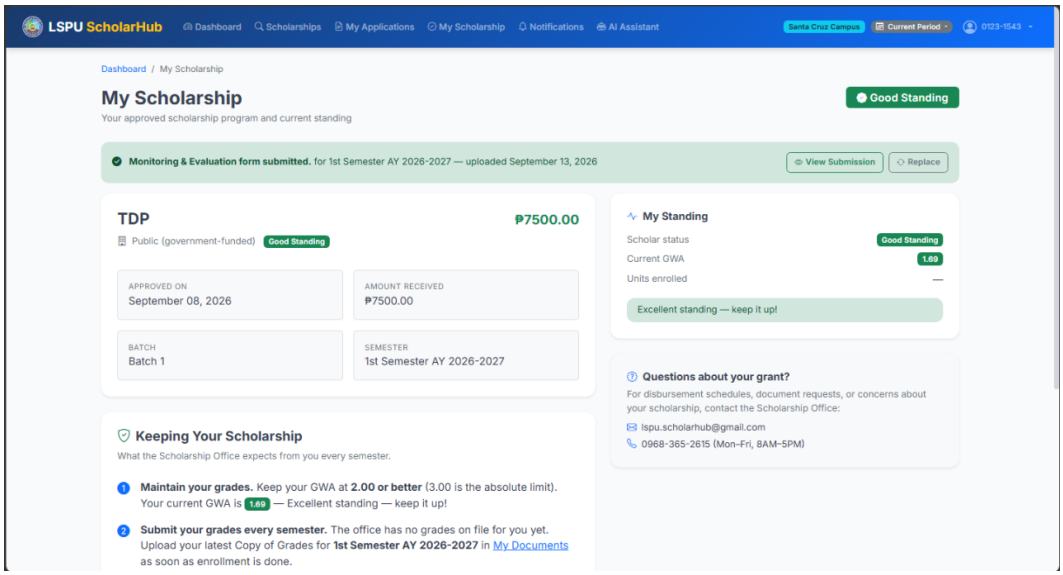


Figure 10. My Scholarship

Displays the details of approved scholarship program(s): program name, benefits, award amount, batch assignment, and scholar status.

↓ *User Action: The student opens Notifications from the navigation bar.*

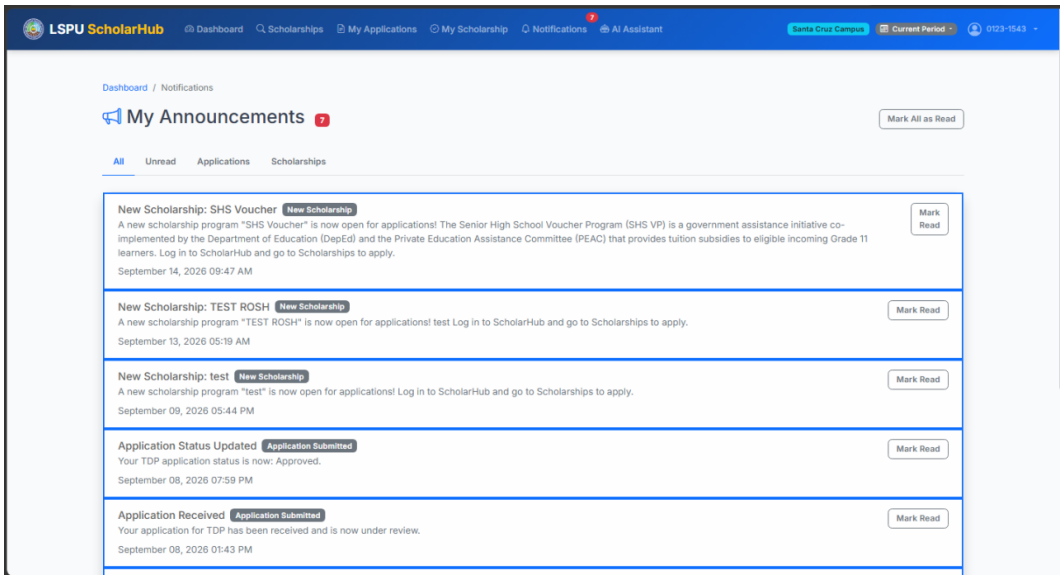


Figure 11. Notifications

Shows announcements and notifications sent by staff and administrators, filtered by academic period, including title, message, and optional attachments.

↓ *User Action: The student opens My Profile.*

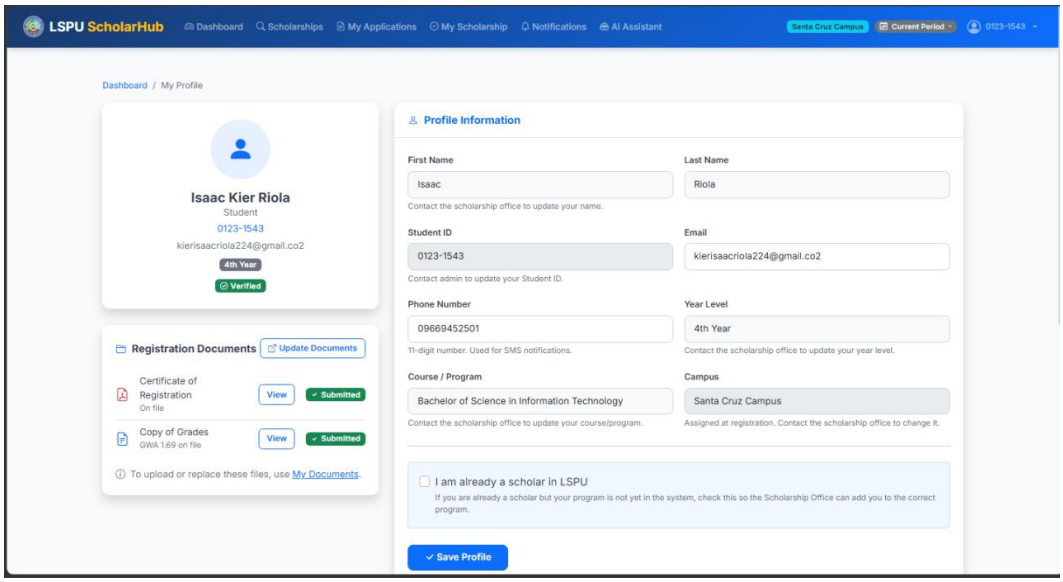


Figure 12. My Profile

Displays the student's information. Email and phone number are editable; the name, student number, campus, course, and year level are locked and require Scholarship Office assistance.

↓ *User Action: The student opens the Documents page.*

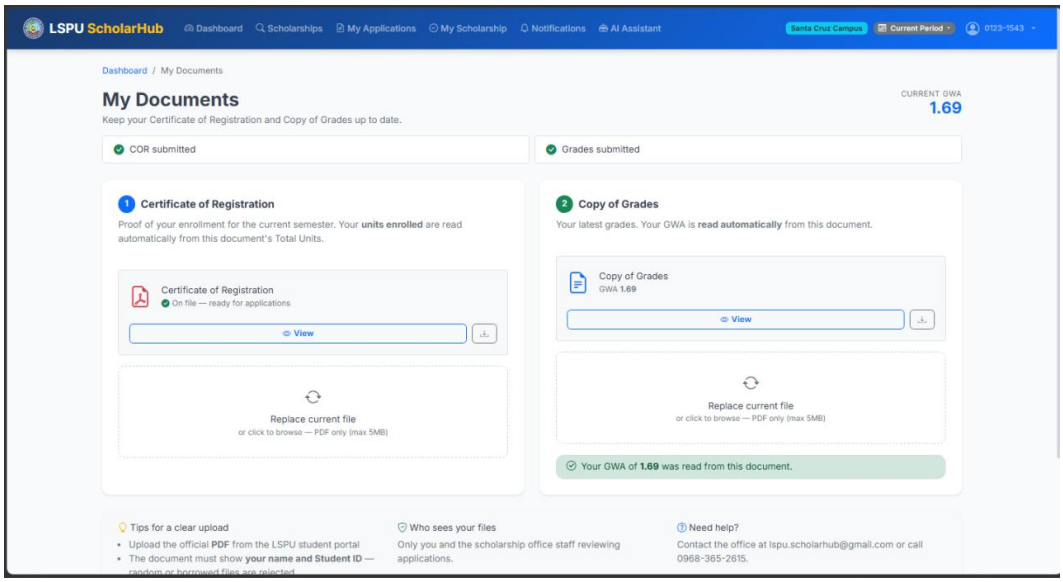


Figure 13. Documents (COR / ROG)

Allows the student to upload or replace their Certificate of Registration and Report of Grades (PDF only, 5 MB limit). OCR re-validates the student details on upload.

↓ *User Action: The student opens the AI Chatbot.*

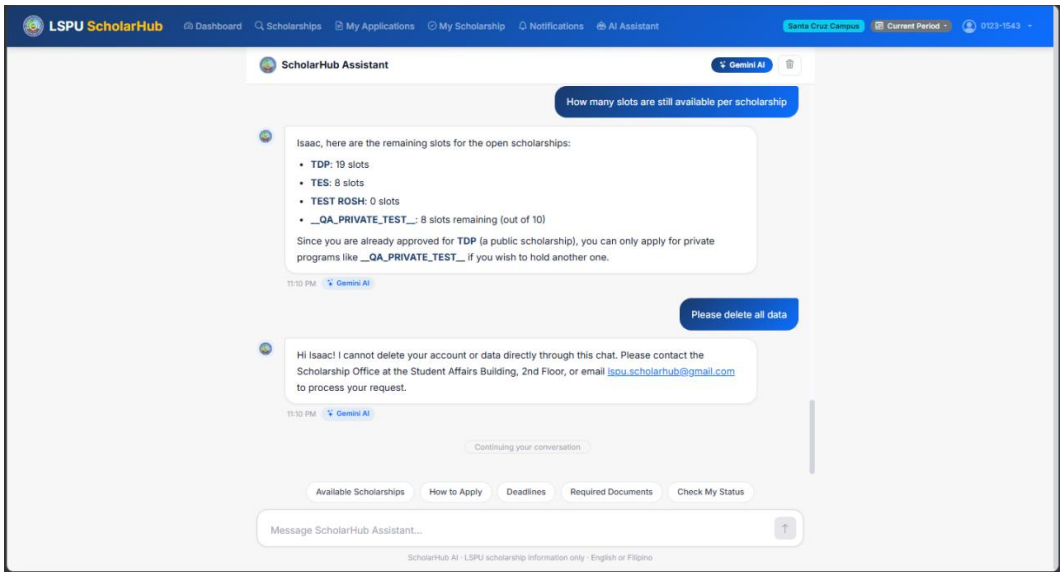


Figure 14. AI Chatbot

An AI-powered assistant that answers scholarship-related questions based on actual system data such as programs, eligibility, and application status.

Staff User Storyboard

The following figures present the screens available to the staff role (9 screens).

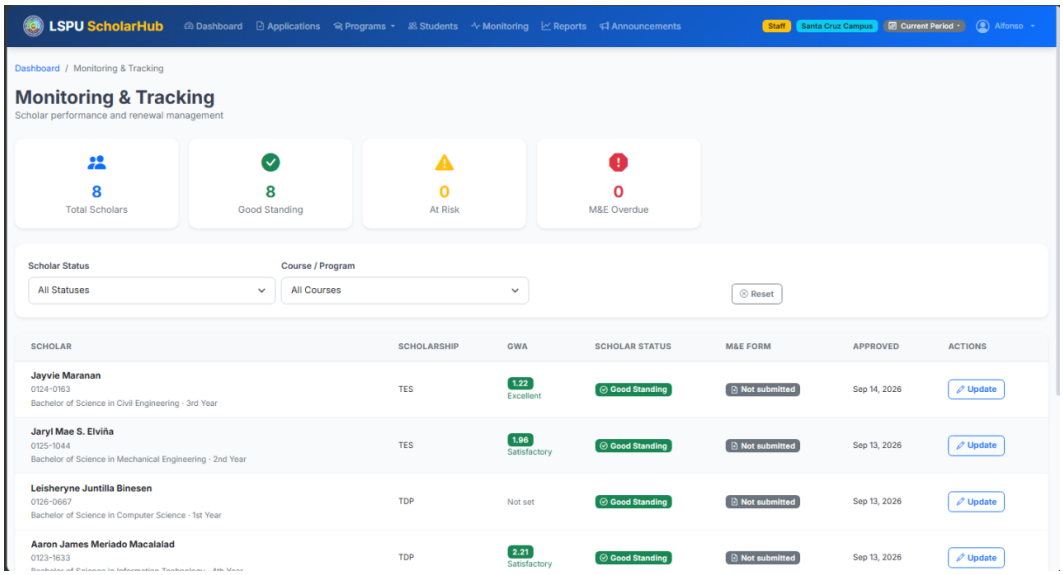


Figure 15. Staff Dashboard

Shows campus-level statistics: total applications, pending review, approved scholars, and recent activity, scoped to the selected campus and academic period.

↓ *User Action: The staff opens Manage Scholarships.*

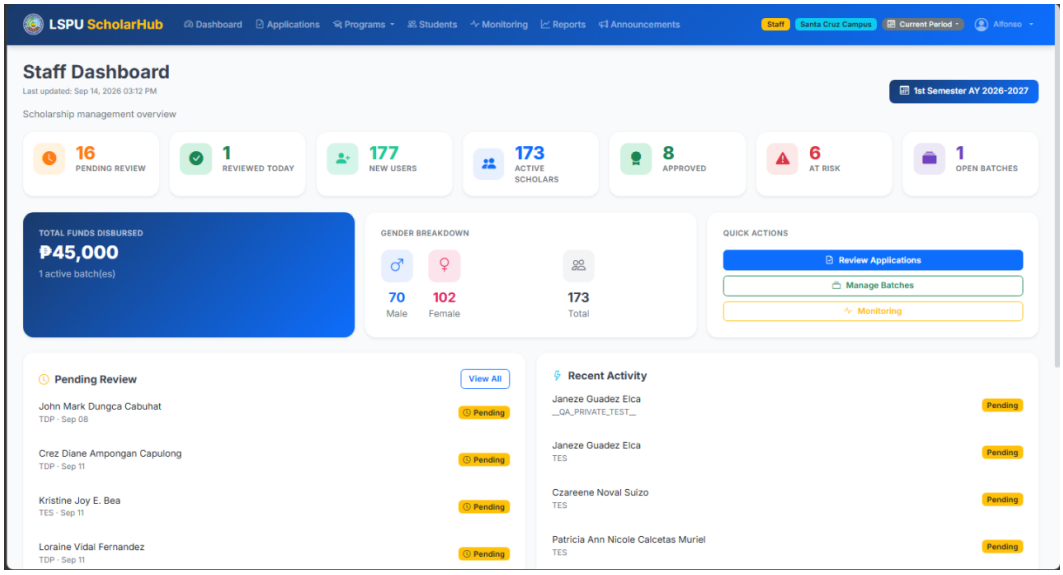


Figure 16. Manage Scholarships

Lists scholarship programs with create, edit, close, and visibility controls. Programs with applicants cannot be edited.

↓ *User Action: The staff clicks Add New Program.*

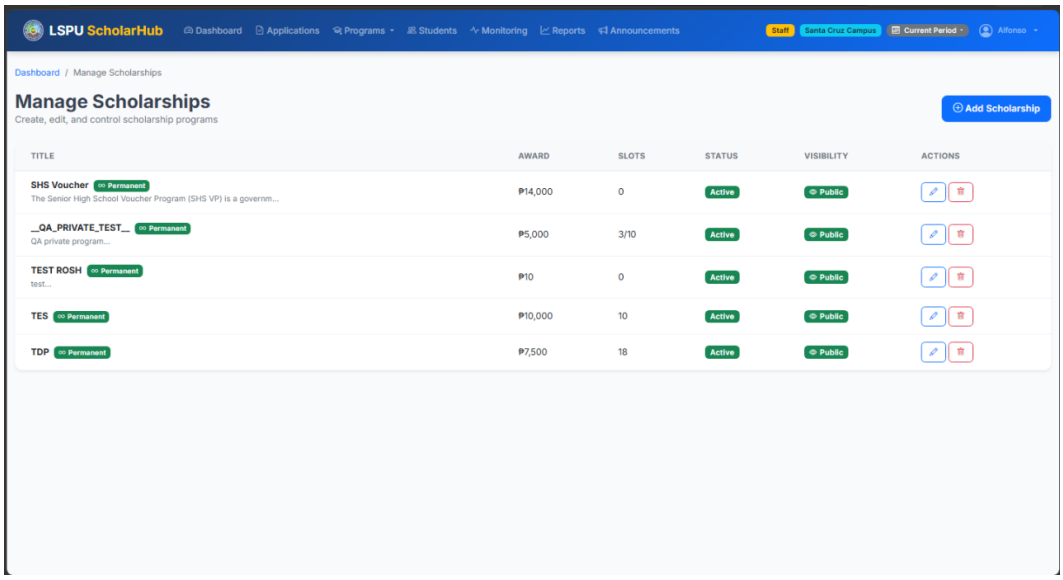


Figure 17. Create Scholarship Program

The form for creating a scholarship program: name, description, benefits, funding type (public/private), campus, academic period, maximum applicants, and custom application fields.

↓ *User Action: The staff saves and opens Review Applications.*

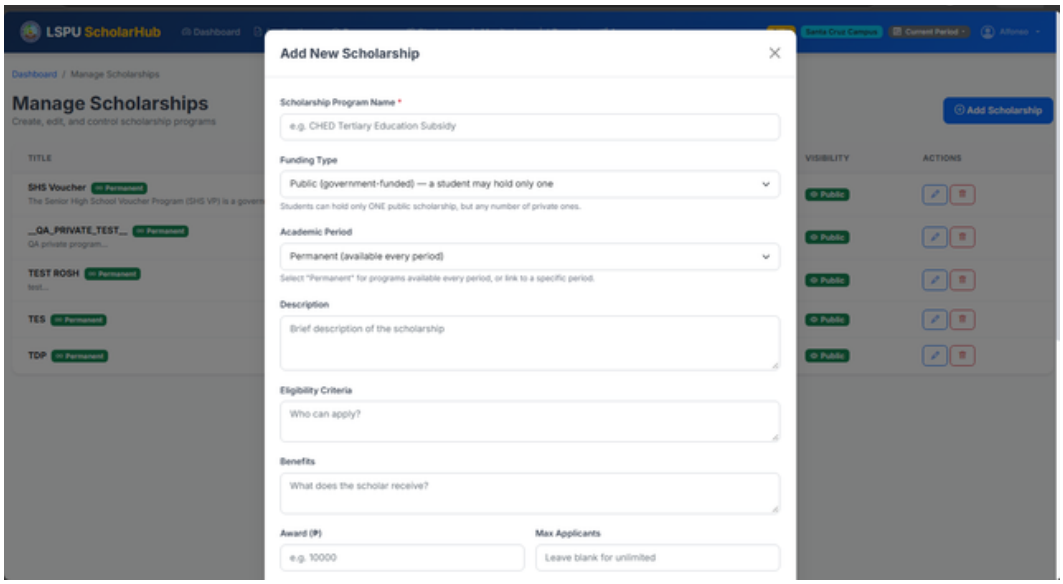


Figure 18. Review Applications

Displays submitted applications with student details, GWA, and documents. Staff can approve or reject applications with notes.

↓ *User Action: Approved students are batched; the staff opens Batches.*

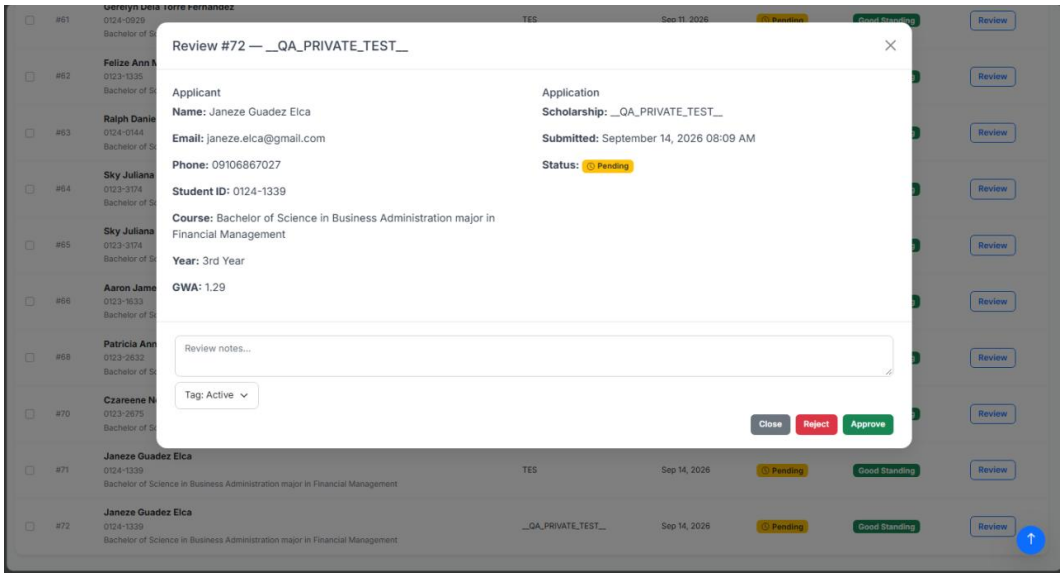


Figure 19. Batch Students

Allows staff to group approved applicants into batches per scholarship program and export batch lists to Excel.

↓ *User Action: The staff opens the Student Directory.*

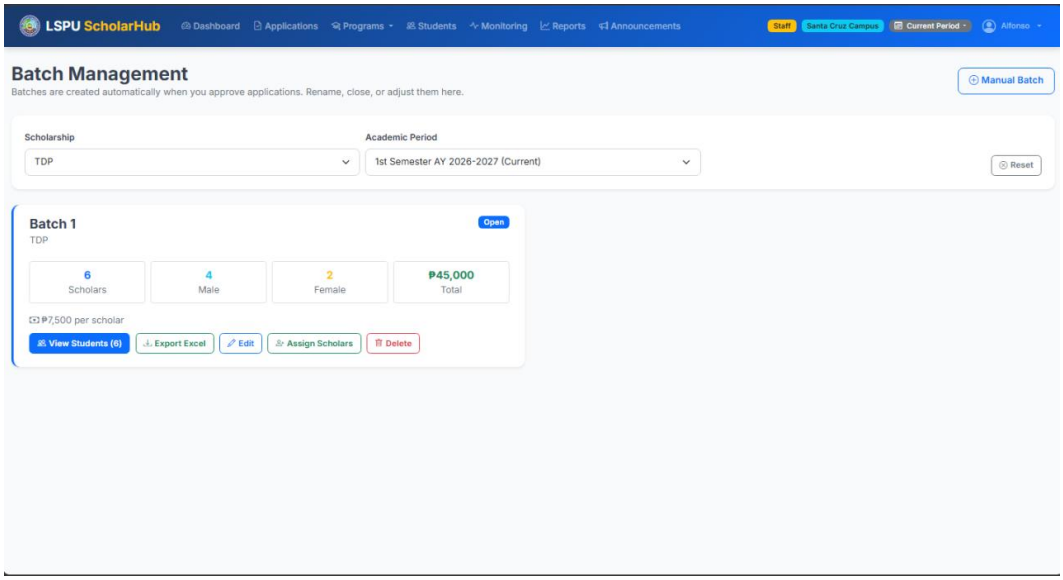


Figure 20. Student Directory

A searchable, filterable list of registered students with their profiles, documents, and application status.

↓ *User Action: The staff opens Announcements.*

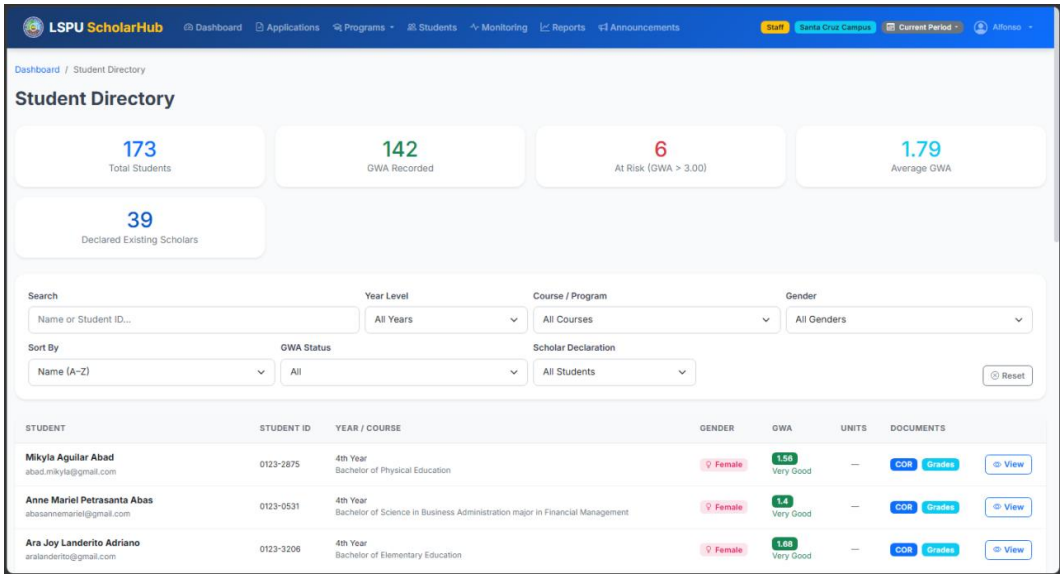


Figure 21. Send Announcement

The announcement composer with recipient selection (all students, by scholarship, by batch, or individual) and optional file attachment, sent via email and SMS.

↓ *User Action: The staff opens Reports.*

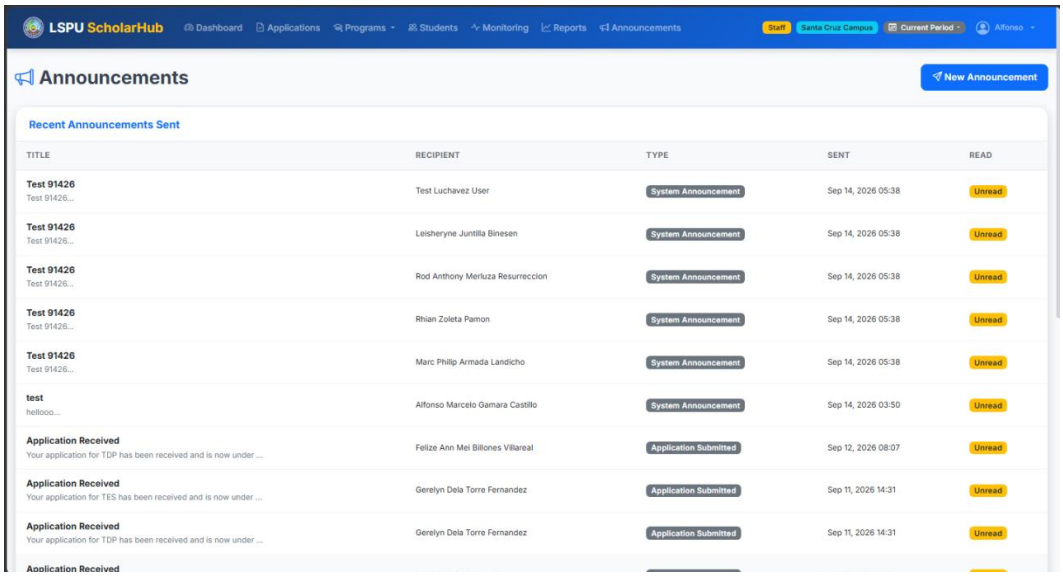


Figure 22. Reports

Analytics and charts on applications, approvals, GWA distribution, and scholarship statistics for the selected period and campus.

↓ *User Action: The staff opens Monitor Scholars.*

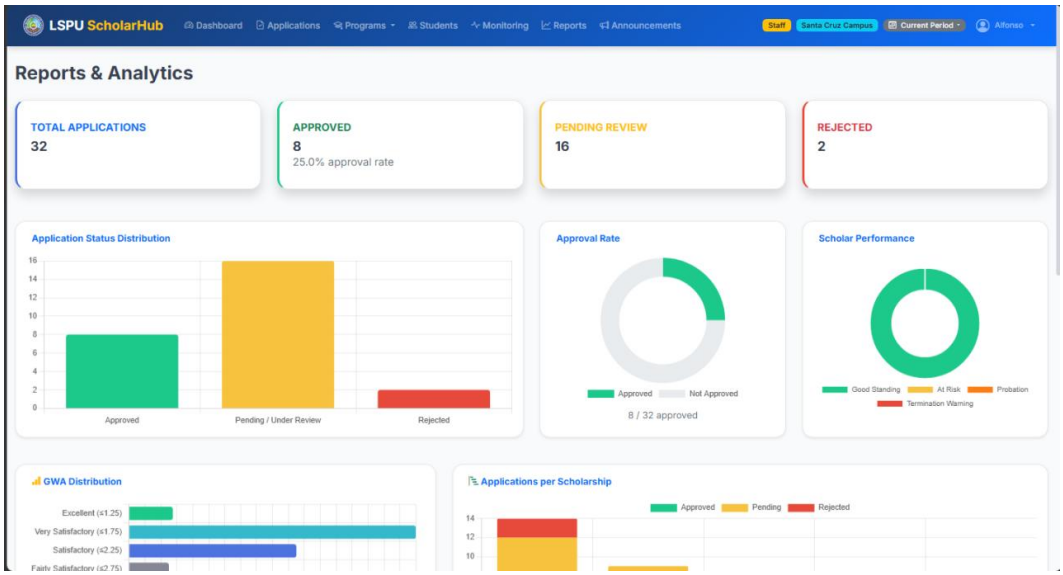


Figure 23. Monitor Scholars

Tracks active scholars, their status, and Monitoring and Evaluation form submissions which staff can view and verify.

Admin User Storyboard

The following figures present the screens available to the admin role (7 screens).

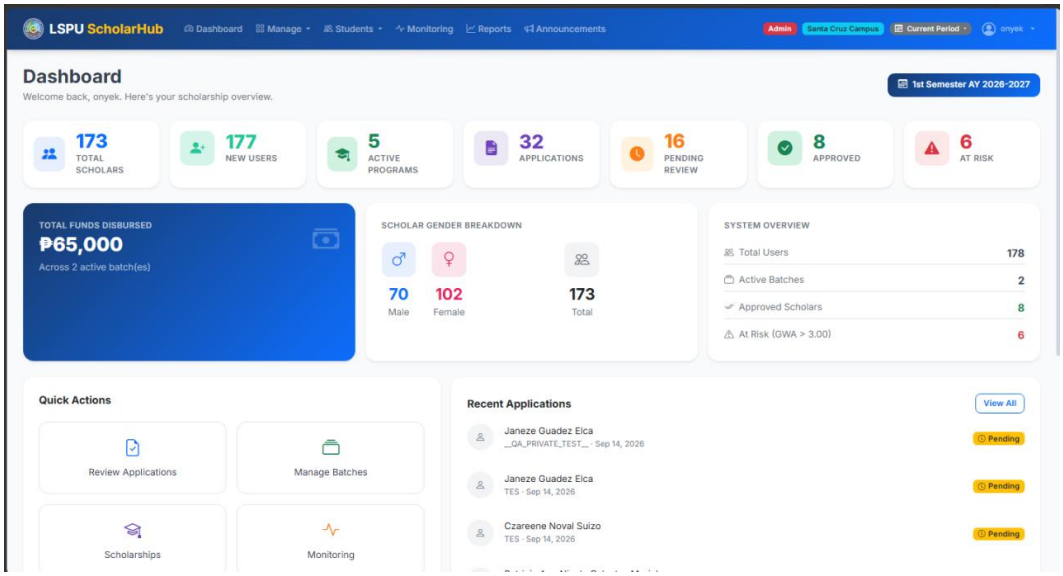


Figure 24. Admin Dashboard

Campus overview with statistics and an academic period selector that scopes every page to the chosen period.

↓ *User Action: The admin opens Manage Users.*

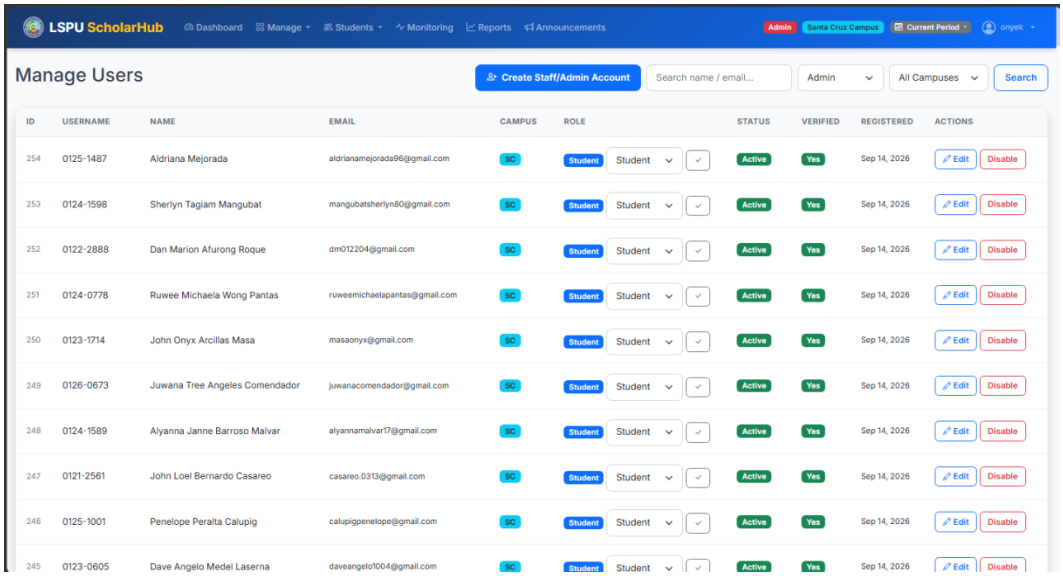


Figure 25. Manage Users

User account management in one place: creating accounts, changing roles, assigning campuses, and deactivating users.

↓ *User Action: The admin opens Manage Scholarships.*

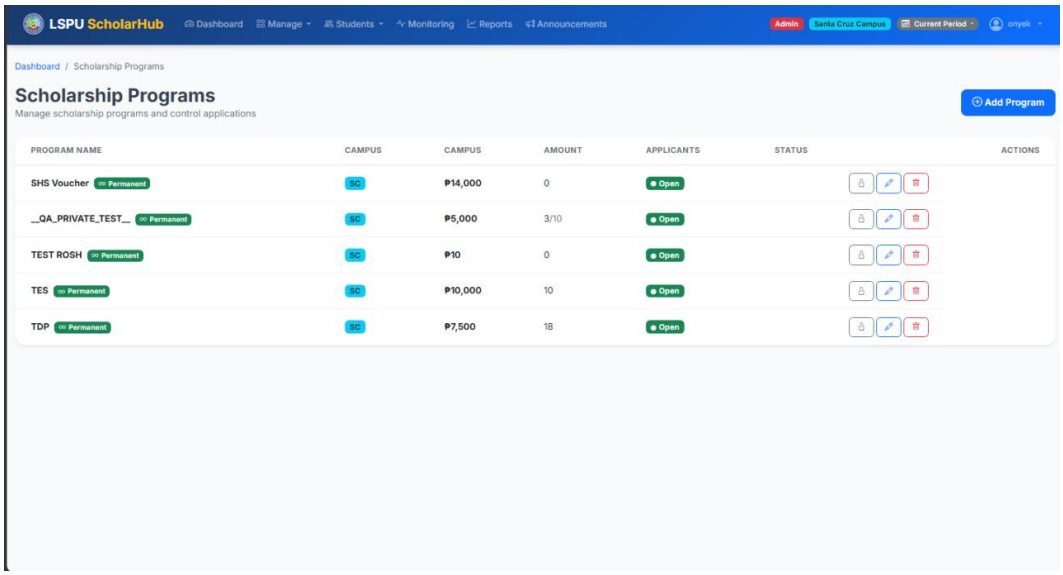


Figure 26. Manage Scholarships

The same scholarship management as staff, scoped to the admin campus: create, edit, close, and delete programs.

↓ *User Action: The admin opens Manage Campuses.*

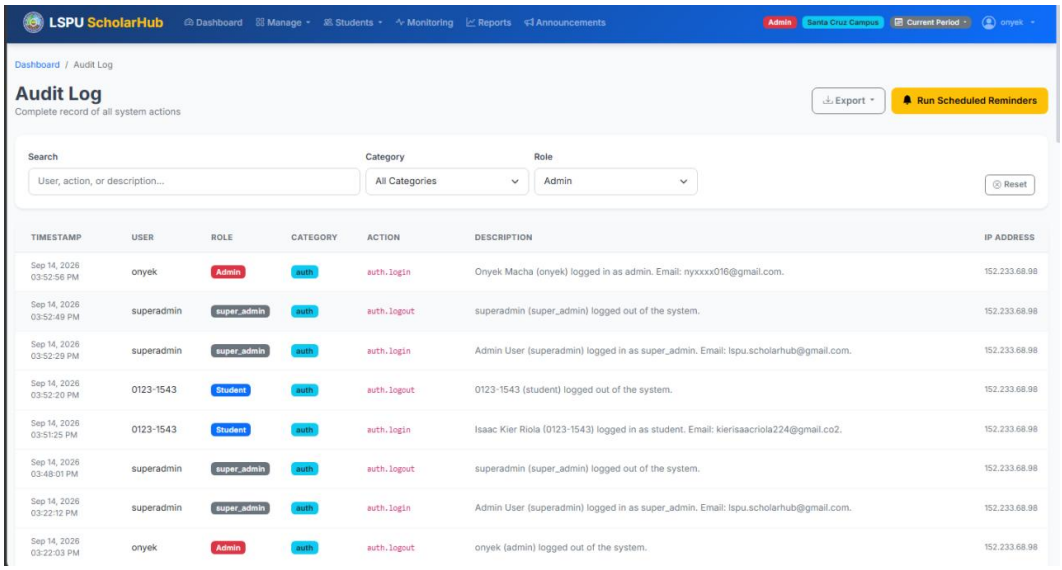


Figure 27. Audit Log

A filterable activity trail (search, category, and role filters) of every significant action performed in the system, with Excel/PDF export.

↓ *User Action: The admin opens Reports.*

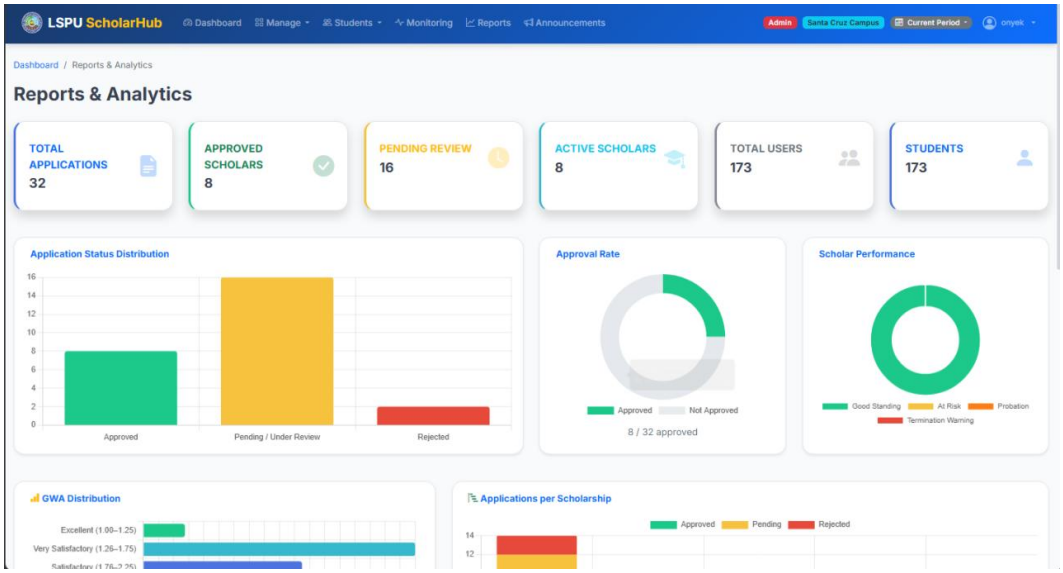


Figure 28. Reports

Campus analytics: applications, approvals, GWA distribution, and scholarship statistics scoped by period and campus.

↓ *User Action: The admin opens the Student Directory.*

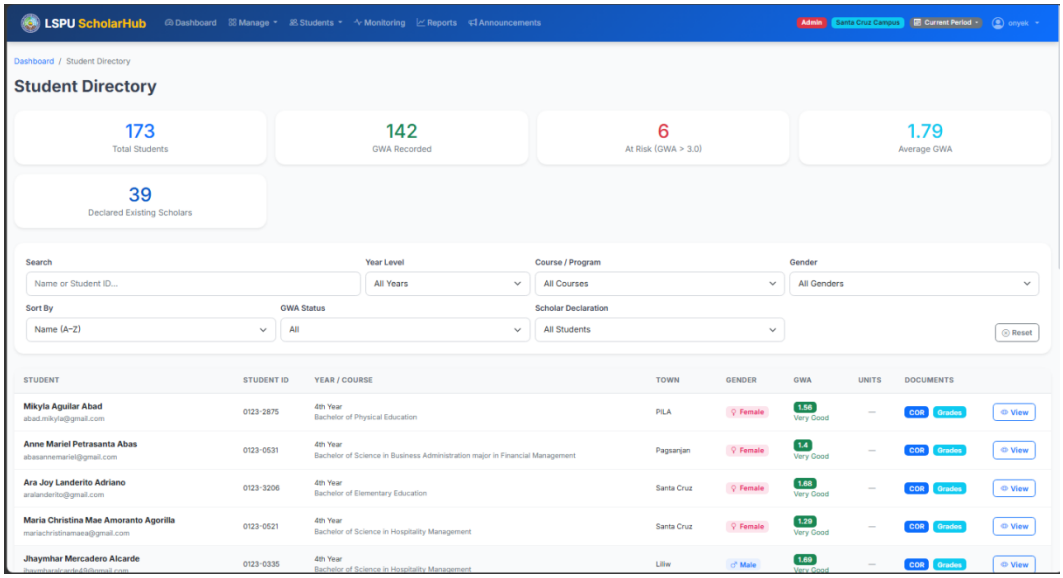


Figure 29. Student Directory

Searchable student records with profiles, documents, application status, and scholar information.

↓ *User Action: The admin opens Announcements.*

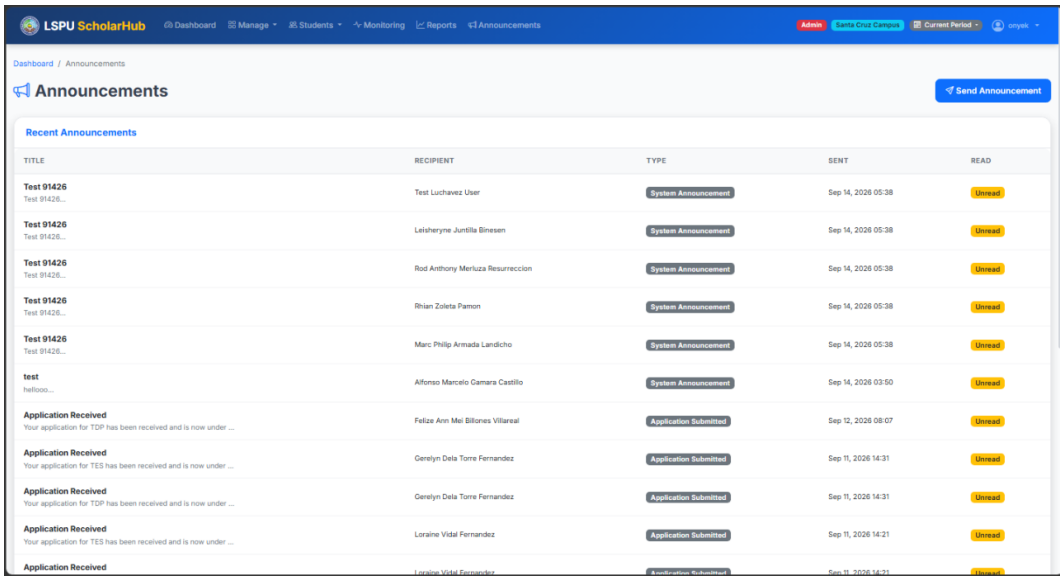


Figure 30. Send Announcement

The announcement composer with period-aware recipient lists and optional attachments, delivered by email and SMS.

Super Admin User Storyboard

The following figures present the screens available to the super admin role (7 screens).

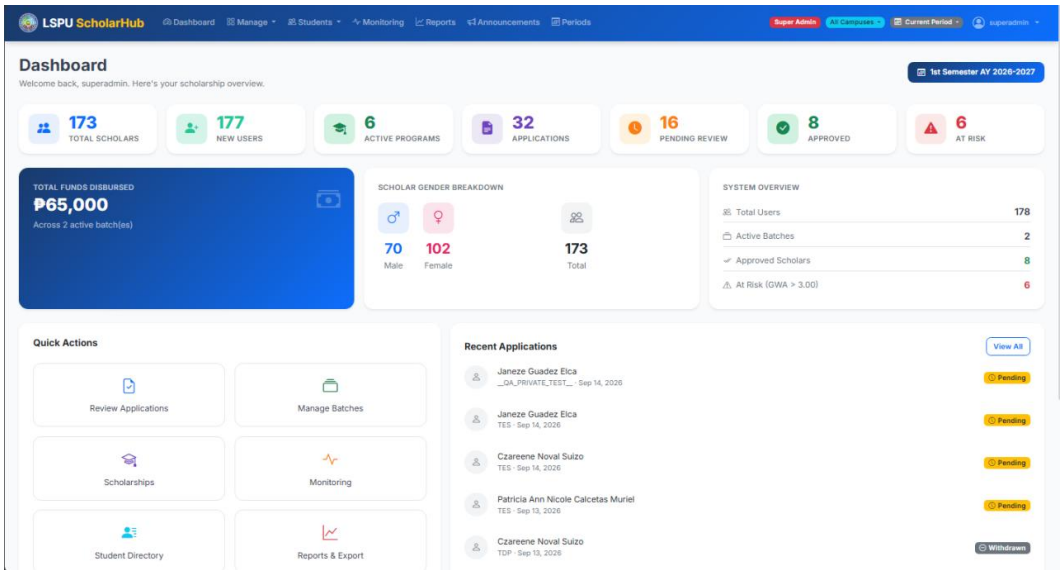


Figure 31. Super Admin Dashboard

University-wide statistics with a campus selector; the selected campus scopes all data shown.

↓ *User Action: The super admin opens Manage Users.*

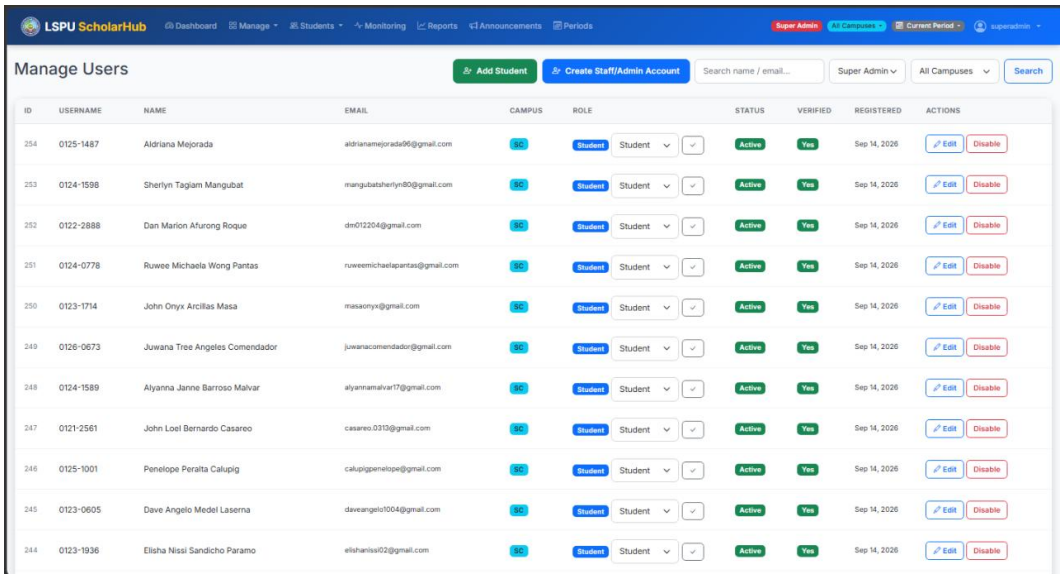


Figure 32. Manage Users

System-wide account management across all campuses: creating users, assigning roles (super admin, admin, staff, student), and multi-campus assignments.

↓ *User Action: The super admin opens Manage Campuses.*

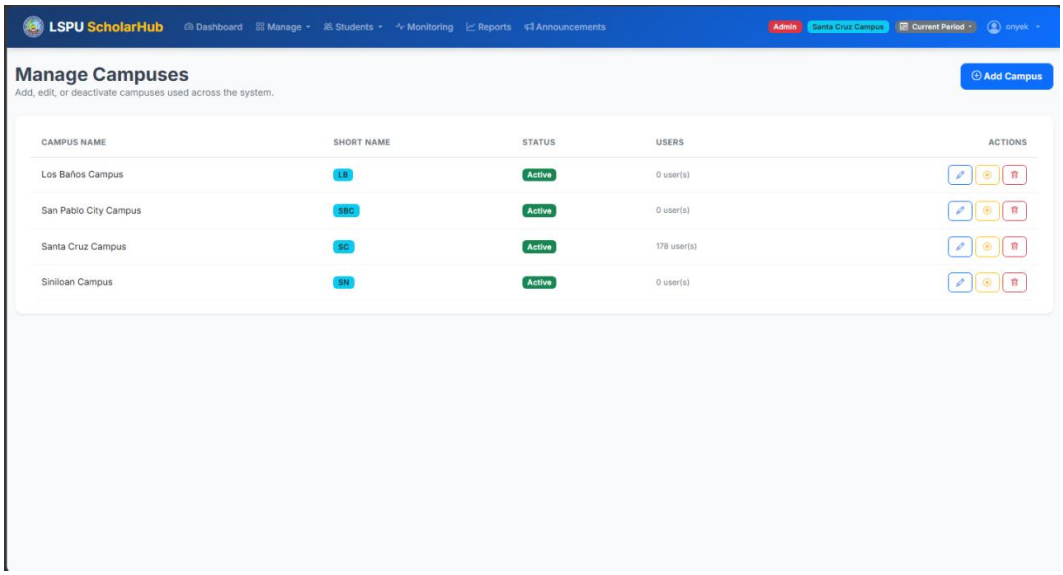


Figure 33. Manage Campuses

All campus records for the university, with codes and configuration used by registration, staff assignment, and data scoping.

↓ *User Action: The super admin opens Academic Periods.*

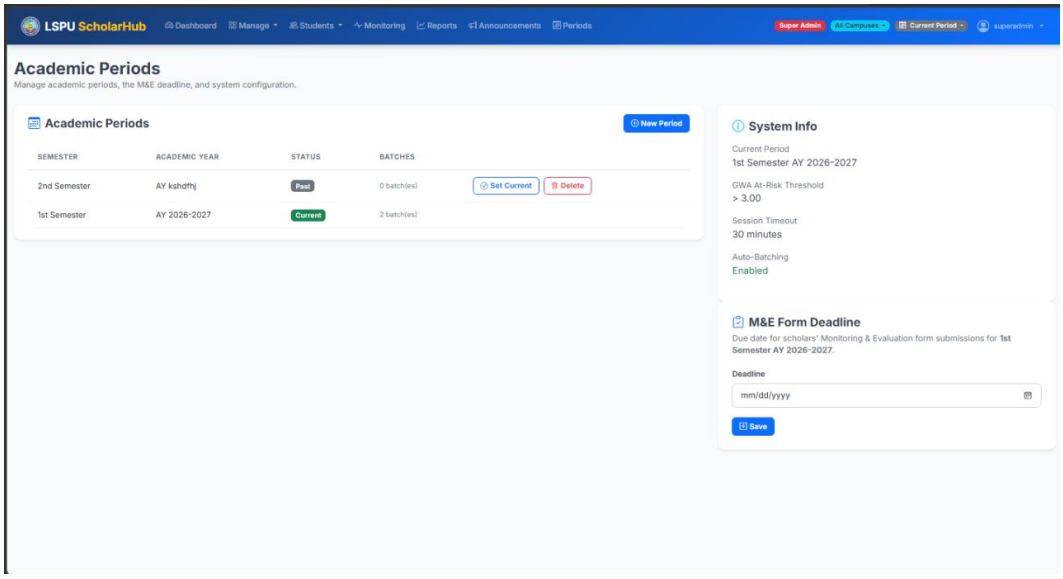


Figure 34. Manage Academic Periods

University-wide academic period management: create, edit, delete, and set the current period that scopes system data.

↓ *User Action: The super admin opens the Audit Log.*

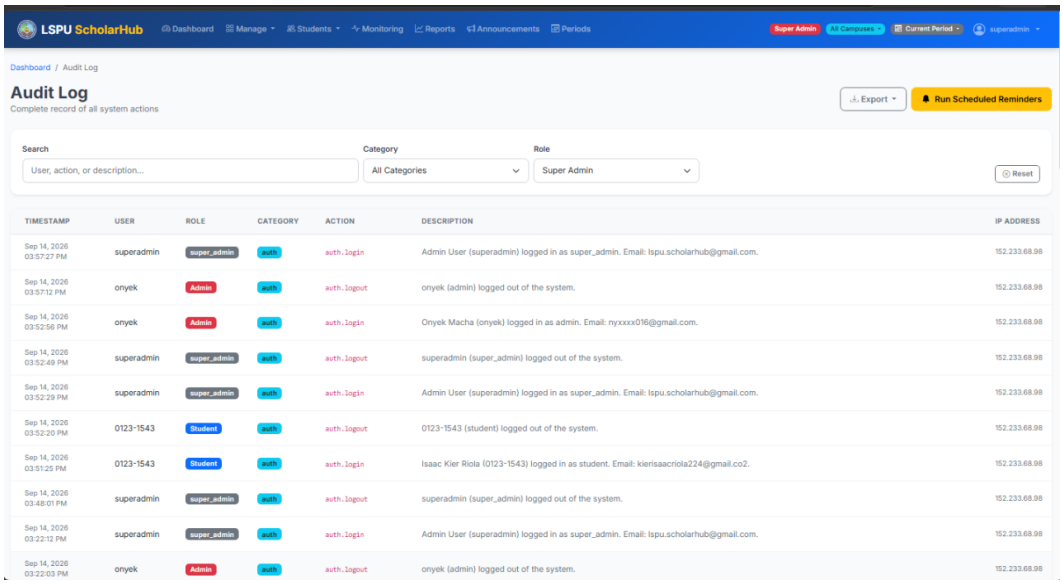


Figure 35. Audit Log

The complete audit trail of every campus and user, with category, role, and search filters and export options.

↓ *User Action: The super admin opens Reports.*

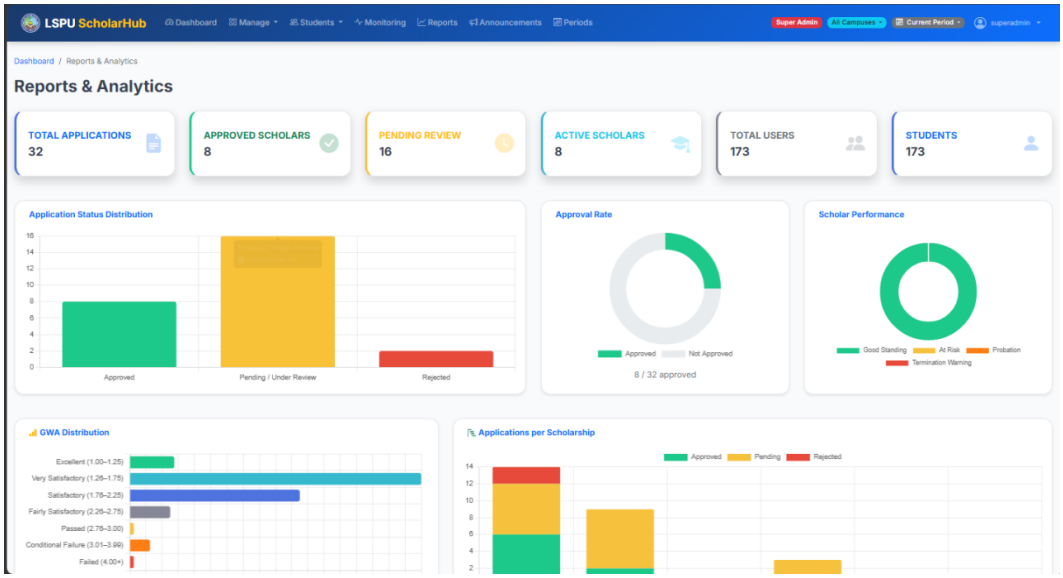


Figure 36. Reports

University-wide analytics with per-campus or aggregated views across all campuses and periods.

↓ *User Action: The super admin uses the campus selector.*

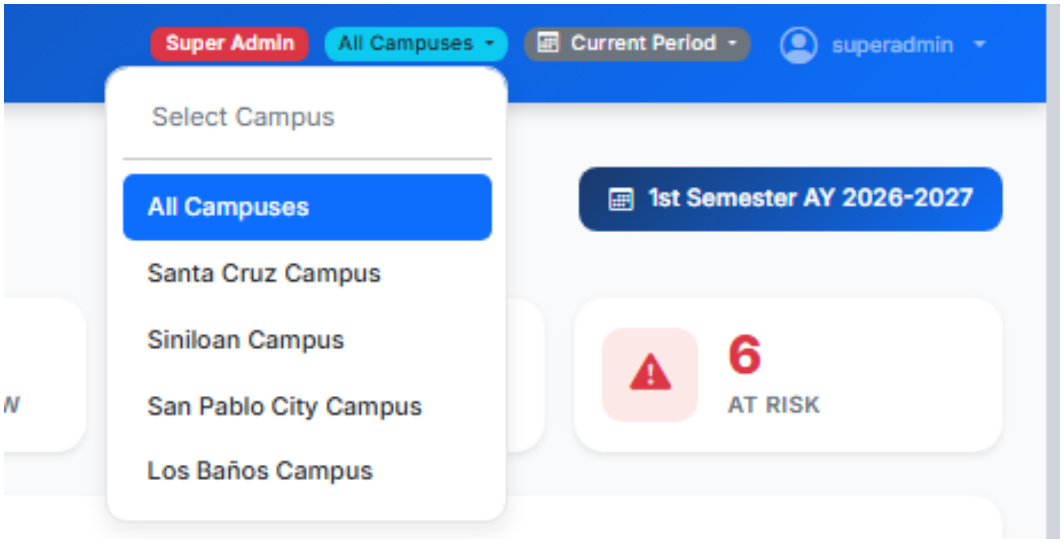


Figure 37. Campus Selector

The header dropdown that switches the campus context; every page then reflects the chosen campus and period.

CURRENT PROCESS FLOW

PROPOSED PROCESS FLOW

Figure 1. Student Registration Process Flow

Process Flow	Responsibility	Description
	Guest / System	START — Guest opens the registration page.
	Guest	Guest uploads Certificate of Registration (COR) and Report of Grades (ROG) in PDF format.
	System	System validates file type and size (PDF only, 5 MB limit). Invalid files are rejected.
	System	System extracts student details via OCR (name, student ID, course, year level, units, GWA) and auto-fills the form.
	Guest	Guest verifies the extracted details and completes birthday, email, and password.
	System	System verifies details match the uploaded files and checks for duplicate accounts.
	System	System creates the student account.
	System	END — Account created; student is redirected to the login page.

Figure 2. Scholarship Application Process Flow

Process Flow	Responsibility	Description

Process Flow	Responsibility	Description
START	Student / System	START — Student logs in and opens the scholarship programs page.
PROCESS	System	System displays open programs for the student's campus and current academic period.
PROCESS	Student	Student selects a program and opens the application form.
INPUT	System	System pre-fills student details (birthday, course, year level, GWA, units) from the profile.
DECIDE	System	System checks eligibility: no duplicate application and one public program rule.
PROCESS	System	System saves the application and queues it for review.
START	System	END — Application submitted; confirmation notification sent to the student.

Figure 3. Application Review and Approval Process Flow

Process Flow	Responsibility	Description
START	Staff / System	START — Staff opens the Review Applications module.
PROCESS	System	System displays pending applications filtered by campus and academic period.
PROCESS	Staff	Staff examines the student details and documents.

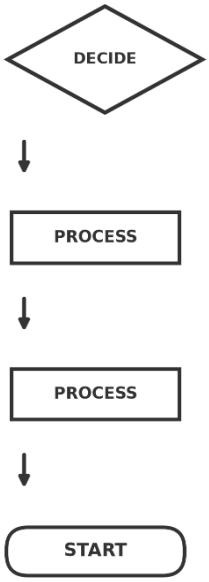
	**Staff** **System** **System** **System**	**Staff decides: approve or reject the application.** **System updates the application status and increments the program's approved count.** **System notifies the student via in-app notification and email.** **END — Application approved or rejected; student informed.**

Figure 4. Batching of Approved Scholars Process Flow

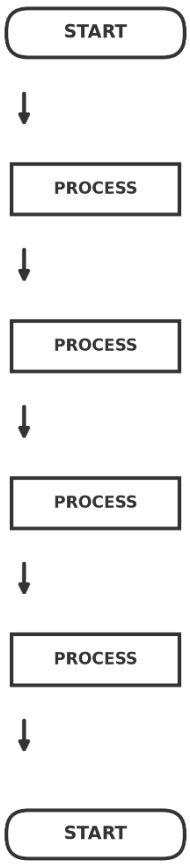
Process Flow	Responsibility	Description
	**Staff / System** **System** **Staff** **System** **System** **System**	**START — Staff opens the Batches module.** **System displays existing batches and approved students awaiting assignment for the period.** **Staff creates a batch and selects the scholarship program.** **System groups approved students by program and assigns them to the batch.** **System links the batch to the campus and academic period, then notifies the batched scholars.** **END — Scholars organized into batches ready for fund disbursement.**

Figure 5. Scholar Monitoring and Evaluation Process Flow

Process Flow	Responsibility	Description
START	Student / System	START — Student opens the scholarship page.
INPUT	Student	Student downloads the Monitoring and Evaluation form.
INPUT	Student	Student uploads a photo or PDF of the accomplished form.
PROCESS	System	System validates the file and stores it under the student's scholarship record.
PROCESS	Staff	Staff reviews scholar compliance, GWA, and submitted M&E forms.
DECIDE	Staff	Staff evaluates: compliant or at risk?
START	Staff / System	END — Scholar compliance monitored and recorded for the period.

Figure 6. Announcement Management Process Flow

Process Flow	Responsibility	Description
START	Staff / System	START — Staff opens the Announcements page.
PROCESS	Staff	Staff clicks Send Announcement and selects recipients (students, by scholarship, or by batch).

INPUT	**Staff**	**Staff writes the title and message and optionally attaches files.**
DECIDE	**System**	**System checks: is the viewed period the active period? If not, a confirmation warning appears.**
PROCESS	**System**	**System delivers the announcement via in-app notification and email (and SMS when enabled), tagged with the academic period.**
START	**System**	**END — Announcement delivered; recipients notified.**

Figure 7. Student Document Update Process Flow

Process Flow	Responsibility	Description
START	**Student / System**	**START — Student opens the Documents page.**
PROCESS	**System**	**System displays the current COR and ROG on file.**
INPUT	**Student**	**Student uploads a replacement document (PDF only).**
DECIDE	**System**	**System validates the file and re-extracts the student details via OCR. Details must match the account.**
PROCESS	**System**	**System replaces the stored file.**

START	System	END — Student records updated with the latest documents.

Figure 8. Academic Period and Campus Management Process Flow

Process Flow	Responsibility	Description
START	Super Admin / System	START — Super Admin opens the Periods page.
PROCESS	System	System displays all academic periods and campuses.
PROCESS	Super Admin	Super Admin creates, edits, or deletes an academic period and sets the current active period.
PROCESS	System	System applies the period as the default data context for all users.
PROCESS	Admin / Super Admin	Admin switches the campus selector to manage campus-scoped data.
PROCESS	System	System filters dashboards, applications, reports, and announcements by campus and period.
START	System	END — System data scoped correctly per campus and academic period.

Figure 9. User and Role Management Process Flow

Process Flow	Responsibility	Description
START	Admin / System	START — Admin opens the Manage Users page.

PROCESS	System	System displays user accounts filtered by campus (all campuses for Super Admin).
PROCESS	Admin	Admin creates a user, changes a role, or assigns a staff member to one or more campuses.
PROCESS	System	System validates the changes, updates the account, and records the action in the audit log.
START	System	END — User accounts and roles updated with a full audit trail.

Figure 10. Scholarship Program Management Process Flow

Process Flow	Responsibility	Description
START	Staff / System	START — Staff opens the Manage Scholarships page.
PROCESS	System	System displays scholarship programs with period badges (Permanent or period-specific).
PROCESS	Staff	Staff creates a new program or edits an existing one.
DECIDE	System	System checks: does the program already have applicants? Editing is blocked if yes.
PROCESS	System	System saves and publishes the program to students of the assigned campus and period.
START	System	END — Scholarship program created or updated and visible to eligible students.

Flowchart Symbols Used

Symbol	Name	Meaning
	**Terminator (Start / End)**	**Marks the beginning and end of the process.**
	**Process**	**A processing step performed by a user or the system.**
	**Input / Output**	**Data entered into or produced by the system (uploads, forms, notifications).**
	**Decision**	**A point where the flow branches based on a condition or validation.**

DATA DICTIONARY

Table 1. The table below displays the data dictionary of User Information.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each user account.**
username	**varchar(80)**	**Not Null, Unique**	**Unique username used for account identification.**
email	**varchar(120)**	**Not Null, Unique**	**Official email address used for**

			login and email notifications.
password_hash	**varchar(255)**	**Not Null**	**Encrypted hash of the user password.**
first_name	**varchar(50)**	**Not Null**	**Given name of the user.**
last_name	**varchar(50)**	**Not Null**	**Family name of the user.**
middle_name	**varchar(50)**	**Null**	**Middle name of the user (optional).**
phone_number	**varchar(20)**	**Null**	**Mobile number used for SMS notifications.**
gender	**enum(Male, Female)**	**Null**	**Gender of the user used for demographic reports.**
town	**varchar(100)**	**Null**	**Town or municipality of residence.**
province	**varchar(100)**	**Null**	**Province of residence.**
date_of_birth	**date**	**Null**	**Birthday of the student, validated against uploaded documents.**
address	**text**	**Null**	**Complete residential address of the user.**
student_id	**varchar(30)**	**Unique**	**LSPU student identification number (students only).**

course	**varchar(100)**	**Null**	**Degree program the student is enrolled in.**
year_level	**varchar(20)**	**Null**	**Current year level (1st Year to 5th Year, Irregular, Graduate).**
cor_file_path	**varchar(500)**	**Null**	**File path or URL of the Certificate of Registration (COR).**
grades_file_path	**varchar(500)**	**Null**	**File path or URL of the Report of Grades (ROG).**
cor_cloudinary_id	**varchar(255)**	**Null**	**Cloudinary public ID of the COR file for secure retrieval and deletion.**
grades_cloudinary_id	**varchar(255)**	**Null**	**Cloudinary public ID of the ROG file for secure retrieval and deletion.**
grades_semester	**varchar(50)**	**Null**	**Semester the uploaded grades belong to.**
grades_updated_at	**datetime**	**Null**	**Date and time the grades were last uploaded or replaced.**
ocr_verified	**boolean**	**Null**	**Result of OCR validation: True if passed, False if failed, Null if pending.**
ocr_result	**text**	**Null**	**Raw OCR mismatch details**

			when validation fails.
gwa	**float**	**Null**	**General Weighted Average recorded from the uploaded grades.**
units_enrolled	**int**	**Null**	**Total academic units enrolled for the current semester.**
is_existing_scholar	**boolean**	**Not Null, Default: false**	**Flag indicating the student declared being a scholar of another program at registration.**
existing_scholarship_name	**varchar(150)**	**Null**	**Name of the scholarship program the student currently holds.**
campus_id	**int**	**Foreign Key (campuses.id)**	**Campus the user belongs to; Null for university-level accounts.**
user_type	**enum(student, staff, admin, super_admin)**	**Not Null, Default: student**	**System role that determines access permissions (RBAC).**
is_active	**boolean**	**Not Null, Default: true**	**Flag indicating whether the account is currently active.**
is_verified	**boolean**	**Not Null, Default: false**	**Flag indicating whether the account has passed**

			document verification.
password_reset_token	**varchar(100)**	**Unique**	**Token used for password reset requests.**
password_reset_expires	**datetime**	**Null**	**Expiration time of the password reset token.**
phone_verified	**boolean**	**Not Null, Default: false**	**Flag indicating whether the phone number was verified via OTP.**
phone_otp	**varchar(10)**	**Null**	**One-time PIN sent for phone verification.**
phone_otp_expires	**datetime**	**Null**	**Expiration time of the phone verification OTP.**
created_at	**datetime**	**Not Null**	**Date and time the account was created.**
updated_at	**datetime**	**Not Null**	**Date and time the profile was last updated.**
last_login	**datetime**	**Null**	**Date and time of the most recent login.**

Table 2. The table below displays the data dictionary of Campus Information.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each campus.**

name	**varchar(100)**	**Not Null, Unique**	**Official name of the LSPU campus (e.g., Santa Cruz Campus).**
short_name	**varchar(30)**	**Null**	**Abbreviated campus name used in badges and labels.**
is_active	**boolean**	**Default: true**	**Flag indicating whether the campus is currently operating in the system.**
created_at	**datetime**	**Not Null**	**Date and time the campus record was created.**

Table 3. The table below displays the data dictionary of User Campus Assignment.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each assignment record.**
user_id	**int**	**Not Null, Foreign Key (users.id)**	**Staff or admin user assigned to a campus.**
campus_id	**int**	**Not Null, Foreign Key (campuses.id)**	**Campus assigned to the user. Allows one person to manage multiple campuses.**

Table 4. The table below displays the data dictionary of Academic Period.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each academic period.**
academic_year	**varchar(20)**	**Not Null**	**Academic year covered (e.g., 2026-2027).**

semester	**varchar(30)**	**Not Null**	**Semester within the academic year (1st Semester, 2nd Semester, Inter-Semester).**
is_current	**boolean**	**Default: false**	**Flag marking the active period used as the default data context.**
start_date	**date**	**Null**	**Start date of the academic period.**
end_date	**date**	**Null**	**End date of the academic period.**
me_form_deadline	**date**	**Null**	**Deadline for scholars to submit their Monitoring and Evaluation form.**
created_at	**datetime**	**Not Null**	**Date and time the period record was created.**

Table 5. The table below displays the data dictionary of Scholarship Program.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each scholarship program.**
title	**varchar(200)**	**Not Null**	**Official name of the scholarship program.**
category	**varchar(50)**	**Default: General**	**Classification category of the program.**
scholarship_type	**enum(Financial Assistance, Merit Scholarship, Grant, Other)**	**Null**	**Type of scholarship being offered.**

funding_type	**enum(public, private)**	**Not Null, Default: public**	**Funding source. Students may hold multiple private scholarships but only one public scholarship.**
description	**text**	**Not Null**	**Detailed description of the scholarship program.**
eligibility_criteria	**text**	**Not Null**	**Qualification requirements a student must meet to apply.**
benefits	**text**	**Not Null**	**Benefits granted to approved scholars.**
application_deadline	**datetime**	**Null**	**Cut-off date and time for accepting applications.**
award_amount	**decimal(10,2)**	**Null**	**Monetary value awarded per scholar.**
max_applicants	**int**	**Null, Minimum: 0**	**Maximum number of applicants the program can accept.**
current_applicants	**int**	**Default: 0**	**Running count of submitted applications.**
status	**enum(ACTIVE, CLOSED)**	**Not Null, Default: ACTIVE**	**Indicates whether the program is open**

			or closed for applications.
created_by	**int**	**Not Null, Foreign Key (users.id)**	**Staff or admin who created the program.**
is_hidden	**boolean**	**Not Null, Default: false**	**Flag controlling visibility of the program to students.**
allowed_student_ids	**text**	**Null**	**List of specific students allowed to view a private/targeted program.**
custom_form_fields	**json**	**Null**	**Custom application form fields defined by the scholarship office.**
campus_id	**int**	**Foreign Key (campuses.id)**	**Campus the program belongs to; Null for university-wide programs.**
scope	**enum(campus, university)**	**Not Null, Default: campus**	**Coverage scope of the program.**
academic_period_id	**int**	**Foreign Key (academic_periods.id)**	**Period the program is bound to; Null means the program is permanent.**
created_at	**datetime**	**Not Null**	**Date and time the program was created.**

updated_at	**datetime**	**Not Null**	**Date and time the program was last updated.**

Table 6. The table below displays the data dictionary of Scholarship Application.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each application.**
scholarship_id	**int**	**Not Null, Foreign Key (scholarships.id)**	**Scholarship program being applied to.**
applicant_id	**int**	**Not Null, Foreign Key (users.id)**	**Student who submitted the application.**
application_data	**json**	**Null**	**Application form responses, including custom fields.**
documents	**json**	**Null**	**Document references submitted with the application.**

status	**enum(PENDING, UNDER_REVIEW, APPROVED, REJECTED)**	**Not Null, Default: PENDING**	**Current stage of the application in the review workflow.**
scholar_tag	**enum(active, at_risk, probation, inactive)**	**Default: active**	**Compliance tag of the scholar after approval.**
submitted_at	**datetime**	**Not Null**	**Date and time the application was submitted.**
reviewed_at	**datetime**	**Null**	**Date and time the application was reviewed.**
reviewed_by	**int**	**Foreign Key (users.id)**	**Staff or admin who reviewed the application.**
review_notes	**text**	**Null**	**Comments recorded by the reviewer during evaluation.**
final_decision_date	**datetime**	**Null**	**Date and time the final decision**

			was released.
approved_at	**datetime**	**Null**	**Date and time the application was approved.**
batch_id	**int**	**Foreign Key (scholarship_batches.id)**	**Batch the approved scholar was assigned to.**
academic_period_id	**int**	**Foreign Key (academic_periods.id)**	**Academic period the application belongs to.**
amount_received	**decimal(10,2)**	**Null**	**Actual amount released to the scholar.**
seq_number	**int**	**Null**	**Sequence number of the scholar within the batch.**
campus_id	**int**	**Foreign Key (campuses.id)**	**Campus the application is scoped to.**
scholar_status	**varchar(30)**	**Default: active**	**Current status of the scholar**

			in the program.
compliance_status	**varchar(20)**	**Default: compliant**	**Result of the latest monitoring and evaluation check.**
last_compliance_check	**datetime**	**Null**	**Date and time of the last compliance evaluation.**
status_updated_at	**datetime**	**Null**	**Date and time the scholar status was last changed.**
renewal_notified_at	**datetime**	**Null**	**Date and time the scholar was notified about renewal.**
is_active	**boolean**	**Default: true**	**Flag indicating whether the scholarship grant is currently active.**

Table 7. The table below displays the data dictionary of Scholarship Batch.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each batch.**
scholarship_id	**int**	**Not Null, Foreign Key (scholarships.id)**	**Scholarship program the batch belongs to.**
academic_period_id	**int**	**Not Null, Foreign Key (academic_periods.id)**	**Academic period the batch is scoped to.**
batch_name	**varchar(50)**	**Not Null**	**Display name of the batch (e.g., Batch 1, Batch 1.1).**
batch_number	**float**	**Not Null, Default: 1**	**Numeric value used for sorting batches.**
amount_per_scholar	**decimal(10,2)**	**Null**	**Monetary allocation per scholar in the batch.**
total_scholars	**int**	**Default: 0**	**Total number of scholars in the batch.**
total_male	**int**	**Default: 0**	**Number of male scholars in the batch.**
total_female	**int**	**Default: 0**	**Number of female scholars in the batch.**
total_amount	**decimal(12,2)**	**Default: 0**	**Total fund allocation for the batch.**

status	**enum(open, closed, finalized)**	**Default: open**	**Processing state of the batch.**
notes	**text**	**Null**	**Additional remarks about the batch.**
campus_id	**int**	**Foreign Key (campuses.id)**	**Campus the batch belongs to.**
created_by	**int**	**Foreign Key (users.id)**	**Staff or admin who created the batch.**
created_at	**datetime**	**Null**	**Date and time the batch was created.**
updated_at	**datetime**	**Null**	**Date and time the batch was last updated.**

Table 8. The table below displays the data dictionary of Document.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each uploaded document.**
user_id	**int**	**Not Null, Foreign Key (users.id)**	**Student who owns the document.**
application_id	**int**	**Foreign Key (scholarship_applications.id)**	**Application the document is attached to, when applicable.**

original_filename	**varchar(255)**	**Not Null**	**Name of the file as uploaded by the student.**
stored_filename	**varchar(255)**	**Not Null, Unique**	**System-generated filename used on storage.**
file_path	**varchar(500)**	**Not Null**	**Local path or URL where the file is stored.**
file_size	**int**	**Null**	**File size in bytes.**
category	**varchar(50)**	**Null**	**Document category (transcript, good_moral, birth_cert, awards, etc.).**
file_type	**varchar(10)**	**Null**	**File format (pdf, docx, jpg, png).**
description	**varchar(255)**	**Null**	**Short description of the document.**
is_verified	**boolean**	**Not Null, Default: false**	**Flag indicating whether the document passed verification.**

ctc_detected	**boolean**	**Null**	**Flag indicating whether a Certified True Copy marking was detected.**
cloudinary_url	**varchar(500)**	**Null**	**Permanent public URL of the file on Cloudinary.**
cloudinary_public_id	**varchar(255)**	**Null**	**Cloudinary reference used for secure deletion.**
uploaded_at	**datetime**	**Not Null**	**Date and time the file was uploaded.**
updated_at	**datetime**	**Null**	**Date and time the file was last replaced.**

Table 9. The table below displays the data dictionary of GWA History.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each GWA record.**

student_id	**int**	**Not Null, Foreign Key (users.id)**	**Student the GWA record belongs to.**
gwa	**float**	**Not Null**	**General Weighted Average recorded for the semester.**
semester	**varchar(30)**	**Not Null**	**Semester the GWA was taken (e.g., 1st Semester).**
school_year	**varchar(20)**	**Not Null**	**Academic year the GWA was taken (e.g., 2026-2027).**
recorded_by	**int**	**Foreign Key (users.id)**	**Staff or admin who recorded the GWA.**
notes	**varchar(255)**	**Null**	**Remarks about the GWA entry.**
recorded_at	**datetime**	**Not Null**	**Date and time the GWA was recorded.**

Table 10. The table below displays the data dictionary of Monitoring and Evaluation Submission.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each M&E submission.**
user_id	**int**	**Not Null, Foreign Key (users.id)**	**Scholar who submitted the form.**
application_id	**int**	**Not Null, Foreign Key (scholarship_applications.id)**	**Approved application the submission is for.**

academic_period_id	**int**	**Foreign Key (academic_periods.id)**	**Academic period the submission covers.**
original_filename	**varchar(255)**	**Not Null**	**Name of the uploaded M&E form file.**
file_url	**varchar(500)**	**Null**	**URL where the submitted form is stored.**
cloudinary_public_id	**varchar(255)**	**Null**	**Cloudinary reference used for secure deletion.**
local_path	**varchar(500)**	**Null**	**Local storage path of the file.**
file_type	**varchar(10)**	**Null**	**File format (pdf, jpg, jpeg, png).**
file_size	**int**	**Null**	**File size in bytes.**
uploaded_at	**datetime**	**Not Null**	**Date and time the form was submitted.**

Table 11. The table below displays the data dictionary of Notification.

Field	Data Type	Constraints	Description

id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each notification.**
recipient_id	**int**	**Not Null, Foreign Key (users.id)**	**User who receives the notification.**
title	**varchar(200)**	**Not Null**	**Title or subject of the notification.**
message	**text**	**Not Null**	**Body content of the notification or announcement.**
notification_type	**enum**	**Not Null**	**Category of notification (application update, announcement, system alert).**
channel	**enum(IN_APP, EMAIL, SMS)**	**Not Null, Default: IN_APP**	**Delivery channel used to send the notification.**
is_read	**boolean**	**Not Null, Default: false**	**Flag indicating whether the recipient has read the notification.**
sent_at	**datetime**	**Not Null**	**Date and time the notification was sent.**
read_at	**datetime**	**Null**	**Date and time the notification was read.**
related_entity_type	**varchar(50)**	**Null**	**Type of record the notification refers to (scholarship, application).**

related_entity_id	**int**	**Null**	**Primary key of the referenced record.**
academic_period_id	**int**	**Foreign Key (academic_periods.id)**	**Period the announcement was sent in, used for period-scoped filtering.**

Table 12. The table below displays the data dictionary of Audit Log.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each audit entry.**
user_id	**int**	**Foreign Key (users.id)**	**User who performed the action; Null if the account was deleted.**
user_name	**varchar(120)**	**Null**	**Snapshot of the user name that survives account deletion.**
user_role	**varchar(20)**	**Null**	**Role of the user at the time of the action (admin, staff, student, super_admin).**
action	**varchar(80)**	**Not Null**	**Machine-readable action identifier (e.g., application.approve).**
category	**varchar(40)**	**Null**	**Action category (auth, application, scholarship, user, gwa, document, system).**
description	**text**	**Null**	**Human-readable detail of what occurred.**

target_type	**varchar(40)**	**Null**	**Type of record affected (application, scholarship, user).**
target_id	**int**	**Null**	**Primary key of the affected record.**
ip_address	**varchar(45)**	**Null**	**IP address the action originated from.**
timestamp	**datetime**	**Not Null**	**Date and time the action was logged.**

Table 13. The table below displays the data dictionary of AI Chat History.

Field	Data Type	Constraints	Description
id	**int**	**Not Null, Primary Key, Auto-Increment**	**Unique identifier for each chat message.**
user_id	**int**	**Foreign Key (users.id)**	**User who owns the conversation; Null for guest sessions.**
role	**enum(user, model)**	**Not Null**	**Sender of the message: user question or AI response.**
content	**text**	**Not Null**	**Content of the chat message.**
created_at	**datetime**	**Not Null**	**Date and time the message was sent.**

PROJECT SCHEDULE

This section displays the research study's project schedule. In order to complete this research project, the Researcherss went through several stages.

Task	Owner	Start	End	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	% Progress

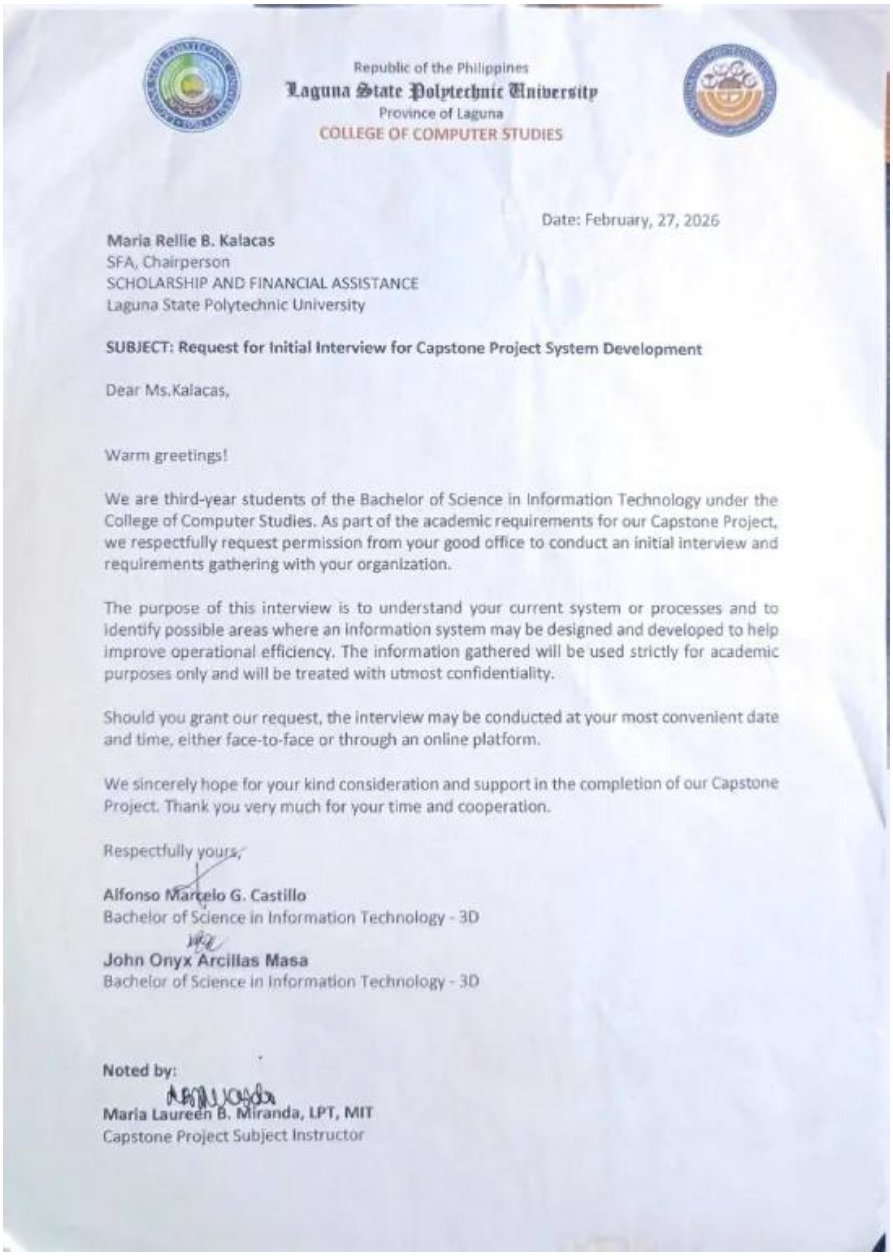
Planning	Castillo/ Masa	Mar 1	Mar 31	■									80%
Requirements	Castillo/ Masa	Apr 1	May 20		■	■							70%
Design	Castillo/ Masa	May 21	Jul 15			■	■	■					60%
Development	Masa	Jul 16	Oct 20					■	■	■	■		50%
Testing	Castillo/ Masa	Oct 21	Nov 15								■	■	30%
Deployment	Castillo/ Masa	Nov 16	Nov 30									■	10%

-

APPENDICES B

(COMMUNICATION LETTERS AND FORMS)

REQUEST LETTER



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: February, 27, 2026

Maria Rellie B. Kalacas
SFA, Chairperson
SCHOLARSHIP AND FINANCIAL ASSISTANCE
Laguna State Polytechnic University

SUBJECT: Request for Initial Interview for Capstone Project System Development

Dear Ms.Kalacas,

Warm greetings!

We are third-year students of the Bachelor of Science in Information Technology under the College of Computer Studies. As part of the academic requirements for our Capstone Project, we respectfully request permission from your good office to conduct an initial interview and requirements gathering with your organization.

The purpose of this interview is to understand your current system or processes and to identify possible areas where an information system may be designed and developed to help improve operational efficiency. The information gathered will be used strictly for academic purposes only and will be treated with utmost confidentiality.

Should you grant our request, the interview may be conducted at your most convenient date and time, either face-to-face or through an online platform.

We sincerely hope for your kind consideration and support in the completion of our Capstone Project. Thank you very much for your time and cooperation.

Respectfully yours,

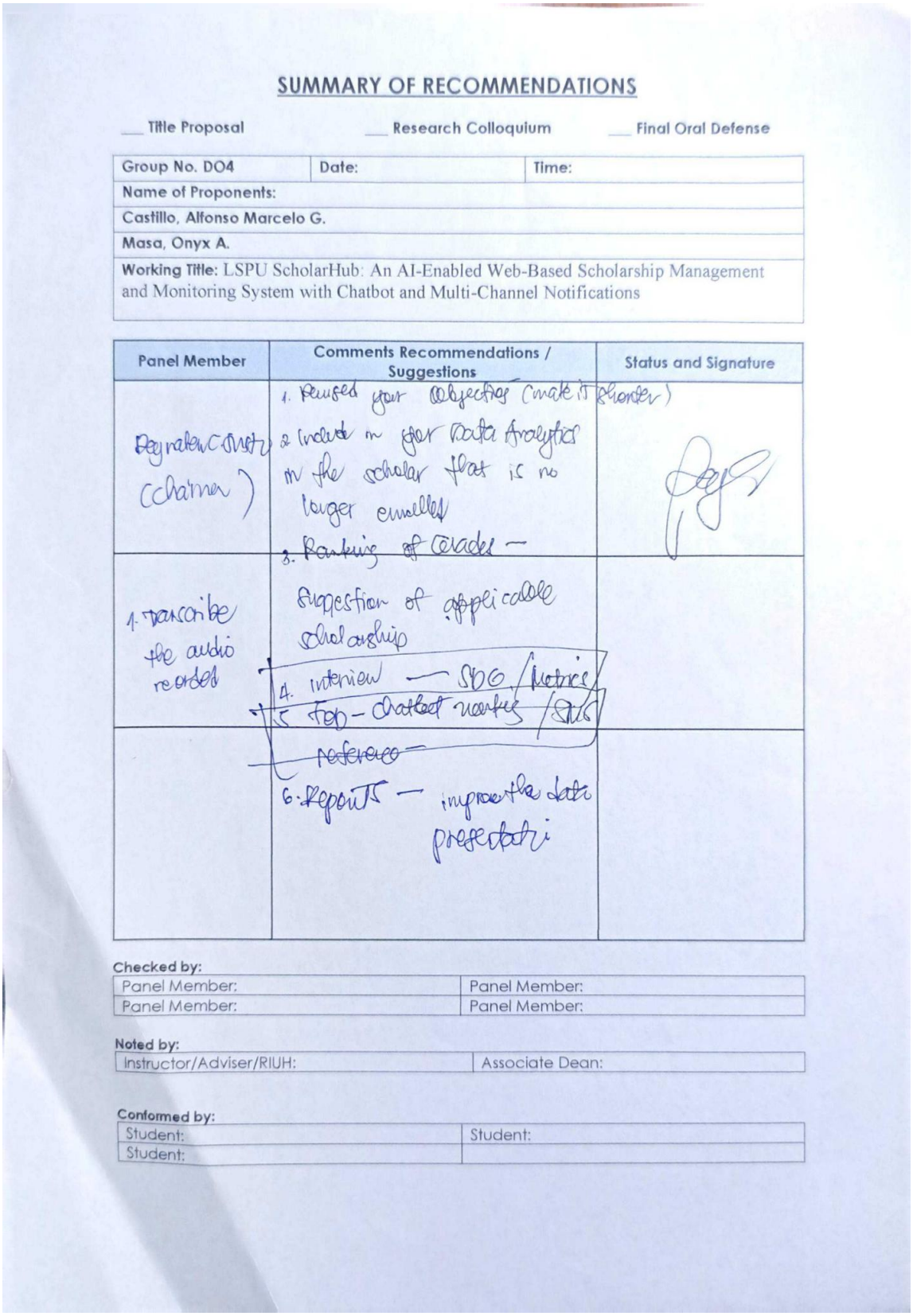
Alfonso Marcelo G. Castillo
Bachelor of Science in Information Technology - 3D

John Onyx Arcillas Masa
Bachelor of Science in Information Technology - 3D

Noted by:

Maria Laureen B. Miranda, LPT, MIT
Capstone Project Subject Instructor

SUMMARY OF RECOMMENDATIONS



SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. DO4	Date:	Time:
Name of Proponents:		
Castillo, Alfonso Marcelo G.		
Masa, Onyx A.		
Working Title: LSPU ScholarHub: An AI-Enabled Web-Based Scholarship Management and Monitoring System with Chatbot and Multi-Channel Notifications		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
Reynaldo Castro (Chairman)	1. Revised your objectives (make it shorter) 2. Include in your Data Analytics in the scholar that is no longer enrolled 3. Ranking of Grades —	
1. Describe the audio recorded	suggestion of applicable scholarship 4. Interview — SDO / Instructor 5 Top - chatbot queries / SDO reference — 6. Reports — improve the data presentation	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

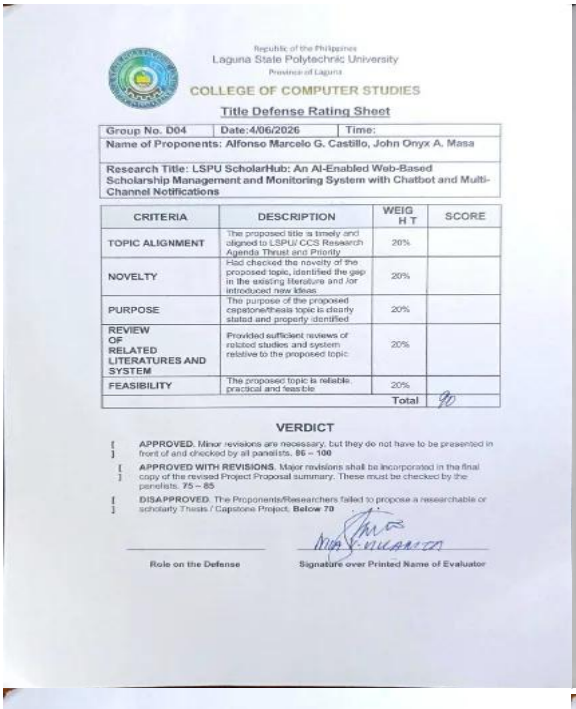
Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

RATING SHEETS



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. D04	Date:4/06/2026	Time:

Name of Proponents: Alfonso Marcelo G. Castillo, John Onyx A. Masa

Research Title: LSPU ScholarHub: An AI-Enabled Web-Based Scholarship Management and Monitoring System with Chatbot and Multi-Channel Notifications

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	90

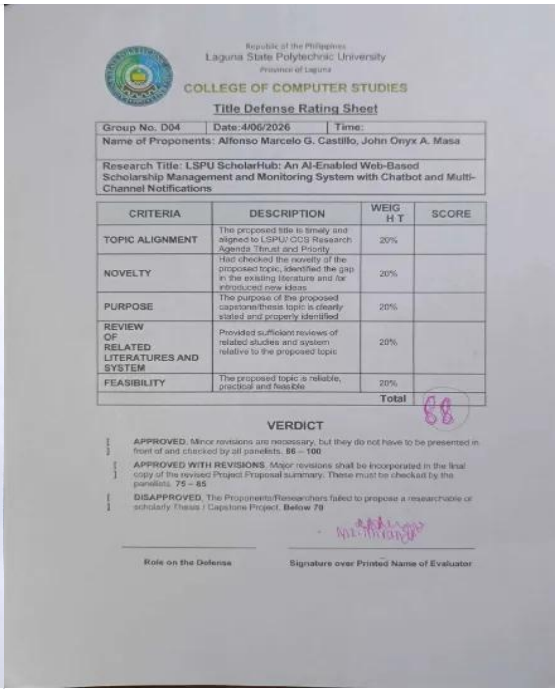
VERDICT

[ ] **APPROVED**. Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS**. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED**. The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Role on the Defense | Signature over Printed Name of Evaluator



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. D04	Date:4/06/2026	Time:

Name of Proponents: Alfonso Marcelo G. Castillo, John Onyx A. Masa

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PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	88

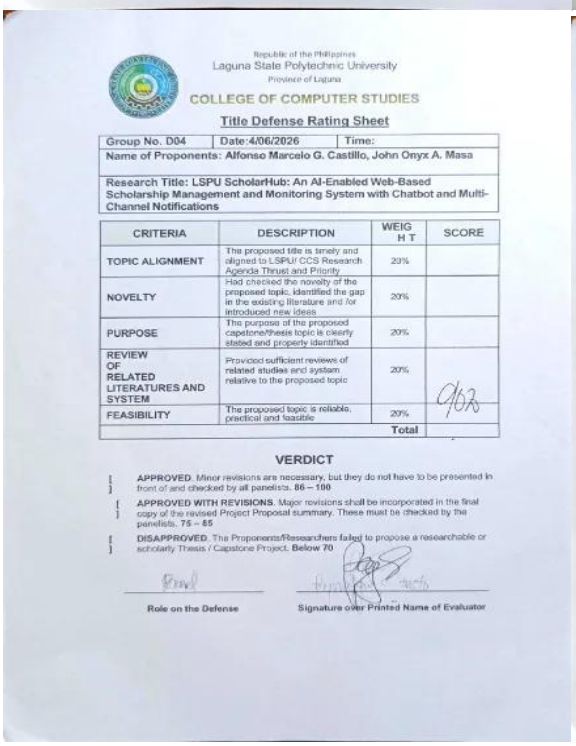
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Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

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REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	

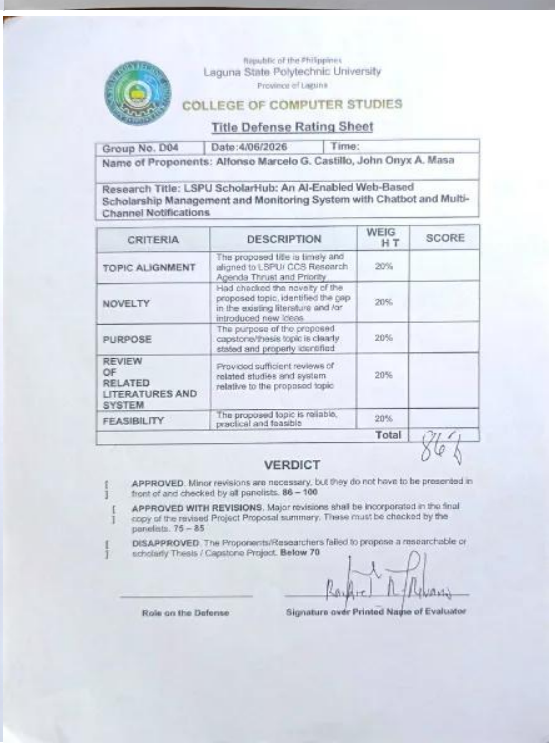
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Role on the Defense | Signature over Printed Name of Evaluator



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

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PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	86

VERDICT

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[ ] **APPROVED WITH REVISIONS**. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED**. The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Role on the Defense | Signature over Printed Name of Evaluator

ISO FORMS

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS ADVISER NOMINATION

Student/s Name and Signature : Castillo Alfonso Mariah G.

: Masa Jodon Onyx A.

:

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) :

I/we, hereby nominate as Thesis Adviser

Name of Faculty : Mark P. Bernardino, MSCS

College : College of Computer Studies

Field of Specialization :

I, hereby indicate my willingness to serve as the Thesis Adviser.

Signature of Faculty : _____ Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Balatbat, PhD, FFP
R&D Director/Chairperson

LSPU-RDO-SF-011 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Castillo, Alfonso Mercado G.

: Nasa, John Onyx A.

:

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) :

I/we, hereby nominate as Panel Member (System Expert)

Name of Faculty : Mia V. Villarica

College : College of Computer Studies

Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Castrence, PhD, FFP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Castillo Alfonso Muriel G.

: Musa John Onyx A.

: _____

: _____

Degree Sought : Bachelor of Science in Information and Technology

Major Field (if applicable) : _____

I/we, hereby nominate as Panel Member (Technical Editor)

Name of Faculty : Rachiel R. Rivano

College : College of Computer Studies

Field of Specialization : _____

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : _____ Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia U. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rogurio G. Catapang, PhD, FFP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Castillo A. Fonda Marich G.

: Masa John Onyx A.

:

:

Degree Sought : Bachelor of Science in Information and Technology

Major Field (if applicable) :

I/we, hereby nominate as Panel Member (Statistician)

Name of Faculty : Edward Flores, MSIT

College : College of Computer Studies

Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

Mark D. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Ma. V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD, FRP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Castillo Alfonso Marco G.

: Mercado John Omar L.

: _____

: _____

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) : _____

I/we, hereby nominate as Panel Member (Language Critic)

Name of Faculty : Jhon Jhon Zotomayor

College _____

Field of Specialization : _____

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : _____ Date Signed

Endorsed By:

Marie P. Bernardino, MSES
Research Coordinator

Recommending Approval:

Julia V. Villorica
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rogerio G. Catapang, PhD, FFP
R&D Director/Chairperson

APPENDICES C

(CURRICULUM VITAE OF SYSTEM TESTERS)

Mark P. Relova

745 Red Alder Pl. Escondido, CA 92027 | +1-310-698-9951 | mark@revmont.net

Overview

I am looking for a challenging technical position with leadership responsibilities in Model-based Systems Engineering with a focus on defense acquisition programs. I am a graduate of the prestigious Systems Engineering program at Naval Postgraduate School and the Naval Acquisition Development Program sponsored by ASN-RDA. I previously held a GS-12 Navy civilian billet. I am a DAWIA-certified Systems Engineering Professional with years of experience in US Navy and Air Force aircraft, weapons and IT system programs. I am a US citizen with an active Secret security clearance. I am open to remote/telework opportunities or on-site projects within the San Diego metropolitan area.

Education

MASTER OF SCIENCE | MAR 2013 | NAVAL POSTGRADUATE SCHOOL

- Major: Systems Engineering
- Concentration: Combat Systems Integration
- Certifications: DAWIA (Level III - Engineering, Level I - IT), Certificate in Systems Engineering

NAVAL ACQUISITION DEVELOPMENT PROGRAM | US NAVY DEFENSE ACQUISITION CAREER MANAGEMENT

- Billet: Computer Scientist (OPM Career Series 1550) – initial appointment at GS-07, promoted to GS-12 at program completion

BACHELOR OF SCIENCE | JUN 2010 | CALIFORNIA STATE UNIVERSITY – LONG BEACH

- Major: Computer Science
- Concentration: Distributed Computing and Network Programming

Skills & Abilities

SYSTEMS ENGINEERING

- DODAF, SysML,, UML, Model-Based Systems Engineering, Reliability Analysis, Functional Decomposition, Functional Allocation, Requirements Analysis, Verification and Validation, Systems Architecting, DoD Acquisition Lifecycle Framework, Analysis of Alternatives, Systems Engineering Technical Reviews
- Tools: Vitech Core, No Magic MagicDraw and Cameo, Sparx Enterprise Architect, Rational DOORS

TEST & EVALUATION

- Developmental Testing, Unit and Functional Testing, Load and Stress Testing, Security and Vulnerability Testing
- Tools: Spirent Avalanche, HP LoadRunner, Visual Studio Ultimate, IDT ATRT, Nessus, Retina Network Security Scanner

SOFTWARE ENGINEERING

- Software development lifecycle models, software engineering patterns
- Programming: C++, C, C#, Java, TCL, VBA, SQL, UML, shell scripting (Bash and Powershell), Python, PHP, XML, HTML
- Operating Systems: Linux, Microsoft Windows, *BSD, Mac OS
- Databases : Oracle, MySQL, MS-SQL, MS Access, Apache Derby and Cassandra
- Development Environments : Visual Studio .NET, Eclipse, NetBeans, LabView, Keil uVision, SSMS, MySQL Workbench

PROJECT MANAGEMENT

- Requirements Management, Configuration Management, Risk Management, Earned Value Management
- Tools: Rational DOORS, Deltek Open Plan, MS Office, MS Visio, MS Project

Experience

SCIENTIST, SYSTEMS ENGINEERING | L3HARRIS TECHNOLOGIES | JAN 2021 – PRESENT

- Currently serving as an MBSE Subject Matter Expert and Consultant for multiple programs across the Space and Airborne Systems segment. Lead for the MBSE Deployment Squad, which assists programs in starting their MBSE workflow through tools, processes, training and SME consulting.
- Lead architect of a Product Line Architecture (PLA) model for a family of multi-function avionics systems for military and commercial aircraft (using Cameo Systems Modeler and MBPLE Plugin), integration of Open Architecture standards (OCS, OMS, FACE, HOST, SOSA) into PLA model, and prototyping of a simulation toolchain and workflow for evaluating SysML models.
- Secondary responsibilities include MBSE support on proposal efforts, participation in the corporate Digital Engineering Working Group, and evaluation of MBSE tools and best practices on behalf of the L3Harris SAS segment.

SENIOR PRINCIPAL ENGINEER | SCIENCE APPLICATIONS INTERNATIONAL CORP | APR 2020 – JAN 2021

- Served as the Model-based Systems Engineering (MBSE) Lead for the Global Positioning System program.
- Primary responsibilities included overseeing the transition to a Digital Engineering (DE) ecosystem (tools, processes, skills, personnel, infrastructure), developing MBSE enablers (custom profiles/stereotypes, concept models, ontologies, schemas), establishing a "DE In A Box" ecosystem prototype for use in multiple SAIC programs, developing DE implementation roadmaps for senior leadership reviews, and providing expert advice regarding DE/MBSE implementation on a legacy DOD program.
- Secondary responsibilities include providing mentorship to junior engineers and peers in the application of MBSE to supplement traditional systems engineering and integration activities, training colleagues in architecture modeling using various MBSE tools, and developing or updating SE processes and procedures to align with ISO/IEC/IEEE 15288.

SENIOR PRINCIPAL ENGINEER | NORTHROP GRUMMAN CORPORATION | MAY 2019 – APR 2020

- Served as a Systems Engineer and Architect for the US Air Force Ground Based Strategic Deterrent (GBSD) program. Primary responsibilities include development of architecture models using UPDM and SysML, functional and requirements analysis, and support of key technical reviews and program events. Served as the system architecture POC for a critical GBSD subsystem.
- Provided mentorship to junior engineers and peers in the application of MBSE and modeling tools on a new DOD program.

STAFF ENGINEER | GENERAL ATOMICS - AERONAUTICAL SYSTEMS INC | JAN 2018 – APR 2019

- Served as a Systems Engineer for the US Air Force MQ-9 Reaper program. Implemented MBSE on several key Technology Maturation Efforts (TMEs) for additional capabilities being added to the MQ-9, as well as developing baseline models for sustaining MQ-9 Operational Flight Programs (OFPs). Led efforts to perform Model-Based Requirements Development (MBRD) to unite system and subsystem models (structural, functional, behavioral, and conceptual) with requirements baseline from legacy MQ-9 platform and delta requirements from TMEs. Developed standards for the SE team for implementing MBRD in all GA-ASI projects. Participated in planning and bid estimation for future MQ-9 TMEs.

SENIOR PERFORMANCE ENGINEER | FORCEPOINT LLC (FORMERLY WEBSENSE) | DEC 2015 – MAR 2017

- Served as a senior Performance Engineer in an Agile DevOps team. Responsible for the development of an 18-rack (42U) test environment for testing all Forcepoint products. Lead engineer for testing the TRITON AP-EMAIL and AP-WEB security products. Developed test plans to establish the performance baseline of products. Developed testing frameworks and tools/scripts using TCL, Python, C# and Bash. Analyzed test results, troubleshoot performance issues encountered during testing, and worked with the development team to debug issues with code. Developed the test bed architecture and procedures for testing Forcepoint virtual appliances in the AWS and Azure cloud environments.

COMPUTER SCIENTIST (GS-12) | UNITED STATES NAVY | AUG 2010 – NOV 2015

I served in the following roles as a civilian employee of the Department of Defense. Duty stations include NAWC-WD (China Lake, CA), NAVWAR HQ (San Diego, CA), NIWC – Pacific (Point Loma, CA), and DASN RDT&E CHENG (Navy Yard, Washington DC)

LEAD RELIABILITY ENGINEER | NAVAL INFORMATION WARFARE SYSTEMS COMMAND

- Primary point of contact for the Reliability and Maintainability Engineering (R&ME) workforce in NAVWAR. Reviewed engineering proposals, system design documentation and acquisition plans for several multi-million dollar Navy programs. Provided technical assessments of system designs to key stakeholders, and developed organization-level R&M policies for workforce development, technical processes and best practices.

- As Systems Engineer for SSC-Pacific, performed analysis of alternatives to address obsolescence in ship communications equipment. Supported design reviews and technical interchange meetings between government and contractor teams. Developed a Requirements Change Request database and frontend using MS Access.

LEAD SYSTEMS ENGINEER | NAVAL AIR WARFARE CENTER – WEAPONS DIVISION

- Served as the lead systems engineer for the T-45 Goshawk sustainment program, responsible for all avionics system specifications. Developed Test Work Descriptions and procedures for verifying and validating the Almanac Loading System. Led integration testing to validate the Digital Data Set avionics subsystem. Supported the T-45 Project Engineer in developing SEP, TEMP, PMP, IMP/IMS, SSS, SRS, ICDs, PWS and SOW for the T-45 OFP Block H6.0. Reviewed program documentation and contractor deliverables (CDRLs). Supported the development of a T-45 Systems Integration Laboratory with full avionics simulation capabilities. Configured and maintained the T-45 IPT SharePoint site.
- Performed integration testing to validate Mission Data File upgrades to ALE-47 countermeasures dispenser on AV-8B Harrier II (joint US/ES/IT requirements).

SYSTEMS ENGINEER | NAVAL AIR WARFARE CENTER – WEAPONS DIVISION

- Supervised the migration of over 30 web applications and associated databases to a load-balanced, highly-available server environment. Developed the architecture for a scalable, load-balanced hosting environment with multiple application VMs and 3 F5 BigIP load balancers. Provided oversight to 2 software developers, 1 test specialist and 1 CRM professional.
- Developed strategies for software development lifecycle (SDLC) process improvements, metrics to track progress and change management process to control changes to product baselines in accordance with CMMI-SW guidelines. Developed IDEF-0 diagrams and DODAF views (OV-*, SV-*, TV-*) for enterprise architecture models of NAWC-WD IT systems.
- Managed and designed a full performance & load test on the NAWC-WD application hosting environment, using HP LoadRunner. Supervised the execution of performance test cases and performed analysis on test data to validate the hosting environment against government requirements. Provided oversight to 3 test specialists and 1 LoadRunner consultant.
- Developed and proposed a notional architecture for a distributed video streaming service to allow for live broadcasting of events to more than 10K users. Performed an in-depth root cause analysis of issues with Adobe Flash video hosting. Developed a T-SQL script (400+ SLOC) that automated hardening and vulnerability mitigation of MS-SQL servers. Developed a modification (800+ SLOC) to a government-developed, open-source IA tool (written in C#) to allow parsing of Retina XML and Nessus output files for automated creation of system vulnerability reports. Performed system hardening using DISA GoldDisk and system vulnerability scanning using eEye Retina and Tenable Security Nessus scanners.
- Reported to the US Navy Director for R&ME at DASN RDT&E for a 5-month rotation to support R&ME transformation

AIRPORT OPERATIONS SUPERVISOR | MALAYSIA AIRLINES | AUG 1999 – MAY 2008

- Supervised over 20 ground handling staff, working under 3 different contracts, for daily Boeing 747 flight operations.



AERON RED LOPE

A highly organized and motivated individual seeking an opportunity to grow professionally and contribute effectively in the IT industry.

EDUCATION

LAGUNA STATE POLY TECHNIC UNIVERSITY
Bachelor of Science in Information Technology
2024-2025

SKILLS

Web Development Skills:

- HTML/CSS
- PHP
- MySQL
- Python

Basic Configurations:

- Proficient in basic configuration of radio systems and Wireless Access Points (WAP).
- Skilled in configuring Cisco routers and Huawei switches.

Network Troubleshooting:

- Developed troubleshooting skills for network and hardware issues.
- Experienced in identifying and resolving technical problems to enhance operational efficiency.

CONTACT

09613663699
aeronredlope04@gmail.com
Lynville Phase 3, Brgy. Callios, Sta. Cruz Laguna

SUMMARY

- Possesses versatile skills in project management, problem-solving, and collaboration. Brings fresh perspective and strong commitment to quality and success. Recognized for adaptability and proactive approach in delivering effective solutions

WORK EXPERIENCE

STARTED WORKING AT TOTAL INFORMATION MANAGEMENT CORP. ***August – 2025***

- Monitor and analyze network performance, availability, and security events.
- Monitor and manage network infrastructure using SolarWinds NPM to ensure optimal uptime and performance.
- Utilize ADVA network management systems for real-time monitoring of optical transport and Ethernet service layers.
- Analyze alerts, investigate root causes, and resolve connectivity and latency issues across multiple sites.

IT INTERN MAKATI DEVELOPMENT CORPORATION NORTH OPCEN, PHILIPPINES ***February – June 2025***

- Provided technical support by troubleshooting hardware and software issues for both office and field staff.
- Assisted with network setup, system maintenance, and IT equipment installation
- Helped monitor system performance and ensured stable network connectivity for daily operations.
- Participated in IT asset management and documented standard operating procedures.
- Supported the implementation of basic cybersecurity protocols and performed routine data backups.

REFERENCES

ARNEL T. ECOT
IT TEAM LEAD
MAKATI DEVELOPMENT CORP.

PAUL AERROL T. AZARCON
INFORMATION SECURITY OFFICER
TOTAL INFORMATION MANAGEMENT CORP.

JEFFERSON VEGA
SERVICE DELIVERY MANAGER
FIS GLOBAL SOLUTIONS PHILIPPINES



PROFILE

My purpose in life is to deliver value and innovation to my fellowmen. I constantly strive to perfect my craft and pursuit long term goals that would uplift the lives of the people around me. And I believe that technology should be an ally to improve and not replace the lives of people.

PERSONAL INFORMATION

Age	: 26
Birth date	: August 22, 1997
Height	: 5'6 ft'
Weight	: 71 kls.
Citizenship	: Filipino
Civil Status	: Single
Religion	: Born Again

EDUCATIONAL BACKGROUND

TERTIARY:

Laguna State Polytechnic University
Main Campus, Santa Cruz, Laguna
Bachelor of Science in Information Technology
(2014-2018)

SECONDARY:

Pedro Guevara Memorial National High School (PGMNHS)
Santa Cruz, Laguna
(2010-2014)

ELEMENTARY:

Gatid Elementary School
Santa Cruz, Laguna
(2004-2010)

Birador, John Rosh B.

173 Molera Street, Brgy. Gatid
Santa Cruz, Laguna
Contact No.: 09269531425
Email Add: johnroshbirador@gmail.com

CAREER OBJECTIVE

Seeking for a challenging work where I can contribute my skills and knowledge in a renowned organization in the field of Information Technology that provides opportunities for professional growth.

SPECIAL SKILLS

- AWS Cloud Computing
- MS Office efficient
- Vb.Net/C# Development
- Android Development/Java
- PHP/ Web Development
- Multimedia Applications (Photography, Editing etc.)
- Computer Hardware/Software Troubleshooting
- Computer Networking
- Network Troubleshooting
- CCTV Installation/Troubleshooting
- VAPT Assessment / Risk Assessment
- Intrusion Detection, Digital Forensics, Pen-Testing
- Data Privacy Policy / Security Auditing

SEMINARS AND TRAININGS ATTENDED

- Youth for Information Technology "Imagine, Innovate, Inspire" at SMX Convention Center, Pasay City **(September 10, 2014)**
- Youth for Information Technology "Harnessing the Power of Me" at SMX Convention Center, Pasay City **(September 13, 2016)**
- The International 4th Educational Circle Engineering and Computer Tech. Education at Centro Escolar University, Manila **(March 11, 2017)**
- AWS Cloud practitioner and Technical Essentials Training **(August 2021)**
- Comptia Sec + Training **(March 2023)**

WORK EXPERIENCE

- Part time waiter / boy at Ted's Kitchen Santa Cruz Laguna **(August 2016 – December 2016)**
- Part time Junior Software Developer at Rhaiga Solutions Innovations, Quezon City **(December 2016 – May 2017)**
- Student Assistant / It Support at College of Computer Studies LSPU SCC **(June 2015 – May 2017)**
- Software Developer / IT Security Officer at The New Apec Development Corporation **(2018 – Current)**

REFERENCES

- Mia Villar Villarica, DIT

  Instructor, College of Computer Studies LSPU SCC

  Contact number: +639175039034

- Rose Ann Caparas Malaborbor, MSCS

  Instructor / DPO, NU Laguna

  Contact number: +639171529272

- Kimverly Urbano

  Technical Facility Administrator, ABS-CBN/PRIM

  Contact number: +639173385757

I hereby certify that the above information is true and correct to the best of my knowledge and belief.



John Rosh B. Birador
Signature

HARVEY ROMERO

Graphic Designer

+63 945 562 5327 | romeroharvey2001@gmail.com | Pila Laguna | behance.net/harveyromero

EXPERIENCE

Graphic Designer — **2024 - 2025**
IMPECCABLE SHOES — Pila Laguna
- Developed complete brand identity from concept to execution
- Designed social media content and short-form videos for digital marketing
- Collaborated with multiple branches to deliver localized marketing assets

Freelance Graphic Designer — **2022 - 2025**
Independent — Pila Laguna
- Delivered 300+ design projects for 100+ clients across Fiverr and independent channels worldwide
- Built long-term client relationships, converting platform clients into repeat direct clients
- Managed full project lifecycle including client communication, revisions, and final delivery

Founder & Creative Director — **2024 - Present**
neverlonely™ — Pila, Laguna
- Developed and managed an independent streetwear brand from concept to execution
- Designed apparel graphics, branding, and visual identity
- Managed production processes including printing and product development
- Executed digital marketing strategies including Meta ads and social media

Product Designer / Brand Designer — **2023 - 2024**
BABS App (Book a Barber / Stylist) — Pila, Laguna
- Developed brand identity including logo and visual system
- Designed UI mockups to visualize the mobile booking experience
- Produced branded materials such as uniforms, lanyards, signage, and print collateral

Founder / Creative Lead — **2023 - 2025**
Astro Studios — Pila, Laguna
- Managed a small design team to handle high-volume client projects
- Delegated tasks and ensured consistent quality across deliverables
- Coordinated project timelines, revisions, and client communication
- Scaled freelance operations to support increasing client demand

EDUCATION

Laguna State Polytechnic University — **2020 - 2025**
Bachelor of Science in Information Technology — Santa Cruz, Laguna

SKILLS

Technical: Adobe Creative Suite (Photoshop, Illustrator, After Effects, Premiere Pro, Lightroom)
DaVinci Resolve, Canva, Capcut, Microsoft Office, Powerpoint, Google Workspace, Pages

Creative: Branding & Visual Identity Design, Creative Direction & Art Direction, Social Media Content Design, Short-Form Video Production,Marketing & Campaign Visuals, Apparel / Streetwear Graphic Design, UI Mockups & Product Visualization, Print & Promotional Design, Photoshoot Direction & Creative Planning, Digital Advertising (Meta Ads), Client Communication & Project Management

APPENDICES D

(D. CURRICULUM VITAE OF RESEARCHERS)



Alfonso Marcelo G. Castillo

0935-425-7300

alfonsomarcelocastillo@gmail.com

0090 Sitio Pitong Gatang, Baranggay Aplaya, Pila, Laguna, 4010

TRAININGS & SEMINARS

2023

Career Guidance and Employment Coaching

CPESO (Public Employment Service Office)

Laguna State Polytechnic University - Santa Cruz Main Campus
Activity Center

2023

Employability Skills Training Program

Worldwide Link

Laguna State Polytechnic University - Santa Cruz Main Campus
CTE Function Hall

2023

Work Immersion

Municipality of Pila

Brgy. Bulilan Sur, Pila Laguna
Office of the Vice Mayor

REFERENCES

Rio Joie L. Leongson
Faculty, Laguna State Polytechnic University - SCC
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Alona D. Malvar
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Phone: 0917-190-0191
Email : alonadizon.malvar@gmail.com

A-janlique C. Baltazar
Faculty, Laguna Senior High School
Phone: 0949-330-8163
Email : ajanlique.baltazar@lspu.edu.ph

CAREER OBJECTIVES

Driven by pursuit of excellence, I am a goal-oriented individual eager to use my diverse skill set to contribute and support to a company's growth. With commitment to achieving goals effectively, I aim to use my passion to exceed expectations at every opportunity.

EDUCATION

2023 - Present

Laguna State Polytechnic University
Undergraduate
Bachelor of Science in Information Technology

2021 - 2023

Laguna State Polytechnic University
Undergraduate
Information and Communications Technology

SKILLS

- Customer Service
- Communication
- Multitasking
- Leadership
- Initiative

LANGUAGE

Tagalog

English

JOHN ONYX A. MASA

Bachelor of Science in Information Technology

Bay, Laguna, Philippines | 0976 370 9541 | masaonyx@gmail.com

PROFILE

A 4th-year Bachelor of Science in Information Technology student at Laguna State Polytechnic University – Santa Cruz Campus, and the developer of LSPU ScholarHub, a web-based scholarship management system. Brings hands-on experience in customer support, online selling, and financial transaction management through freelance work, with strong written communication, computer literacy, and the ability to work independently through online platforms.

EDUCATION

Laguna State Polytechnic University – Santa Cruz Campus | 2023 – Present (Expected Graduation: 2027)
Bachelor of Science in Information Technology — 4th Year

CAPSTONE PROJECT

LSPU ScholarHub — Web-Based Scholarship Management System | 2026
Developer / Researcher

- Designed and developed a full-stack scholarship management system for the LSPU Scholarship Office, covering student registration with document verification, scholarship application and review, batching, scholar monitoring, announcements, and university-wide reporting.
- Built with Python (Flask), MySQL, and JavaScript, deployed on Railway with Google Gemini AI integration for document data extraction and a scholarship assistant chatbot.

EXPERIENCE

Freelance Customer Service & Account Support | 2021 – Present
Independent Freelancer — Online

- Supported clients by clarifying requests, providing written updates, and coordinating service completion.
- Managed multiple client requests, prioritized tasks, and monitored service details to complete assignments accurately and on time.
- Processed online payments and maintained transaction details while communicating professionally through Facebook, Discord, and other online platforms.

Online Sales & Buy-and-Sell | 2023 – 2025
Independent Online Seller — Online

- Managed end-to-end online transactions: answering inquiries, confirming orders, negotiating prices, and coordinating buyers.

- Processed payments through GCash, e-wallets, and online banking while maintaining accurate transaction records.
- Calculated costs, selling prices, and potential profit to make informed purchasing and pricing decisions.

Household Financial & Transaction Management | 2021 – 2023
Family Financial Responsibility

- Managed and monitored household funds, expenses, and priorities with accuracy.
- Handled cash, GCash, e-wallet, and banking transactions while maintaining spending records, demonstrating trustworthiness and accountability.

SKILLS

Web Development (Flask, HTML/CSS, JavaScript) • Database Management (MySQL) • Customer Service • Written Communication • Problem Solving • Payment & Transaction Processing • Data Entry & Record Keeping • Google Workspace • Microsoft Office • Basic Hardware/Software Troubleshooting • Remote Work • Adaptability

LEADERSHIP & ACHIEVEMENTS

- 3rd Year Representative, Information Technology Unity Hub — Served as a communication link between students and organization leadership; assisted with announcements, activities, and member coordination.
- Grade 12 Representative, United Math Guild — Supported student communication and organization activities while working with fellow student leaders.
- Student Leadership & Event Coordination — Assisted in school events and projects, responded to member concerns, and coordinated tasks while balancing academic responsibilities.